INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.

A Capstone Project
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ABSTRACT

In this study, the development and evaluation of the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., in San Pedro, Laguna. The project resolved issues like slow stock monitoring, delayed reporting etc. related to business. manual maintenance scheduling. Agile Scrum Methodology is used, scanners for staff, and an inventory system. The researchers created a web platform for admins, a mobile app with barcode/QR scanners for staff, and an inventory system. Staff will be scanned by code, and a Retrieval-Augmented Generation (RAG) AI will also be utilized. Assistant which generates text responses and generates PDF or Excel reports. The Ten (10) IT experts reviewed the system in a system testing. Accumulated 100% passing rate of and approved the BlackBox test cases showing that the system is suitable for further testing. The researchers then used a Purposive Sampling design. The number of users was confirmed using actual company records, a final sample size of 144 was the resulting respondents. This number represents 3 administrators, 16 staff and 125 customers chosen through systematically random sampling. The user of the questionnaire was a combination of the Acceptance Evaluation Questionnaire and Technology Acceptance Quality metrics used in Model (TAM) and ISO/IEC 25010. The statistical analysis made the following outstanding results in the system, where an overall average weighted mean was obtained of 4.86 and interpreted as Highly Acceptable. It also got a very low standard deviation of 0.35, which proves that the answers of the respondents were highly consistent and shared positive assessment feedback from the intended users. In conclusion, these findings prove that the developed system successfully met all technical requirements and is completely ready for real-world deployment.

Keywords: Inventory Management; Vehicle Maintenance; Decision Support System; Agile Methodology; QR Code Technology; Retrieval-Augmented Generation; ISO/IEC 25010 Software Quality Metrics.

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DEFINITION OF TERMS

To ensure clarity and consistency in the use of language throughout this study, it is important to define key terms and concepts that may have different interpretations or meanings. This section provides a comprehensive list of technical and operational terminologies used in this research, along with their corresponding definitions and explanations.

Technical Terms

These are terms related to the technologies, tools, and frameworks used in the development of the system.

CSS	It is used to format a webpage's layout, and you may modify the color, font, size and space between parts.
Firebase Authentication	A security framework used to manage user sign-ins and identity verification, ensuring that only authorized students can access the mobile app.
Flutter SDK	An open-source UI software development kit used to create the mobile application's interface and logic for both Android and iOS devices.
HTML	It defines the structure of a Web page and is the standard markup language for building Web pages.
Laptop/Mobile	It is an electronic device used for storing and processing data.
Visual Studio Code	The integrated development environment (IDE) used by the researchers for writing, debugging, and maintaining the source code of the entire system.

Operational Terms

These terms are defined based on how they are used and applied within the context of the system.

Agile Methodology	The iterative development process used to design, test, and refine the hardware and software components of the system.
Barcode Technology	The hardware and software features are used by employees to scan products and instantly monitor spare parts inventory levels.
Decision Support System (DSS)	The system component that generates low-stock alerts and preventive maintenance schedule notifications.
Descriptive Statistics	The mathematical method used to interpret the numerical evaluation data gathered from the system's stakeholders.
Integrated System	The web and mobile-based platform developed to unify vehicle services and inventory monitoring.
Inventory Management	The digital process of spare parts using QR codes and barcode technology.
ISO/IEC 25010 Standard	The software evaluation baseline used to construct the survey questions regarding the system's operational quality, speed and utility.
Likert Scale	The 5-point grading table was used by respondents to rate their agreement on the system's overall performance.
Preventive Maintenance Scheduling (PMS)	The feature that tracks vehicle history and alerts users of upcoming maintenance dates.
QR Code Technology	The digital code system is scanned by mobile devices to display accurate, real-time vehicle service history.
Retrieval-Augmented Generation (RAG)	The backend data-retrieval process that feeds real-time service histories and inventory metrics into the AI chatbot to ensure accurate responses.

CHAPTER I

INTRODUCTION AND BACKGROUND

In today's industry, vehicle maintenance and inventory matter a lot, for operational efficiency and reliability. Integrated solutions can be used to manage fleets and spare parts, while gathering service data that improves performance (IBM, n.d.). But manual monitoring can result in inaccurate data, delays in service, and higher costs. Based on interviews conducted with the operations head at JA Noble Enterprise Inc., an automotive service center in San Pedro, Laguna and direct observation of their current workflows, the company faced problems with their paper-based processes and spreadsheets that were not linked. The operations head stated that their current POS system only records sales and could not keep track of a vehicle's service history, could not record the materials used for each job and could not provide real-time visibility of stock levels. Workflow observations showed that the lack of a central system led to inefficient communication, frequent stockouts of important parts, and an inability for managers to pull useful reports to make decisions.

To overcome these problems, this study proposed the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc. The system is a web and mobile-based application that uses QR code and barcode technology in monitoring services and inventory in real-time. It has a Decision Support System (DSS) to provide low-stock alerts and reorder recommendations, and automated notifications for Preventive Maintenance Scheduling (PMS). A secure mobile portal allows customers to view their vehicles service history and check maintenance due dates. The system also uses Retrieval-Augmented Generation (RAG) to produce useful management reports. This proved to be a way to

increase transparency, accuracy and efficiency, at JA Noble Enterprise Inc. It consolidates all these tasks into one place, making Operations smoother and more reliable.

1.1 Project Context

The automotive service industry requires inventory management and maintenance work to keep customers happy and to provide quality service. With more people wanting faster and more reliable auto services, companies are increasingly adopting likely turn to digital solutions that streamline processes, cut down on mistakes made by hand and make operations run more smoothly overall.

JA Noble Enterprise Inc. is a Caltex gas station and auto repair shop located in San Pedro, Laguna. Based on interviews and observations of how the company operates, the establishment still uses manual methods to handle inventory and keep track of vehicle maintenance. Paperwork orders were used to record service tasks and spreadsheet software was used to update inventory records. The interview findings indicated that these disconnected processes led to delays in service completion and created differences between what's recorded and what is actually in stock. Also, the existing point-of-sale system only records sales. It did not track the materials used for each service job or connect services to the vehicles that were being serviced.

Based on the observations and interviews conducted, there were some difficulties in the current process. Staff sometimes had difficulty communicating with each other, and checking previous vehicle service records was not always easy. There was also no system in place for scheduling preventive maintenance, making it harder to keep track of upcoming services. In addition, monitoring the current inventory levels

could be difficult, especially when the information was not updated immediately. These problems could lead to errors in reports and might also cause delays when management needs information to make decisions.

In response to these challenges, the proposed project aimed to develop an Integrated Vehicle Maintenance and Inventory Management System with Decision Support. The proposed system aimed to digitize service and inventory tasks through a web and mobile-based platform. It utilized QR code and barcode technology for accurate data capture. The system also included a Decision Support System (DSS) module that assisted in determining reorder points for parts and scheduling maintenance. Additionally, it featured an AI Assistant powered by Retrieval-Augmented Generation (RAG) technology which could answer queries, find relevant data, and create concise summaries to help with managerial analysis. This project was designed to improve the capabilities of JA Noble Enterprise Inc. by offering a centralized data-driven solution that supported faster, more accurate, and more reliable workflows.

1.2 Research Objectives

This study aimed to develop, test, and evaluate an Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., with regard to functional correctness using systematic testing and user acceptance through the Technology Acceptance Model (TAM).

Specifically, this study aimed to achieve the following objectives:

1. Design and develop a web and mobile-based Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., featuring core modules for vehicle maintenance

tracking, QR/barcode-enabled inventory management, automated low-stock and preventive maintenance alerts (DSS), a RAG-powered AI assistant for intelligent queries and reporting, and a secure customer portal for service history access.

2. Evaluate the functional performance of the developed system across its major modules, including the Vehicle Maintenance Management, Mobile-Based Inventory Management with QR/Barcode Scanning, Decision Support System (DSS) for stock replenishment and PMS scheduling, and Retrieval-Augmented Generation (RAG) for the AI Assistant, by measuring relevant indicators to confirm that all key features are working as intended.
3. Determine the perceived software quality and user acceptance of the developed system after system demonstration and hands-on use, based on the Technology Acceptance Model (TAM) in terms of Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience.

1.3 Conceptual Framework

The conceptual framework translates the objectives of the study into a systematic model that guides the design, development, testing, and evaluation of the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc. The framework is composed of five main components: Inputs, Process, Evaluation, Outputs, and System Components, connected through a Development Path and a Feedback Loop.

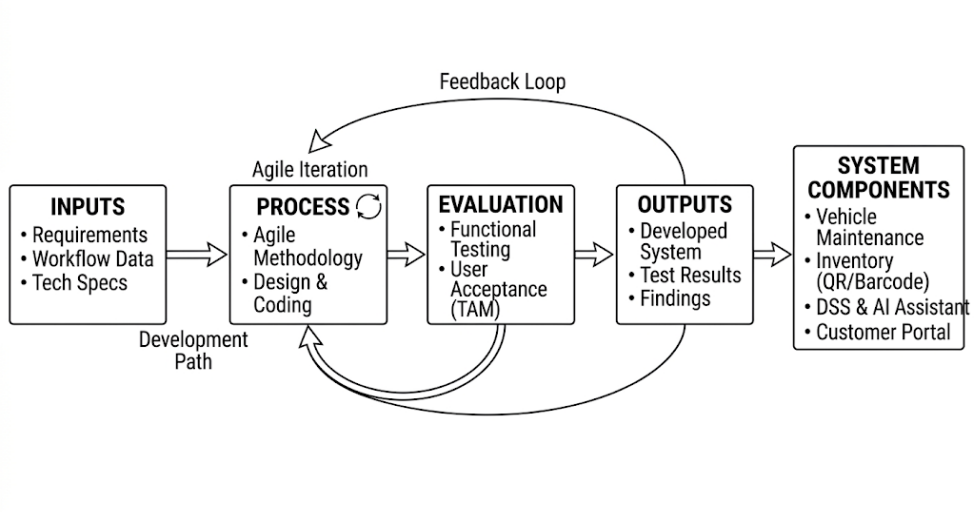


Figure 1. Conceptual Framework of the Study

The figure was generated using an artificial intelligence (AI) image-generation tool based on the researchers' conceptual framework and specifications.

Figure 1 shows the framework of the study showing the flow and links among the main phases of the proposed system. The proposed system is the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc. The framework is divided into five parts: Inputs, Process, System Components, Evaluation and Outputs all of which feed into the Integrated Vehicle Maintenance and Inventory Management System with Decision Support.

The process begins with the Inputs stage. In this phase, the researchers gathered data, including the workflow analysis of JA Noble Enterprise Inc. The researchers also collected system requirements that were identified through interviews and observations, and the researchers identified the technology specifications needed to close the operational gaps in vehicle maintenance and inventory management for JA Noble Enterprise Inc.

Following the inputs is the Process stage, which uses Agile Methodology. In this phase, the system was refined through iterative sprints. This means the researchers worked in cycles that include planning, coding, and gathering feedback. These cycles

helped keep the system adaptable and make sure the system matches what users need even as the users' needs change over time.

The System Components represent the functional modules that were created during the development process. The System Components include the Vehicle Maintenance Management module, the Mobile-Based Inventory Management module that uses QR and barcode scanning, the Decision Support System (DSS) which assists with stock replenishment and preventive maintenance scheduling, the Retrieval-Augmented Generation (RAG) AI Assistant, and the secure mobile customer portal.

Once the components were integrated, the system goes through Evaluation. This Evaluation has two parts: Functional Testing, which checks whether the system works correctly, is dependable, and performs according to rules, such as real-time monitoring and automated alerts. The other part is User Acceptance, where the intended users of the system, like administrators, staff, and customers, check how easy it is to use, how fast it works, and how good the experience is using the Technology Acceptance Model (TAM).

The final stage is the Outputs. This represents the tangible deliverables of the study, which include the fully developed and tested Integrated Vehicle Maintenance and Inventory Management System, the documented functional test results, the statistical findings from the user evaluation, and actionable recommendations for system deployment.

The framework has a feedback loop that takes the Evaluation stage and sends it back, to the Process. This keeps the system moving. This means that any bugs found during testing or suggestions for improvement, from user acceptance, are taken into account and worked on again. The system keeps getting better over time because of this

cycle. As a result, the study makes sure that the proposed system actually solves the real-world problems faced by JA Noble Enterprise Inc. It delivers a combined and automated solution that fits their needs.

1.4 Project purpose

The proposed project, the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Stock Replenishment and Preventive Maintenance Scheduling, is significant because it addresses the systemic inefficiencies in vehicle service and inventory management at JA Noble Enterprise Inc. by leveraging a centralized digital platform. Its importance is reflected across several key stakeholder groups:

For Administrators. The system provides a centralized repository and analytical dashboard to support evidence-based decision-making. By utilizing the Decision Support System (DSS) for low-stock alerts and preventive maintenance scheduling, it ensures transparent management of inventory and service operations while maintaining high data integrity. It replaces slow manual processes with automated reporting, allowing management to monitor business performance and generate data-driven reports efficiently.

For Staff. The platform offers a streamlined digital environment that replaces manual paperwork and spreadsheets. Inventory managers and mechanics benefit from QR code and barcode scanning, which reduces manual data entry and minimizes errors in inventory records. The automated check-in processes and service monitoring tools help staff track vehicle histories and manage maintenance tasks efficiently, ensuring smoother daily operations and preventing delays.

For Customers. The system provides transparent and efficient service experience. Through the Customer Portal and automated check-in processes, customers gain real-time visibility into their vehicle's service status and history. This ensures they are kept informed about maintenance schedules and the availability of parts for their vehicle repairs, enhancing trust and satisfaction with the service center.

For Future Researchers. This study serves as a foundational reference for integrating web and mobile technologies with Decision Support Systems and AI assistants (RAG) in automotive service settings. It offers a model that can be replicated or expanded by other institutions or businesses looking to modernize their inventory and maintenance workflows.

By emphasizing these benefits, the Integrated Vehicle Maintenance and Inventory Management System demonstrates how technology can modernize operational workflows and improve overall experience for administrators, staff, and customers at JA Noble Enterprise Inc.

1.5 Scope and Limitations of the Study

This section delineates the scope, boundaries, and inherent limitations of the study. It establishes the parameters of the investigation by specifying the system components, processes, target users, and institutional context covered by the research, as well as the factors that may constrain the interpretation and generalizability of its findings.

The study, entitled "Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Stock Replenishment and Preventive Maintenance Scheduling for JA Noble Enterprise Inc.," encompasses three primary

areas: (1) the design and development of the system architecture and core functionalities; (2) the technical testing of the system modules including the Decision Support System and AI Assistant; and (3) the evaluation of the system's functional performance, software quality, and user acceptance among selected administrators, staff, and customers. The study is institutionally and geographically delimited to JA Noble Enterprise Inc., a Caltex Service Station and Automotive Service Provider located in San Pedro, Laguna, Philippines. Accordingly, the findings are interpreted within the specific operational and technological context of JA Noble Enterprise Inc. and are not intended to represent the vehicle maintenance and inventory management practices of other service centers or automotive businesses without further validation.

The scope of the study is defined by the following areas, which are aligned with the research objectives and establish the specific parameters, system components, and evaluation activities covered by the investigation:

- Preliminary investigation and requirements analysis through consultation with JA Noble Enterprise Inc. to identify existing operational practices, information requirements, system specification, and challenges in vehicle maintenance and inventory management.

- Document and process analysis to examine relevant vehicle service and inventory-related documents, including service records, stock levels of spare parts, lubricants, and filters, as well as existing procedures for vehicle check-in, maintenance scheduling, inventory tracking, and reporting. The findings from these activities will serve as the basis for defining the functional requirements and technical specifications of the proposed system.

- Development of a multi-platform system with a relational database to support

secure vehicle service record management, inventory tracking, document storage and retrieval, and administrative information management.

- Integration of role-based authentication for administrators, staff, and customers, enabling customers to securely access their vehicle's service history and upcoming Preventive Maintenance Scheduling (PMS) dates via a mobile portal, while administrators can query all records and reports through the AI Assistant.

- Implementation of a Decision Support System (DSS) to aid decision-making through low-stock alerts, reorder suggestions, and PMS notifications.

- Integration of Retrieval-Augmented Generation (RAG) to generate text responses and export smart management reports in PDF and Excel formats using predefined templates, with short-term context maintained within the same chat session to ensure all responses are strictly grounded in database information to prevent fabricated answers.

- System testing and performance evaluation to assess the functionality of the major system modules.

- User evaluation among selected administrators, staff, and customers following system demonstration and hands-on use to determine the perceived software quality and user acceptance of the system.

While the Integrated Vehicle Maintenance and Inventory Management System provides a technological platform for managing vehicle maintenance and inventory-related processes and information, the researchers acknowledge the following limitations:

- The system is limited to cars and trucks and does not include motorcycles, buses, or heavy machinery.
- The system does not cover non-maintenance tasks such as refueling or car washing.
- Due to the high cost of direct SMS gateway integrations, the system uses in-app push notifications and email alerts for PMS reminders instead of SMS.
- The system does not include online payment systems, external point-of-sale systems, supplier networks, automated purchasing platforms, predictive maintenance algorithms, computer vision for damage detection, or full ERP features such as accounting or human resource management.
- The AI Assistant is restricted to answering questions within the defined scope and does not respond to unrelated queries. It may misunderstand words, abbreviations, slang, or unclear questions.
- The AI Assistant only works with text and cannot create images, audio, or video files. It is a reading system and cannot change, delete, or update any database records through user chats.
- The AI Assistant's responses depend completely on the completeness and accuracy of the English-written system database.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the related literature and studies that back up the creation of the system that is being suggested. It has details collected from books, journals, articles, websites and other trustworthy sources that are connected to the study. Specifically, the goal of this chapter is to offer knowledge to build a stronger base for research and help the people doing the research understand the ideas, tools and ways that are connected to the study.

2.1 Related Literature

Several studies and books have looked at car maintenance keeping track of parts, tools for making choices and similar tools. Collectively, these studies give ideas about creating a system that handles car maintenance, parts tracking, when to do regular checks and when to get more supplies.

2.1.1 Vehicle Preventive Maintenance Management System

For a fleet to operate smoothly and for machines to have a long life, it is essential to plan and manage the preventive maintenance of vehicles in a way that preempts any trouble on the road. In the past, vehicle service was handled by more conventional means: paper logs and physical cards. But such methods are prone to human error and can be slow, with records going missing and regular part repairs or inspections being overlooked. As a result, an increasing number of companies that run cars and move goods are turning to centralized systems and mobile tools. With these at their disposal they can organize jobs and put out reminders so as to get ahead of potential issues.

The shortcomings of the old ways have made the case for computerized maintenance systems all the more compelling. Dwipa et al. (2024) examined the upkeep of emergency coarsers and aircraft and found the paper checklists in use to be wanting; the inspection process was inconsistent and locating problems was a slow business, not to mention the missing records. To address this, they put together PMCSS, a web application that puts checklists in the form of digital documents requiring a sign off from personnel, thus ensuring emergency vehicles are ready for duty and errors are kept to a minimum. Then there is the matter of data fragmentation in repair shops. Babu and Narayani (2026) noted that keeping to paper makes it difficult to pull together all the information, leading to slow booking and trouble with old files. Their solution is a system for managing those records and tracking car status that has proven to make shop operations far more efficient.

In the field of motorbike and car repairs Kana and Aji (2024) The use of paper for service orders has been noted as a source of by-have pointed. Confusion and lack of sales records and slow report filing. To put an end to this, they put together a web and phone application to automatically Confirm sales and ensure service orders are delivered timely. There's also the issue of record keeping. Veena et al. (2024) noted how paper can be an obstacle when it is only in making reservations and registering that it becomes apparent. Their solution is an app that does away with keeping track of spares and streamline the process, not to be used as a paper of allocation of time and workers to wash and repair cars.

Ramadhan et al. (2026) have also been interested in the bus rentals operation where things can go awry if one is not on top of driver schedules or road time. They devised a system so that managers could have everything in one place to check the log

of the drivers and available buses for hire. In a separate QR code into their project. Ramadhan and Falevi (2025) took their project one step further by adding QR codes. The full service comes with a scan that's provided by either the owner or the mechanic history of the vehicle is at hand, which preempts any delay in locating paper easily handles and expedites maintenance. What these studies have in common is to avoid putting these things on paper and digital records are the way to go. To prevent errors and to ensure proper maintenance. Digital systems are also put to use for planning and the dispatch.

Based on the records of reminders. One is Anisa (2025) who has Created a web system along with WhatsApp to look at car files when called and issue a warning; this will alert both drivers, and the man-in-charge was to them. Besides this, there is the work of Muharom (2024), who established a system for workers to assign work and to record when it was done, and this, in turn, accelerates things and helps minimize errors. The importance of real-time data.

This is clearly visible in the phone app developed, which includes and alerts automation for improved planning This was written by Balia and Ab Ghaffar (2026). Through the use of a computer and car tool, the app can calculate battery level and distance travelled and deduce oil used Adjust intervals and give an alert, without the risk of late maintenance or battery change-over. trouble. To put it simply, these studies suggest that apps that alert users to the potential dangers of a phone call can help them avoid those risks. Can change the maintenance strategy for vehicles.

It's more of a problem when it involves coming more. To fleet planning, where it becomes necessary to manage the availability of cars with the creation of a plan. Wang et al. (2022) have overcome this by creating an Apply model to make predictions

of part-lives to inform service scheduling decisions at reasonable cost and time effective manner. A study from Crespo del Castillo et al. (2024) provides some insights. another approach. They rely on a model that is based on daily work plans and forecasts to see. How to allocate resources, both human, financial and spares, to ensure that they are used optimally to maintain cars in operation.

Planning the modern fleet is no longer a matter of the schedule, it's a matter of. can be used to prevent problems before they occur through tracking tools, sensors and mathematical models. gets out of hand. Abidin and Fadzil (2024) in their review of tracking systems noted that GPS and real-time car sensor information are extremely useful for this type of work. Prediction, of course, is another thing. Foké et al. (2023) put together a system using real data from city buses and after filtering the data, researchers found that there were some problems that are red flags for what may emerge as a big problem. problem. In short, their research indicates that one can use models to put Find out, just before they break. But some machine-driven approaches to problem solving require a great deal of pre-labeled data; this is not always practical. Instead, Giannoulidis and Gounaris (2023) suggested that: really detected in the environment. Invented a technique which could be applied to sensor data to group cars according to what they actually sensed. are doing and detecting faults without the need for any specific data. It is a way to keep vehicles running and costs down. Such innovations look toward a change in the product life cycle. Switching from one set of established schedules to more creative, flexible plans for maintenance.

Yet for all these digital advances, there are still obstacles. Babu and Narayani (2026) have identified the cost of implementation as the need for internet access and data security are big challenges. The human element cannot be ignored: overlooked

either. The tools need to be simple, and workers will have to be trained in the use of. Otherwise, they can fall back into their old habits. Another issue is data quality. Abidin and Fadzil, in 2024 said that the data used must be good. If it is not the plans can be wrong and maintenance might be done soon or too late.

2.1.2 Inventory Management System for SMEs (Small and Medium Enterprises)

The growing complicated ways of working in medium businesses, especially those that depend on inventory and the car industry, have made it more important to have good, reliable and tech-based systems to manage things. In this case, the parts of the process that deal with inventory and keeping things in shape are very important to make sure the business can keep running well and make customers happy. According to Alam, Thakur and Islam (2023) small and medium businesses often have trouble keeping the amount of inventory because they do not have enough resources and still use old ways to check on inventory. Also, Madamidola et al. (2024) said that there has been a change in the ways inventory is managed over time. This change went from doing things by hand to using computer-based systems to make sure everything is done right and fast. Furthermore, adding decision support tools has become very important for companies that want to make the way they get more automatic and also try to predict what will happen next.

At first the way inventory was managed used manual methods that were based on looking at things, writing things down on paper and physically counting the stock. This way of doing things led to not working, having incorrect numbers of items and making decisions about when to get more stock late. According to Cabungcal et al. (2023). These old ways really hurt the money side of a business. Also, Gaborno et al. (2025) said that owners of small businesses often used informal and experience-based

ways to track things, which made it hard to grow and be accurate. So, Ekron, Ongori and Akwee (2023) said that not having ways to control the stock directly made customers unhappy, which was another problem with manual inventory management.

With the advancement of technology, there has been a tendency to put computers to work on inventory in a more orderly fashion. Odasco and Saong (2023) would argue that computerized systems are of real value in how services are rendered; they simplify access to information and help avoid errors in the process. In line with this Gabriel (2025) said that using organized ways to manage inventory in stores helped make the inventory more accurate and the money side of things better. Owajege and Saif (2025) also showed that a good way to keep track of inventory can help small and medium businesses do better by making operations work better and making choices. At this stage the main thing was to get everything even though data was organized better, making choices was still mostly up to people.

Then the inventory management systems turned into systems that had some parts that worked on their own using tools that helped make decisions. Alam et al. (2023) Said that now a lot of inventory systems for medium businesses are done on digital platforms that help with procurement and tracking. In the way Rodrguez, Lopez and Vera (2025) talked about new ways to manage inventory that help make more money for small and medium businesses by keeping a close eye on things and making them better. When it comes to the car industry, Yang (2025) showed how managing inventory in the parts supply chain in the car industry needs to be done with systems that work together and are connected to handle needs and complicated supply issues. Zvinowanda, Ncube and Mafini (2025) explained how the car industry uses ways to

manage inventory with knowledge from systems and organized steps even though the systems still are not fully automatic.

Advanced ways have also tried to deal with smart inventory systems. These kinds of systems can have ways to predict things and help with making decisions. Sayal et al. (2025) Looked at clear inventory systems in the car industry and said that managing uncertainty makes the choices about inventory better. In addition, Puiu, Petre and Boscoianu (2025) used Six Sigma to make car inventory. This means looking at data to make things work better. Johnson (2025) studied how to plan for making cars and said that better planning systems can help reduce the number of cars that are made and stored. Lastly, Ta and Le (2025) found that managing inventory well has an effect on how much money a store makes which helps small and medium businesses do better with better inventory systems.

Even though there has been a lot of progress in making inventory systems there are still many problems with the way these systems work. Madamidola et al. (2024) said that in most of the inventory systems there is still not full connection between different parts like getting things, tracking them and helping with decisions. Also, Alam et al. (2023) And Gabriel (2025) said that small and medium businesses still use old ways to do inventory tasks. Rodrguez et al. (2025) said that even the newer systems do not have the ability to predict what will happen which is needed to make getting more stock automatic. Sayal et al. (2025) also said that with the new smart models, it is still hard for many companies, especially small and medium businesses, to use them because of how complicated they are.

The studies reviewed in this literature show that the inventory system has changed a lot, going from doing things by hand to using computers and then to systems

that have a lot of ways to help make choices. One finds that the system's various components are not as well integrated as they should be, and there are issues with the way predictions are made and with decisions on inventory and maintenance. It is clear from this that small to medium-sized enterprises in the car sector, for instance, require a properly functioning system that can tie everything together.

What the studies make clear is that vehicle maintenance is in the process of changing. The old way of putting pen to paper is being left behind for digital systems that put all the relevant data in one spot and can be accessed from a phone. These tools will even handle the scheduling on their own and keep an eye on things and that can predict when a car needs work. These things are often done on their own, which shows the need for a system that brings all the parts together like having one place, for car maintenance records and a way to plan when the work should be done for car service businesses.

2.1.3 Decision Support System (DSS)

For JA Noble Enterprise Inc, the vehicle maintenance and inventory management system is of considerable import. It is what allows the company to act in a proactive and data-driven manner, instead. Doing something by rote. The result is equipment that can be relied on, with less risk, and a reduction in the costs associated with downtime. This review is intended to provide the groundwork for a system Often designed specifically for JA Noble Enterprise Inc and based on appropriate research focused on decision Vehicle support, inventory management and vehicle maintenance.

Historically, such work was left to man's observation and judgment. This sometimes requires excessive maintenance, and low stock levels. Sayal et al. (2025)

states that the size of the study groups ranged from 1 to 10 participants. Older inventory systems were not sufficiently flexible to meet the needs of the industry. On the other side, companies have resorted to management systems that can process operational data. Kozin (2023) put forward a decision support system for To illustrate the value of using data to make better decisions, let's look at service workshops.

Web and mobile tech have come a long way, and interactive products like these are becoming more popular. SUV Maintenance Prediction System by Naga Raju & Nikhitha (2025); and it employs machine learning to give recommendations in line with technical specification of the manufacturer and driving behavior. They've grown into platforms that infuse intelligence into the overall supply chain and performance. JA Noble Enterprise Inc is a case in point.

Also, a newly emerging literature is exploring the possibility of: The replacement of the traditional time-based maintenance with predictive and condition-based maintenance. Implementation of predictive and condition-based maintenance instead of the traditional time-based maintenance. schedules. Sala et al. (2022) have put forth a framework for factories to ensure the health of assets can be monitored, and Bhat and Aravind (2025) argue that for the betterment of India's economy.

Mixing techniques to order and optimize maintenance jobs. All this suggests an increased need for predictive analytics to maintain as more relies on it operating costs down and the work in balance. The goal of inventory management scientists is to put in place systems capable of withstanding the vagaries of an unpredictable market. For example, Sayal et al. (2025) tested the fuzzy models on the Tested other inventory systems and concluded that they are not just as effective but also more economical. In addition, the work of Sá et al. (2025) developed a decision support system that included

safety stock and time buffers as a means of uncertainty management on supply and demand side. Khalil et al. (2026) have focused their work in the area of fleet maintenance applied programming to figure out part requirements, thereby reducing inventory costs while maintaining service reliability.

Despite technological advancements, gaps remain in existing studies regarding the development of an integrated vehicle maintenance and inventory management system tailored to the operational needs of JA Noble Enterprise Inc. A maintenance and inventory management system. Dwivedy et al. (2025) pointed out this finding was that most readiness models for Industry 4.0 are too generic to be applicable to businesses in developing countries. Crespo del. Castillo et al. (2024) have drawn a conclusion that attention to lack of models such as predictive maintenance Balancing workload and/or reducing inventory costs. To address this, the aim of the current work is to design a framework specific to the SME, one that combines live scheduling and automatic reorder points and prioritization.

The literature review indicates that there is a definite trend towards processes that become more 'data driven' with the backing of machine learning and AI to better Manage vehicle and inventory at JA Noble Enterprise Inc. While the shift manual to automated has accelerated decisions with greater speed and precision. Integrating such tools is not an easy task for a medium-size company, finite resources. In the end, the results indicate that the most effective way to increase is to: efficiency and contain long term costs is to integrate spare parts management with maintenance scheduling.

2.1.4 Retrieval-Augmented Generation (RAG) for AI Assistants

The older vehicle maintenance and inventory computer systems were developed using a rule-based process and a static database with little modification. evolution. While such methods were good at the business of data entry and in the case of storage, they were unable to provide the depth and impact of analysis needed. For effective decision making.

As Gupta et al. (2024) noted, these traditional setups were content to just save and retrieve data; they didn't do much to synthesize information from various sources or make sense of a dynamic Faced with the situation in real time, decisions to be made slowly and by hand. But with the introduction of artificial intelligence, all that has changed. AI models can now access knowledge bases and use their knowledge in a manner that is now possible. systems a greater sense of context and adaptability. An example of this is the rise of retrieval-augmented generation (RAG). Where in the past, retrieval and generation were two discrete activities but now they are combined. single, integrated process.

Huang and Huang (2024) expressed it in two steps: documents are extracted from the available knowledge, and used as the foundation for a response, which serves to limit errors and keep the output right. But the current RAG systems have gone beyond their basic beginnings, as Sharma (2025) explains. pipeline model. With improved ranking and reliability, they are designed to deliver superior results. This has been supported by preliminary studies on passages. RQs that can comprehend actual queries (retrieval and dense retrieval), and Keyword matching never could have done this. And while long-context language models are able to take in vast quantities of text, Li et al. (2024) make the You will find a variety of cases that for an enterprise system that has

to deliver accurate answers from its databases, a retrieval-based approach is still indispensable.

In a nutshell, when someone is putting a data structure, there is no room for error in data handling. Apply the same process to the system and this applies just as much in business. Khan et al. (2024) point out that RAG systems make frequent use of PDFs since they are where you will find operational details such as inventory records, maintenance reports and manuals. However, it is certainly not easy to work with these files. The design of the tables, and the intermingling of text with them, is simple. When using other materials, it can be difficult to extract the information. Researchers have put forward text segmentation and chunking as a means to retrieve documents more easily without affecting the semantic content. There's also the issue of a government. Kesuma (2025) states that an enterprise RAG system should enable every output to be traced back to its source. That traceability is what makes facts credible on which to make decisions about restocking or maintenance schedules, without relying on conjecture.

More recent research has aimed to improve the capabilities of RAG systems Agile and adaptable to variability. Yan et al. (2024) have come up with a Corrective Retrieval Augmented Generation model that will re-run retrieval if the quality of the documents in hand is lacking, thus nipping any errors from No irrelevant or missing information sources in the bud before a response is released. Li et al. (2024) have explored combining LCRLMs with Retrieval for the purposes of reasoning and speed. Yet Yu et al. (2024) would argue that it can be a difficult task to evaluate, especially in the case of Evaluating uniformity and correctness of the type of conditions encountered in actual business operations.

Evaluation and research have been given more attention, Benchmarking with regard to the challenges. Narita et al. (2026) have Emphasized that in today's context, today's assessments are not suited to what enterprises need, Many wanting multi step action, or the ability to track evidence. To address this, Walden et al. (2026) have put forward automated tools for the intent of report checking. The lack is for a scalable and reliable method for measuring performance. One can also make the case that a pressing One of the problems that arises with RAG systems is whether or not the error is a result of the RAG system's failures. retrieval or on the generation side.

There is an evolution in literature: One had simple There are now some context-aware decision-making tools available in retrieval-based systems. making. Combining retrieval and generation has been the trend, and this is serving to improve the precision and value to a company. But it has its drawbacks. difficulties. Even ensuring reliability in real time is not a trivial matter and data structures are ill-suited for this purpose. Are likely to be complex, and decisions could be more transparent. Governance and evaluation need to be tightened to develop truly 1.4.2. integrated systems. Transitions to the next generation would do much to raise the bar on to be more efficient and accountable, and to be available and traceable.

2.1.5 Mobile QR/Barcode Scanning

The inventory has undergone a significant change in how it's managed. Where one was able to use hand labels, it’s now digital. Solutions such as barcodes and QR codes are now the norm. The latter concept was introduced by Denso Wave of Toyota in the year 1994; Chikkamanju and Ambika. Covered by the other two mirrors, it was designed to monitor car parts during their manufacturing process. made. This was in a

way an innovation for product and inventory control, providing companies with the ability to store and access information with a small device. digital tag.

Historically, warehousing and inventory management were a problem of. Manual counting and paper records. These are considered "old school" methods. Today, they're not only time-consuming but they're also prone to errors and the potential for documents to be misplaced and operations to be put on hold (Al Battashi, 2024).

A manual system is simply not able to keep up when inventory in high-volume sectors like, for example, sectors with a high number of web pages, levels rise too slowly and are not reliable. As auto parts or equipment. The solution has been automation, with barcode systems, using mechanical methods to accomplish tasks. As Punith et al. (2024) put it, this kind of automation reduces the amount of work that needs to be done in order to get huge quantities of things moving, accounting to an organization's need for efficiency and data that is accurate.

Then there's mobile technology, that has sped up the speed of product identification and tracking. With mobile apps one can Scan barcodes or QR codes – use a dedicated scanner for inventory control. Patil et al. (2022) have an app that will showcase an item's service history and other information from a central database is revealed after scanning its QR code does a similar thing with an inventory system linked to a database (2023) Through QR codes, real-time stock visibility. Patel et al. (2022) note that these centralized databases are faster in everything they do, from selling to Buying but preventing the loss of assets. From the findings that Kevin (2022) found, he noted that, for medium-sized enterprises, it is enough to use a scanning app in the workshop on the smartphone. There is no need for the hardware of a traditional scanner, making the whole process more accessible and cost effective.

The current fashion is to combine web and mobile for cloud storage and time sync. Wairooy et al., (2024) have built together an integrated warehouse solution using Flutter and Firebase. The former allows cross-platform apps and the latter for live databases, and in combination they serve to prevent losses and cut down on errors in management.

Research by Yaqin et al. (2024) shows that a Django Python QR system can put an end to 20% of inventory discrepancies. In a separate Data entry rate was 43% lower using barcode systems compared to undertaking. The accuracy of 98.5%, which further reduced the chance of error (Sahetapy and Suhirman) 2025\). The applications of such technology have increased, and now it's not only for keeping an eye on the usage, maintenance and condition, tracking items are used. inventory. In workshops, Sermonia and Pabelona (2025) verified this. Cloud-based solutions and QR tagging for repair & time inventory.

Yet Traditional methods are not an asset in specialized automotive applications. As per Saputra et al. (2024), there are numerous inventory applications that are lacking in. user features, but there is no means to cope with return requests or to keep people. Responsible for items borrowed equipment. Even if the system is fast in recording. Integration is frequently desired and/or required. Maintenance histories will not be available in one. Correlated with stock order or repair history.

2.1.6 Mobile and Web Push Notifications

There has been considerable change in the automotive repair and maintenance sector this century. With the advent of digital tools, customers expect to have a line of communication with companies in real time. Luna Alexander has pointed out that such

dialogue is essential for customer satisfaction and a company's operation requires it. Therefore, service providers are avoiding the old approaches and opting for automated systems, looking to eliminate and control interactions more effectively and prevent errors.

One can observe how the system of automotive services has developed from simple record-keeping to mobile integrated platforms. In the old days, centres would still rely on traditional methods for some things such as confirmation, etc. or to give a service update. This was a method that would have been subject to delays, misunderstandings, not to mention the extra work it put on administrative staff. Those earlier systems would have been able to process reservations and calculate the fees, they faced challenges of lack of real-time tracking and specific dedicated applications. Garud et al. have noted a recent move in the direction of Vehicle Service Automation App ecosystems, which make use of Firebase and Android Studio to provide a more user-friendly interface for both parties. But there's the issue of whether the customer will be notified of the push notification, Maintains tasks on the schedule and in the loop.

Web Push Notifications are a good example of that; they allow one to send an individual message to a certain user even if the site is closed, browser instance will remain. Carboneri et al. report that they can get higher engagement than SMS or email. But timing is everything. Gavilan and Martinez-Navarro warn that a bad time or The "reminder" feature that is already enabled, is only going to irritate the person taking the reminder. And Nguyen Tran et al. demonstrated by a delay between some action and the onset of the change of state, such as 15-30 minutes, is enough to trigger the change of state. However, there is a difference when it comes to getting a response. Li et al also

discovered that immediate can decrease engagement with notifications ranging from 1 to 6 minutes. can make users more receptive.

The modern design of automotive service systems is based on twofold. The goal of being helpful not only to the user but the administrator. A customer, for instance, will be able to put in a registration for his vehicle, book a service and keep an eye on its progress as it happens; once the work is done, he can leave some feedback. The administrator has other tools available for them to use. handle appointments, change service statuses or pull up customer data for a more tailored experience. Then of course there are the automated functions. Alexander has pointed out that reminders reduce no shows by 25-60%. The number, cent. translates to more reliable revenue projections and more efficient use of resources. More advanced systems will even self-monitored and include maintenance targets to remind users when maintenance is needed. As for restocking, alerts can be triggered using the inventory and decision support systems can be programmed to do so, that parts are available for all planned jobs.

However, the current selections come with their drawbacks, hence the need for greater integration at JA Noble Enterprise Inc. the employers, and the students' readiness to meet that requirement. The level of digital literacy required by the employers and the students' preparedness to meet that requirement, coverage, language options. There may be some platforms that can be a barrier to access. Carboneri et al. have drawn the following conclusions: Also, attention to session mismanagement is important, as this type of issue. are not properly handled between browsers or sessions, opening the door to for privacy reasons or attempted delivery failure. There is also a digital divide in the use of digital resources for there are differences in digital literacy.

For the big dealership networks, inventory is managed while the big dealership networks have their CRM. Smaller and middle-sized companies will typically remain in the older processes. In most cases, research on maintenance scheduling and service decisions can be found that are approached from either a maintenance or a service perspective. discovery on their own, little consideration was thought to how they might be together with the inventory reminder structure.

The automotive service industry is heading towards an automated A system based on real-time data and optimized notification strategies. The aim of the proposed system for JA Noble Enterprise Inc is to solve the existing problems by integrating the two features of customer reminder for Remind the contractor to replenish the inventory on a maintenance and administrator reminder, or single platform. The System will provide timing optimization techniques, backed by research, which will help to minimize fatigue and enhance service compliance. Through this holistic solution, efficiency and customer retention can be enhanced via a data-based communication system.

The literature reviewed has collectively pointed to the evolution in the field of automotive and SME vehicle maintenance, inventory management, decision support systems and supporting digital technologies. Initial research focuses on time-based maintenance and inventory processes, which can be ineffective, unreliable and time-consuming, and in most cases, these processes are not in real time. These have evolved over time into condition and predictive maintenance systems, with analytics, optimization and new technologies like machine learning and IoT monitoring. The same applies to inventory management systems where there is a progression from tracking to computerized and semi-automated systems, and increased focus on

accuracy, efficiency and demand forecasting. Despite all these developments, many systems are still deployed as standalone systems, with the maintenance schedule and inventory replenishment processes not being integrated.

The integration of decision support systems for optimizing efficiency and uncertainty minimization has been a common subject in the literature for automotive and SME applications. The studies demonstrate that there is a clear benefit to combining models, predictive algorithms and optimization techniques to optimize maintenance scheduling, minimize downtime and control inventory. Furthermore, technologies like QR/barcode systems and cloud platforms enhance the ability to track in real-time and ensure that data is readily accessible and operations can be coordinated. But there is a gap in creating a unified platform that integrates the maintenance data, inventory levels and intelligent decision making. Factors like automation, interoperability, and real-time decision support are still a major challenge in achieving operational efficiency, particularly for SMEs.

2.2 Related Systems and Technology Gap Analysis

A total of ten (10) existing systems were analyzed. The table below shows the comparison of features between related systems and the proposed system.

Table 1. Comparison of Related Systems and the Proposed System

System/Features	Pm Scheduling	Decision Support System	Inventory Management System	Ai Assistant	Mobile Qr/Barcode Scanning	Web & Mobile Push Notification
Preventive Maintenance Checklist Smart System (PMCSS)	✓	✗	✗	✗	✗	✗
Mobile Application for Vehicle Maintenance and Scheduling	✓	✗	✗	✗	✗	✗

System/Features	Pm Scheduling	Decision Support System	Inventory Management System	Ai Assistant	Mobile Qr/Barcode Scanning	Web & Mobile Push Notification
System						
Mobile QR Code Vehicle Service Management Application	✓	✗	✗	✗	✓	✗
Web-Based Vehicle Maintenance Scheduling System with WhatsApp Gateway	✓	✗	✗	✗	✗	✓
Mobile-Based Preventive Maintenance Information System	✓	✗	✗	✗	✗	✗
Real Time Scheduling of an Automotive Maintenance	✓	✓	✗	✗	✗	✓
Optimizing the Maintenance Schedule for a Vehicle Fleet: A Simulation-Based Case Study	✓	✓	✗	✗	✗	✗
Dynamic Fleet Management: Integrating Predictive and Preventive Maintenance with Operation Workload Balance to Minimize Cost	✓	✓	✗	✗	✗	✗
Decision Support System for the Management of a Vehicle Service Workshop	✗	✓	✗	✗	✗	✗
Mobile Based Inventory Management System Using QR Code	✗	✗	✓	✗	✓	✗
Mobile Application Development for Inventory Management Using Barcode Technology	✗	✗	✓	✗	✓	✗
Development of Mobile QR Warehouse Management Application Based on Flutter and Firebase	✗	✗	✓	✗	✓	✗
Cloud-Based Inventory System with Quick Response (QR) Tagging for Technology Workshop	✗	✗	✓	✗	✓	✗
Proposed System	✓	✓	✓	✓	✓	✓

Legend: Check (✓) – Fully functional and Cross (✗) – No function

As shown in Table 1, most existing systems lack the combination of preventive maintenance scheduling (PMS), decision support system (DSS), inventory management

system, AI assistant, mobile QR/barcode scanning, and web and mobile push notifications in one integrated platform. The related systems mainly focus on specific functions. Some systems focus on preventive maintenance scheduling and decision support for maintenance activities, while others focus on inventory management with QR or barcode scanning for easier recording and monitoring of items. Meanwhile, notification features are mainly used for maintenance reminders and scheduling updates.

The comparison shows that none of the existing systems provides all six identified features in one platform. The features are instead distributed among different systems, with each system addressing only certain operational needs. An AI assistant is not included in the reviewed systems, while inventory management and QR/barcode scanning are mainly found in separate inventory-focused applications. This indicates that existing systems provide useful solutions for specific functions but do not provide a complete and integrated approach to vehicle maintenance and inventory management. The proposed system addresses this gap by combining PMS, DSS, inventory management, AI assistant, mobile QR/barcode scanning, and web and mobile push notifications in a single integrated platform.

Synthesis

The reviewed studies show how different technologies and functions are used for vehicle maintenance, inventory management, preventive maintenance scheduling (PMS), decision support systems (DSS), QR/barcode scanning, RAG/AI, and mobile and web push notifications. These studies address different operational needs, with some studies focusing on one function such as maintenance, inventory, or scheduling and some combine different technologies to meet specific operational situations or needs.

Among the systems reviewed these functions are used in applications and across various operational settings. The proposed system takes what already exists and builds on it by integrating vehicle maintenance, inventory management, DSS for stock replenishment, QR/barcode scanning, AI-assisted information access, and web and mobile push notifications to meet the operational needs of JA Noble Enterprise Inc.

CHAPTER III

RESEARCH METHODOLOGY

This chapter explains the plan; the way the researc hers collected data and the steps that the researchers took to build and test the Integrated Vehicle Maintenance and Inventory Management System. The researchers see that it gives a structure so that the building process is open and sound. Also, it explains how the researchers added the Decision Support System (DSS) and Retrieval-Augmented Generation (RAG) to make inventory control better and to schedule maintenance for JA Noble Enterprise Inc.

3.1 Research Design

The research design describes the plan used to look into the problem being studied and check if the goals of this project are met. It serves as a guide, for getting the data needed choosing the people to study, doing tests to see if users accept the product and looking at the data in a way. For this study the researchers used a Developmental Research Design that follows a way and also uses both numbers and words to get the information. The developmental approach controls the steps of finding problems in a company checking the software that was made and making sure it works well for JA Noble Enterprise Inc. In this context, descriptive and inferential statistics serve as the primary methods to summarize, describe, and analyze the quantitative data gathered from the user acceptance testing phase. This approach ensures a strict boundary between the research validation framework and the technical development execution of the software.

3.2 Population of the Study

The population of the study refers to the group of people who are relevant to the evaluation of the capstone project. In this case the target population includes the administrators, shop staff and active customers of JA Noble Enterprise Inc. This business operates at their Caltex service station in San Pedro, Laguna. These groups are the fit for the study because they are the ones who interact with the business' vehicle services and inventory processes every single day. Their daily involvement makes them ideal participants for assessing how well the system works in real-world conditions.

The group chosen for the study matches the goals of the research the users the system is built for and the people who will benefit from it. The administrators and shop staff are essential because they need tools for managing data, tracking inventory levels and receiving alerts for maintenance schedules. Their input helps ensure the system supports the core functions required by the business. On the hand active customers are important because they will use the mobile features of the system. This includes checking their vehicle's service status and getting notifications. Their feedback helps test how easy and useful the system is for end users.

The total number of people in the population is 344. To carry out the evaluation, the researchers selected a sample of 144 participants. This sample includes 3 administrators and 16 staff members. These individuals were chosen through enumeration, meaning every one of them was included because their roles are vital, to the system's success. The remaining 125 participants are customers, selected from a total of 325 services. This ensures that the sample reflects the user base and that the system can be tested fairly and accurately. Collecting feedback from this group allows the researchers to measure the system's performance, its speed, and how well users

accept it before it is officially launched. This step is crucial to making sure the system works well in practice and meets the needs of all the users.

3.3 System Testers

The research team invited 10 IT experts to test the system. The researchers was excited to watch the IT experts evaluate the performance and functionality of the system, which is called the Integrated Vehicle Maintenance and Inventory Management System with Decision Support, for JA Noble Enterprise Inc. The IT experts supplied feedback. The feedback from IT experts helps to make the system reliable, efficient, and free of issues.

Table 2. System Testers Profile

Tester	Educational Background / Credentials	Current Position / Affiliation	Years of Relevant Experience
Janssen Reyes Espino	BSIT GRADUATE	MISO - IT Staff at Municipality of Santa Cruz, Laguna	2025-Present
John Marvin A. Tabo	BSIT GRADUATE	System Software and Hardware Divion at Municipality of Santa Cruz, Laguna	2022-Present
Jonas T. Santos	BSIT GRADUATE	MISO - IT Staff at Municipality of Santa Cruz, Laguna	2025-Present
Kevin Adlaon	BSIT GRADUATE	System Development Division at Municipality of Santa Cruz, Laguna	2022-Present
Mharjun V. Tobias	BSIT GRADUATE	System Development Division at Municipality of Santa Cruz, Laguna	2025-Present
Robin Paolo Bona	BSIT GRADUATE	System Development Division at Municipality of Santa Cruz, Laguna	2019-Present
Wilmar T. Morata	COMPUTER PROGRAMMING UNDER-GRADUATE	MISO - IT Staff at Municipality of Santa Cruz, Laguna	2025-Present
John Marie P. Nerpio	BSIT GRADUATE	Administrative Aide I / Encoder at Municipality of Luisiana, Laguna	2025-Present

Tester	Educational Background / Credentials	Current Position / Affiliation	Years of Relevant Experience
Justin Jhay Lopez	BSIT GRADUATE	System Development Division at Municipality of Santa Cruz, Laguna	2025-Present
Jerico Dia	BSIT GRADUATE	Information Systems Analyst 1 at Laguna State Polytechnic University	2024-Prsent

3.4 Respondents for User Evaluation

The respondents for user evaluation include the admin, staff, and customers of JA Noble Enterprise Inc. in San Pedro, Laguna. The research and the system are designed to help everyone involved in the business by making vehicle maintenance and inventory monitoring much easier and more organized.

Table 3. Distribution of Respondents for User Evaluation

Respondent Group	Target Population	Sample Size	Percentage
Administrator	3	3	2.08%
Staff	16	16	11.11%
Customer	325	125	86.81%
Total	344	144	100%

3.5 Sampling Design

This section discusses the structural framework and techniques deployed to systematically select the evaluation of participants for the Integrated Vehicle Maintenance and Inventory Management System with Decision Support. To achieve empirical validity, the researchers utilized a Purposive Sampling Design as the overarching selection mechanism. As established in recent methodological reviews by Nikolopoulou (2022) and contemporary sampling overviews, purposive sampling is a non-probability sampling method where units are deliberately selected based on

predefined criteria, professional qualifications, or specific operational traits that directly align with the core objectives of the study. By implementing this design, the researchers guaranteed that the evaluation data was gathered exclusively from key stakeholders actively engaged in the ecosystem of JA Noble Enterprise Inc., specifically grouping them into station administrators, workshop staff, floor mechanics, and active customers. This approach prioritizes information-rich cases and users with direct contextual experience, ensuring that the gathered software metrics are both relevant and anchored on genuine operational understanding.

Under this purposive framework, a Mixed Sampling Approach was executed to address the distinct population sizes of the stakeholder groups. For the internal user roles, consisting of three (3) administrators and sixteen (16) workshop staff and mechanics verified from the official human resource employee roster, Total Enumeration was applied due to the compact and census-viable nature of the group. For the external stakeholder group, a Systematic Random Sampling technique was deployed to objectively select the final sample size of 125 customers from the verified baseline population of 325 active clients found in the company's central service log records, aligned with modern systematic selection models. The researchers calculated a fixed sampling interval by dividing the total customer population by the target sample frame derived from Slovin's Formula with a 7% margin of error. Starting from a randomly selected initial ledger entry, every subsequent customer profile was extracted at this precise step interval until the target of 125 unique respondents was achieved, effectively eliminating researcher selection bias.

$$n = \frac{N}{1 + Ne^2}$$

Equation 1. *Slovin's Formula*

Where:

- n = Sample size
- N = Total population
- e = Margin of error
- 1 = Constant

To analyze the data gathered from the system evaluation survey the researchers will utilize statistical tools to ensure accuracy and reliability. Slovin's Formula was employed with a 7% margin of error to determine the sample size for the customer group using a proportional sampling approach to ensure statistical adequacy. Conversely total enumeration will be applied to the administrator and staff groups due to their population sizes and critical roles in system operations meaning their entire populations will serve as respondents without sampling.

3.6 Data Collection Instrument

This section provides the details of the instruments and the systematic approach used by the researchers to collect the qualitative and technical data for the requirements analysis, the structural design and initial development stage of the proposed system. Structured fact-finding mechanisms were heavily used to make sure the data collected were empirically grounded, logically consistent and adequate to answer the specific operational objectives of the study. These instruments were carefully developed to collect more granular data from the primary stakeholders and everyday consumers about the legacy workflows of the automotive service, identified transactional bottlenecks, back-end system requirements, and structural database relationships of JA Noble Enterprise Inc. The data collection framework was effectively designed to focus on the initial phase of data collection and requirement elicitation so that the system developed later is closely connected to the real environment of the enterprise.

The first step in this study was the fact-finding program to gather primary raw data on the actual preventive maintenance scheduling procedures and the autohouse and warehouse operations of Caltex service station and automotive service center (ASC) of JA Noble Enterprise Inc., San Pedro, Laguna. These base line activities were the key empirical foundations for defining the exact touch points and documents handled when customers check-in, the baseline functional limits of the proposed mobile and web platform, and the active service lifecycle.

They collected key functional information by conducting synchronous video conferencing discussions and thorough interviews with the business operations manager using a semi-structured approach. The researchers did not use this interview as a "quick overview", but rather carefully tracked how services orders move and how they get to the point where components are issued from the customer's arrival to the final issuance of the components. Special attention was given to mechanical tracking loops, service history logging, managing low stocks and allocation of resources. This narrative exploration enabled the researchers to capture the recording, cross-referencing and sharing of critical data fields (part serial numbers, fluid capacities, vehicle plate metrics, and previous vehicle history markers) throughout the management dashboard and the shop floor.

These fact-finding consultations also thoroughly examined the key operational issues, difficulties and constraints faced within the legacy framework. The researchers noted systemic weaknesses, such as high levels of real-time stock reporting issues, a high rate of human mistakes as a result of manual appointment scheduling, easily spotted discrepancies between parts inventories, and a large communications gap between administrative encoders and active mechanics in the shop. Data gathered from

the mapping of these operational risks were mathematically and logically analyzed to identify manual subprocesses capable of being structurally centralized, automated or augmented. This directly supports the use of an algorithmic Decision Support System (DSS) and the notifications when stock levels are low.

In addition, the researchers conducted in-depth on-site observations directly at the Caltex service station, to verify the workflows claimed during the remote interviews. The team physically identified the type of documents and parameters used by the staff, with a specific focus on work orders, manual ledger pages and legacy spreadsheet architecture. These are direct observations of the physical and used directly to design the schema of the system's database, rules for input validation, and the relational structures of tables. By learning about the practical use of hardware barcodes and vehicle identification tags, researchers could design the particular hardware-level QR and barcode scanning functions of the mobile app. Finally, these story-driven approaches allowed the team to get a true and reality-based estimate of the automotive repair experience, making the data processing, Retrieval-Augmented Generation (RAG) report formats and workflows for the user actual.

3.7 Project Design

The researchers fully. Developed a functional multi-platform system that combines a web-based management panel, a mobile application and an integrated AI Assistant. The system was built with QR code and barcode scanning capabilities to automate parts monitoring and vehicle history logging. The mobile application was designed to deliver real-time service status updates and interactive notifications to users. A centralized relational database supports the system to record service transactions, maintain vehicle maintenance records and manage inventory stock levels.

During the development phase the researchers followed an approach including planning, system design, software integration and comprehensive testing of functionality and usability based on the ISO/IEC 25010 standard. After completing the system, it was implemented at JA Noble Enterprise Inc. In San Pedro, Laguna for a pilot deployment. This allowed administrators, shop staff and customers to use the system in a world automotive environment. Feedback gathered during this phase was used to assess the system's performance and ensure it met the goals of improving business efficiency and enabling data-driven decision-making. The project design includes three components: the web management panel, the mobile application and a Retrieval-Augmented Generation (RAG) powered AI Assistant.

3.7.1 System Development Methodology

The researchers used the Agile Development Methodology, the Scrum Framework to build the Integrated Vehicle Maintenance and Inventory Management System. This approach was selected because it supports iterative development. Of building the full system at once the team worked in small, focused cycles known as "sprints." This allowed continuous improvements to be made, in inventory tracking and maintenance features, based on feedback from JA Noble Enterprise Inc.

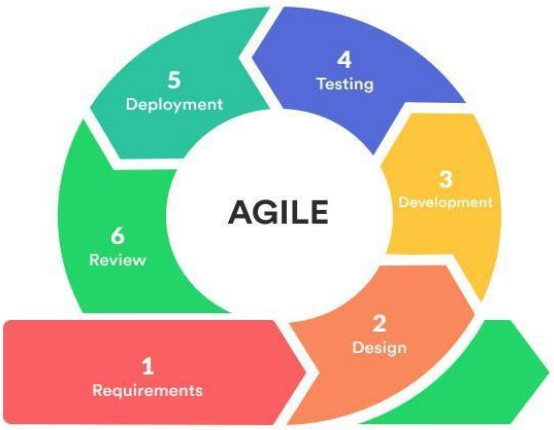


Figure 2. The Agile Software Development Life Cycle (SDLC) Model

The specific Agile SDLC model deployed throughout this project. During the execution of this model, the development stages were directly translated into actual project activities to fulfill the operational demands of the enterprise:

Phase 1 – Requirements: The actual project activities in this initial stage involved gathering detailed workflow specifications from the managers and workshop staff of JA Noble Enterprise Inc. The researchers focused on finding the structural goals of the project. They looked closely at where manual tasks were slowing things down. They also made sure to list all the details needed for system connections. These included automated maintenance scheduling, barcode scanning and a read-only AI chatbot dashboard.

Phase 2 – Design: In this stage, the actual activities focused on building the platform's architecture and designing the user interface layouts. The researchers created relational database schemas to manage data relationships such as linking spare parts to specific vehicle models. They also developed UI wireframes with care to make the system easy to use. The goal was to reduce confusion, for floor mechanics, administrators and customers by keeping the interface simple and intuitive.

Phase 3 – Development: During the development phase, the researchers executed active programming sprints to build and integrate system features piece by piece. The actual project activities involved coding the backend infrastructure developing the portal, for QR/Barcode scanning features and embedding Retrieval-Augmented Generation (RAG) engines to handle conversational AI query processing and automated management report generation.

Phase 4 – Testing: The actual activities, in this stage focused on using a black-box testing methodology to check the functional integrity of each newly added module.

The researchers carefully examined how accurate the barcode scanning module was. They also looked at how fast data synchronization worked across platforms. At the time they tested the reliability of the RAG-powered AI Assistant when comparing it with authorized corporate records. All of this was done to make sure the system behaved exactly as described in the Functional Specification Document (FSD).

Phase 5 – Deployment and Feedback: Once a stable Release Candidate version was stable the project moved to launching a pilot deployment, at the Caltex service station of JA Noble Enterprise Inc. Selected staff, mechanics and administrators used the web and mobile apps every day, which let researchers gather real-world usage feedback and record runtime behaviour for adjustments.

Phase 6 – Maintenance: After the pilot deployment the work shifted to improving the software and stabilizing the platform. The researchers looked closely at system logs to find and fix runtime errors as soon as they appeared. They also made changes to how indexes were processed which helped the AI Assistant respond faster. At the time they strengthened the rules in the database to make sure transactions worked smoothly and that data stayed consistent across every web and mobile device.

3.7.2 Hardware and Software Requirements

The hardware and software requirements of the platform system were identified based on its cross-platform architecture, dual web-and-mobile functionality, and artificial intelligence capabilities. The selected resources are strictly required to provide adequate support for the development, operation, maintenance, and successful deployment of the system while ensuring responsive real-time performance, secure data processing, and integration of the system's advanced analytical and hardware

components.

Table 4. Hardware and Software Requirements

Category	Requirement	Specification/Description	Purpose in the System
Hardware	Development Workstation	Laptop equipped with an AMD Ryzen 5 (7000 Series) processor and 16GB of RAM	Used by the developers to write code, manage microservices, compile application builds, and maintain local environments.
Hardware	Mobile Testing Device	Samsung Android smartphones running a compatible Android OS	Utilized during the testing phase to verify mobile accessibility, responsive user interfaces, and the performance of native camera operations.
Hardware	Integrated Scanning Modules	Built-in high-resolution mobile camera hardware	Serves as the physical capture input required for scanning barcodes and quick-response (QR) codes during field operations.
Software	Workstation Operating System	Windows 11 Operating System	Provides the primary hosting environment required to run integrated development environments (IDEs) and development tools.
Software	Mobile Development Framework	Flutter Framework powered by the Dart programming language	Serves as the multi-platform framework used to build and compile the native user interface for the mobile application.
Software	Web Backend Framework	Python programming language with the Flask web framework	Handles server-side processing, core business logic, application routing, and primary service configurations.

Category	Requirement	Specification/Description	Purpose in the System
Software	Centralized Database System	Cloud Firestore deployed via Firebase	Serves as the production database engine used to manage, query, store, and synchronize system records across all application clients.
Software	High-Performance AI Framework	FastAPI (Version 0.115.0) with Uvicorn (Version 0.30.6)	Powers the specialized artificial intelligence backend microservice, enabling asynchronous request handling and fast endpoint response loops.
Software	Large Language Models (LLM)	Qwen 2.7B / Qwen 32B Large Language Models integrated via the Groq SDK	Delivers the advanced machine intelligence, contextual analysis, text processing, and automated core operations of the application.
Software	Data Validation Layer	Pydantic validation library	Enforces strict, type-safe data validation constraints on incoming system inputs and backend payload schemas.
Software	Cryptographic Security Components	Standard Python cryptography libraries	Implements data encryption mechanisms to guarantee secure data transmission and protect sensitive database transactions.
Software	Automated Reporting Tools	ReportLab and openpyxl library frameworks	Used to dynamically compile and format operational data into downloadable PDF documents and structural Excel spreadsheets.

3.7.3 System Components and Modules

The Integrated Vehicle Maintenance and Inventory Management System is designed as a multi-platform application composed of interconnected modules that support the major operational requirements of JA Noble Enterprise Inc. Each module performs a specific function while contributing to the overall objective of centralizing automotive service data, automating stock tracking, streamlining parts management, improving scheduling mechanics, and supporting administrative decision-making. The major system components and modules are described as follows:

1. AI-Powered Decision Support and Reporting Module (RAG). This module facilitates the processing of high-level administrative queries and structural report generation through an advanced Retrieval-Augmented Generation (RAG) engine. The system automatically pulls real-time parameters directly from the relational database to compile data-driven summaries and export management profiles into predefined PDF and Excel templates. To assist executive decision-making, the assistant maintains short-term context within active chat sessions, ensuring that all text responses are strictly grounded in verified database facts to prevent information fabrication. Final operational choices and financial procurement validations remain subject to review and authorization by ownership.

2. Centralized Vehicle and Maintenance Database Module. This module serves as the primary repository for all transactional and historical information within the enterprise. It organizes and maintains customer account parameters, structural vehicle records, active work orders, historical Preventive Maintenance Scheduling (PMS) timelines, and spare parts inventory thresholds within a

centralized relational database. By consolidating scattered shop floor files into a structured repository, the module eliminates dependence on fragmented files and legacy manual accounting records, enabling fast, concurrent data updates and reliable data retrieval for administrators and active mechanics.

3. Automated Stock Replenishment and Decision Support (DSS) Module**.** This module provides a proactive system interface for tracking maintenance materials such as lubricants, filters, and spare parts. It dynamically evaluates stock levels against predefined safety thresholds to trigger automated low-stock replenishment alerts and calculate suggested reorder quantities. By introducing smart rule-based tracking, the module minimizes human forecasting mistakes, eliminates warehouse stockouts of crucial mechanical components, and prevents the over-purchasing of capital-intensive automotive materials.

4. Mobile-Based Inventory and QR/Barcode Scanning Module. Developed specifically for shop floor staff and mechanics, this module handles physical items and digital parts issuance through native device hardware integration. It features quick-response (QR) code scanning for instant tracking of vehicle plate profiles and maintenance histories, paired with automated barcode scanning to execute inventory parts lookups, log item quantities, and document parts consumption on open work orders. This digital transition minimizes manual keyboard input error, accelerating the physical parts identification cycle directly from the Caltex service station bays.

5. PMS Module. This module manages the automated distribution of time-sensitive operational alerts and vehicle status updates to multiple end-users. It processes preventative maintenance timelines and active repair updates to

dispatch targeted push notifications and structured email reminders concerning upcoming PMS due dates and active vehicle check-out alerts. The system ensures that customers remain informed of their vehicle's service lifecycle, while mechanics receive immediate alerts regarding parts allocations or workflow updates.

6. Role-Based Access Control (RBAC) and Security Module. This module regulates system resource accessibility according to the explicit permissions and professional responsibilities assigned to users. It enforces rigid permission boundaries for shop administrators, staff, and customers, thereby protecting confidential business records and administrative dashboards. This security boundary restricts customer access strictly to their personal vehicle history profiles via a secure mobile portal, ensuring that critical backend configurations, stock adjustments, and human resource modules can only be accessed by verified management roles.

Collectively, these components establish an integrated system architecture in which digital parts tracking, database synchronization, proactive scheduling, intelligent decision support, automated communication, and access control operate as interconnected functions to optimize business performance for JA Noble Enterprise Inc.

3.7.4 Database Structure

This section presents the logical design of the system’s database using an Entity-Relationship Diagram (ERD), as shown in Figure 3.2 The system utilizes Firebase Cloud Firestore as its centralized cloud-based database for storing and managing essential operational data, including user accounts, physical parts inventory, vehicle

maintenance records, transaction histories, and notification logs. The database organizes information into collections and documents with logical relationships to support real-time synchronization between the mobile application and web admin panel. This centralized architecture helps maintain data consistency, preserve historical maintenance records, and support automated low-stock monitoring and Retrieval-Augmented Generation (RAG)-driven AI text analysis.

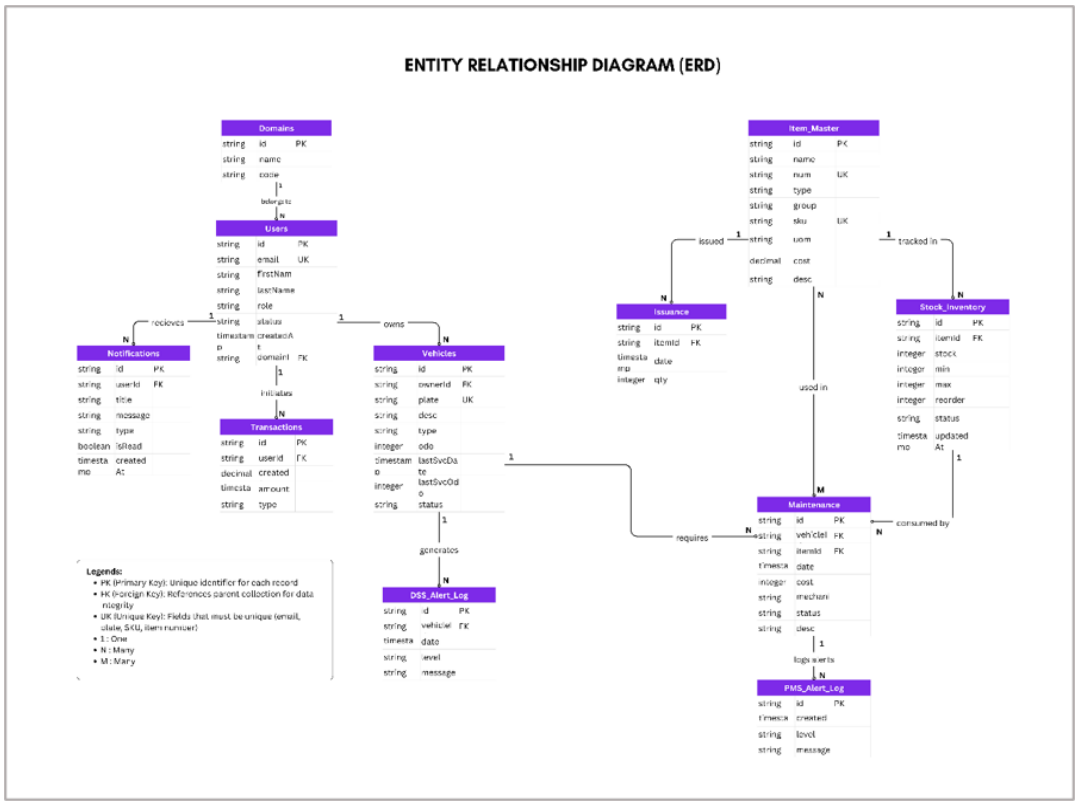


Figure 3. Entity-Relationship Diagram (ERD)

To ensure data integrity and normalized operational workflows, the relational schema directly supports the major functional features evaluated in this study:

- User Account and Role Management: The Users collection stores user account information, including names, email addresses, and assigned roles. It supports access management for administrators, staff, mechanics, and customers.

- Parts Inventory Management: The Item_Master and Stock_Inventory collections manage part details, stock quantities, minimum stock levels, and inventory status. These collections support inventory monitoring and real-time updates when staff record stock changes through the mobile application.

- Vehicle Maintenance and Transaction Tracking: The Vehicles, Maintenance, Transactions, and Issuance collections organize vehicle information, maintenance activities, transaction records, and issued materials. Their logical relationships help preserve maintenance histories and track the materials consumed during service activities.

- Automated Low-Stock Monitoring and Notifications: The Stock_Inventory, Notifications, DSS_Alert_Log, and PMS_Alert_Log collections support stock monitoring, alert generation, and notification history. These functions help identify items that require replenishment and record automated alerts.

- AI-Assisted Information Retrieval: Operational data stored in Firebase supports the system's RAG-driven AI text analysis through the backend developed using FastAPI and Uvicorn. Relevant inventory and maintenance records can be retrieved as contextual information to help the AI Assistant answer user inquiries.

- Real-Time Synchronization and Data Consistency: Firebase Cloud Firestore serves as the shared data source for the mobile application and web admin panel. Changes made by authorized users can be synchronized across connected platforms, while appropriate validation and transaction handling help reduce inconsistent records and conflicting inventory updates.

3.7.5 User Interface Design

The user interface design of the platform is divided into two operational environments engineered specifically to accommodate different user roles within the enterprise. The design structure focuses entirely on providing clear data displays and clean navigational paths, ensuring that both administrative and field users can efficiently interact with the system's features without operational confusion.

The mobile application user interface is built natively using the Flutter framework. This environment provides shop staff, mechanics, and customers with a touch-based layout optimized for immediate service updates and quick navigation. It integrates a dedicated, connected interface utilizing the device's native camera hardware to execute scanning operations. The view is structured around interactive alerts that immediately notify users regarding vehicle service updates and material status changes.

The website frontend is constructed using a standard combination of HTML, CSS, and JavaScript. This environment provides enterprise administrators and managers with a clean and clear layout designed for comprehensive data-based readings. The desktop interface functions as a centralized space for viewing data-driven summaries and managing system-wide settings, ensuring that high-level administrative functions remain completely separated from the mobile service floor activities.

3.7.6 System Communication and Interaction

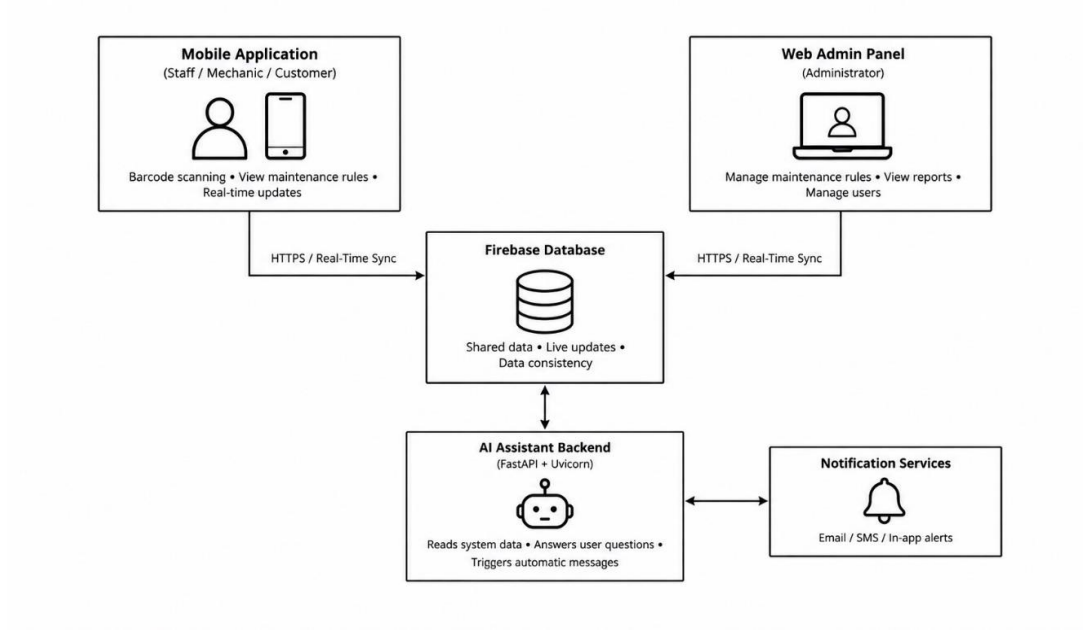


Figure 4. System Communication and Interaction

The system employs a layered communication architecture that defines how information is exchanged among the user interfaces, cloud database, backend AI service, and notification services. As illustrated in the architecture diagram, the architecture is organized into three major layers: the User Interface Layer, the Database Layer, and the Backend and Auxiliary Services Layer. This separation establishes clear communication boundaries among system components and provides a structured mechanism for processing requests, managing real-time data, integrating service frameworks, and delivering updates to users.

The User Interface Layer consists of two primary client applications: the Mobile Application, developed using Flutter, and the Web Admin Panel, engineered for administrative workstations. The Mobile Application serves staff, mechanics, and customers who perform activities such as executing barcode scanning, viewing maintenance rules, and submitting real-time updates. In contrast, the Web Admin Panel provides authorized personnel with functions for managing maintenance rules, viewing

generated reports, and executing user management. Requests originating from either interface are transmitted through secure communication channels utilizing HTTPS/Real-Time Sync connections, which connect the client applications directly to the central backend infrastructure.

The Firebase Database functions as the centralized repository and primary intermediary between the client applications and the system's backend data flows. It performs essential database-level controls, including shared data hosting, live updates, and data consistency tracking, before distributing synchronized data to the connected environments. This arrangement prevents client applications from experiencing data delays and provides a centralized point for managing and controlling application synchronization. The database subsequently communicates with the platform's external processing services, ensuring that data changes made by an admin on the web app or a mechanic scanning a material barcode on a phone are transmitted instantly across devices.

The AI Assistant Backend provides the communication pathway between the Firebase Database and the system's processing core. When an administrator or a floor technician submits an interaction or a query, the request is transmitted through the cloud data layer to the backend service, which communicates using FastAPI and Uvicorn for asynchronous data processing. The processed interaction is then returned through the system layer to the requesting user environment. The AI assistant is therefore integrated as a support mechanism within the system architecture while maintaining the FastAPI backend as the controlled intermediary for reads, system queries, and automated message triggers.

The Notification Services component communicates with the system core to

manage the automated distribution of platform-related communications. The notification service processes system events and coordinates with internal drivers to handle the delivery of email notifications, short message service (SMS) delivery, and localized in-app alerts. Through this component, the system can disseminate critical maintenance reminders, system-wide notifications, and rule-based messages to intended recipients. Centralizing notification processing within the application architecture allows communication events to be generated dynamically from database transactions while maintaining a consistent mechanism for delivering messages through multiple channels.

The communication flow of the platform follows a structured, bidirectional pathway in which user requests are securely transmitted from the user interfaces through the HTTPS network connections to the Firebase Database and the FastAPI backend service, with processed responses subsequently returned to the requesting interface. The layered structure ensures that each component has a defined communication responsibility while allowing the system to integrate real-time cloud data, automated notification dispatching, and intelligent assistant reporting. By establishing controlled service boundaries among the mobile app, web admin panel, real-time database, and FastAPI endpoints, the architecture provides a structured foundation for the stable operation of the system.

3.8 System Testing and Evaluation Procedure

System testing is a part of the software development life cycle or SDLC. This phase checks the system to confirm it meets the required standards. Torkjazi and Raz (2024) say that today's software development needs both test and evaluation or DT&E to be done by the development team. This makes sure the system meets rules. At the

time operational test and evaluation or OT&E is carried out by real users. This helps see how the system performs in the world. Following this idea the researchers used two evaluation methods for the web-based and mobile application. One method was for system specialists. The other was for users. The system testing and evaluation process showed that the system works as expected and matches the design. A full introduction and demonstration of the system, to users helped assess their computer literacy. It also helped collect feedback on how acceptable users found the system.

3.8.1 Testing Plan

The testing plan follows an order that is split into two main evaluation phases to look at different parts of the system. The first phase is the Technical Evaluation Phase. During the Technical Evaluation Phase, system experts, developers and IT specialists check the platforms compliance, core responsiveness, internal security settings, overall performance and architectural reliability.

The second activity is the Operational Evaluation Phase. The Operational Evaluation Phase acts as the Operational Test and Evaluation (OT&E). During the Operational Evaluation Phase, the actual managers, administrators, maintenance staff and customers of JA Noble Enterprise Inc. Work with the software through guided testing scenarios to check the software real-world system performance, user-facing outputs and overall usability, before final deployment.

3.8.2 Testing Data

The testing data shows the datasets, the inputs that are used in operations and the information parameters that the researchers used to run the test cases in exact system conditions. Of using random text that doesn't make sense the testing data has real entries

that fit the limits of the multi-platform system to show how things work in real life for cars. For the user accounts situation, the testing data has things like user registration fields, different lengths for customer names and correct login details. To make sure the system's checks and handling of errors work well some incorrect data was added to the system. This included emails that are not written right like missing the part after the @ sign and passwords that're do not correct, on purpose.

3.8.3 Testing Procedure

The testing process followed an approach. This organized approach combined step-by-step checks by developers with user feedback to fully examine the software. During development the research team used an Agile Scrum method. They tested parts of the software in small sections known as sprints. These parts include QR code scanning features; they also contain logic for detecting stock levels in the DSS system. The researchers see that they provide backend AI functions for embedding data. I believe these backend AI functions help embed data. The backend AI functions are highlighted in Dela Cruz et al., 2023. Each new part of the software has to go through required checks These checks looked at how the software worked across platforms, how well the software responded to user actions, in the background and whether the database structure remained correct.

Once a stable version of the software was ready called a Release Candidate it was tested in a real-world setting. The test took place at the Caltex service station of JA Noble Enterprise Inc. In San Pedro, Laguna. This location served as the working environment for the trial. A group of 144 people took part in the test. The group included three administrators, sixteen shop staff members and 125 regular customers. All participants used either computers or Samsung Android smartphones to interact

with the system. The researchers led these users through tasks using black box testing techniques. This meant they only focused on what the users saw and experienced like how barcodes scanned or how quickly notifications appeared without asking them to look at the internal code (Myers, Sandler, & Badgett, 2024). To measure how well the system performed participants completed an evaluation form after using the software. The form was designed based on the ISO/IEC 25010 standard, which provides guidelines for assessing software quality. The results from the forms were. Analyzed mathematically. This analysis provided measurable data. The test confirmed that the software met the business needs of the enterprise. The software performed reliably in a real-life scenario.

3.8.4 Software Testing Method

Black Box Testing is a way to test software when the inside of the system is hidden from the tester. In Black Box Testing the tester does not look at the code or design. Instead, the tester checks how the application works by sending data into it and comparing the results with what should happen. According to Myers, Sandler and Badgett (2024) Black Box Testing works well to prove that the software does what users expect even if the inner workings are unknown.

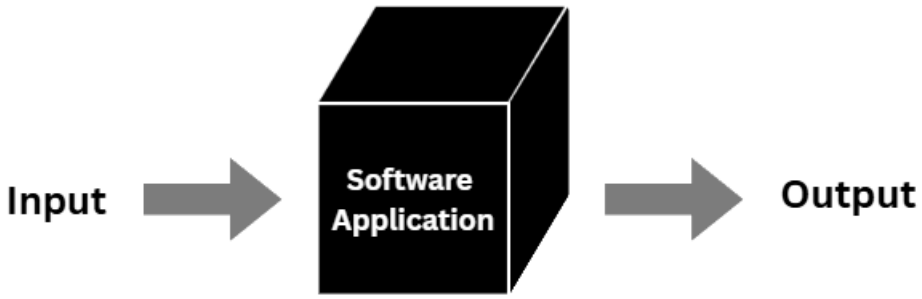


Figure 5. Black Box Testing

The Black-Box Testing procedure used by the researchers to evaluate the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise INC. platform. This testing approach was applied to check how the system responds to various user inputs and scenarios related to fleet operations and resource monitoring.

3.8.5 Test Cases and Test Scenarios

Test cases were documented in a structured format containing the feature or requirement being tested, test conditions, expected results, actual results, and test status. The detailed test cases and scenarios (TC_01 through TC_05), which map directly to the system requirements of the Integrated Vehicle Maintenance and Inventory Management System including user account management, vehicle maintenance tracking, barcode scanning, and administrative overview controls are presented in the APPENDIX.

Five test cases were developed based on the functional requirements of the system: (TC_01) Core User Account and Authentication Functionalities, (TC_02) Customer Dashboard and Service Booking Management, (TC_03) Staff Dashboard and Vehicle Maintenance Operations, (TC_04) Barcode and QR Code Scanning Functionality, and (TC_05) Comprehensive Administrator Dashboard and Report Generation. Each test case included preconditions, step-by-step procedures, expected results, and pass/fail indicators.

3.8.6 Testing Instrument

A system testing checklist and a test case form were used to record and check how well the system worked. The instrument was designed to note the features or rules

being tested the test conditions, the expected result, the actual result and the test status. The checklist covered the functions of the system such as vehicle maintenance management, mobile inventory that scans QR or barcodes decision support, for stock restocking and preventive maintenance Retrieval-Augmented Generation (RAG) tasks, data sync and report creation. The testing instrument was used during Blackbox testing, where the researchers checked the system by giving it data and seeing the output without looking at the source code. Each test case was run under the defined conditions the observed result was compared with the expected result, and the test status was recorded to show whether the feature passed or failed based on the set rules. Black-box testing is a way to check software when the tester does not know the workings of the system. Blackbox testing is a method to confirm that software meets user needs no matter how the inside of the program is built.

3.8.7 Validation of Testing Instrument

The main tool that the researchers use to record and check how the system works is a Test Case Sheet. The Test Case Sheet is made for the Integrated Vehicle Maintenance and Inventory Management System with Decision Support, at JA Noble Enterprise Inc. This Test Case Sheet carefully lays out each scenario so that the researchers can confirm the application behavior as its technical design says.

As you can see in the layout of the Test Case Sheet, the Test Case Sheet records metadata and the steps to run the test, such as:

Thesis Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.

Tester Name:		**Date Tested:**		**Test Case (Pass / Fail):**	

PRECONDITIONS:	• The system tester should know how to handle and use basic functionalities of the system. • The system tester should know how to navigate the system. • The Laptop/Smartphone device must have access to the internet.
INSTRUCTIONS:	• Testers must indicate his/her name and basic information before rating the testing sheet. • Testers should rate the functionality of the system whether it is passed or failed with the expected results given. • Testers should indicate the actual result if they fail to the given expected results. • The system tester should indicate "non-existed" to the actual result column if the stated steps details are not present in the application.

USER ACCOUNT TEST CASE SCENARIO

Test Case ID:	TC_01	**Test Case Description:**	Test the account registration, login, and forgot password.

Step No.	**Step Details**	**Expected Results**	**Actual Results**	**Status**	
				Pass	**Fail**
1	Navigate to the Customer Registration page .	Registration form opens.			
2	Enter the required information and click the "Create Account" button.	The account is created successfully and redirects to the Login page.			

Figure 6. Validation of Testing Instrument

The standardized Test Case Sheet deployed to systematically evaluate and document the performance of the system. The instrument is divided into key operational sections to ensure objective data collection. The top section captures critical metadata, including Tester Information and Metadata (Tester Name, Date Tested, and overall Pass/Fail rating). Below this, the Preconditions block outlines the baseline requirements, ensuring testers have sufficient system knowledge and that their laptop or smartphone has an active internet connection. The Instructions section guides testers on rating rules, specifying how to log findings and when to mark a feature as "non-existed." The core of the instrument features the Test Case Scenario Details, which isolates the specific module under review (such as TC_01: User Account Test Case Scenario). Finally, the performance is recorded using an Evaluation Matrix, a tabular tracking system that matches Step Numbers, Step Details, and Expected Results with the Actual Results and a binary Pass or Fail status.

3.8.8 Interpretation of Test Results

Test results were recorded as pass or fail for each individual step across all test cases. A feature was considered fully functional when all its test steps received a pass status. Failed steps were documented with the actual result observed and compared against the expected result to identify discrepancies or required debugging. The overall system pass rate was computed by dividing the total number of passed test steps by the total number of test steps and expressing it as a percentage. A module or feature was considered functionally acceptable when it achieved a pass rate of 80 percent or higher across all test cases.

3.8.9 System User Evaluation

The system user evaluation will be carried out to formally find out how the intended users see the integrated platform. The researchers look at software metrics and how acceptable the platform feels overall. To reach the project goals the researchers will use a mix of usability testing and user questionnaires. These tools are designed for user roles: administrators, workshop staff, mechanics and customers. During hands-on usability testing respondents will get to use the features of the web panel and the mobile app. They can log inventory. Scan barcodes or QR codes in real time. After using the software user questionnaires will be given. These questionnaires will collect numbers that show how users feel about usability, functionality, efficiency, accessibility and overall satisfaction. This evaluation method makes sure the data that the researchers gather truly shows how well the system works and how ready users are before launch it officially.

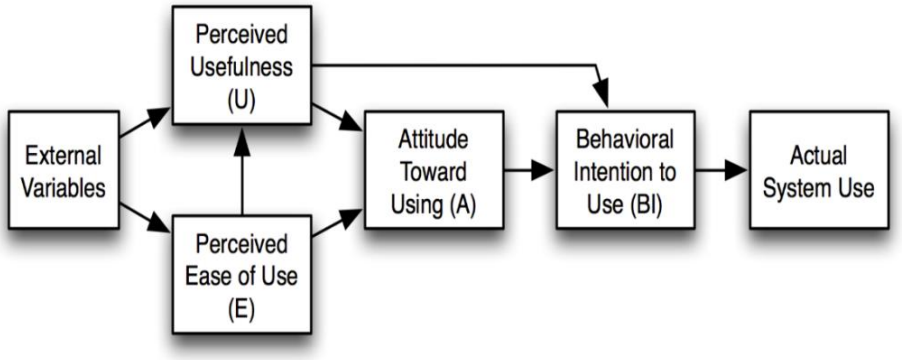


Figure 7. Technology Acceptance Model (TAM)

The researcher designed questionnaires based on TAM constructs to gather feedback from users after system deployment. This feedback provides enhancements to improve user experience, ensuring higher acceptance and engagement with the platform.

3.8.10 User Evaluation Instrument

The user evaluation uses a questionnaire that mixes ISO/IEC 25010 quality metrics with the Technology Acceptance Model (TAM). It measures software quality and how acceptable the system is, for administrators, workshop staff, mechanics and customers. The instrument looks at Perceived Usefulness and Perceived Ease of Use. It is. Approved by the research adviser before starting using it.

Table 5. Likert Scale and Interpretation

Scale	Mean Range	Verbal Interpretation
5	4.21 - 5.00	Highly Acceptable
4	3.41 - 4.20	Acceptable
3	2.61 - 3.40	Moderately Acceptable
2	1.81 - 2.60	Unacceptable
1	1.00 - 1.80	Highly Unacceptable

The Likert Scale was used by the researchers to measure the level of satisfaction of the respondents. This table shows the scale, range and verbal interpretation of the Likert Scale Level of Satisfaction from Highly Unacceptable to Highly Acceptable. According to Al-Refai et al. Using a weighted mean with intervals provides a way to classify Likert-scale data, which helps reduce personal bias. This organized approach is supported by Simera (2024) who says that giving each description a range allows us to analyze responses in an objective way. The organized approach gives clarity and the range assigned to each description provides a clear way to measure and compare results. This is especially useful in studies that evaluate software, were clear and fair interpretation of feedback's important. In this study a weighted mean ranging from 4.21 to 5.00 is interpreted as Highly Acceptable while 3.41 to 4.20 is Acceptable and 2.61 to 3.40 is Moderately Acceptable. Scores from 1.81 to 2.60 are classified as Unacceptable and 1.00 to 1.80 as Highly Unacceptable giving a clear metric to assess whether the system meets the operational needs of the stakeholders.

3.8.11 Instrument Source and Validation

The tool used in this study comes from the known models called the Technology Acceptance Model or TAM and the ISO/IEC 25010 quality standards that were talked about earlier. The questions were changed to match the jobs of the Integrated Vehicle Maintenance and Inventory Management System. For the checking process the survey was carefully looked at. Then checked, improved and officially agreed on by the research adviser. This was done to make sure the questions were clear worked well and measured what they were supposed to measure before giving them to the admin, staff and customers.

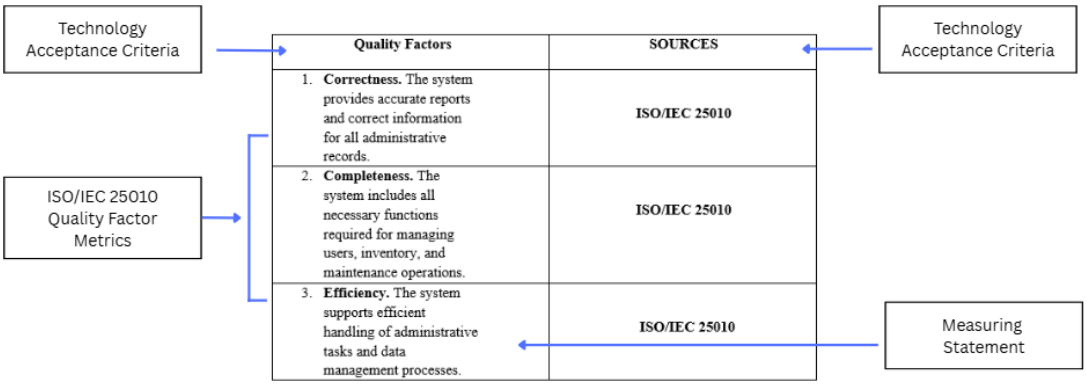


Figure 8. The Structural Format of the Validation.

The tool the researchers used during the study evaluation. It contains the technology acceptance criteria, a measuring statement, and the source/reference. The researcher builds a questionnaire for user evaluation based on Technology Acceptance Model (TAM) as a category and ISO/IEC 25010 quality metrics criteria. The researcher's research adviser validated the questionnaire before being distributed to the target respondents for user evaluation.

3.8.12 User Evaluation Procedure

The user evaluation will be done carefully by the researchers with all the groups of people who're the target respondents. These groups include administrators, people who work at workshops, mechanics and customers. The evaluation will happen in a three-step process that has been planned. First the platform that was created will be shown to the people who are being evaluated. This will be done with a presentation and a tour of the software. The researchers will lead this. Will show the main parts of the software. These parts include a web-based control panel for administrator's tools that can scan barcodes and QR codes in time automatic records for maintenance work and talks with an AI assistant. Second, the people being evaluated will get access to the system. They will be given test areas where they can try the software on their own.

They will use this time to check how useful, clear and easy to learn the website and mobile parts of the software are. They will try these on devices and web browsers. Third, after they finish testing, their opinions will be gathered away. This will be done using a questionnaire that has been tested and is reliable. The questionnaire will be sent online using a Google Forms link. In places where there is no internet the questionnaire will be given out in paper form. This way all the information will be collected completely. Kept safe for later analysis.

3.8.13 Ethical Considerations

Ethical considerations were strictly observed throughout the research process. The evaluation instrument will be administered with adherence to official protocols and ethical research standards. First the researchers will obtain clearance and written permission from the management of JA Noble Enterprise Inc. To legally carry out the evaluation study at their facility. Once approval is granted the validated questionnaires, incorporating the Technology Acceptance Model (TAM) and ISO/IEC 25010 metrics, will be given to the target respondents, including administrators, staff and customers. The researchers will personally oversee the distribution process. They will clearly explain the purpose of the survey. Assure participants that their personal information and assessment scores will remain strictly confidential. The questionnaire will be delivered in printed form with paper sheets provided to each respondent. Immediately after participants complete the survey the researchers will collect all the forms. Compile the data to ensure a 100% response rate, making sure no fields are left blank before statistical analysis begins.

3.9 Statistical Treatment

To analyze the data collected from the system evaluation survey the researchers applied statistical tools. These tools were chosen to make sure the results were accurate and reliable. As Bhandari (2022) points out picking the statistical methods is important. It helps ensure that conclusions are valid and not just due to chance. Instead, the findings should reflect patterns in the data. Creswell & Creswell (2022) also highlight that using measurable numerical data allows researchers to describe results in a way. This helps make sure the findings truly match what is happening in the world. The study used the formulas:

The weighted mean, commonly used for Likert-scale evaluation:

$$\bar{x}_w = \frac{\sum fx}{N}$$

Equation 2. *Weighted Mean*

Where:

- $\bar{x}_w$ = Weighted Mean
- f = Frequency of responses
- x = Like Scale Weight
- N = Total number of respondents

Equation 3*. Standard Deviation*

$$SD = \sqrt{\frac{\sum(x-\bar{x})^2}{N-1}}$$

Where:

- SD = Standard Deviation
- x = Individual response Mean
- $\bar{x}$ = Mean
- N = Number of respondents

To analyze the data gathered from the system evaluation survey the researchers

will utilize statistical tools to ensure accuracy and reliability. The Weighted Mean was computed to measure the tendency and provide an overall evaluation for each survey item across these stakeholder groups by multiplying the frequency of responses by their respective Likert scale weights. Lastly the Standard Deviation will be applied to determine the variability of the responses and assess the consistency of the feedback, around the computed mean ensuring a mixed sampling approach that enhances the validity of the final evaluation.

CHAPTER IV

RESULT AND DISCUSSION

This chapter presents all data which has been collected during the evaluation process of the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Ja Noble Enterprise Inc. It includes the findings from the conducted black box testing and user acceptance evaluation. The evaluation used the responses provided by the Admin, Staff, Customers, and IT Experts who demonstrated the system.

4.1 System Architecture and Development Results

This section presents the system architecture and development results of the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc. It describes how the existing manual processes for vehicle maintenance, inventory management, service booking, and preventive maintenance scheduling were transformed into an integrated web and mobile application. The following sub sections discuss the system overview, system architecture, hardware and software requirements, database structure, user-interface design and system modules, and functional and non-functional requirements that define the completed system and its operation.

4.1.1 System Architecture

This part shows the overall structural design of the system and explains how its features and functions work together to support the system's tasks. It lists the core layers of the application, the cloud database and services, the backend servers, user devices and the outside AI service that helps run the system. The architecture shows how the

web and mobile apps talk to the backend and cloud services to manage data in one place, provide system features and add AI-powered help.

The system is a cloud-based service and inventory management platform made for JA Noble Enterprise Inc. It has three parts: a mobile app built with Flutter; a web app built with Flask and an AI backend built with Python FastAPI. These parts use Firebase as the cloud platform with Cloud Firestore as the main database. This setup lets the web and mobile apps share data. Stay in sync across the system.

The system uses role-based access control for three types of users: Administrator, Staff and Customer. The Administrator does tasks and setups using the Flask-based web app. This app is made to be used on a desktop or laptop. Staff and Customers use the Flutter-based app to handle daily tasks like managing service work, checking vehicle details, watching maintenance schedules and handling bookings. Both the web and mobile apps have dashboards that match each user's role. This feature allows users to move between platforms whenever they want.

The system is split into three parts. The first part is the presentation layer, which contains the app and the web app. These are the parts where users see and interact with the system. The second part is the application and intelligence layer, which contains the web app and the AI backend. These parts are still unfinished. These parts handle the logic and smart features of the system. The third part is the data layer, which manages all the information stored and used by the system. Each layer works together to make sure everything runs smoothly. parts handle requests, run the systems logic, do AI tasks and help make reports. The third is the cloud services and data layer which include Firebase services like Firebase Authentication and Cloud Firestore. These provide user login. Store all the data in the cloud.

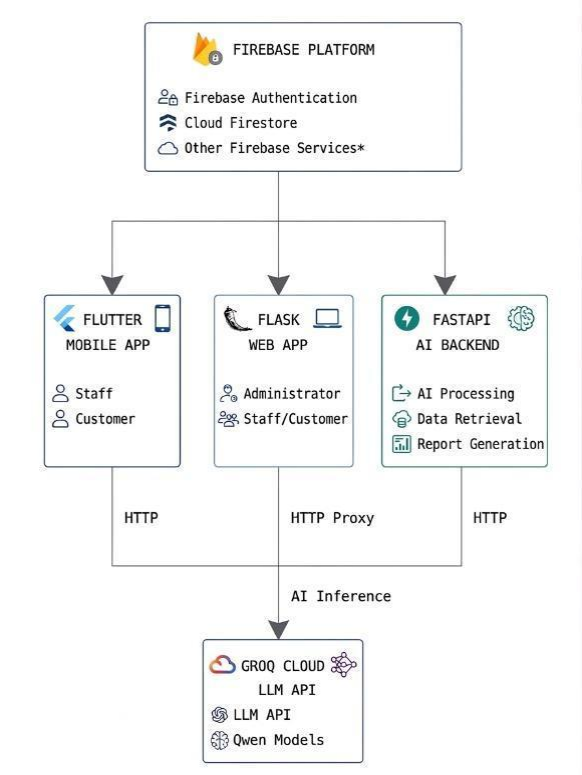


Figure 9. System Architecture Diagram

Image generated using Claude (September 2026) with the prompt: "A clean, modern vector system architecture diagram on a white background, featuring minimalist corporate tech styling, simple flat icons, light grey cards, subtle shadows, and a blue and grey color scheme. Use clearly labeled components, connecting lines, and arrows to illustrate the Firebase platform, Flutter mobile app, Flask web app, FastAPI AI backend, and Groq Cloud LLM API in a structured layout."

The communication flow between the main parts of the system that was developed. Firebase acts as the cloud platform that connects the mobile application, the web application and the FastAPI AI backend. AI-related requests from the application and the web application are handled by the FastAPI backend and then sent to Groq Cloud for AI inference using the Qwen models that are set up. This design makes it possible to have data that's easy to access from one place while also helping the system, with AI-powered features.

4.1.2 Hardware and Software Requirements

This section presents the hardware and software resources needed to run the

completed system. The requirements are split into hardware requirements, software requirements and deployed system resources. The minimum requirements show the specs that the system needs to work. The recommended requirements suggest performance and easier use. The deployed system requirements explain the cloud services and application parts that are used to keep the system running

Table 6. Hardware Requirements

Component	Minimum Requirement	Recommended Requirement	Intended User
Mobile Device	▫ Android 11 or later ▫ 4 GB RAM ▫ quad-core processor 32 GB storage	▫ Android 13 or later ▫ 6 GB RAM or higher ▫ octa-core processor ▫ 64 GB storage or higher	Staff / Customer
Desktop / Laptop	▫ Dual-core processor ▫ 4 GB RAM ▫ 128 GB storage ▫ 1366 × 768 display	▫ Intel Core i5/AMD Ryzen 5 or equivalent ▫ 8 GB RAM or higher 256 GB SSD or higher ▫ Full HD display	Administrator
Internet Connection	Stable connection with at least 5 Mbps	Stable connection of 10 Mbps or higher	All users

The hardware that is needed to use and run the system. Staff and Customers mostly need a device that can run the Flutter-based mobile application and can scan QR codes or barcodes using its camera. Administrators mostly use a desktop or laptop computer to access the web-based application. Everyone needs an internet connection because the system uses cloud-based services for logging in getting data and using artificial intelligence features. The suggested specifications give power and space for the system to work better.

Table 7. Software Requirements

Software / Service	Minimum / Required Version	Recommended / Implemented Configuration	Intended User
Operating System – Mobile	Android 11 or later	Android 13 or later	Staff / Customer
Web Browser	Modern browser supporting HTML5, CSS3, and JavaScript	Latest version of Google Chrome, Microsoft Edge, or equivalent	Administrator
Internet Access	Stable connection with at least 5 Mbps	Stable connection of 10 Mbps or higher	All users

The software that users need to use the system. Staff and customers must have an Android operating system to run the mobile app. Administrators on the hand need a modern web browser to use the web application. Users do not need to install any backend frameworks, databases or AI libraries on their devices because all these parts are set up and handled by the systems backend infrastructure.

Table 8. Deployed System Software and Cloud Requirements

Software / Service	Implemented Configuration	Function in the Completed System
Firebase Authentication	Firebase Authentication	Manages user authentication and access to the system
Cloud Firestore	Firebase Cloud Firestore	Stores system data such as users, vehicles, bookings, maintenance records, and inventory
Flask Web Application	Python Flask	Provides the web-based system interface and web application functions
FastAPI AI Backend	FastAPI 0.115.0 with Uvicorn 0.30.6	Provides the AI backend, API endpoints, data processing, and AI-related functions
Firebase Admin SDK	Firebase Admin SDK 6.5.0	Allow the FastAPI backend to securely access Firebase services and Firestore data
Groq Cloud	Groq API	Provides access to the

Software / Service	Implemented Configuration	Function in the Completed System
		large-language-model service used by the AI assistant
Qwn LLM Model	Qwen 3.6 27B	Process natural-language requests and generates AI-assisted responses
Pydantic	`Pydantic ≥2.11.7`	Validates and structures data used by the FastAPI backend
ReportLab	Python ReportLab	Generates PDF reports
Render	Cloud deployment platform	Hosts the deployed application/backend services
Firebase Cloud Messaging (FCM)	Firebase Cloud Messaging	Delivers mobile/web push notifications and alerts, if implemented
Cloud Storage	Firebase Cloud Storage	Stores system files or uploaded resources, if implemented

The software and cloud services that support the deployed system. Firebase Authentication and Cloud Firestore provide centralized authentication and data storage, while the Flask web application provides a web-based interface. The FastAPI AI backend handles AI-related processing and communicates with Firebase through the Firebase Admin SDK to retrieve authorized system data. Groq Cloud provides an external large-language-model service, with the Qwen3.6 27B model used to process natural-language requests. Render provides the cloud hosting environment for the deployed Flask and FastAPI components. Together, these resources support the operation of the completed system without requiring users to install backend technologies on their own devices.

4.1.3 Database Structure

This section presents the database structure of the developed system, including

its major entities, relationships, and data components. It shows how these elements are organized to support the system's core functions, such as vehicle maintenance, service booking, inventory management, and user management.

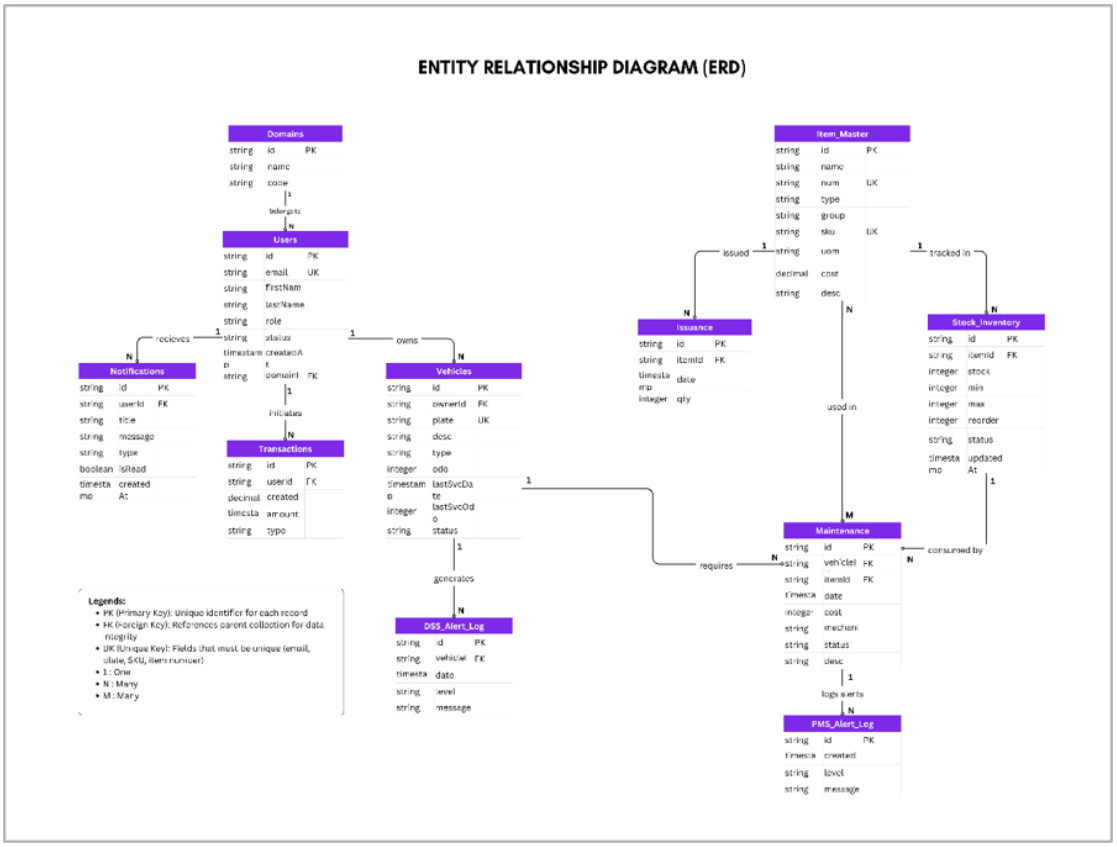


Figure 10. Database Schema

This figure presents database schema of the developed system, illustrating the resulting database structure and the relationships among its major data entities. The database consists of entities for Users, Domains, Vehicles, Transactions, Notifications, Item Master, Stock Inventory, Issuance, Maintenance, DSS Alert Log, and PMS Alert Log. These entities contain the data required to support the different functions of the developed system.

The Users entity serves as a central component by storing user information and connecting users to their assigned domains, vehicles, notifications, and transactions.

The Vehicles entity stores vehicle information and is connected to Maintenance and DSS Alert Log to support vehicle maintenance monitoring and preventive maintenance scheduling. For inventory management, Item Master contains the details of inventory items, while Stock Inventory, Issuance, and Maintenance support the tracking of available stocks and materials used or issued during maintenance activities.

The relationships among these entities allow related information to be connected and retrieved across the system. Wherein a user may own multiple vehicles, inventory items may be tracked through stock records and issuances, and vehicle maintenance records may reference the materials used during a service. The DSS Alert Log and PMS Alert Log store alerts generated for stock replenishment and preventive maintenance scheduling, respectively. The ERD also identifies the Primary Keys (PK), Foreign Keys (FK), and Unique Keys (UK) used to maintain data integrity and establish relationships among the entities.

4.1.4 User-Interface Design and System Modules

The user-interface design of the Integrated Vehicle Maintenance and Inventory Management System (IVMIMS) is organized according to the roles of administrators, staff, and customers of JA Noble Enterprise Inc. The web-based administrative dashboard allows administrators to manage user accounts, vehicle maintenance records, service bookings, inventory, and reports. The staff mobile application provides access to service tasks, customer and vehicle information, and maintenance records. Meanwhile, the customer mobile application allows customers to manage their vehicle information, book services, view maintenance history, receive notifications, and interact with the AI assistant for system-related inquiries. Each interface is designed to provide users with the functions and information relevant to their responsibilities,

making the system easier to navigate and use.

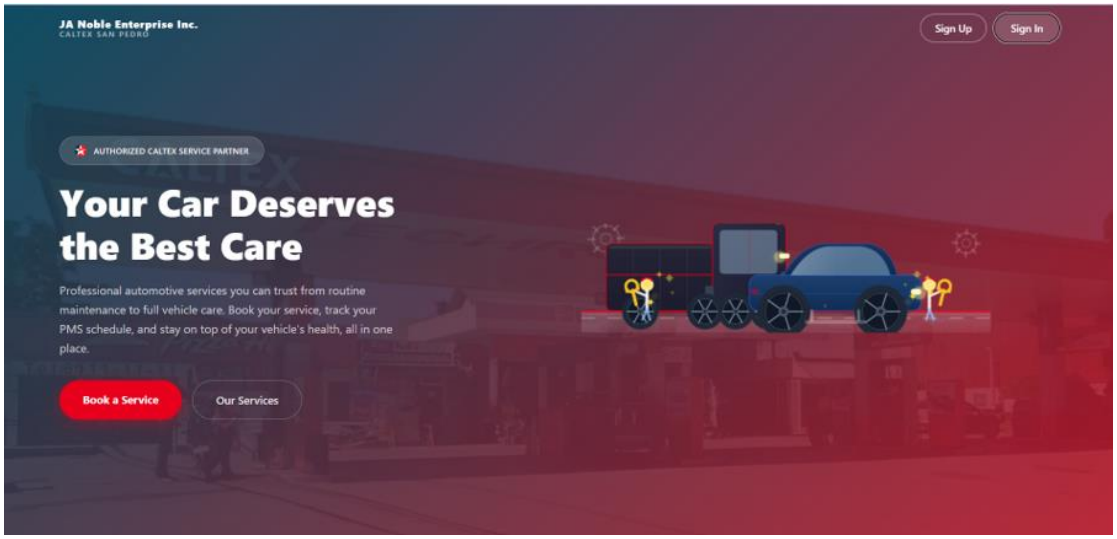
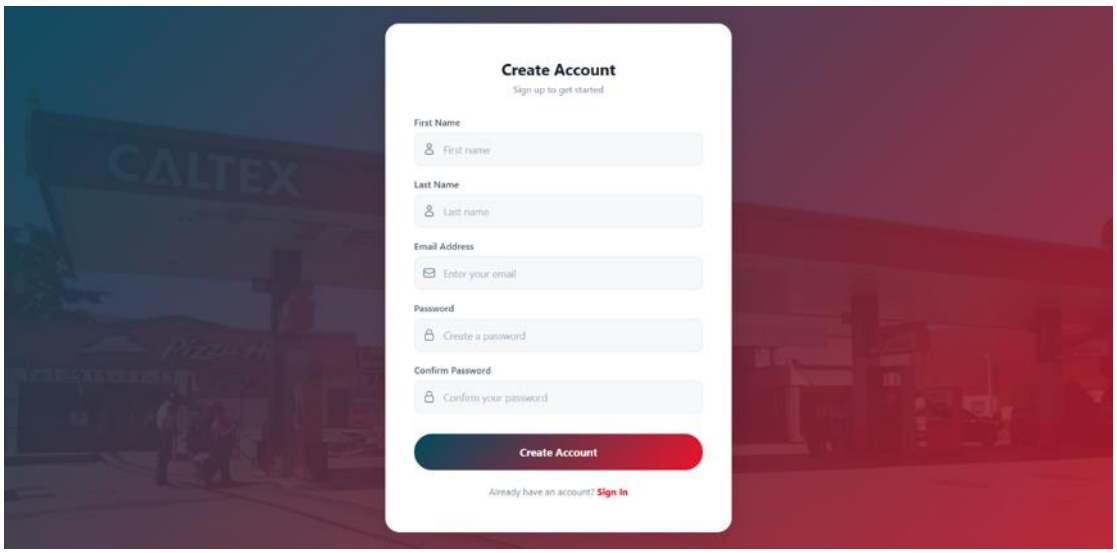


Figure 11. System Website Landing Page

Figure 11 presents the system website landing page where it displays the company information, Sign Up and Sign In options, service information, and booking features. The page also highlights the company's services and benefits, promotes the AutoPro mobile application through a QR code, and provides contact, navigation, and legal information in the footer.



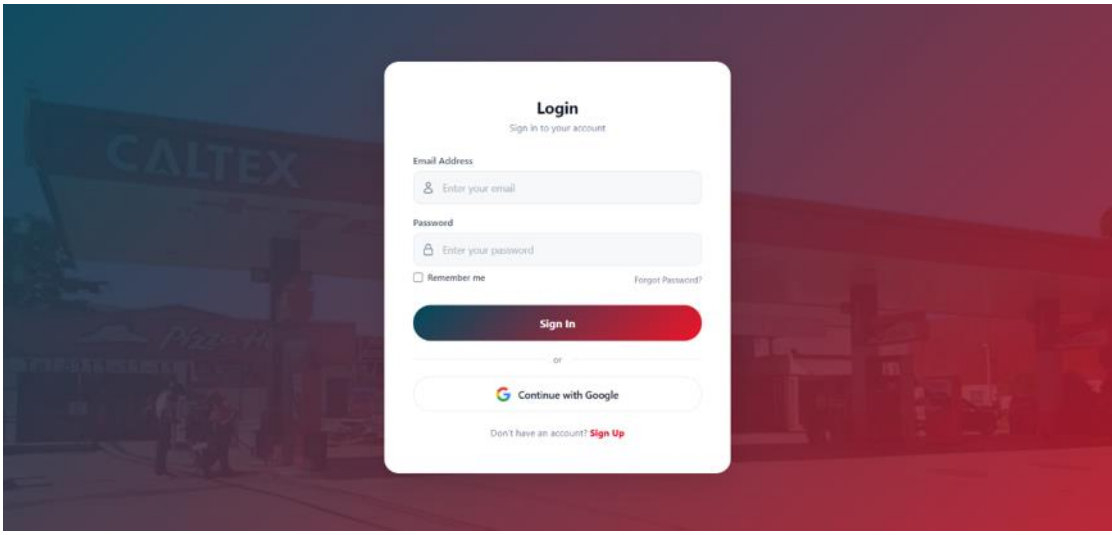


Figure 12. Website Login and Registration Page

Figure 12 shows the Website Login and Registration Page, where users can create an account or sign in to access the system. It allows authorized users to securely access the features and services available on the website.

4.1.4.1 Administrator Web Interfaces

The administrator web interface serves as the main platform for managing the operations of JA Noble Enterprise Inc. Administrators can manage user accounts, customer and vehicle records, service bookings, maintenance records, inventory, and system reports. The dashboard also provides access to the AI assistant for retrieving system information and assisting with administrative tasks. Access to administrative functions is restricted according to user roles to help protect sensitive business information and ensure that only authorized users can perform administrative operations.

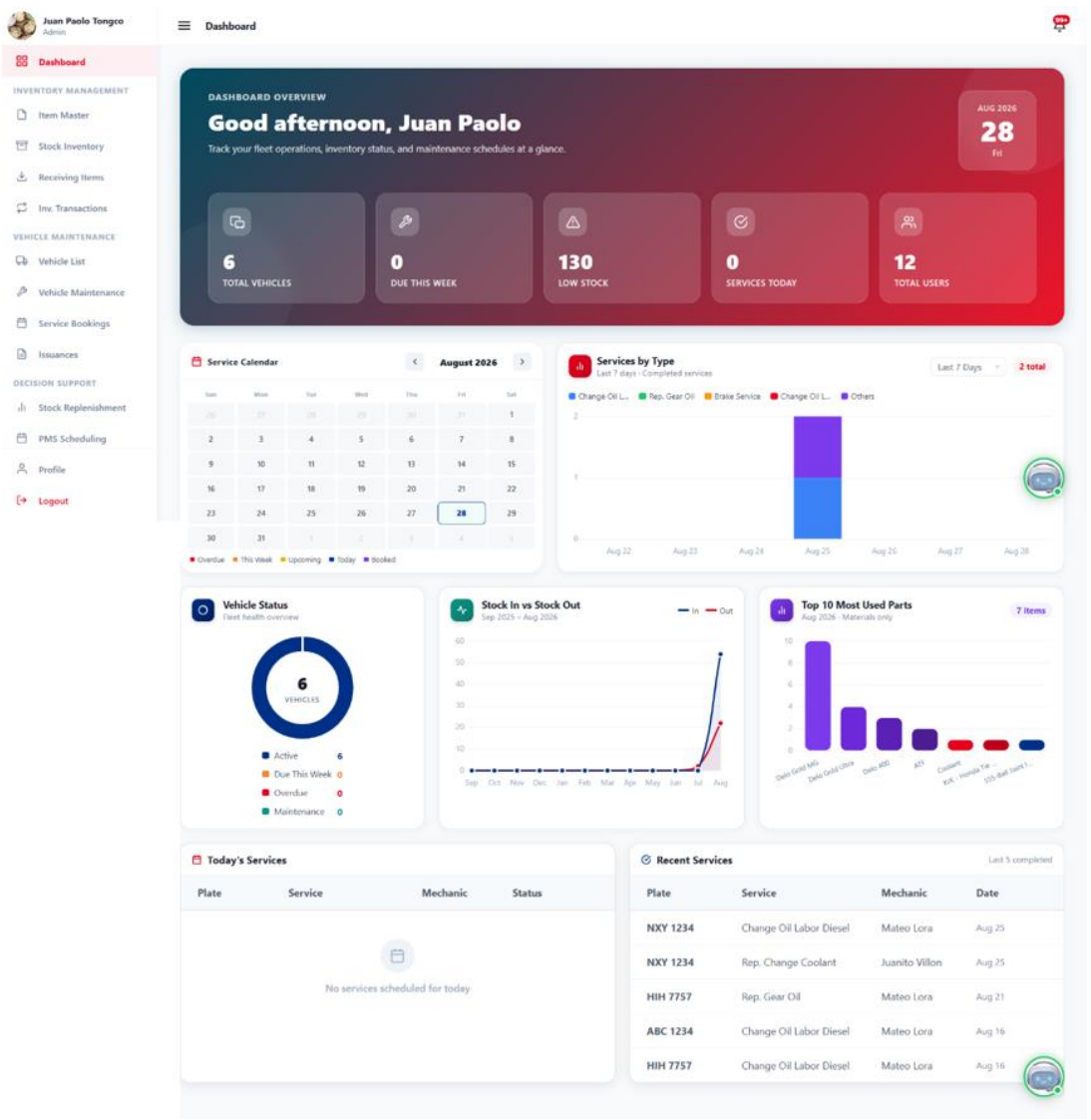


Figure 13. Admin Dashboard Page

Figure 13 shows the Admin Dashboard Page, which provides an overview of the system's operations. It displays key information and graphs such as total vehicles, services, stock levels, and users, along with service schedules, vehicle status, inventory data, frequently used parts, and recent services.

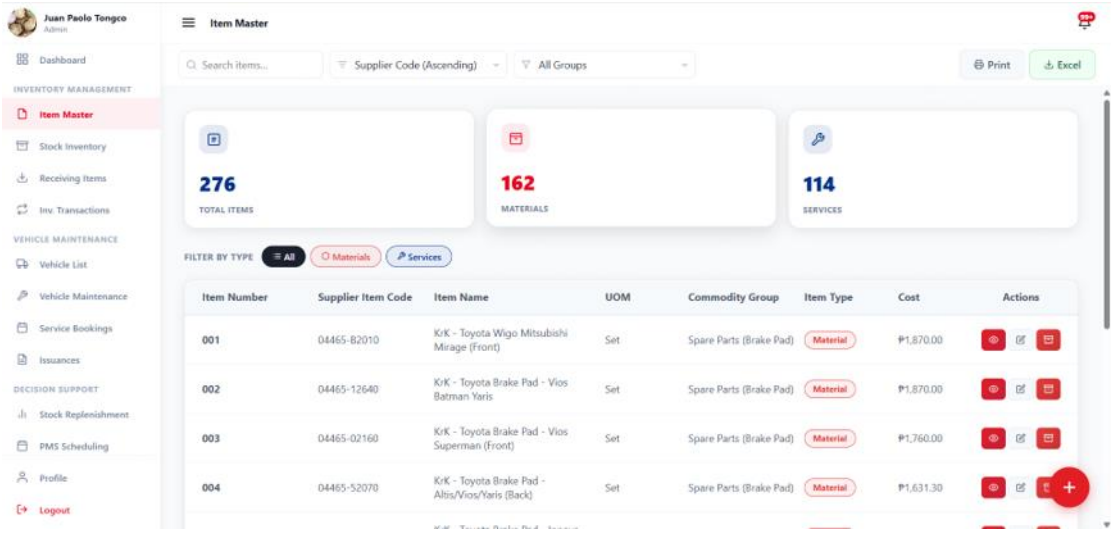


Figure 14. Item Master Page

Figure 14 shows the Item Master Page, where the administrator can manage inventory items, including materials and services. It provides options to search, filter, add, view, edit, and delete item records.

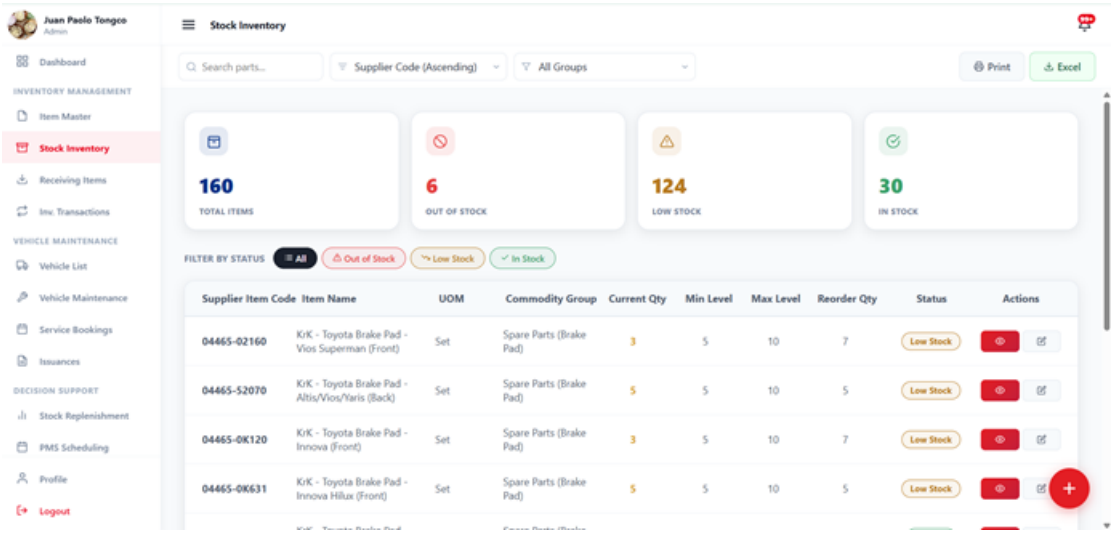


Figure 15. Stock Inventory Page

Figure 15 shows the Stock Inventory Page, which displays the current inventory levels of materials. It allows the administrator to monitor stock status and manage inventory records.

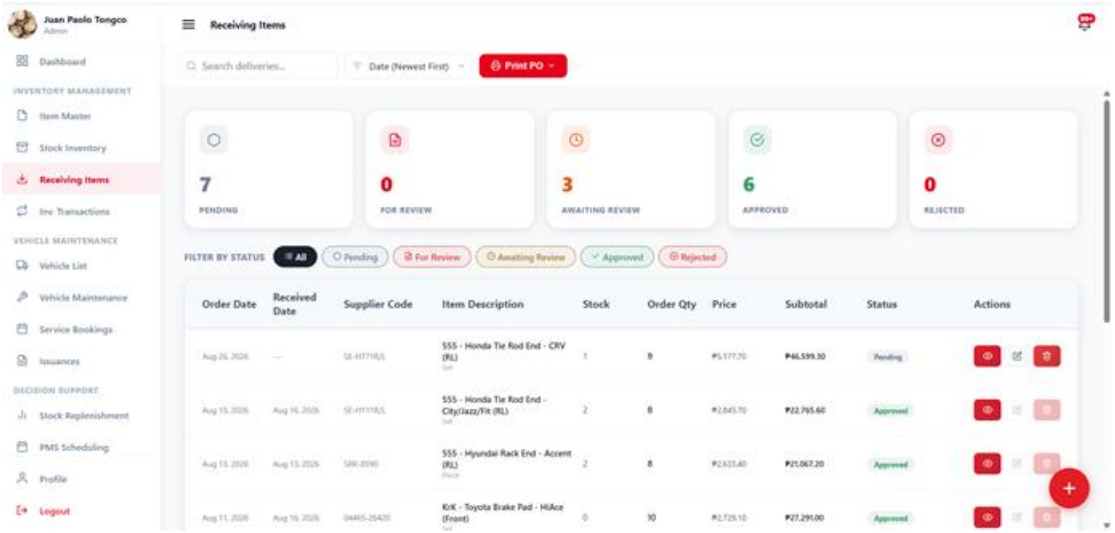


Figure 16. Purchase Orders Page

Figure 16 shows the Purchase Order Page, where the administrator can manage and monitor incoming inventory items from suppliers. It displays the status of each receiving transaction.

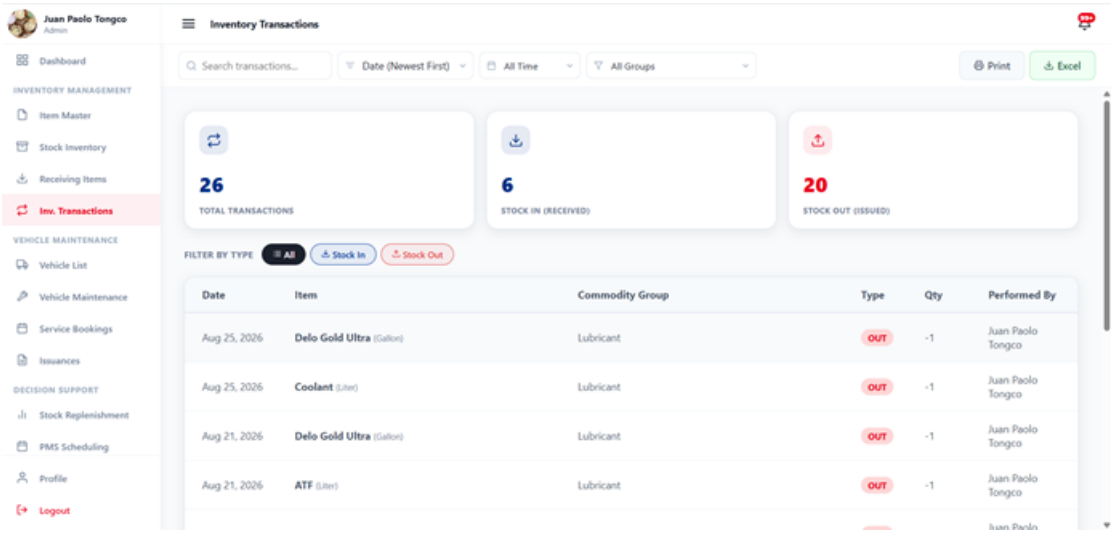


Figure 17. Inventory Transactions Page

Figure 17 shows the Inventory Transactions Page, which records stock-in and stock-out transactions. It allows the administrator to monitor the movement and history of inventory items.

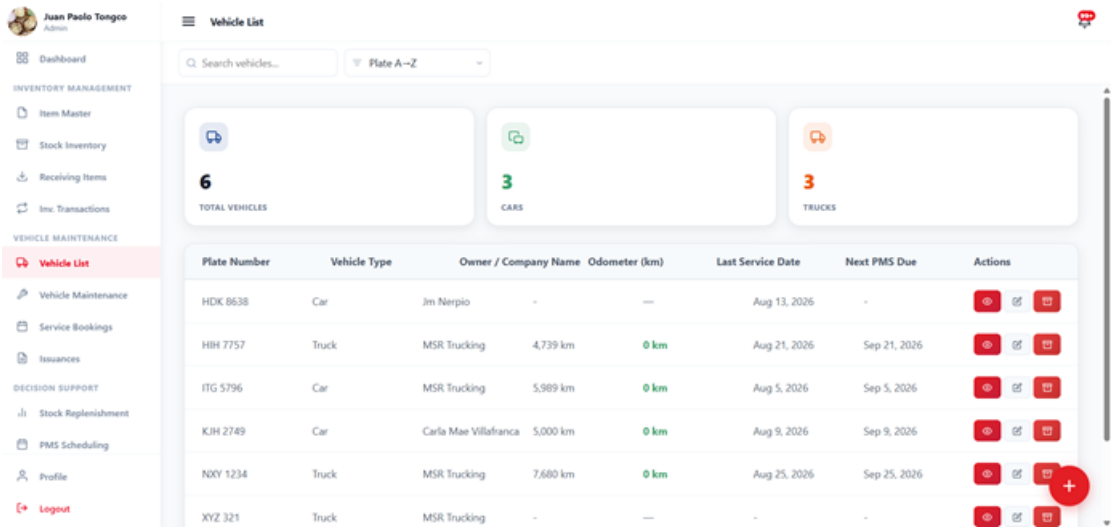


Figure 18. Vehicle List Page

Figure 18 shows the Vehicle List Page, where the administrator can manage registered vehicle records. It displays vehicle information and allows viewing, editing, and managing vehicle details.

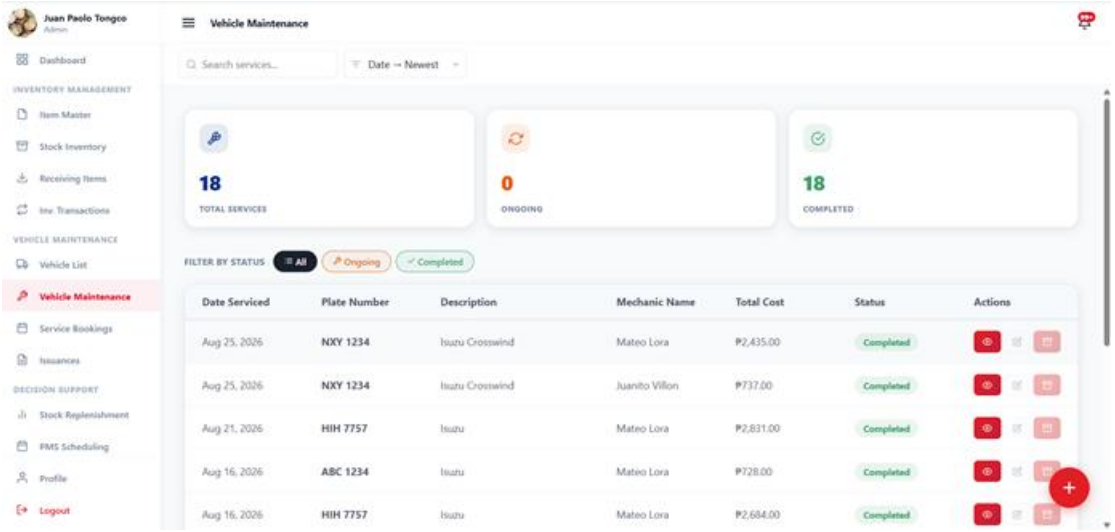


Figure 19. Vehicle Maintenance Page

Figure 19 shows the Vehicle Maintenance Page, which displays maintenance records and their status. It allows the administrator to monitor ongoing and completed vehicle maintenance services.

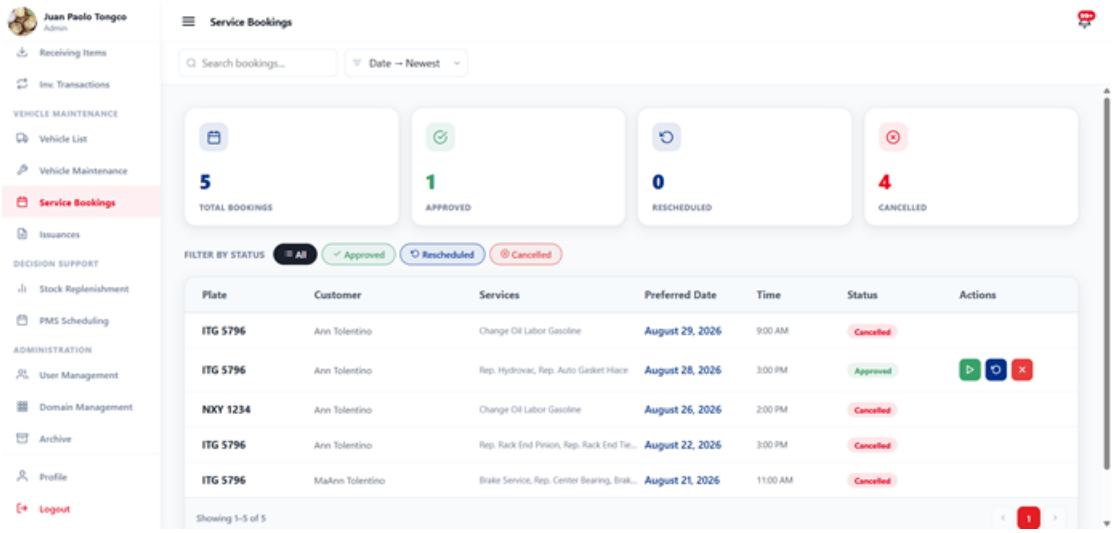


Figure 20. Service Bookings Page

Figure 20 shows the Service Bookings Page, where the administrator can manage customer service appointments. It displays booking details and their corresponding status.

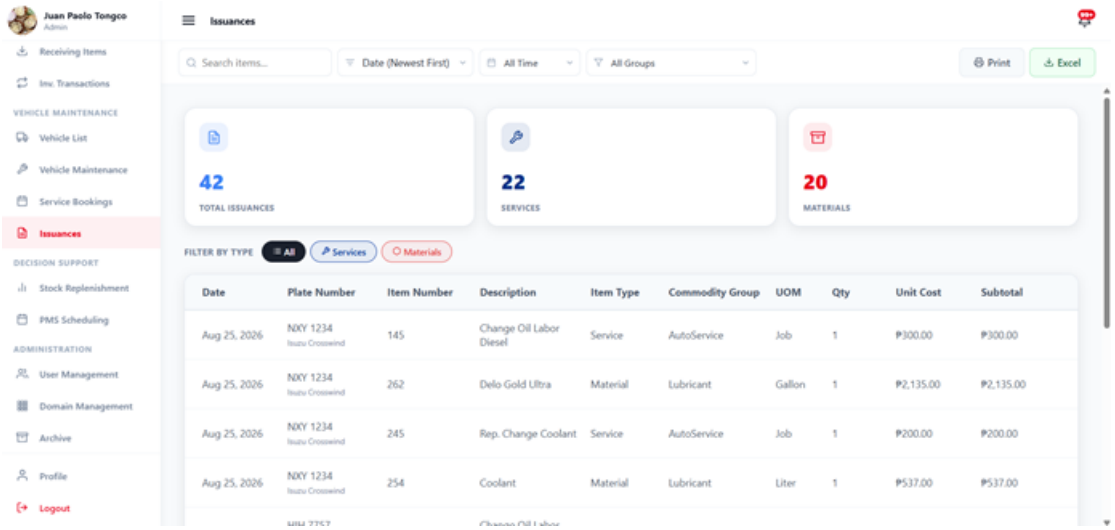


Figure 21. Issuances Page

Figure 21 shows the Issuances Page, which records inventory items and services issued for vehicle maintenance. It displays details such as the issued item, quantity, cost, and related vehicle.

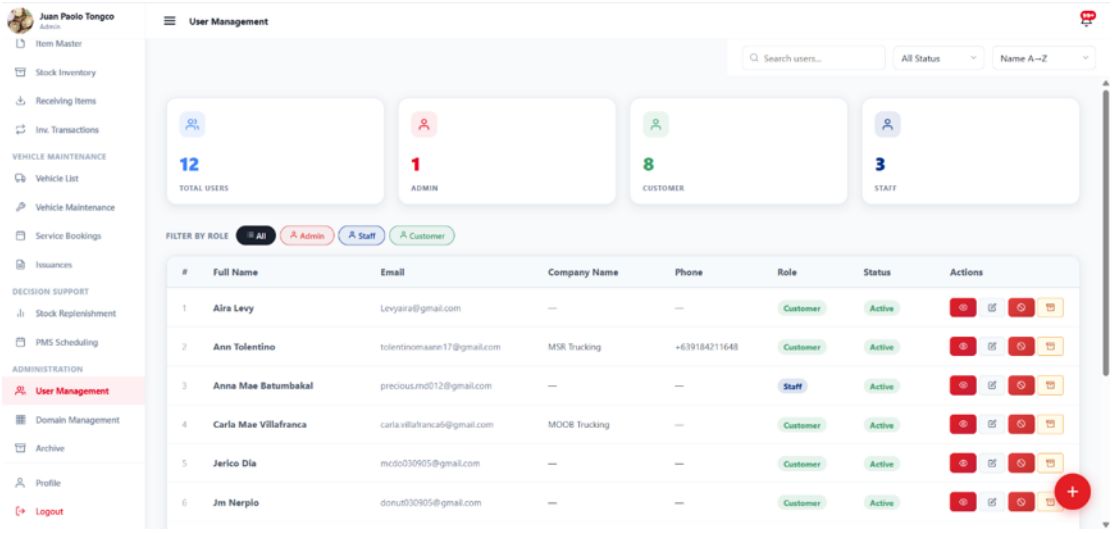


Figure 22. User Management Page

Figure 22 shows the User Management Page, where the administrator can manage system users, including administrators, staff, and customers. It allows the management of user accounts and their access.

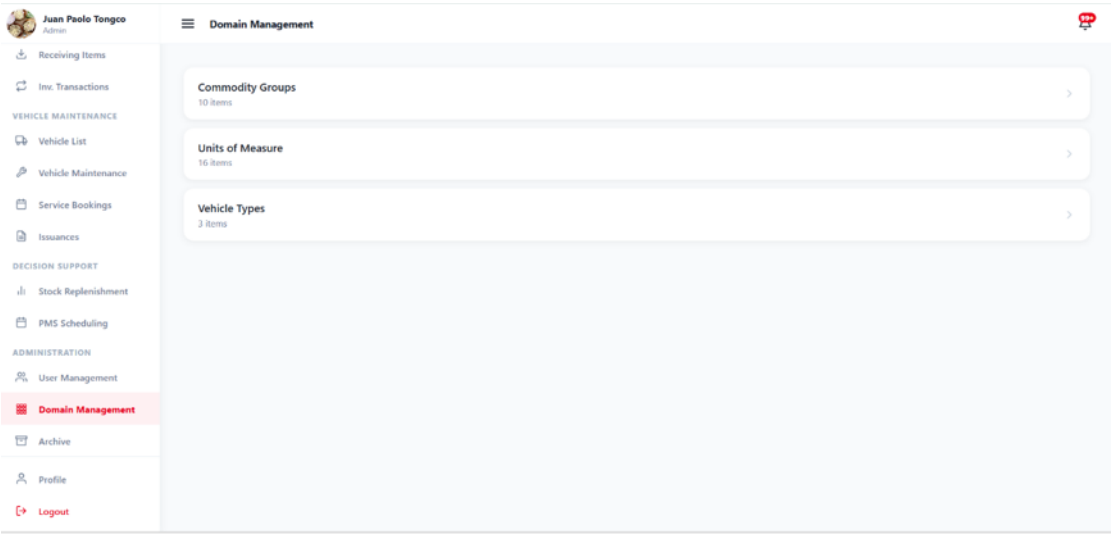


Figure 23. Domain Management Page

Figure 23 shows the Domain Management Page, where the administrator can manage system categories such as commodity groups, units of measure, and vehicle types.

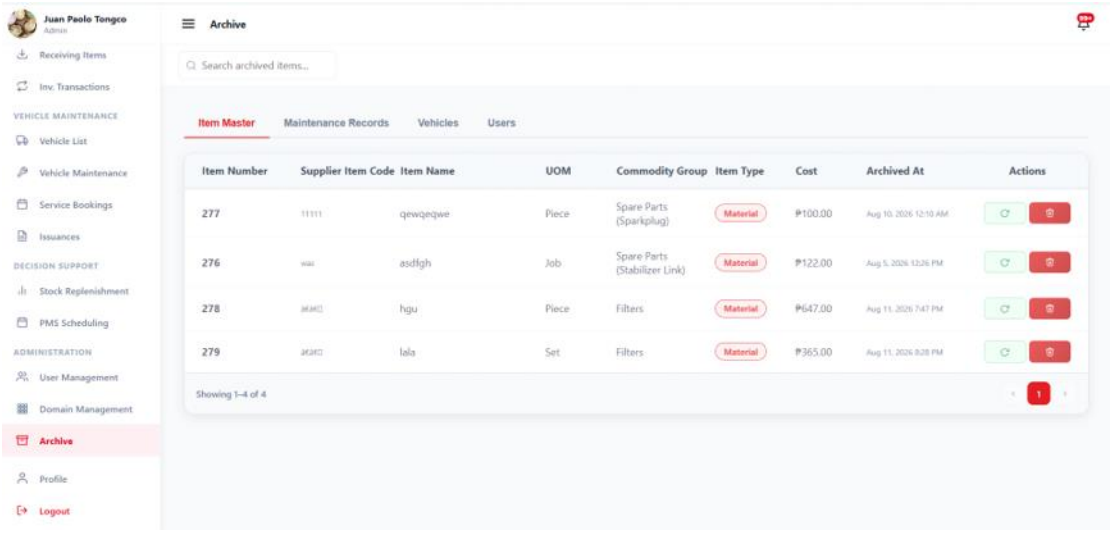


Figure 24. Archive Page

Figure 24 shows the Archive Page, which stores archived system records, including items, maintenance records, vehicles, and users. It allows the administrator to view and manage archived data.

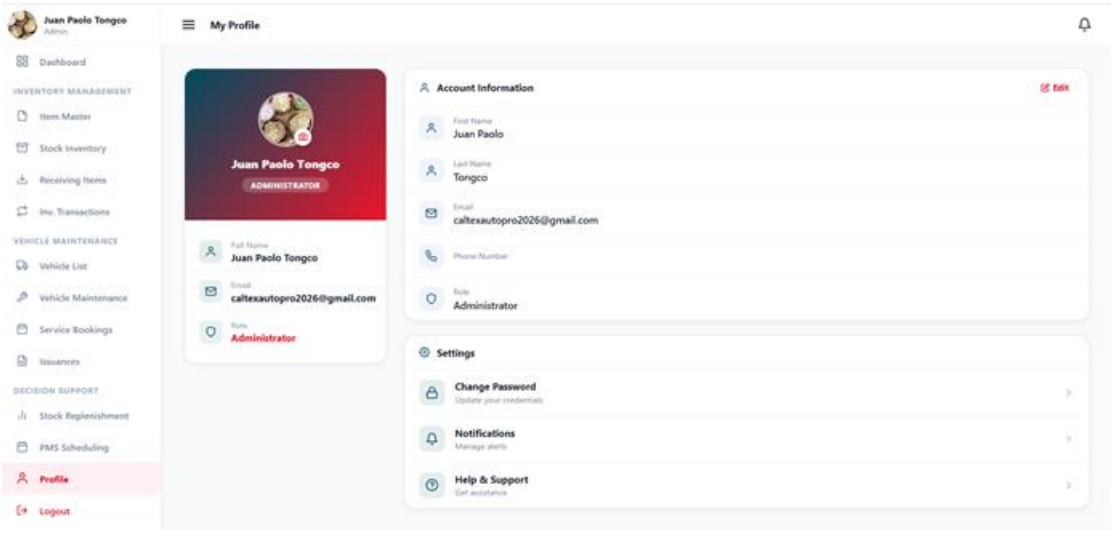


Figure 25. Admin Profile Page

Figure 25 shows the Admin Profile Page, where the administrator can view and manage personal account information, change the password, and access other account settings.

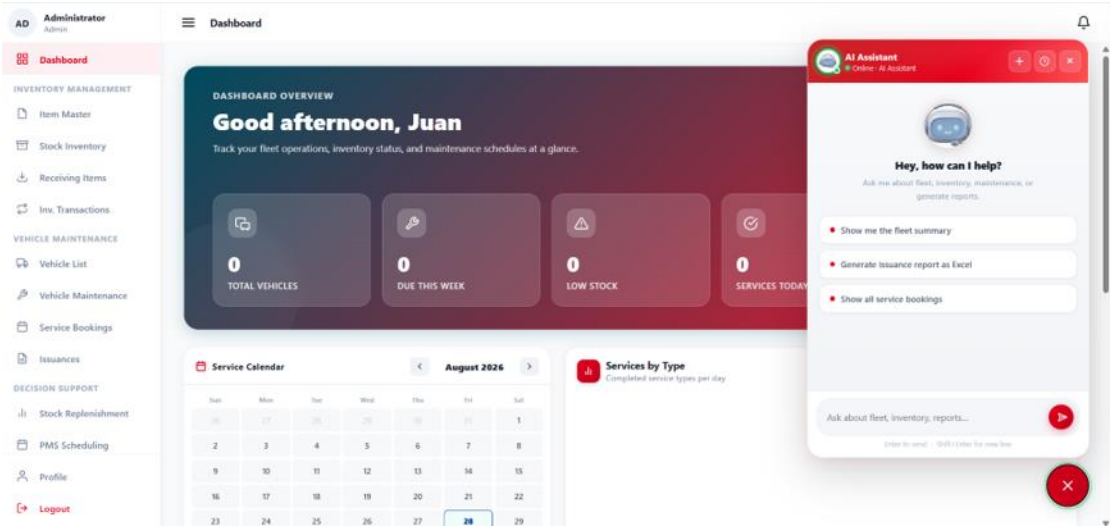


Figure 26. Admin AI Assistant Page

Figure 26 shows the Admin AI Assistant, which allows the administrator to interact with the system using natural language queries. It assists in retrieving system information and generating relevant reports or summaries.

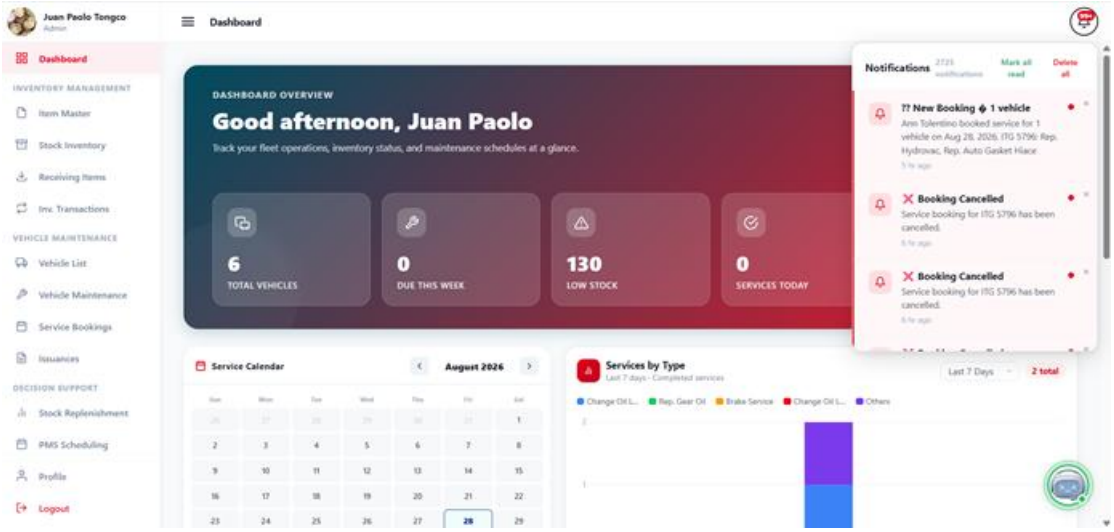


Figure 27. Admin Notifications Page

Figure 27 shows the Admin Notifications Page, which displays important system alerts and updates, such as new service bookings and other relevant activities that require the administrator's attention.

4.1.4.2 Staff Mobile Interfaces

The staff mobile interface provides employees with access to the functions needed to perform their daily operational tasks. Staff can view assigned service bookings, access customer and vehicle information, update maintenance records, and perform inventory-related tasks based on their assigned permissions. The application also supports scanning QR codes or barcodes for identifying relevant vehicle or inventory records, where applicable. Its mobile interface is designed to help staff access and update operational information efficiently while carrying out their responsibilities.

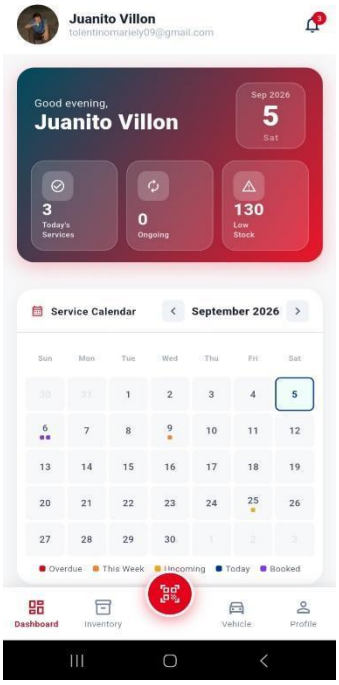


Figure 28. Staff Dashboard Page

Figure 28 shows the Staff Dashboard Page, which provides an overview of inventory, vehicle maintenance, and other operational information.

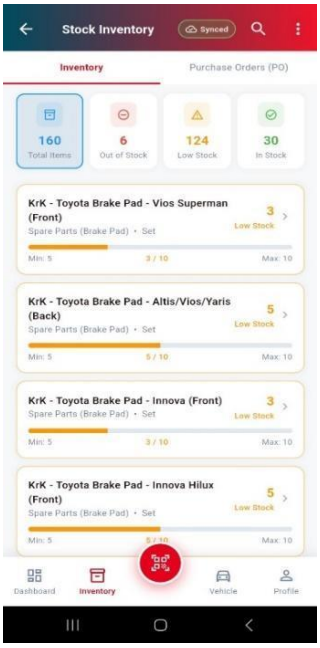


Figure 29. Stock Inventory Page

Figure 29 shows the Stock Inventory Page, where staff can monitor available stock items and view their inventory details.

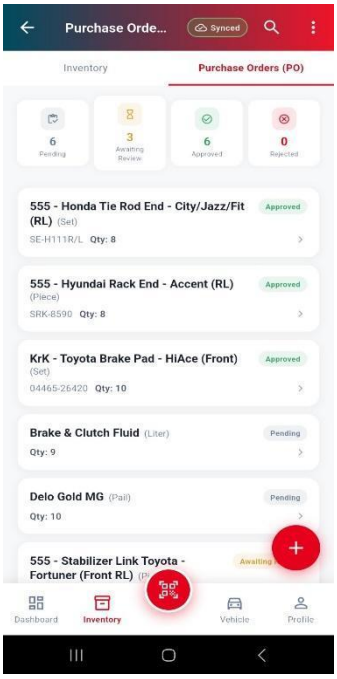


Figure 30. Purchase Order Page

Figure 30 shows the Purchase Order Page, where staff can create purchase orders, receive items, and submit received items for review and approval.

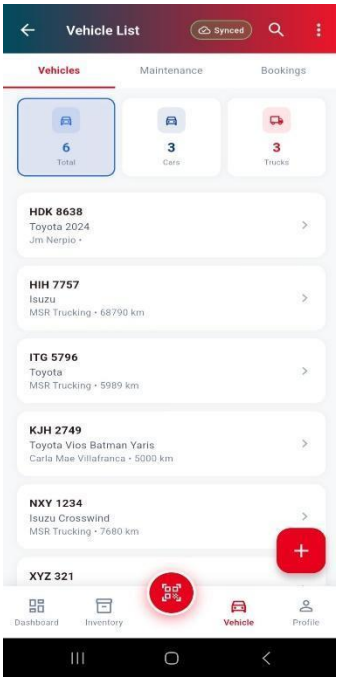


Figure 31. Vehicle List Page

Figure 31 shows the Vehicle List Page, where staff can view, add, and update registered vehicle information.

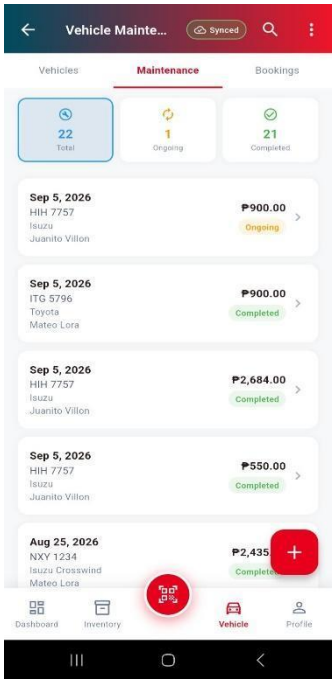


Figure 32. Vehicle Maintenance Page

Figure 32 shows the Vehicle Maintenance Page, where staff can create, update, and monitor vehicle maintenance records and service status.

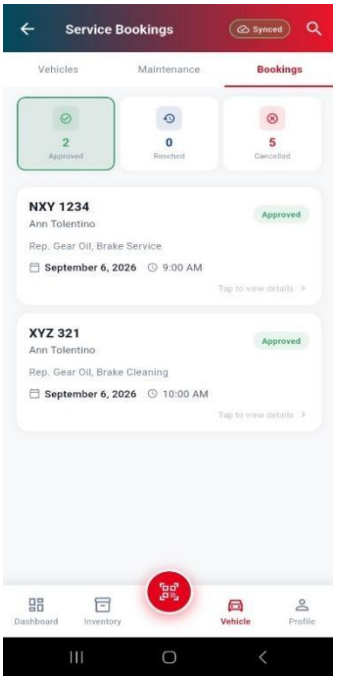


Figure 33. Service Bookings Page

Figure 33 shows the Service Bookings Page, where staff can view and manage customer service bookings, including starting, rescheduling, and cancelling scheduled services.

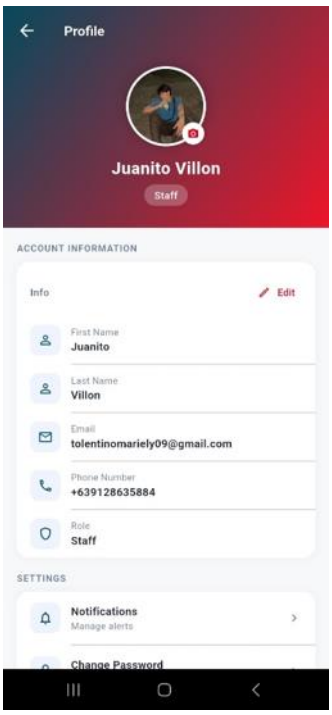


Figure 34. Staff Profile Page

Figure 34 shows the Staff Profile Page, where staff can view and update their

account information, manage their password, and access account settings.

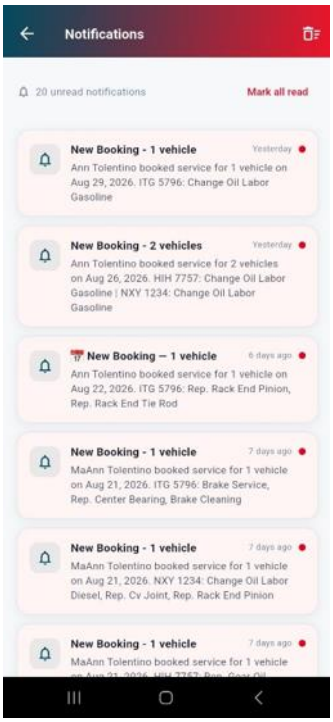


Figure 35. Staff Notifications Page

Figure 35 shows the Staff Notifications Page, which displays important alerts and updates related to system activities and service bookings.

4.1.4.2 Customer Mobile Interfaces

The customer mobile interface allows customers to access vehicle maintenance services through their mobile devices. Customers can manage their vehicle information, book maintenance services, view their service and maintenance history, monitor booking details, and receive notifications regarding their bookings and vehicle maintenance. The application also includes an AI assistant that helps customers with system-related inquiries and provides information about their own records and services. The interface is designed to make service booking and vehicle record monitoring more convenient and accessible.

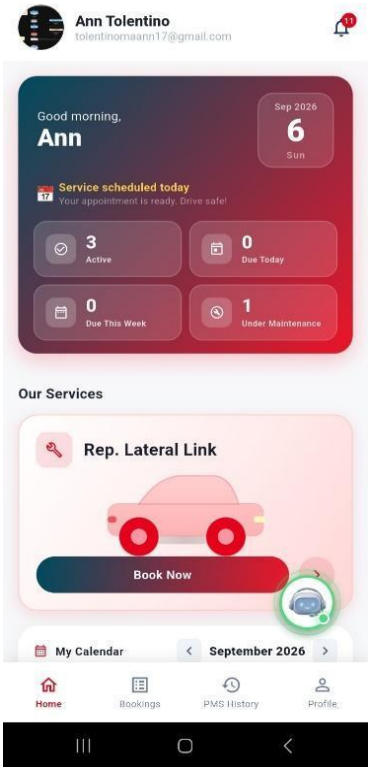


Figure 36. Customer Dashboard Page

Figure 36 shows the Customer Dashboard Page, which provides an overview of the customer's registered vehicles and their maintenance and PMS status.

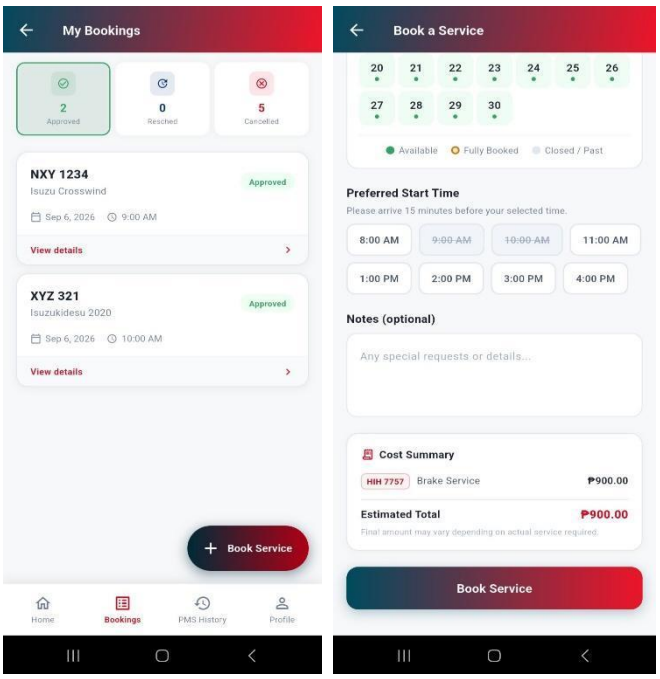


Figure 37. Bookings Page

Figure 37 shows the Bookings Page, where customers can book vehicle services, view booking details, and manage or cancel their service appointments.

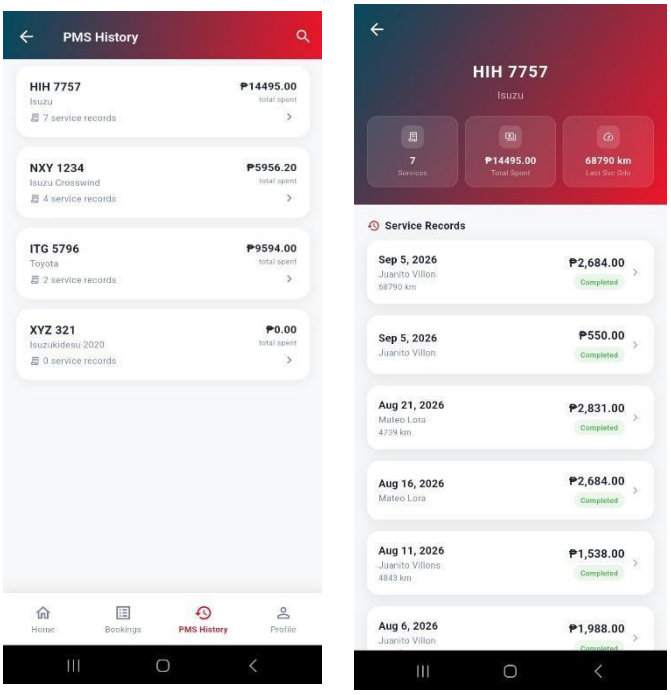


Figure 38. PMS History Page

Figure 38 shows the PMS History Page, where customers can view their registered vehicles and access their previous vehicle maintenance and service records.

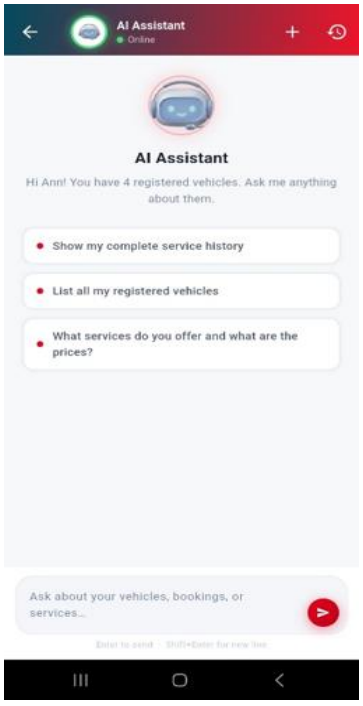


Figure 39. Customer AI Assistant Page

Figure 39 shows the Customer AI Assistant Page, which allows customers to ask system-related questions about their vehicles, service history, and available automotive services. The assistant only provides information related to the authenticated customer's account.

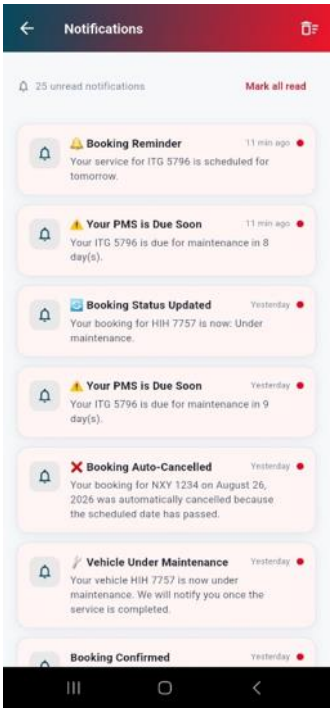


Figure 40. Customer Notifications Page

Figure 40 shows the Customer Notifications Page, which displays important updates regarding service bookings, maintenance schedules, and other relevant activities.

4.1.5 Functional and Non-Functional Requirements

This section presents how the completed system addresses the identified functional and non-functional requirements. Each requirement is matched with its corresponding implemented feature, testing evidence, and resulting outcome to demonstrate that the system met the specified requirements and performed as intended.

Table 9. Functional Requirements and Corresponding Results

Id	Functional Requirement	Implemented Feature	Testing Evidence	Result
FR-01	The system should authenticate users and provide access based on their assigned roles.	User authentication and role-based access control	Table 4.18 – Registration and Login Testing	Passed
FR-02	The system shall allow authorized users to manage vehicle records.	Vehicle List	Table 4.19 – Vehicle List Testing	Passed
FR-03	The system shall allow authorized users to record and update vehicle maintenance information.	Vehicle Maintenance	Table 4.37 – Admin Vehicle Maintenance Management Testing	Passed
FR-04	The system shall allow staff to update vehicle repair or service statuses.	Vehicle Maintenance	Table 4.37 – Admin Vehicle Maintenance Management Testing	Passed
FR-05	The system shall allow users to manage and monitor inventory records.	Item Master, Stock Inventory, and Inventory Transactions	Table 4.26 – Staff Inventory Management Testing	Passed
FR-06	The system shall allow users to scan QR codes or barcodes to identify inventory items.	Barcode/QR Scanning	Table 4.31 – Staff Barcode/QR Scanning Testing	Passed
FR-07	The system shall update inventory quantities based on recorded inventory transactions.	Stock Inventory, Receiving Items, Inventory Transactions, and Vehicle Maintenance	Inventory and Maintenance Module Testing	Passed
FR-08	The system should monitor inventory levels and identify items that reach their defined stock thresholds.	Stock Inventory and Stock Replenishment	Stock Replenishment Module Testing	Passed
FR-09	The system should generate low-stock alerts and reorder suggestions through the Decision Support System (DSS).	Stock Replenishment	Stock Replenishment Module Testing	Passed
FR-10	The system shall support preventive maintenance scheduling	PMS Scheduling	Table 4.40 – Admin PMS Scheduling Testing	Passed
FR-11	The system shall allow authorized administrators to view and generate operational reports.	Issuances and AI Assistant	Table 4.44 – Admin AI Assistant Testing	Passed
FR-12	The system shall provide an AI	AI Assistant	Table 4.23 – Customer AI Assistant Testing	Passed

Id	Functional Requirement	Implemented Feature	Testing Evidence	Result
	Assistant that retrieves relevant information using the Retrieval-Augmented Generation (RAG) mechanism.		and Table 4.44 – Admin AI Assistant Testing	
FR-13	The system shall allow the AI Assistant to respond to user queries based on available system information.	AI Assistant	Table 4.23 – Customer AI Assistant Testing and Table 4.44 – Admin AI Assistant Testing	Passed
FR-14	The system shall store and retrieve user, vehicle, inventory, and maintenance records through the centralized database.	User Management, Vehicle List, Stock Inventory, Vehicle Maintenance, and PMS History	Corresponding Module Testing	Passed
FR-15	The system shall synchronize relevant information between the web-based management panel and mobile application through the centralized system database.	Stock Inventory, Receiving Items, Vehicle List, Vehicle Maintenance, and Service Bookings	Corresponding Web and Mobile Module Testing	Passed

Table 9 shows that all fifteen (15) identified functional requirements were addressed through their corresponding implemented features. The testing evidence indicates that the required functions were successfully implemented and passed the conducted tests, demonstrating that the system performed according to the specified functional requirements.

Table 10. Non-functional Requirements and Corresponding Results

Id	Functional Requirement	Implemented Feature	Testing Evidence	Result
NFR-01	Usability: The system should provide an understandable and user-friendly interface appropriate for administrators, shop staff, mechanics, and customers.	Role-specific web and mobile interfaces	Black Box Testing and User Acceptance Testing	Passed
NFR-02	Performance: The system shall process user	Optimized web, mobile, backend, and	Black Box Testing and User Acceptance	Passed

Id	Functional Requirement	Implemented Feature	Testing Evidence	Result
	requests and retrieve system information within an acceptable response time during normal operations.	database operations	Testing	
NFR-03	Reliability: The system shall maintain accurate and consistent records during system operations and transactions.	Controlled database transactions and system validation	Module-Level Testing	Passed
NFR-04	Security: The system shall restrict access to system functions and information based on user authentication and assigned roles.	Authentication and role-based authorization	Module-Level Testing and Black Box Testing	Passed
NFR-05	Compatibility: The system shall support the identified devices and platforms used by the intended users, including computers for the web-based application and Samsung Android smartphones for the mobile application.	Web application and Android mobile application	User Acceptance Testing	Passed
NFR-06	Data Integrity: The system shall maintain the accuracy and consistency of user, vehicle, inventory, and maintenance records.	Centralized database, validation, and controlled data operations	Module-Level Testing	Passed
NFR-07	Availability: The system shall be accessible to authorized users during the intended.	Cloud-hosted web services and database	Black Box Testing and User Acceptance Testing	Passed
NFR-08	Maintainability: The system shall use organized and modular components to facilitate	Modular frontend, backend, database, and AI components	Black Box Testing	Passed

Table 10 shows that all eight (8) identified non-functional requirements were addressed through the implemented system components and validated through the conducted testing activities. The results indicate that the system satisfied the expected usability, performance, reliability, security, compatibility, data integrity, availability, and maintainability requirements.

4.2 System Testing Results

To ensure that the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Ja Noble Enterprise Inc. meets its intended requirements, Software Quality Assurance (SQA) was conducted using black box testing with defined test cases and a pass or fail evaluation scheme. This method focuses on verifying system functionality based on expected outputs without examining the internal code structure.

4.2.1 Black-Box Testing Results - TC-01: User Account Registration, Login, and Password Recovery

Test Case Description: Verify that users can register an account, log in using email and Google authentication, log out, and recover or change their passwords through the system's authentication features.

Table 11. Test Case Results – Registration and Login Process

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Customer Registration page.	10	10		100%
2	Enter the required information and click the “Create Account” button.	10	10		100%
3	Enter an invalid email format (e.g., “user@”) and click Sign In.	10	10		100%
4	Enter a valid email but incorrect password.	10	10		100%
5	Enter valid email and correct password.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
6	Navigate to the Profile tab and click the “Logout” button below.	10	10		100%
7	Sign in using Google Authentication. Click “Continue with Google” button	10	10		100%
8	Check the email inbox/spam for the OTP message. Enter the correct OTP login.	10	10		100%

Table 11 displays the outcome of black box testing which was conducted to evaluate the registration features of the system. The ten (10) testers successfully completed all testing elements which included registration page, creation of account. The registration and login process operate correctly according to system requirements.

Table 12. Test Case Results – Forgot Password

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Click the "Forgot Password" link on the login page.	10	10		100%
2	Enter the registered email and click the “Send OTP” button.	10	10		100%
3	Check the email inbox/spam for the OTP code. Enter the correct OTP received via email and click the "Verify" button.	10	10		100%
4	Open the email and check the inbox or spam, then click	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	"Reset Link".				
5	Enter a new password and click the "Save" button.	10	10		100%
6	Go to the Login page, enter the email and the new password, then click "Sign In".	10	10		100%

Table 12 displays the outcome of the black box testing conducted to evaluate the Forgot Password feature of the system. The ten (10) testers successfully completed all testing functions, which included clicking the "Forgot Password" link, entering a registered email and sending an OTP, verifying the OTP, accessing the reset link through email, creating a new password, and logging in using the new password. The Forgot Password process operates correctly according to the system requirements.

4.2.2 Black-Box Testing Results - TC-02: Customer Dashboard, Bookings, PMS History, and AI Assistant

Test Case Description: Verify that customers can view their dashboard, manage vehicle information, book or cancel services, access maintenance history, receive notifications, manage their profiles, and use the AI assistant while ensuring that customer data remains private.

Table 13. Test Case Results – Customer Dashboard

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in using valid customer credentials.	10	10		100%
2	Customer Dashboard / Homepage loads completely.	10	10		100%
3	Click the vehicle card.	10	10		100%

Table 13 displays the outcome of the black box testing conducted to evaluate the Customer Dashboard features of the system. The ten (10) testers successfully completed all testing elements, which included logging in using valid customer credentials, loading the Customer Dashboard or Homepage completely, and clicking the vehicle card. The Customer Dashboard operates correctly according to the system requirements.

Table 14. Test Case Results – Booking Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the My Bookings tab and click the “Book Service" button.	10	10		100%
2	Select a vehicle, click the “Select Service” button, choose services per vehicle, and pick a preferred date and time. And then click the “Book Service” button.	10	10		100%
3	Click "View Details" in the booking card.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
4	Click the “Cancel Booking” button.	10	10		100%
5	Select a cancellation reason (optional), then click "Yes" in the dialog.	10	10		100%

Table 14 displays the outcome of black box testing conducted to evaluate the My Bookings Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the My Bookings tab, booking a service, viewing booking details, cancelling a booking, and selecting a cancellation reason. The My Bookings Management features operate correctly according to the system requirements.

Table 15. Test Case Results – PMS History Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the PMS History tab.	10	10		100%
2	Click on a specific vehicle card (e.g., NXT 1234 - Mitsubishi Mirage 2017).	10	10		100%
3	Click on a specific service record card.	10	10		100%
4	Click the “Notifications” bell in the header.	10	10		100%
5	Click the Calendar widget button located at the bottom	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	right corner.				
6	Click on a specific highlighted date.	10	10		100%
7	Click the Close (X) button or click outside the modal.	10	10		100%

Table 15 displays the outcome of black box testing conducted to evaluate the PMS History Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the PMS History tab, viewing vehicle and service records, checking notifications, accessing the calendar, selecting highlighted dates, and closing the calendar modal. The PMS History Management features operate correctly according to the system requirements.

Table 16. Test Case Results – Customer AI Assistant

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the “AI Assistant" chat.	10	10		100%
2	Enter the query: "Show my service history." Then click the submit button.	10	10		100%
3	Enter the query: "Show my vehicles." Then click the submit button.	10	10		100%
4	Enter the query: "Show all inventory stock levels." Then click the submit button.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
5	Enter the query: "Generate an inventory report." Then click the submit button.	10	10		100%
6	Enter the query: "How much is an Oil Change service?" Then click the submit button.	10	10		100%
7	Enter an empty message and attempt to submit it.	10	10		100%
8	Enter invalid or random characters only (e.g., @#$%^&*).	10	10		100%
9	Click Clear Chat, then enter the query: "What was my previous question?"	10	10		100%
10	Enter the query: "What is the capital of Japan?" Then click the submit button.	10	10		100%
11	Enter the query: "Show another customer service history." Then click the submit button.	10	10		100%
12	Enter the query: "Show the bookings of customer Juan Dela Cruz." Then click the submit button.	10	10		100%
13	Click the “Chat History” button.	10	10		100%
14	Verify the displayed chat history.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
15	Click on a previous query from Chat History.	10	10		100%
16	Click the “Delete All Chats” button.	10	10		100%
17	Click “Cancel” in the confirmation dialog.	10	10		100%
18	Click the “Delete All Chats” button again and click “Confirm/Yes.”	10	10		100%
19	Verify the “Chat History” panel after deletion.	10	10		100%
20	Close the Chat History panel and open it again.	10	10		100%

Table 16 displays the outcome of black box testing conducted to evaluate the Customer AI Assistant features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the AI Assistant, submitting different queries, testing invalid and restricted queries, clearing chats, accessing and managing chat history, and deleting chat records. The Customer AI Assistant operates correctly according to the system requirements.

Table 17. Test Case Results – Customer Profile Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Profile tab.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Click the “Edit” button.	10	10		100%
3	Modify any fields and then click the “Save” button.	10	10		100%
4	In the settings section, click “Change Password”.	10	10		100%
5	Enter current password, new password, confirm new password and then click the “Save” button.	10	10		100%
6	In the settings section, click the "Notifications"	10	10		100%
7	Navigate back to settings and click” Help & Support”.	10	10		100%
8	Click “Logout”	10	10		100%
9	Click “Yes" in the dialog.	10	10		100%

Table 17 displays the outcome of black box testing conducted to evaluate the Customer Profile Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing and editing the profile, updating profile information, changing the password, accessing notifications and Help & Support, and logging out of the system. The Customer Profile Management features operate correctly according to the system requirements.

4.2.3 Black-Box Testing Results – TC-03: Staff Dashboard, Inventory, and Vehicle Maintenance Management

Test Case Description: Verify that staff can access their dashboard, manage inventory-related tasks, create purchase orders, receive items, update vehicle and maintenance records, process service bookings, scan items, receive notifications, and manage their accounts according to their assigned permissions.

Table 18. Test Case Results – Staff Dashboard

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Enter valid staff credentials (email address and temporary password) that were created by admin and sent to the registered email. Click the "Sign In" button.	10	10		100%
2	Enter the New Password, confirm the new password, then click the “Update” button.	10	10		100%
3	Logout and sign in again using the new password	10	10		100%

Table 18 displays the outcome of black box testing conducted to evaluate the Staff Dashboard features of the system. The ten (10) testers successfully completed all testing elements, which included logging in using valid staff credentials, updating the temporary password, and logging in again using the new password. The Staff Dashboard operates correctly according to the system requirements.

Table 19. Test Case Results – Staff Inventory Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Inventory Management tab.	10	10		100%
2	Click on a specific stock item card to view details.	10	10		100%

Table 19 displays the outcome of black box testing conducted to evaluate the Staff Inventory Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the Inventory Management tab and viewing the details of a specific stock item. The Staff Inventory Management features operate correctly according to the system requirements.

Table 20. Test Case Results – Staff Purchase Orders Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Purchase Orders tab and click the “Plus/Add” button then click the “Create Purchase Order”.	10	10		100%
2	Select the items that need replenishment and click the “Save” button.	10	10		100%
3	Click the “Plus/Add” button, then click the “Edit Purchase Order” button.	10	10		100%
4	Modify Order Quantity and click the "Update" button.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
5	Wait for the admin to approve the Purchase Order request.	10	10		100%
6	Click the “Plus/Add” button then click the "Scan Item" button.	10	10		100%
7	Enter Actual Qty, select Condition (OK/Defective/Shortage), add Remarks (optional), and click the "Submit" button.	10	10		100%
8	Wait for the admin to review and verify the received items.	10	10		100%

Table 20 displays the outcome of black box testing conducted to evaluate the Staff Receiving Items Management features of the system. The ten (10) testers successfully completed all testing elements, which included creating and editing purchase orders, selecting items for replenishment, scanning and receiving items, submitting received item details, and verifying updated inventory stock. The Staff Receiving Items Management features operate correctly according to the system requirements.

Table 21. Test Case Results – Staff Vehicle List Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Vehicle List tab and click the “Add” button.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Enter Plate Number, Description, select Vehicle Type, Owner Name or Company Name, Current Odometer (km), check "Vehicle was just serviced", enter Last Service Date, Last Service Odometer (km), Service Frequency (months), and click "Save" button	10	10		100%
3	Click the card then the "Edit" button.	10	10		100%
4	Modify any fields and click the "Update" button.	10	10		100%

Table 21 displays the outcome of black box testing conducted to evaluate the Staff Vehicle List Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the Vehicle List, adding a vehicle, entering vehicle information, editing vehicle details, and updating vehicle records. The Staff Vehicle List Management features operate correctly according to the system requirements.

Table 22. Test Case Results – Staff Service Booking

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Service Bookings tab.	10	10		100%
2	Click the card then click the "Start Service" button for a booking with today's scheduled date.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
3	Click "Yes" in the dialog.	10	10		100%
4	Complete the service form and click the "Save" button.	10	10		100%
5	Return back to the Service booking tab and click another booked service card then click the "Reschedule" button.	10	10		100%
6	Select new date and time and click “Confirm” button	10	10		100%
7	Select another Booked Service card, then click the “Cancel” button.	10	10		100%
8	Click "Yes" in the dialog.	10	10		100%
9	Return back to the Staff Dashboard and click the “Notifications” bell in the header.	10	10		100%

Table 22 displays the outcome of black box testing conducted to evaluate the Staff Service Booking features of the system. The ten (10) testers successfully completed all testing elements, which included accessing service bookings, starting a scheduled service, completing the service form, rescheduling a booking, and confirming a new service schedule. The Staff Service Booking features operate correctly according to the system requirements.

Table 23. Test Case Results – Staff Profile Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Profile tab.	10	10		100%
2	Click the “Edit” button.	10	10		100%
3	Modify any field and then click the “Save” button.	10	10		100%
4	In settings section, click the “Change Password”	10	10		100%
5	Enter current password, new password, and confirm new password and then click the “save” button	10	10		100%
6	In settings section, click the "Notifications"	10	10		100%
7	Return back to the settings and click “Help & Support”.	10	10		100%
8	Return back and click the “Logout” button below.	10	10		100%
9	Click “Yes" in the dialog.	10	10		100%

Table 23 displays the outcome of black box testing conducted to evaluate the Staff Profile Management features of the system. The ten (10) testers successfully completed all testing elements, which included managing profile information, accessing Help & Support, and logging out of the system. The Staff Profile Management features operate correctly according to the system requirements.

4.2.4 Black-Box Testing Results – TC-04: Barcode and QR Code Scanning

Test Case Description: Verify that the barcode and QR code scanner can identify inventory items, display item details, access stock information, process received items and automatically populate material information when recording vehicle maintenance services.

Table 24. Test Case Results – Barcode/QR Scanning

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Click the Scanner icon/button located in the center of the bottom navbar.	10	10		100%
2	Point the camera at a valid barcode or QR code on an item. Ensure the code is clearly visible within the scanning frame.	10	10		100%
3	Click the "View Stock" button in the action modal.	10	10		100%
4	Close the View Stock modal, scan the item again, and click "Receiving Items" (for items with pending delivery).	10	10		100%
5	Enter Actual Quantity Received, select Condition (OK/Defective/Shortage), add Remarks(optional), and click the "Submit" button.	10	10		100%
6	Close modals, scan another item, and click "Use in Maintenance".	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
7	Complete the service form with the required details and click the "Save" button.	10	10		100%

Table 24 displays the outcome of black box testing conducted to evaluate the Staff Barcode/QR Scanning features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the scanner, scanning a valid barcode or QR code, viewing stock information, receiving items, and using items for vehicle maintenance. The Staff Barcode/QR Scanning features operate correctly according to the system requirements.

4.2.5 Black-Box Testing Results – TC-05: Administrator Dashboard and System Management

Test Case Description: Verify that administrators can manage inventory, stock replenishment, purchase orders, vehicle maintenance, service bookings, preventive maintenance scheduling, user accounts, system settings, archived records, and reports, while ensuring that the AI assistant retrieves system data and generates reports according to user requests and access restrictions.

Table 25. Test Case Results – Admin Dashboard and Item Master

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Log in using valid admin credentials.	10	10		100%
2	Observe the Admin Dashboard in the main screen / homepage.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
3	Navigate to the Item Master tab and click the “Add” button.	10	10		100%
4	Enter the Supplier Item Code, OEM, Item Name, Description, Commodity Group, UOM, Cost, and Type, then click the “Save” button.	10	10		100%
5	Enter the Current Quantity, Minimum Level, and Maximum Level. Verify that the Reorder Quantity is automatically calculated, then click the “Save” button.	10	10		100%
6	Click the “Edit” button of the current added item.	10	10		100%
7	Modify any field and click the “Update” button.	10	10		100%
8	Click the “Archive” button.	10	10		100%
9	Click "Ok" in the dialog.	10	10		100%

Table 25 displays the outcome of black box testing conducted to evaluate the Admin Dashboard and Item Master features of the system. The ten (10) testers successfully completed all testing elements, which included logging in as an admin, accessing the dashboard, adding an item, entering item and stock details, calculating the reorder quantity, editing item information, and archiving an item. The Admin Dashboard and Item Master features operate correctly according to the system

requirements.

Table 26. Test Case Results – Admin Stock Inventory Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Stock Inventory tab.	10	10		100%
2	Click the “Edit” button.	10	10		100%
3	Modify any field and click the “Update” button.	10	10		100%

Table 26 displays the outcome of black box testing conducted to evaluate the Admin Stock Inventory Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the Stock Inventory tab and editing stock information. The Admin Stock Inventory Management features operate correctly according to the system requirements.

Table 27. Test Case Results – Admin Purchase Orders Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Purchase Orders tab and click the “Plus/Add” button then click the “Create Purchase Order” button.	10	10		100%
2	Select items that need replenishment and click “Save” button	10	10		100%
3	Click the “Edit” button.	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
4	Modify Order Quantity and click the "Update" button.				
5	Click the “Plus/Add” button then click the “Review Purchase Order” button.	10	10		100%
6	Review and verify the item and then click the “Approve” button.	10	10		100%
7	Click the “View” button to open the item details, then click the “Receive Item” button.	10	10		100%
8	Enter Actual Qty, select Condition (e.g., OK), add Remarks, and click the "Submit" button.	10	10		100%
9	Click the "Review & Verify" button.	10	10		100%
10	Verify details are correct and click the "Approve & Update Stock" button.	10	10		100%

Table 27 displays the outcome of black box testing conducted to evaluate the Admin Purchase Order Management features of the system. The ten (10) testers successfully completed all testing elements, which included creating and editing purchase orders, selecting items for replenishment, receiving items, and verifying and updating stock. The Admin Receiving Item Management features operate correctly according to the system requirements.

Table 28. Test Case Results – Admin Inventory Transactions Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Inventory Transactions tab.	10	10		100%
2	Filter the records by Type, Date Range, or Commodity Group.	10	10		100%
3	Select Sort by: Date (Newest First) / Date (Oldest First) / Item Name A→Z / Item Name Z→A / Qty (Highest) / Qty (Lowest)	10	10		100%
4	Enter the search keyword in the search box.	10	10		100%
5	Click the “Print" button, set Print settings, and click Print button again.	10	10		100%
6	Click the “Excel" button to export the file in excel format.	10	10		100%

Table 28 displays the outcome of black box testing conducted to evaluate the Admin Inventory Transactions Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing inventory transactions and filtering records by type, date range, or commodity group. The Admin Inventory Transactions Management features operate correctly according to the system requirements.

Table 29. Test Case Results – Admin Vehicle List Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Vehicle List tab and click the “Add” button.	10	10		100%
2	Enter Plate Number, Description, select Vehicle Type, Owner Name or Company Name, Current Odometer (km), check "Vehicle was just serviced", enter Last Service Date, Last Service Odometer (km), Service Frequency (months), and click "Save" button	10	10		100%
3	Click the "Edit" button.	10	10		100%
4	Modify any fields and click the "Update" button.	10	10		100%
5	Click the "Archive” button.	10	10		100%
6	Click "Ok" in the confirmation dialog.	10	10		100%

Table 29 displays the outcome of black box testing conducted to evaluate the Admin Vehicle List Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the vehicle list, adding vehicles, entering vehicle information, and editing vehicle records. The Admin Vehicle List Management features operate correctly according to the system requirements.

Table 30. Test Case Results – Admin Vehicle Maintenance Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to Vehicle Maintenance tab and click “Add”	10	10		100%
2	Enter Plate Number, Mechanic Name, Service Date, Odometer Reading (km), select Vehicle Issues (optional)(e.g., Engine Problem, Brake Issue), add custom issue if needed, add row in Services Rendered with Service Name, add row in Materials Used with Material Name, and click "Save" button	10	10		100%
3	Click the "Edit" button.	10	10		100%
4	Modify any fields and click "Update" button	10	10		100%
5	Click the “View” button to open the details modal, then click the "Mark as Completed" button.	10	10		100%
6	Click "OK" in the dialog.	10	10		100%

Table 30 displays the outcome of black box testing conducted to evaluate the Admin Vehicle Maintenance Management features of the system. The ten (10) testers successfully completed all testing elements, including adding vehicle maintenance records, entering maintenance details, editing records, and updating maintenance information. The Admin Vehicle Maintenance Management features operate correctly

according to the system requirements.

Table 31. Test Case Results – Admin Service Bookings Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Service Bookings tab.	10	10		100%
2	Click the "Start Service" button for a booking with today's scheduled date.	10	10		100%
3	Click "Ok" in the dialog.	10	10		100%
4	Complete the service form and click the "Save" button.	10	10		100%
5	Return back to the Service booking tab and click the "Reschedule" button.	10	10		100%
6	Select new date and time, then click the “Confirm” button	10	10		100%
7	Click the “Cancel” button.	10	10		100%
8	Click "Ok" in the dialog.	10	10		100%

Table 31 displays the outcome of black box testing conducted to evaluate the Admin Service Booking features of the system. The ten (10) testers successfully completed all testing elements, which included accessing service bookings, managing booking details, starting or completing services, and rescheduling bookings. The Admin Service Booking features operate correctly according to the system requirements.

Table 32. Test Case Results – Admin Issuances

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Issuances Tab.	10	10		100%
2	Select filter by type / date from to / commodity group.	10	10		100%
3	Select Sort by: Date (Newest First) / Date (Oldest First) / Item Name A→Z / Item Name Z→A / Qty (Highest) / Qty (Lowest) / Cost (Highest) / Cost (Lowest)	10	10		100%
4	Enter the search keyword in the search box.	10	10		100%
5	Click the “Print” button.	10	10		100%
6	Click the “Export” button (Excel).	10	10		100%

Table 32 displays the outcome of black box testing conducted to evaluate the Admin Reports Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing reports, selecting report types and date ranges, generating reports, and downloading reports. The Admin Reports Management features operate correctly according to the system requirements.

Table 33. Test Case Results – Admin User Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the User Management tab and click	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	the “Add user” button.				
2	Enter first name, last name, email address, and select the role. If the role is a customer, enter the company name and click the “Save” button.	10	10		100%
3	Click the “Edit” button.	10	10		100%
4	Modify any user details and click the "Update" button.	10	10		100%
5	Click the “Deactivate” button.	10	10		100%
6	Click "OK” in the dialog.	10	10		100%
7	Click the "Activate" button.	10	10		100%
8	Click "Ok" in the dialog.	10	10		100%
9	Click the "Archive” button.	10	10		100%
10	Click "Ok" in the confirmation dialog.	10	10		100%

Table 33 displays the outcome of black box testing conducted to evaluate the Admin User Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing user management, adding users, entering user information, editing user details, and managing user accounts. The

Admin User Management features operate correctly according to the system requirements.

Table 34. Test Case Results – Admin Domain Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Domain Management tab.	10	10		100%
2	Click on any card (e.g., Commodity Groups / Unit of Measurements / Vehicle Types) and then click the "Add" button.	10	10		100%
3	Enter Name and click the “Save” button.	10	10		100%
4	Click the “Edit” button.	10	10		100%
5	Modify the name and click the "Update" button.	10	10		100%
6	Click the “Delete” button.	10	10		100%
7	Click “OK” in the dialog.	10	10		100%

Table 34 displays the outcome of black box testing conducted to evaluate the Admin Profile Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing and editing profile information, changing the password, accessing Help & Support, and logging out of the system. The Admin Profile Management features operate correctly according to the system requirements.

Table 35. Test Case Results – Admin Archive Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the Archive tab.	10	10		100%
2	Click the “Restore” button.	10	10		100%
3	Click "OK" in the dialog.	10	10		100%
4	Navigate to the respective module (e.g., Item Master, Vehicle Maintenance) and verify the restored item.	10	10		100%
5	Return to the Archive tab and click the "Delete Permanently" button	10	10		100%
6	Confirm permanent deletion by clicking "Yes "	10	10		100%

Table 35 displays the outcome of black box testing conducted to evaluate the Archive Management features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the Archive tab, restoring archived items, verifying restored records, and permanently deleting archived items. The Archive Management features operate correctly according to the system requirements.

Table 36. Test Case Results – Admin AI Assistant

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Navigate to the AI Assistant	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	chat bubble on the dashboard.				
2	Enter a valid report query (e.g., "Generate the issuance report for the month of July 2026 in PDF.") then click the send button.	10	10		100%
3	Click the Download PDF or Download Excel button.	10	10		100%
4	Enter an empty message and attempt to submit it.	10	10		100%
5	Enter invalid or random input (e.g., $%^&*). Then click the send button.	10	10		100%
6	Enter a query outside the automotive management system (e.g., "Who won the FIFA World Cup in 2018?") then click the send button.	10	10		100%
7	Request a report with filters that return no records (e.g., "Generate the issuance report for December 2035") then click the send button.	10	10		100%
8	Click the Clear Chat button.	10	10		100%
9	Enter the query: “Show all low-stock inventory items." then click the send button.	10	10		100%
10	Enter the query: "How many registered customers are in the system? Then click the	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	send button.				
11	Enter the query: “Show all pending service bookings." Then click the send button.	10	10		100%
12	Click the “Chat History” button.	10	10		100%
13	Verify the displayed chat history.	10	10		100%
14	Click the “Delete All Chats” button.	10	10		100%

Table 36 displays the outcome of black box testing conducted to evaluate the Admin AI Assistant features of the system. The ten (10) testers successfully completed all testing elements, which included accessing the AI Assistant, generating and downloading reports, submitting valid and invalid queries, retrieving system information, viewing chat history, and deleting chats. The Admin AI Assistant operates correctly according to the system requirements.

4.2.6 Testing Summary

The overall testing results show that the implemented system successfully met its intended functional requirements. Black-box testing was conducted across the system’s customers, staff, and admin modules, with each test involving ten (10) IT experts. The results consistently showed a 100% pass rate and 0% failure rate for the tested functions. The tested areas included registration and login, customer and staff management, vehicle and maintenance management, service bookings, inventory

management, barcode/QR scanning, preventive maintenance scheduling, decision support functions, and AI Assistant features. These results indicate that the major system functions operated as expected and complied with the defined requirements. Although all tested functions passed, the IT experts provided several comments and suggestions for further improvement of the system. They recommended that the booking cancellation process requires customers to provide a reason for cancellation and that cancellations should not be allowed within 24 hours before the scheduled service date. They also suggested limiting the supported languages to English and Filipino and clearly stating this limitation in the system's scope. Improvements to the user interface were also recommended, particularly in refining the color palette to create a more consistent and visually appealing design. Additionally, the booking process, particularly the automatic approval of bookings, was suggested for further improvement to ensure that the approval process is appropriate and clearly defined. Lastly, it was suggested that the offline functionality of the mobile application be optional, depending on the system requirements and whether offline access is necessary for the intended users. The recommendations also identify some limitations and areas that were not fully covered by the system. Key areas like the booking cancellation policy, the automatic booking approval process, the list of ai assistant supported languages, the design consistency of the user interface and offline access options may need work or additional features. While these issues did not stop the tested functions from passing the black-box evaluation they point to opportunities for improvements. These comments and suggestions provide additional areas for improvement despite the system achieving a 100% pass rate during blackbox testing. The feedback may serve as a basis for future enhancements to further improve the system's usability, accessibility, functionality, and overall user experience.

Table 37. Consolidated Black Box Testing Results

Testing Result	Outcome
Number of tested functional areas	26
Testers per functional area	10
Passed	100%
Failed	0%
Overall Result	**Passed**

Overall, the testing confirms that the system functions were successfully implemented and performed according to their expected outputs. The consistent passing results across the tested modules demonstrate that the system is ready for its intended use and further evaluation.

4.2.7 RAG/AI Assistant Evaluation Results

The RAG/AI Assistant was evaluated in black-box testing. The tests used predefined test cases for the Customer interface and the Admin interface. The evaluation focused on data retrieval by the RAG/AI Assistant. The evaluation also focused on role-based access control, for the RAG/AI Assistant. The evaluation also focused on how the RAG/AI Assistant handled requests. The evaluation also focused on how the RAG/AI Assistant handled requests. The evaluation also focused on how the RAG/AI Assistant handled failures. The evaluation also focused on the core functionality of the RAG/AI Assistant. A test case was considered passed when the actual behavior of the RAG/AI Assistant matched the expected result.

Table 38. AI Assistant Evaluation Criteria with Results

Evaluation Criterion	Tested Function	Result	Status
Data Retrieval	Retrieval of customer service history, vehicles, service information, low-stock inventory, customer count, and pending bookings	The AI Assistant retrieved the requested authorized information from the system data.	Passed
Role-Based Access Control	Customer and Admin access to authorized information	Customer and Admin requests were processed according to their assigned access permissions.	Passed
Unauthorized Request Handling	Customer requests for another customer's records, inventory, and administrative reports	Restricted requests were denied and protected information was not displayed.	Passed
Unsupported Query Handling	Questions unrelated to the Automotive Service and Inventory Management System	The AI Assistant identified the requests as outside its supported scope and responded appropriately.	Passed
Failure and Input Handling	Empty input, random characters, and requests with no matching records	Invalid inputs were handled without system failure, while requests with no matching records returned an appropriate response.	Passed
Ai Assistant Functionality	Report generation, report download, chat clearing, chat history, and chat deletion	The tested AI functions performed according to their specified requirements.	Passed

Table 38 results show that the RAG/AI Assistant that was built did the retrieval and functional tests the way it was supposed to do. When only authorized people asked questions the RAG/AI Assistant gave back the system information. The RAG/AI Assistant kept Customer users from seeing other customers' records and from seeing data because of the role limits. The RAG/AI Assistant also managed questions and bad input without giving out protected data or crashing. The RAG/AI Assistant handled tasks, such as making reports and pulling operational details and it gave the expected answers during the tests. These findings confirm that the RAG/AI Assistant and its access-control rules worked as the specifications required.

4.3 System User Evaluation Results

The researchers used the Technology Acceptance Model (TAM) questionnaire to assess how well the system performed and how effective it was. The questionnaire provided criteria related to Quality Factors, Perceived Ease of Use, Attitude Towards Using, Behavioral Intention of Use, and Experience to determine the level of acceptability of the system based on the respondents' perceptions.

After researchers created the questionnaire, it was subjected to validation by the research adviser, Mrs. Justo, to ensure its clarity, relevance, and appropriateness. After researchers validated the instrument, they began preparing it for distribution to the selected respondents.

A 5-point Likert scale was used to measure the respondents' level of agreement, ranging from Not Acceptable to Highly Acceptable, corresponding to their level of acceptance of the system. The Admin, Staffs and Customers who demonstrated the system were selected to complete the questionnaire which helped researchers obtain feedback about system usability and effectiveness.

4.3.1 Respondent Profile

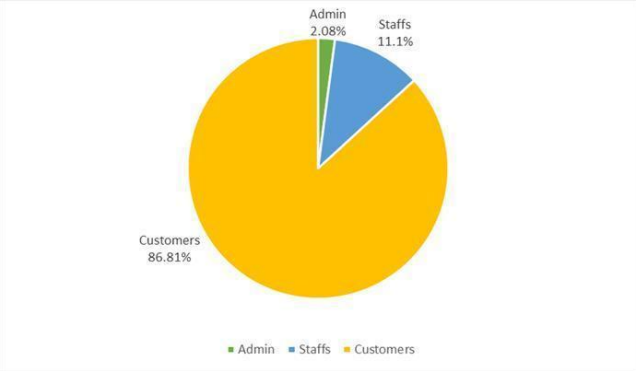


Figure 41. Percentage of Respondents Based on Classification

Figure 41 shows the demographic distribution of the respondents who participated in the evaluation of the system, with a total of 144 respondents from the Administrators, Staffs and Customers. The demographic profile was categorized based on the role of the respondents as indicated in the survey questionnaire.

The graph shows that the majority of the respondents are Customers, with 125 participants, which is equivalent to 86.81% of the total population. Followed by Staffs with 16 participants, representing 11.1% of the respondents. And with the least participants with 3 respondents equivalent to 2.08% represent the Administrators. The results show that most respondents are the customers, which constitutes the primary user group that will assess the system.

4.3.2 Evaluation Results by Criterion

This section presents the results of the user acceptance survey conducted to evaluate the system. The findings are presented in tabular form, showing the computed weighted mean and standard deviation for each indicator. The results are organized according to the different components of the Technology Acceptance Model, namely Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use and Experience. Each section is followed by its corresponding interpretation

based on the Likert scale.

Table 39. Level of Acceptability of the System in Terms of Quality Factors

Quality Factors	Weighted Mean	Standard Deviation	Verbal Interpretation
Q1	4.84	0.37	Highly Acceptable
Q2	4.92	0.28	Highly Acceptable
Q3	4.90	0.30	Highly Acceptable
Average	4.89	0.31	Highly Acceptable

Table 39 above shows the level of acceptability of the system in terms of Quality Factors. Q1 obtained a mean of 4.84 with a standard deviation of 0.37, interpreted as "Highly Acceptable." Q2 got a mean of 4.92 with a standard deviation of 0.28, also interpreted as "Highly Acceptable." Q3 obtained a mean of 4.90 with a standard deviation of 0.30, interpreted as "Highly Acceptable." Overall, the Quality Factors obtained an average of 4.89, interpreted as "Highly Acceptable," indicating a high level of acceptability of the system in terms of its quality factors.

Table 40. Level of Acceptability of the System in Terms of Perceived Ease of Use

Perceived Ease Of Use	Weighted Mean	Standard Deviation	Verbal Interpretation
Q1	4.81	0.40	Highly Acceptable
Q2	4.75	0.40	Highly Acceptable
Q3	4.78	0.41	Highly Acceptable
Average	4.78	**0.40**	Highly Acceptable

Table 42 above shows the level of acceptability of the system in terms of Perceived Ease of Use. Q1 obtained a mean of 4.81 with a standard deviation of 0.40, interpreted as "Highly Acceptable." Q2 got a mean of 4.75 with a standard deviation of 0.40, also interpreted as "Highly Acceptable." Q3 obtained a mean of 4.78 with a

standard deviation of 0.41, interpreted as “Highly Acceptable.” Overall, the Perceived Ease of Use obtained an average of 4.78, interpreted as “Highly Acceptable,” indicating a high level of acceptability of the system in terms of its ease of use.

Table 41. Level of Acceptability of the System in Terms of Attitude Towards Using

Attitude Towards Using	Weighted Mean	Standard Deviation	Verbal Interpretation
Q1	4.81	0.40	Highly Acceptable
Q2	4.83	0.40	Highly Acceptable
Q3	4.85	0.35	Highly Acceptable
Average	4.83	**0.38**	Highly Acceptable

Table 41 above shows the level of acceptability of the system in terms of Attitude Towards Using. Q1 obtained a mean of 4.81 with a standard deviation of 0.40, interpreted as “Highly Acceptable.” Q2 got a mean of 4.83 with a standard deviation of 0.40, also interpreted as “Highly Acceptable.” Q3 obtained a mean of 4.85 with a standard deviation of 0.35, interpreted as “Highly Acceptable.” Overall, the Attitude Towards Using obtained an average of 4.83, interpreted as “Highly Acceptable,” indicating a high level of acceptability of the system in terms of its attitude towards using the system.

Table 42. Level of Acceptability of the System in Terms of Behavioral Intention to Use

Behavioral Intention To Use	Weighted Mean	Standard Deviation	Verbal Interpretation
Q1	4.88	0.33	Highly Acceptable
Q2	4.91	0.29	Highly Acceptable
Q3	4.87	0.36	Highly Acceptable
Average	4.88	**0.33**	Highly Acceptable

Table 42 above shows the level of acceptability of the system in terms of

Behavioral Intention to Use. Q1 obtained a mean of 4.88 with a standard deviation of 0.33, interpreted as “Highly Acceptable.” Q2 got a mean of 4.91 with a standard deviation of 0.29, also interpreted as “Highly Acceptable.” Q3 obtained a mean of 4.87 with a standard deviation of 0.36, interpreted as “Highly Acceptable.” Overall, the Behavioral Intention to Use obtained an average weighted mean of 4.88, interpreted as “Highly Acceptable,” indicating a high level of acceptability of the system in terms of its behavioral intention to use the system in the future.

Table 43. Level of Acceptability of the System in Terms of Experience

Experience	Weighted Mean	Standard Deviation	Verbal Interpretation
Q1	4.88	0.32	Highly Acceptable
Q2	4.90	0.30	Highly Acceptable
Q3	4.91	0.29	Highly Acceptable
Average	4.90	0.30	Highly Acceptable

Table 43 above shows the level of acceptability of the system in terms of Experience. Q1 obtained a mean of 4.88 with a standard deviation of 0.32, interpreted as “Highly Acceptable.” Q2 got a mean of 4.90 with a standard deviation of 0.30, also interpreted as “Highly Acceptable.” Q3 obtained a mean of 4.91 with a standard deviation of 0.29, interpreted as “Highly Acceptable.” Overall, the Experience obtained an average weighted mean of 4.90, interpreted as “Highly Acceptable,” indicating a high level of acceptability of the system in terms of user experience. This result is further supported by the qualitative feedback provided by one of the customer respondents, who stated, “Very Good Application. Must be improved more for better access.” The feedback suggests that while the system provides positive overall experience, there are still opportunities for further improvements, particularly in accessibility and ease of access, to enhance the users’ experience.

4.3.3 Statistical Results

Table 44. Summary of System Evaluation Results

Quality Factors	Weighted Mean	Standard Deviation	Verbal Interpretation
Quality Factors	4.89	0.31	Highly Acceptable
Perceived Ease of Use	4.78	0.40	Highly Acceptable
Attitude Towards Using	4.83	0.38	Highly Acceptable
Behavioral Intention to Use	4.88	0.33	Highly Acceptable
Experience	4.90	0.30	Highly Acceptable
Average	4.86	0.35	Highly Acceptable

Table 44 above shows that all five criteria received a Highly Acceptable verbal interpretation, with weighted means ranging from 4.83 to 4.90. This shows that the respondents generally provided favorable evaluations across all criteria. Among the five criteria, Criterion 5 obtained the highest weighted mean of 4.90 with a standard deviation of 0.30, indicating a consistently high level of agreement among the respondents. Criteria 1 gained an average weighted mean of 4.89, while Criterion 3 obtained 4.83 and Criterion 4 with 4.88. Although Criterion 2 received the lowest weighted mean of 4.78, it was still interpreted as Highly Acceptable, indicating that the respondents continued to view this aspect of the system positively.

The standard deviations ranged from 0.30 to 0.40, showing relatively low variation in the respondents' evaluations. This suggests that the ratings were generally consistent, with respondents having similar perceptions of the system across the five criteria. Overall, the system obtained an average weighted mean of 4.86 and a standard deviation of 0.35, resulting in a verbal interpretation of Highly Acceptable. These findings demonstrate that the completed system was positively evaluated across the

assessed criteria and was generally perceived by the respondents as acceptable for its intended use.

4.4 Discussion of User Evaluation

The results from the user acceptance survey show that the system developed was interpreted as Highly Acceptable with an overall weighted mean of 4.86. This result reflects positive assessment from the respondents across all criteria evaluated. The evaluation covered different aspects and in each one the responses were consistently positive. This indicates that the respondents did not evaluate the system positively based on only one aspect but also recognized its value in terms of its quality factors, ease of use, attitude toward using the system, intention to use it, and overall user experience.

When looking at the Quality Factors the system was seen as effective in supporting tasks related to vehicle maintenance and inventory management. It offered features that match what users needed in their work. The Perceived Ease of Use results suggest that users found the system simple to understand and manage. This is important because the system includes processes and supports different types of users. For a system to be accepted it must not have useful functions but also be straightforward for people to interact with.

The Attitude Towards Using the system showed that users generally viewed it in a light. They felt comfortable with it and saw it helpful. Similarly, the Behavioral Intention to Use results shows that most users said they would continue using the system. They were not just satisfied many expressed a desire to keep using it in the future. The Experience results further confirm this. Users reported feeling engaged and pleased when working with the system’s features.

In summary, the findings support the research objective of determining the level of acceptability of the developed Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise. The overall interpretation of Highly Acceptable indicates that the completed system was positively received by its intended users and is considered suitable for its intended purpose. The findings further demonstrate that the system has the potential to support and improve the vehicle maintenance and inventory management processes of JA Noble Enterprise Inc.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents a summary of the study findings, draws conclusions from the results and gives recommendations for improvements of the Integrated Vehicle Maintenance and Inventory Management System with Decision Support System.

5.1 Summary

This study aimed to design, develop, test, and evaluate the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., a Caltex Service Station and Automotive Service Provider in San Pedro, Laguna. The study addressed the issues associated with the vehicle service center, particularly related to the ability to track the history of vehicles serviced, monitor material requirements for each job, view stock levels in real-time, and utilize manual processes and/or spreadsheets which are slow, error-prone and disorganized resulting in stock shortages, poor staff communications and difficult-to-produce analytical reporting. The major findings of the study, aligned with the specific research objectives, are summarized as follows:

- System Development (Objective 1): The study successfully designed and developed the Integrated Vehicle Maintenance and Inventory Management System with Decision Support as a multi-platform system utilizing a Web Management Panel, Mobile App, and an Integrated AI Assistant. The developed system featured automated check-in processes using QR Code and Barcode Scanning, a Decision Support System (DSS) to provide Low-Stock Alerts and Preventive Maintenance Scheduling (PMS), and an Interactive AI Assistant

utilizing Retrieval-Augmented Generation (RAG) to facilitate Chat-style Data Queries and Automatic Context Summaries. The system utilized a Relational Database to ensure availability for Administrators, Staff, and Customers. An Agile Development Methodology employing the Scrum Framework was utilized to manage the team's work through short sprints covering Requirement Planning, Design, Development, Testing, Deployment, and Feedback/Maintenance.

- System Testing and Functionality (Objective 2): The developed system underwent rigorous Black-Box testing across five (5) distinct test cases covering all major features, including the Vehicle and Mobile Inventory Management modules (with QR/Barcode Scanning), the Decision Support System (DSS) for Low-Stock Alerts and Preventive Maintenance Scheduling (PMS), and the Interactive AI Assistant built on Retrieval-Augmented Generation (RAG). The testing resulted in a 100% functional success rate, marked as passed with zero (0) failed results, confirming that all core functionalities of the system are working as intended.
- User Acceptance Evaluation (Objective 3): The system's acceptance was evaluated using the Technology Acceptance Model (TAM) and the ISO/IEC 25010 software quality standards. The evaluation measured Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. A questionnaire that was validated by the research adviser was administered to 144 respondents following system demonstrations, comprising 3 Administrators (2.08%), 16 Staff (11.11%), and 125 Customers (86.81%) of JA Noble Enterprise Inc. The results demonstrated very high scores for all of the metrics associated with quality, functionality, and usability. A weighted

mean of 4.89 (on a scale from 1–5) was achieved for Quality Factors; 4.78 for Perceived Ease of Use; 4.83 for Attitude Toward Using; 4.88 for Behavioral Intention to Use; and 4.90 for Experience. Since all metrics are well above the minimum acceptable of 4.21 and into the High acceptable area (between 4.21 and 5.00), the overall result indicates that users viewed the system as reliable, simple, efficient, and useful to support inventory management, improve business processes, facilitate better-informed decisions by leveraging data, and simplify maintenance workflows. Overall, these findings indicate users believe the system is reliable, easy to use, and simple. The system has also been able to improve the way users manage their inventory, improve operational efficiencies, and provide them with tools to make more informed data-driven decisions. In addition, the findings demonstrate that the system has met its purpose in solving the issues that currently exist in the operation areas and meet the needs of those who will be utilizing the system.

5.2 Conclusions

Based on the development, testing, and evaluation of the proposed system, the following conclusions are drawn:

- With the successful development of the web and mobile Integrated Vehicle Maintenance and Inventory Management System with all functional requirements for JA Noble Enterprise. Inc. have been met. The system has provided a centralized, automated solution to replace the disjointed paper and spreadsheet processes of the transition to digital operations which was much less burdensome than it was in the past, thanks to the move. Integrating vehicle maintenance, inventory and the customer portal on one platform.

- The testing of the system's core modules has been performed, and the results show that the system is consistent. with technical specifications. Actually, 100% pass rate was achieved in all test cases. This applies to Vehicle Maintenance Management as well the Mobile-Based Soundscape project. The module has been engaged in recording service histories. Inventory Management module which is able to update stock via QR/Barcode scanning. Similarly, Decision Support System (DSS) is proven to put out the right low-stock alerts and PMS notifications, while the RAG based AI Assistant has shown it can pull the necessary. To generate responses and reports for management from database records.
- End-users are open to making use of the system. Based on ISO/IEC 25010 standards and the Technology Acceptance Model, the system came in at a weighted mean of 4.86 or “Highly Acceptable.” It is evident that the tool is dependable and trustworthy to administrators, staff and customers. Help them with their work.
- In putting these tools together, we have created a framework to cut down on manual entry and open up lines of communication between the staff and their customers. There continues to be a need for post-deployment. The following month monitoring was performed to observe the actual impact of running in terms of eradication of data. The high acceptance and successful testing leave few errors or stockouts, there was never any doubt about the preparedness of the system. When time is up, the platform will be set to improve decision making at the Caltex level to be more efficient and data driven In San Pedro, Laguna, you will find franchise stores.

5.3 Recommendations

Based on the findings and limitations of this study, the following recommendations are proposed to enhance the system and guide future research:

- Future researchers implement a pilot in all Caltex service stations in Laguna. The goal is to test the system's viability in a multi-branch auto care environment and identify any limitations of scalability before expanding to other regions of the country and to continue to guide our infrastructure upgrades.
- Future researchers ought to consider SMS gateway or bulk messaging options so that automated alerts can be sent out for maintenance. Having real-time payment gateways like PayMaya, GCash or credit card processors will also be useful because it will allow customers to make an advance payment on their reservations, making the billing process much easier and reducing the chances of no-shows.
- Future researchers need to create API integrations that can be connected with supplier networks and POS to facilitate automatic inventory restocking. These can be integrated into ERP modules for HR, accounting and become a true business management system.
- Future researchers should train the NLP model and should be taught to identify slang & Tagish, as well as the ability to use voice commands. Authorized users may also be provided with the ability to add or edit database records without risk using role based transactional AI.

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APPENDICES

APPENDICES A

(TECHNICAL BACKGROUND)

Appendix A.1 TRANSCRIBE INTERVIEW

Date: February 16, 2026

Respondent: Juan Paolo F. Tongco

Position: Operations Head / Director

Researchers: *Good evening po, may mga question lang po kami regarding po sa business niyo po. So tama po, yung name po ng business niyo po is JA Noble Enterprise INC po. Tama po ba yun?*

Mr. Tongco: *Yes. Yes.*

Researchers: *Bali po, ilang taon na po nag-ooperate yung business niyo po?*

Mr. Tongco: *25 years na*

Researchers: *Okay po. And then, bali, ilan po kaya yung empleyado niyo po or staff po na nagtatrabaho po sa inyo? And then, ano po kaya yung kanilang mga roles po?*

Mr. Tongco: *16*

Researchers: *Then, ano po kaya yung mga roles po nila?*

Mr. Tongco: *So, una sa pinaka mababa, CSR, Customer Sales Representative. So, ito yung mga gas boy. Next, Cashier, and then next supervisor. Tapos, sa taas ng supervisor, merong assistant operation head. Sa tas nun, may operation head. Tapos, yung owners na yung taas nun, yung presidente at saka director. Tapos mayroong tinatawag na Admin head. Tapos meron din kaming Lubes master mechanic. Lubes Admin. Iba pa yun sa Lubes master mechanic. Tapos sa baba niya, meron siyang mechanics. Okay na yun, yun na yun.*

Researchers: *Okay po, thank you po. And then, ilang customers po kaya yung bumibisita or pumupunta po sa inyo kada araw?*

Mr. Tongco: *Marami rin.*

Researchers: *And then, ano po yung current process nyo po sa pagmamanage po ng company niyo po?*

Mr. Tongco: *Currently ngayon, naka-cloud kami lahat. Cloud, Excel file, Dropbox.*

Researchers: *Pero may system po kayo na ginagamit?*

Mr. Tongco: *Meron POS. Yun ang pinaka-complicated na. Yun lang yung may bayad na system, POS. Bali, yung POS may shift report. So, doon namin binabalanse.*

Researchers: *And then, um, gaano po, gaano nyo po kadalas, ah, gaano po kayo kadalas nag-checheck ng stock po?*

Mr. Tongco: *Every shift, every shift. Three times a day.*

Researchers: *Ah, three times po? So, parang daily po halos.*

Mr. Tongco: *Yes, hindi halos daily talaga, three times a day. So, three times a day, pagka-end ng shift, magme-measure sila, tapos genegenerate nila yung report sa POS, lalagay nila yung parehas na result na yun sa Excel sheet. Tapos yung Excel sheet may automated na nakakalculate kung ilan yung variance or walang variance mark.*

Researchers: *May any problem po ba kayo na na-encounter po?*

Mr. Tongco: *Wala naman kaming problem ngayon sa operation. Ang problem namin ngayon sa marketing talaga, kailangan namin yung automated Facebook na messenger, tapos yung kapag may nagpa-repair ng sasakyan, matatrack namin kung kailan siya huling nagpa-service sa amin. So mostly marketing ang*

problema namin. Sa marketing, kailangan namin ng system for marketing.

Researchers: *Bali yung una nyo po palang sinend sa akin, yung project 1 po yun po*
yung gusto nyo pong ipagawa po sa amin?

Mr. Tongco: *Oo, yun yung project 1, yun yung pang marketing namin talaga. Kasi wala kaming visibility online. So, kung baga, ang gasolinahan kasi, hindi lang yan basta sa daanan. Pwedeng may naghahanap online dahil kunwari gustong magpakarga ng sampung truck, daming magchat-chat, or merong magpaparepair ng sasakyan. Ito yung sasakyan ko, ito yung master. Kailangan namin ng automated na parang chatbot*
na magche-check, mag-e-estimate.

Researchers: *So, bali yung pangalawa po, yun na po yung meron po kayo?*

Mr. Tongco: *Hindi, yung pangalawa, hindi rin, wala rin kami nun. Yung pangalawa, asset management yun. Ang asset management kasi, kumbaga, yung asset ng mga customer namin, i-mamanage na. So, kunwari, ikaw may sampung truck, Pinapaservice mo yung tatlo ngayong buwan. Tapos, after 3 months, papaservice mo yung natitira pang tatlo. Tapos, after 6 months, papaservice mo yung natitira pang 4. So, after 2 years, hindi mo na alam ngayon, alin ba dito talagang truck yung una kong pinaservice sa amin? Minsan makakalimutan mo, diba? Tapos, ano ba yung pinalitan dun sa unang truck? So, kailangan namin ng ganun to help our customer na, ah, okay pala. Itong tatlong truck lang pala yung pagpapaservice ko ngayong year. So, that's for customer management, hindi siya for operation. Pero yun nga ang mga kailangan namin talaga na asset management.*

Researchers: *Ano po, mobile-based po yung mangyayari po?*

Mr. Tongco: *Sana, oo mobile-based para madali na lang sa amin. Parang*

pipicturan lang namin yung plaka, makikita na doon na, ah, ito two years ago, ito pinaayos sa amin, tapos ito yung mga ginawa. Chinange oil, ah, continue repair, ganun, pinalitan yung brake pad. Makikita namin na bago pa nga yung brake pad.

Researchers: *So, sa ano po, sa inventory po, eh di bali QR code po ba or barcode po*
yung sa inventory?

Mr. Tongco: *Kung pwede, kaya niyo parehas. Usually, although yung mga parts namin naka-barcode yun. May mga barcode naman. Pero kung pwede sana, kaya niya yung barcode, QR code, tsaka yung pipicturan mo yung text. Kaya niya magbasa ng text. Yung iba kasi walang nakasulat na barcode or QR code. Nakasulat lang. TX-1941. Yun lang ung nakasulat.*

Researchers: *Sige po. Gagawin din po ba naming web?*

Mr. Tongco: *Tingin ko masyado ng complicated pag nag web availability pa. Pero Kung Kaya, di, why not? Kung kita mo yung view sa mobile, kung kaya niyong kita rin sa web, why not, diba? Pero ang tingin ko, masyadong mahirap yun. Yung mobile na lang kasi sa mobile app, marami ng ano, marami ng plugins. Marami ng API na available.*

Researchers: *Kung gagawin po namin kayo ng website or app po, sino po kaya yung mga gagamit?*

Mr. Tongco: *Sorry, medyo choppy ka.*

Researchers: *Ano po, sino po kaya yung gagamit? I mean, may admin po ba? May client?*

Mr. Tongco: *Ang gagamit, three types of users lang. Usually kaming mga admin, And then, yung mga tao ko. And then, kung kaya sana, mayroong parang customer view. Kapag nag-login siya, tapos ang type ng login niya is customer, makikita niya kung ano yung*

mga listahan ng asset niya. Una, una yun, listahan ng asset niya. Tapos, pwede niyang makita yung history ng mga ginawa dun sa kada truck. Kahit viewing lang.

Researchers: *So bale po, admin, then staff, then customer po yung magiging user?*

Mr. Tongco: *Yes yes. Yung pinagkaiba nung admin tsaka staff, yung super admin yun yung may access sa lahat. Yung staff, ito yung may mga pili lang. Hindi siya pwede mag delete ng asset. Hindi siya pwede ma-update. Ang pwede lang insert or viewing. Insert new asset or add new maintenance record. So kunwari may dumating na truck, i-picture niya yung truck. Kung wala pa sa system, i-add niya. And then kung nasa system niya, i-update niya lang. So parang new service, change oil, tapos add new service, or change brake pad.Add new service. Add oil. Add new service. Change filter. So, ganun lang. Pero hindi niya ma-edit yung ibang gawa. Then, yung super admin, yun lahat pwede niyang gawin. Add, Update, Delete.*

Researchers: *Dun po sa preventive maintenance services po, paano po yung paglilist po nung ano? Sa preventive maintenance po?*

Mr. Tongco: *Ang PMS, listahan lang yun ng service na ginagawa sa isang sasakyan. So, una, pag may pumuntang truck sa amin, may dadaan lang, ang preventive maintenance ng isang truck, una, kailangan mong i-change oil. Tapos, next, that's one preventive maintenance service.So, isipin mo lang yun, one preliminary service. Tapos, sinilip ng mechanical, sabihin niya, Sir, kailangan pa ng change ng brake pad. That's second, preventive maintenance service. So, sa asset management, para malinawan kayo, may two types of service, preventive maintenance, tsaka corrective. Corrective, kapag sira na. So, nag-LRT ba kayo, MRT, papunta sa mga... Ano? Sa mga... Kahit saan. Nag-LRT, MRT na ba kayo? Sino sa inyo nakapag-LRT na?*

Researchers: *Parang wala pa po.*

Mr. Tongco: *Wala sa inyo. Sige, sino sa inyo nag-bus na?*

Researchers: *Ako po nag bus na.*

Mr. Tongco: *Anong bus company yung sinasakyan mo?*

Researchers: *Ano po, yung DLTB?*

Mr. Tongco: *DLTB, okay. So, kunwari ako yung may-ari ng DLTB. Kailangan ng maintenance ng mga bus, di ba? Pag hindi ako nag-preventive maintenance, sabihin natin binili ko yung 10 bus ko 20 years ago. Tapos ang isang bus, meron yung mga nag-de-degrade na part. After 10 years, merong sabihin nating gear na masisira.Kapag hindi ako nag PMS, may chance na yung sampung bus ko, kailangan ko ng corrective maintenance. Kasi nung biniyahe ko sila, nasira lahat ng gear nila sabay-sabay. Ang mangyayari, wala akong kita. It takes two weeks to order. Ibig sabihin, two weeks akong walang negosyo kasi hindi ko na-maintain yung bus ko.May mga parts pa lang every 10 years dapat pinapalitan. So, that's corrective maintenance kapag nasira na yung part. So, para maiwasan yun, kailangan mo ng PMS or preventive maintenance. So, yung S-schedule,right, or service na lang. PMS, preventive maintenance service. So, ipapasok mo ngayon yan sa mga nag-check ng bus. So, sisilipin yan ng mekaniko na, oh, itong bus mo, sir, itong bus sampung taon na. Usually, yun ang tinatawag naman sa sampung taon, meter, asset meter. Usually, ang bus na sampung taon na ng meter, kailangan palitan ng ganyang gear. Next, sisilipin ng mekaniko. May isa pang asset meter, yung odometer.So, pag tinignan niyo yung sasakyan sa may dashboard, may gumagalaw doon na kilometro na biyahe ng bus. 30,000 kilometers, 50,000 kilometers. Basta kada biyahe ng bus, gumagalaw yun kada isang kilometro. So, isa ring meter yun. Nasa libro rin ng mga mekaniko, nasabihin nila, bus yung*

bus mo, 30,000 kilometers na kailangan ng palitan itong part na to sa mga ganitong pag inabot na yung ganitong meter. Okay. So, yun naman ang asset meters. So, yun ang PMS. Yun ang preventive maintenance. Copy?

Researchers: *Yes po. Bale, sa pag-inventory po ba, ano po yung problem po na i-encounter niyo po?*

Mr. Tongco: *Sa inventory naman kasi, itong mahirap sa spare parts. So, kunwari, meron kang sampung bus ulit. Sabihin natin doon yung tatlong mong bus, diesel, manual. Yung isa mo pang tatlong bus, gasoline, automatic. Tapos yung natitira mong apat na bus, electric. So, kada part doon, tapos iba-iba sila. Yung isa Toyota, yung isa Mitsubishi, yung isa Honda. Nung pinacheck mo yung Honda, na manual na diesel, kapag nag-PMS yan, iba ang parts niyan. Iba ang in-order. Okay? So lahat-lahat yan may kanya-kanyang timpla kung paano pini-PMS at iba-iba rin yung parts niyan. Ibig sabihin, iba yung gear nung manual na Toyota. Iba pa yun sa gear ng manual na Honda. Iba pa yun sa gear ng bus na electric. At iba pa yung gear kapag diesel at gasoline. Ibig sabihin kami as service provider, hindi namin yan kayang i-imbak sa inventory, lahat ng parts na yan, kung hindi naman gagawin ito. Pero kung meron kaming app na magsasabing, o itong types of bus, by next week, due for PMS na, kasi tatlong taon na silang hindi napapaservice, mache-check namin, ito yung mga kakailangang pyesa para sa mga parts na ito. Yun lang yung bibilin namin. Nag-gets ba yung business value niya?*

Researchers: *Yes po.*

Mr. Tongco: *So gano'n rin, kung kahit hindi kayo service station, pag kayo'y may ari ng bus, sabihin nyo may sarili kayong mekaniko na may maintain nung mga bus nyo, ayaw mo rin mag-imbak ng lahat ng pyesa ng iyon sa isang taon. Gusto mo, mag-imbak ka lang pag alam mo ng malapit na siyang masira. Dun ka lang bibili.*

Alright? So yun yung road ng inventory.

Researchers: *Sa asset maintenance naman po?*

Mr. Tongco: *Yung asset maintenance naman, kada isang asset, listahan lang yun ng mga preventive maintenance service na ginawa dyan sa asset. Sa database, one is too many records lang yun. One asset equals too many preventive maintenance services. So, pag tinignan mo yung asset, lalabas na dun yung record mo ng mga preventive maintenance na nagawa for the past.*

Researchers: *So yun, bali, yun lang po muna yung mga questions po namin. Magchachat na lang po ako sa inyo kapag may mga follow up questions pa po kami.*
Salamat po.

Mr. Tongco: *Okay, thank you din.*

User Storyboard

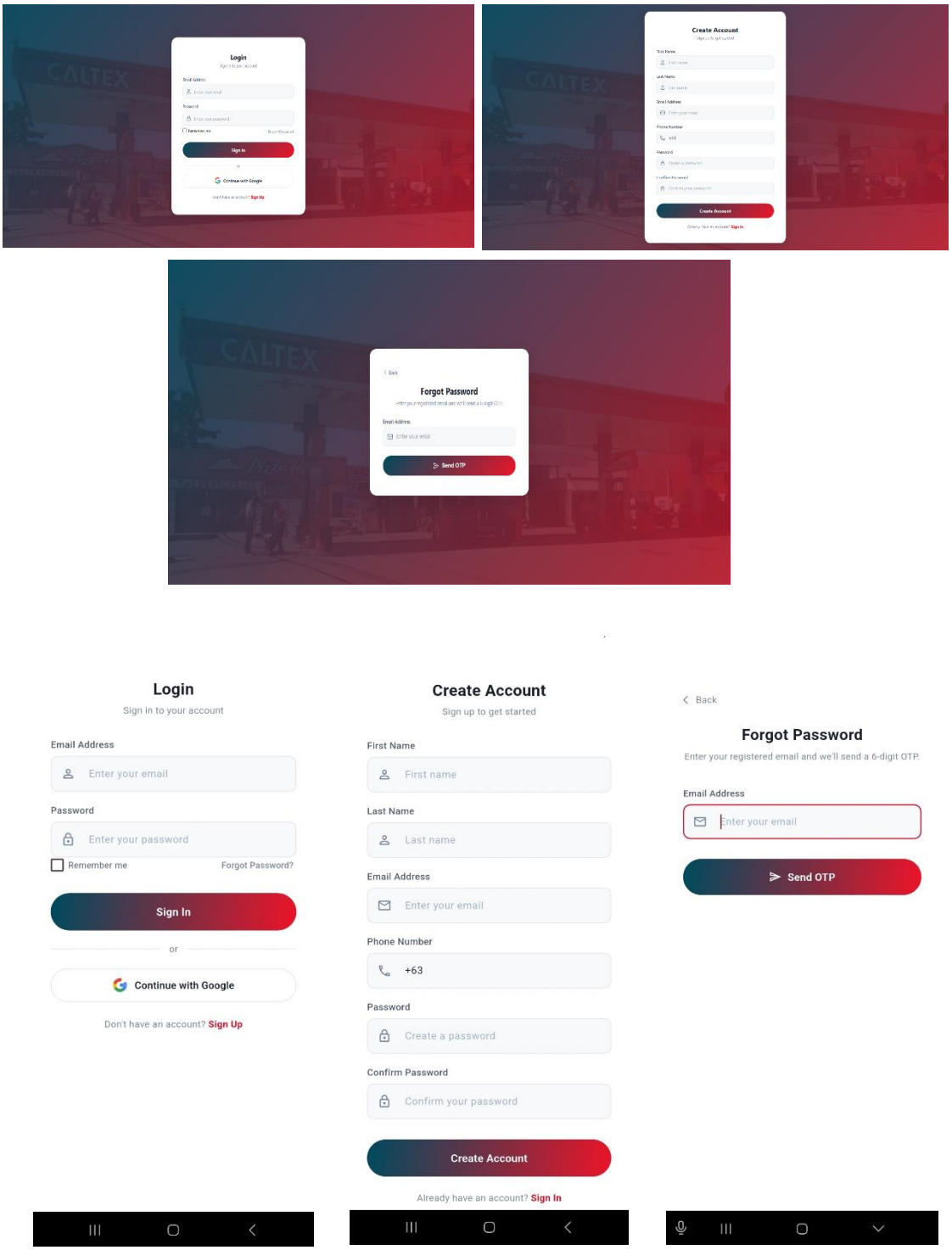

Figure 42. Web and Mobile App Interface for Login, Register, and Forgot password

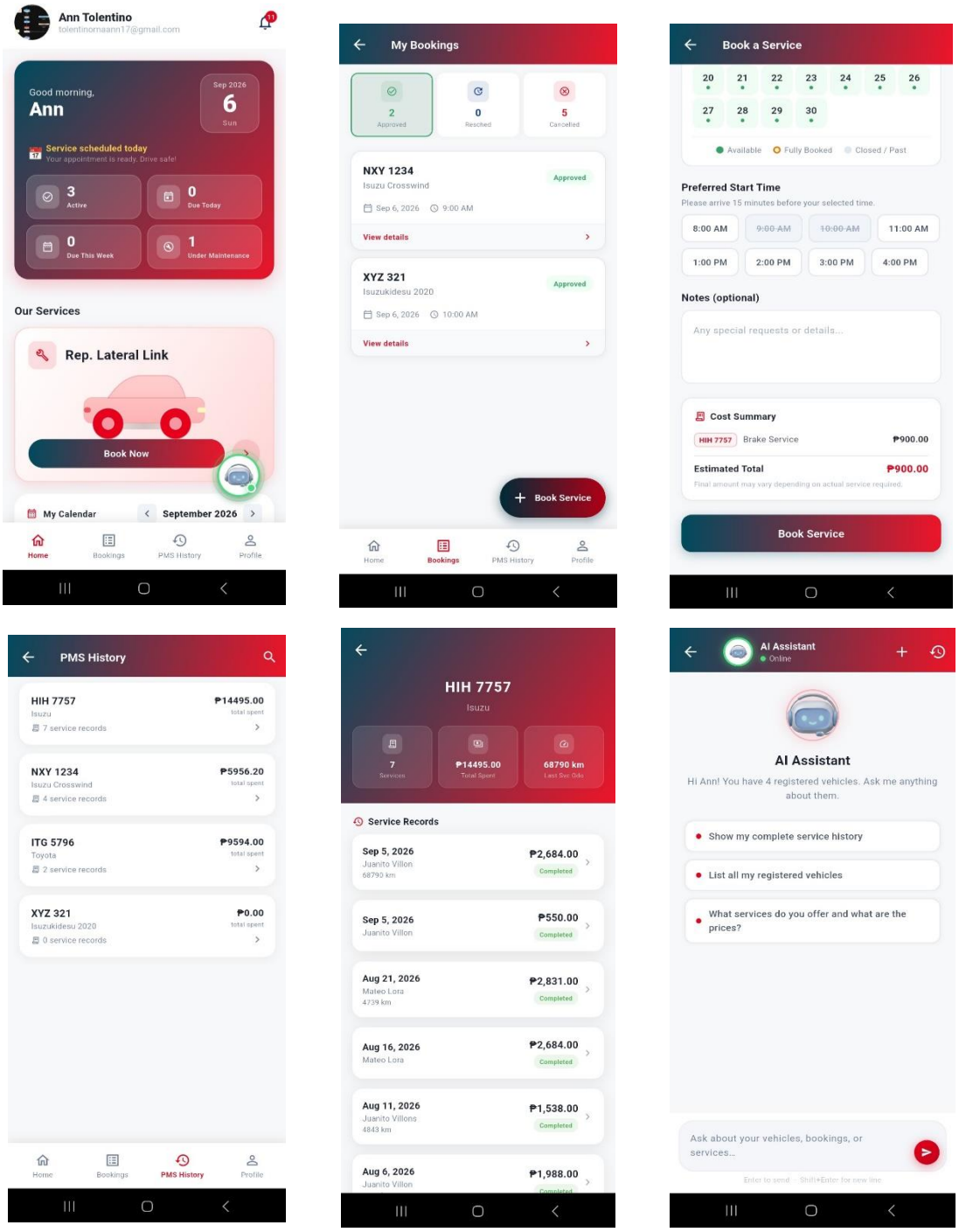

Figure 43. Mobile App Interface for Customer Account

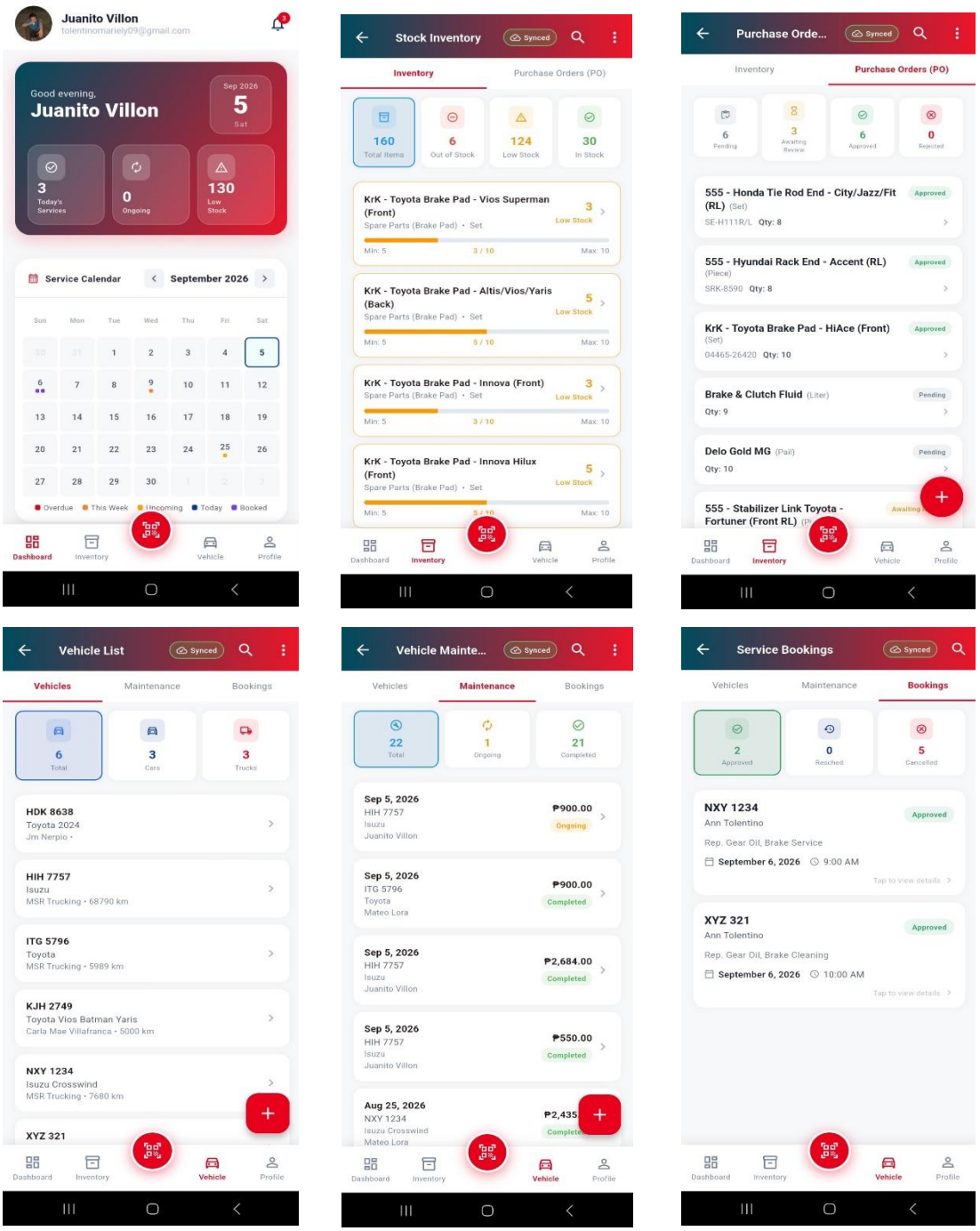

Figure 44. Mobile App Interface for Staff Account

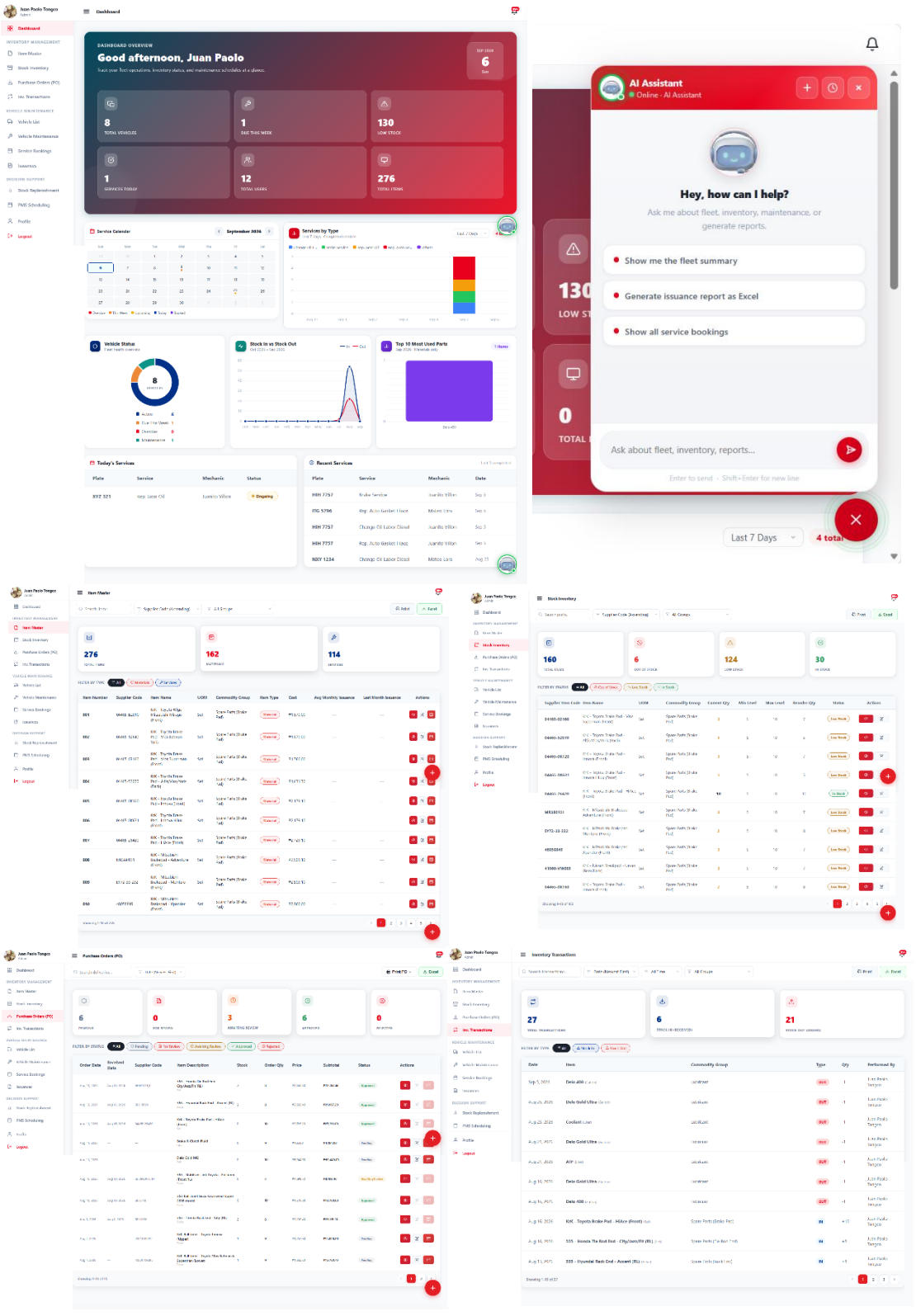

Figure 45. Web Interface for Admin Account

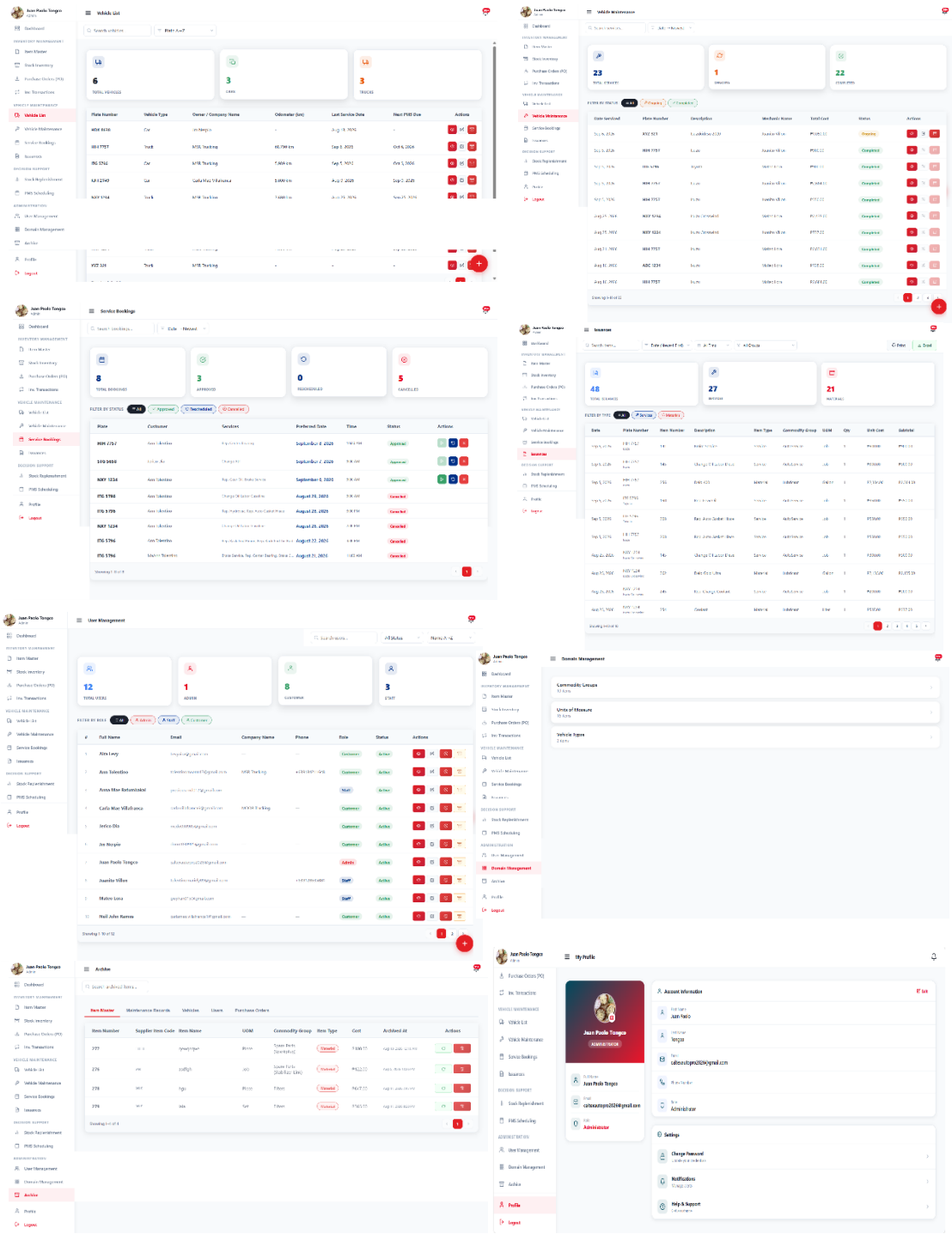

Figure 45. Web Interface for Admin Account

Proposed Process Flowchart

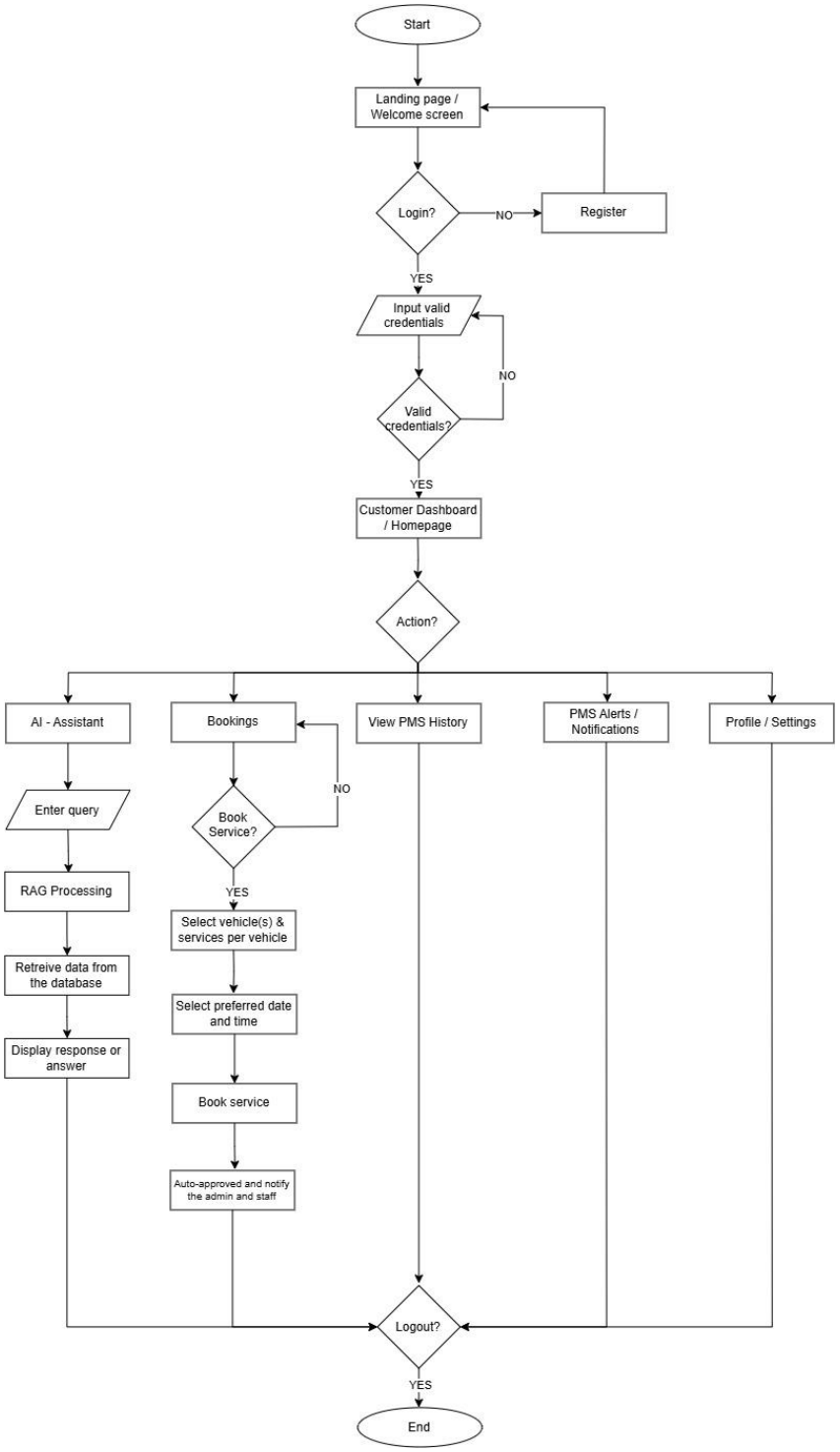

Figure 46. The Proposed Process Flowchart of the Customer of the study entitled "Integrated Vehicle Maintenance and Inventory Management System with DSS for JA Noble Enterprise Inc."

Proposed Process Flowchart

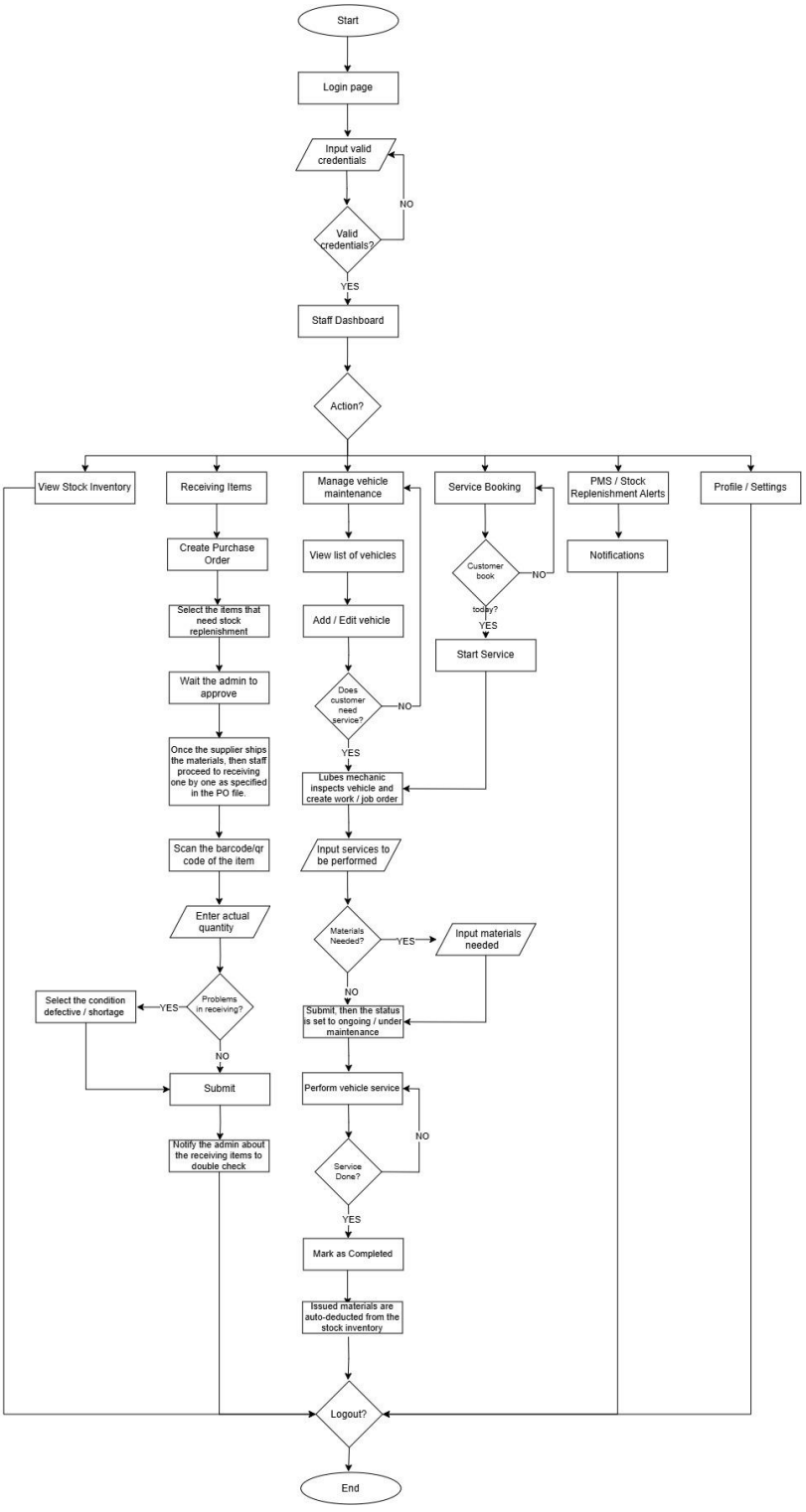

Figure 47. The Proposed Process Flowchart of the Staff of the study entitled "Integrated Vehicle Maintenance and Inventory Management System with DSS for JA Noble Enterprise Inc"

Proposed Process Flowchart

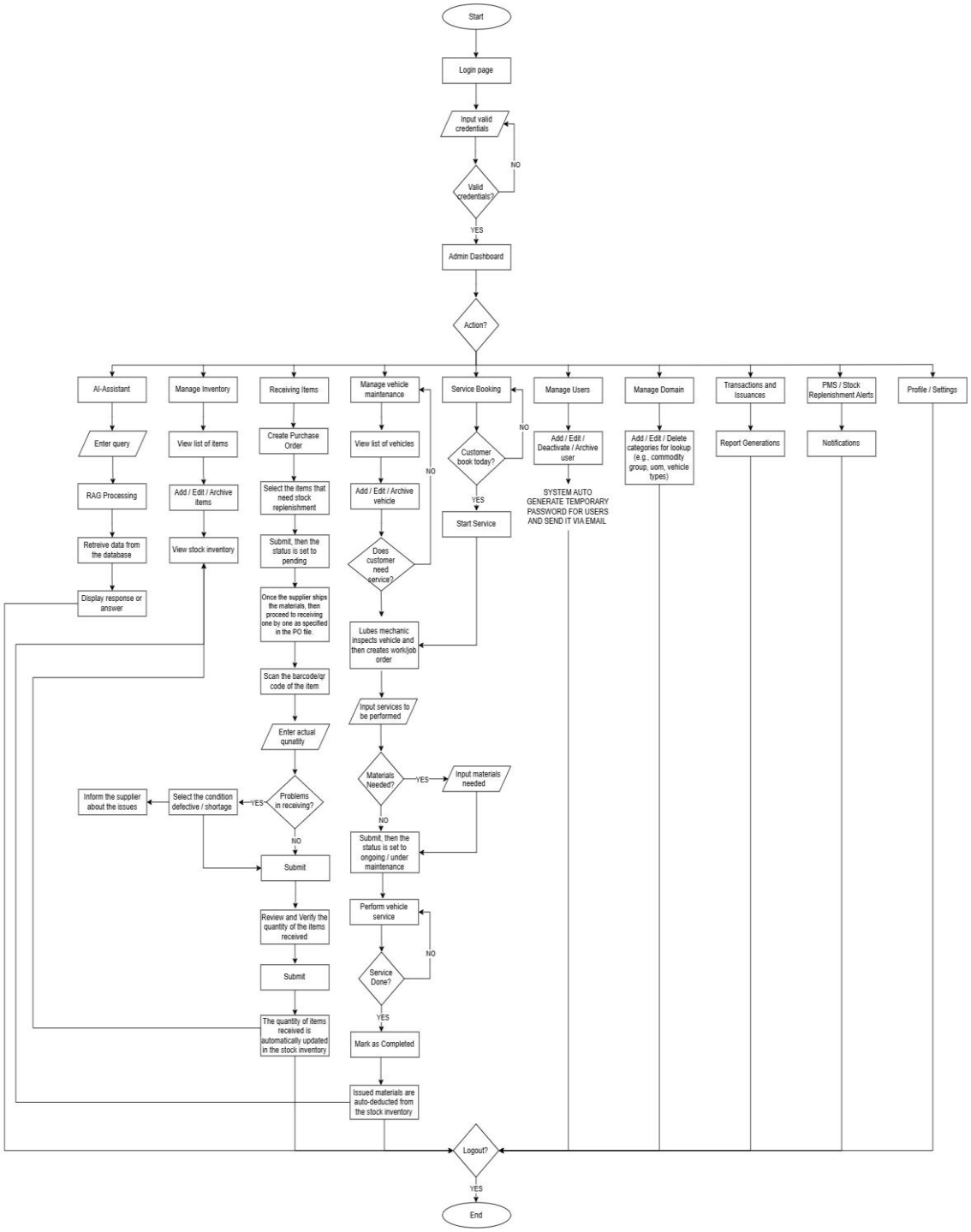

Figure 48. The Proposed Process Flowchart of the Admin of the study entitled "Integrated Vehicle Maintenance and Inventory Management System with DSS for JA Noble Enterprise Inc"

Proposed Used Case Diagram

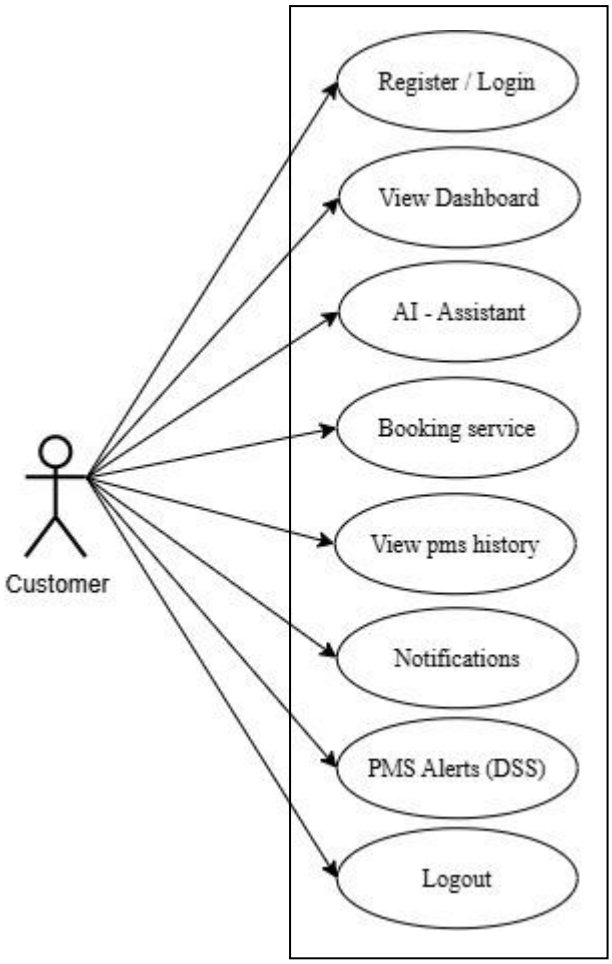

Figure 49. The Proposed Use Case Diagram of the Customer for Integrated Vehicle Maintenance and Inventory Management System with DSS for JA Noble Enterprise Inc.

Proposed Used Case Diagram

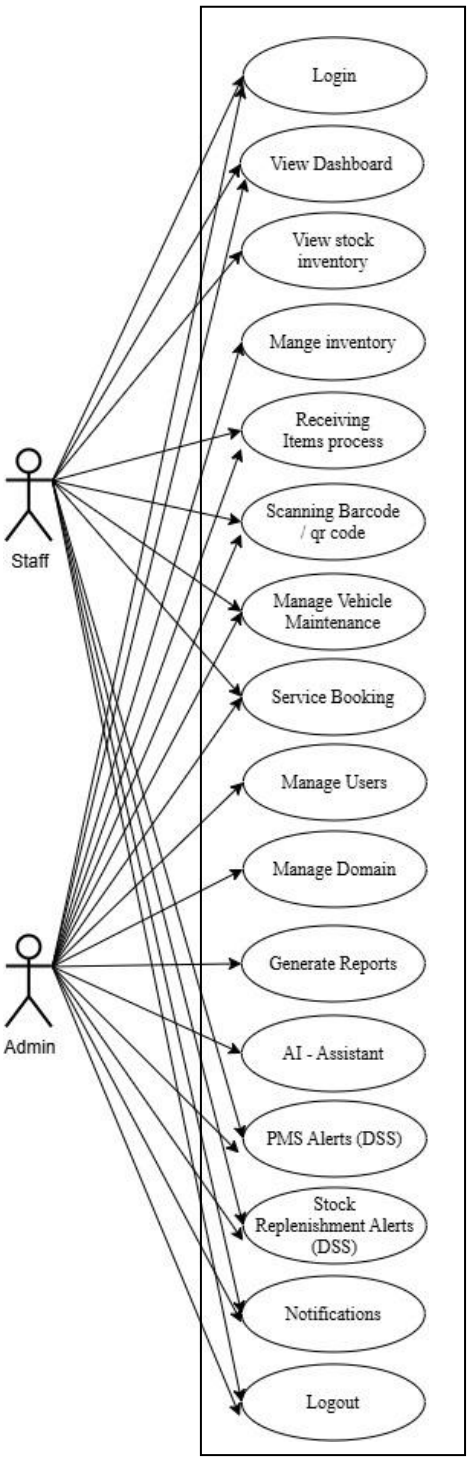

Figure 50. The Proposed Use Case Diagram of the Staff and Admin for Integrated Vehicle Maintenance and Inventory Management System with DSS for JA Noble Enterprise Inc.

System Context Diagram

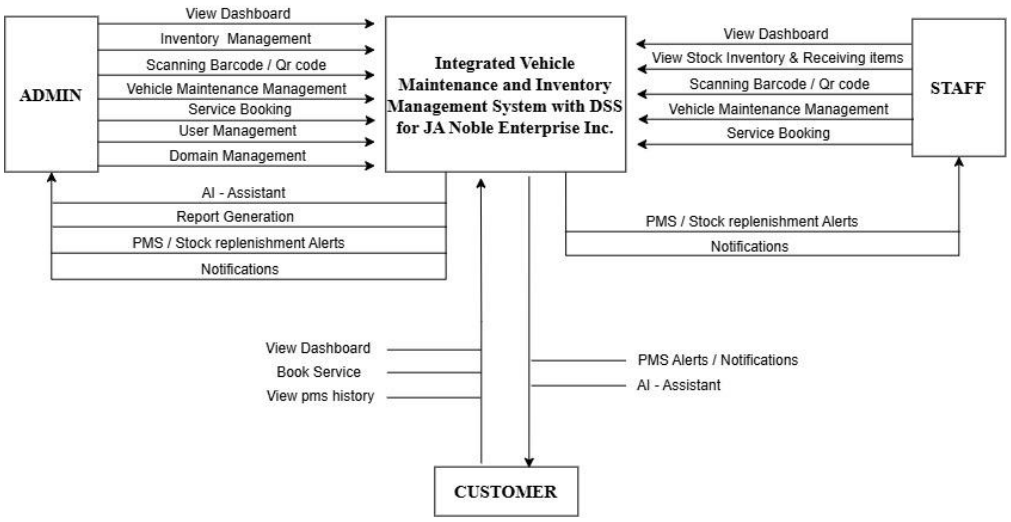

Figure 51. System Context Diagram of the study entitled "Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc."

Data Flow

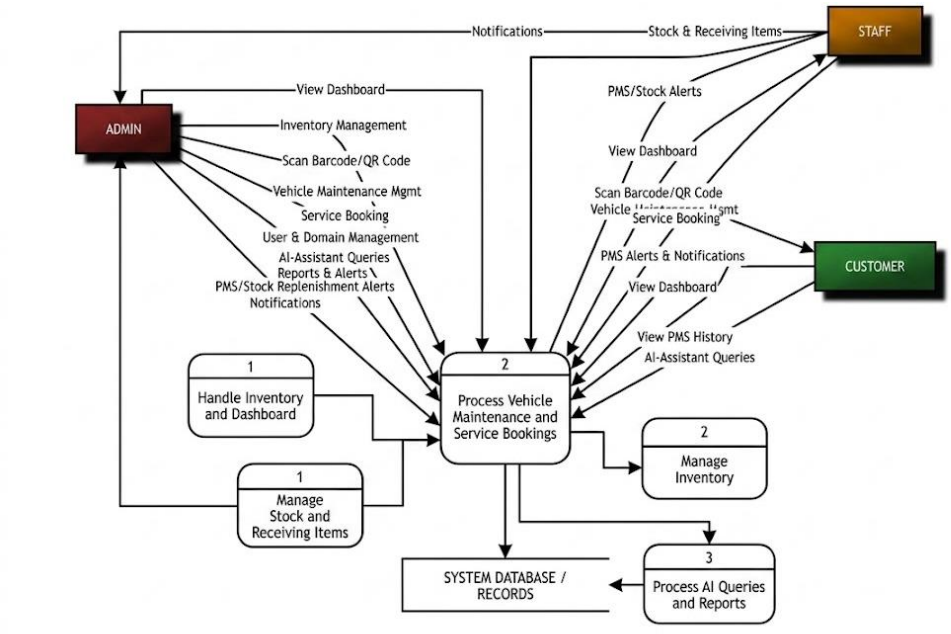

Figure 52. Data Flow Diagram of the study entitled "Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc."

Figure 51 and 52 illustrates the interaction between the system and its external entities, namely the Admin, Staff, and Customer. The diagram shows how data such as inventory management, vehicle maintenance management, service booking, user and domain management, and barcode/QR code scanning are entered into the system by the Admin and Staff. In return, the system generates outputs including dashboard views, report generation, PMS / stock replenishment alerts, notifications, and AI-Assistant support. Furthermore, it details how Customers interact with the system by booking services, viewing dashboards, and checking PMS history, while receiving PMS alerts, notifications, and AI assistance. Overall, the context diagram demonstrates how the Integrated Vehicle Maintenance and Inventory Management System with DSS effectively manages vehicle servicing, stock inventory, and decision support processes for JA Noble Enterprise Inc.

DATA DICTIONARY

Table 1. The table below displays the data dictionary of User Accounts

Field Name	Data Type	Description
uid	string	Firebase Auth UID; mirrors the document ID
firstname	string	User's first name
lastname	string	User's last name
companyName	string	Company or fleet operator name
email	string	User's email address
phoneNumber	string	User's phone number
photoUrl	string	User's profile photo
role	string	Access role; controls feature visibility in the app/system
status	string	Account status
mustChangePassword	sboolean	Set to true for admin-created accounts; forces password change on first login
archived	string	Soft-delete flag
archivedAt	timestamp	Server timestamp when the user was archived
createdAt	timestamp	Server timestamp written on account creation

Table 2. The table below displays the data dictionary of Item Master

Field Name	Data Type	Description
itemNum	string	Unique system-generated identifier for each item
supplierCode	string	Supplier's item code; used as the seed for barcode generation
oem	string	OEM (Original Equipment Manufacturer) part number
itemName	string	Item or service name
desc	string	Detailed description of the item
group	string	Commodity group category
uom	string	Unit of measure
cost	number	Unit cost
type	string	Distinguishes physical materials from labor/service items
barcode	string	Barcode value for scanning
qr	string	QR code value for scanning
archived	boolean	Soft-delete flag
archivedAt	timestamp	Server timestamp when the item was archived
createdAt	timestamp	Server timestamp of document creation

Table 3. The table below displays the data dictionary of Stock Inventory

Field Name	Data Type	Description
itemNum	string	Item number; synced from the item master catalog
supplierCode	string	Supplier's own item code
oem	string	Original equipment manufacturer
name	string	Item name
group	string	Commodity group
uom	string	Unit of measure
stock	number	Current quantity on hand
min	number	Minimum stock level; stock at or below these triggers Low status
max	number	Maximum stock level; stock above these triggers Over status
reorder	number	Reorder quantity; calculated as max - stock
Status	string	Stock level status: Low if stock ≤ min, over if stock > max, else OK
updatedAt	timestamp	Server timestamp of last stock update
createdAt	timestamp	Server timestamp of document creation

Table 4. The table below displays the data dictionary of Receiving Items

Field Name	Data Type	Description
id	string	Unique system-generated identifier for each purchase order
status	string	Overall delivery status
receivedBy	string	Name of the staff who received the delivery
checkedBy	string	Name of the admin who performed double-check
reviewNotes	string	Admin comments from the review step
reviewAt	timestamp	When the admin reviewed the delivery
receivedAt	timestamp	When the delivery was received
createdAt	timestamp	When the delivery order was created
items	array of maps	Embedded list of line items in the delivery (see below)

Table 5. The table below displays the Embedded Map: items[*]

Field Name	Data Type	Description
itemNum	string	Item number
itemName	string	Item name
uom	string	Unit of measure
supplierCode	string	Supplier's item code
orderQty	number	Ordered quantity
actualQty	number	Quantity actually received; filled during receiving step
condition	string	Condition of received item
remark	string	Per-item remarks
stockAtReceive	number	Stock level frozen at receive time for reference

Table 6. The table below displays the data dictionary of Transactions

Field Name	Data Type	Description
id	string	Unique system-Generated identifier for each transaction
itemName	string	Name of the item involved in the transaction
commodityGroup	string	Commodity group of the item
type	string	Direction of stock movement: IN for receiving, OUT for issuance
qty	string	Quantity
date	date	Display date of the transaction
by	string	Full name of the staff or admin who performed the action
createdAt	timestamp	Server timestamp of document creation

Table 7. The table below displays the data dictionary of Vehicles

Field Name	Data Type	Description
id	string	Unique system-generated identifier for each vehicle
ownerId	string	Firebase Auth UID of the owning customer; authoritative foreign key
ownername	string	Customer full name or company name; used for fuzzy owner-to-vehicle joins
plateNumber	string	Vehicle plate number
description	string	Vehicle make, model, and year
type	string	Vehicle type (e.g., Car, Truck)
odometer	number	Current odometer reading
lastServiceDate	date	Date of most recent completed service
lastServiceOdometer	number	Odometer reading at the time of the last service
serviceFrequency	number	How often scheduled PMS should occur
status	string	Current operational status of the vehicle
nextPMSDue	date	Calculated next PMS due date; updated on service completion
completedAt	timestamp	When the latest maintenance was completed
archived	boolean	Soft-delete flag; true means vehicle is archived
archivedAt	timestamp	Server timestamp when the vehicle was archived
createdAt	timestamp	Server timestamp of document creation

Table 8. The table below displays the data dictionary of Maintenance

Field Name	Data Type	Description
id	string	Unique system-Generated identifier for individual vehicle service and repair job records.
vehicleId	string	Firestore document ID of the serviced vehicle
ownerId	string	Firebase Auth UID of the vehicle owner; authoritative FK
plateNumber	string	Vehicle plate number
description	string	Vehicle description captured when the job was created
mechanic	string	Full name of the assigned mechanic or staff member
date	date	Display date of the service
odometer	number	Odometer reading at time of service
cost	number	Total job cost
status	string	Current job status
issues	array of strings	Vehicle issues reported by the customer (e.g., ["Brake Issue", "Oil Leak"])
svcRows	array of maps	Services performed; see embedded map below
matRows	array of maps	Materials used; see embedded map below
bookingId	string	Links back to the originating booking, if the job came from a customer booking

archived	boolean	Soft-delete flag
archivedAt	timestamp	Server timestamp when record was archived
createdAt	timestamp	Server timestamp of document creation

Table 9. Embedded Map: svcRows[*] and matRows[*]

Field Name	Data Type	Description
itemName	string	Item or service name (FK to item_master.name)
qty	number	Quantity (stored as string)
uom	string	Unit of measure
cost	number	Unit cost as a numeric string

Table 10. The table below displays the data dictionary of Service Bookings

Field Name	Data Type	Description
id	string	Unique system-generated identifier for each service booking
customerId	string	Firebase Auth UID of the booking customer
customerName	string	Customer full name captured at booking service
vehicleId	string	Firestore document ID of the vehicle to be serviced
plateNumber	string	Vehicle plate number captured at booking serviced
vehicleDesc	string	Vehicle description captured at booking serviced
services	array of strings	Names of the services requested
preferredDate	timestamp	Customer's preferred service date
preferredTime	timestamp	Customer's preferred time slot
notes	string	Additional notes or instructions from the customer
status	string	Booking lifecycle status
previousDate	timestamp	Originally preferred date before rescheduling
preferredTime	timestamp	Customer's preferred time slot
rescheduledBy	string	Who performed the reschedule
rescheduledAt	timestamp	When the booking was rescheduled
cancelledAt	timestamp	When the booking was cancelled
cancelledBy	string	Who or what triggered the cancellation
cancellationReason	string	Reason for system or customer cancellation
createdAt	timestamp	Server timestamp of booking creation

Table 11. The table below displays the data dictionary of Issuances

Field Name	Data Type	Description
id	string	Unique system-generated identifier for each issuance
maintenanceId	string	Firestore document ID of the maintenance
plateNumber	string	Vehicle plate number
vehicleDesc	string	Vehicle description
itemNum	string	Item number of the issued material or service
itemName	string	Item name
itemType	string	Whether this issuance is for physical material or a service
commodityGroup	string	Commodity Group
uom	string	Unit of measure
qty	number	Quantity issued
unitCost	number	Unit cost
subtotal	number	Total cost
date	string	Display date of issuance
createdBy	string	Full name of the admin or staff who created the record
createdAt	timestamp	Server timestamp of document creation

Table 12. The table below displays the data dictionary of Domains

Field Name	Data Type	Description
id	string	Unique system-generated identifier for each domain
name	string	Display value shown in dropdowns
createdAt	timestamp	Server timestamp of entry creation

Table 13. The table below displays the data dictionary of Notifications

Field Name	Data Type	Description
id	string	Unique system-generated identifier for each notification
title	string	Notification title
message	string	Notification body text
type	string	Notification severity/category for visual styling
targetRole	string	Role of users who should see this notification
targetUid	string	Firebase UID of the specific recipient (single-user notifications)

readBy	map	Tracks which users have read the notification; key is Firebase UID, value is true
isRead	boolean	Convenience read flag; initially false for all; not updated server-side (deprecated in favor of readBy)
createdAt	timestamp	Server timestamp of notification creation

PROJECT SCHEDULE

This section displays the research study's project schedule. To complete this research project, the researchers went through several stages.

TASK	2026								
	MAR	APR	MAY	JUN	JUL	AUG	SEPT	OCT	NOV
Planning									
Requirement									
Design									
Development									
Testing									
Deployment									

Hardware and Software Resources

This section discusses the software and hardware resources that are needed to accomplish the study.

The system needs both hardware and software elements to achieve proper development and operational performance and user accessibility. For hardware, the development environment utilizes personal laptops powered by an AMD Ryzen 5 (7000 Series) processor and 16GB RAM, while operational testing relies on Samsung Android smartphones for mobile access and built-in cameras for barcode or QR code scanning. For software, the development and operation run on the Windows 11 operating system. The website backend is engineered using Python with the Flask web framework, while the mobile application is constructed using the Flutter framework and the Dart programming language. The entire platform is integrated with Cloud Firestore through Firebase as its centralized database. Additionally, advanced technical requirements include FastAPI 0.115.0 and Uvicorn 0.30.6 for the AI backend, the Groq SDK deploying Qwen3.6 27B and Qwen3 32B Large Language Models, Pydantic for data validation, cryptography libraries for secure encryption, and reporting tools such as ReportLab and openpyxl.

Testing and Evaluation Instruments

This section presents the testing and evaluation tools that the researcher used in conducting the user evaluation and testing phase of the study.

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TECHNOLOGY ACCEPTANCE QUESTIONNAIRE

Dear Respondents,

We are student researchers from Laguna State Polytechnic University currently conducting a study entitled "Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc." As part of our study, we developed a website and mobile application designed to improve vehicle maintenance monitoring, inventory management, stock replenishment, and preventive maintenance scheduling. In line with this, we would like to request your participation in evaluating the system by answering this acceptance questionnaire.

Your honest feedback and evaluation are highly valuable to our research. The information and responses gathered will help us assess the functionality, usability, efficiency, reliability, and overall acceptability of the developed system. The results of this study will be used solely for academic and research purposes.

Please be assured that all information you provide will be treated with strict confidentiality and will be handled in accordance with applicable data privacy policies. Participation in this survey is voluntary, and you may answer the questionnaire based on your actual experience and observations while using the system.

We sincerely appreciate your time, effort, and cooperation in supporting our study. Your participation will greatly contribute to the improvement and success of this research.

Thank you for your time and cooperation.

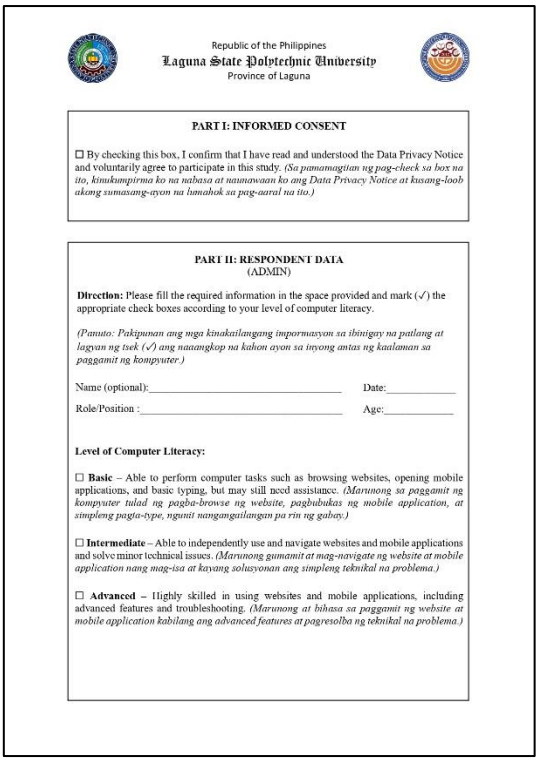
Respectfully,

Rondilla, Precious C.
Researcher

Tolentino, Mariely Ann L.
Researcher

Villafranca, Carla Mae R.
Researcher

Questionnaire for Admin



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PART I: INFORMED CONSENT

☐ By checking this box, I confirm that I have read and understood the Data Privacy Notice and voluntarily agree to participate in this study. *(Sa pamamagitan ng pag-check sa box na ito, kinukumpirma ko na nabasa at nauunawaan ko ang Data Privacy Notice at kusang-loob akong sumasang-ayon na lumahok sa pag-aaral na ito.)*

PART II: RESPONDENT DATA
(ADMIN)

Direction: Please fill the required information in the space provided and mark (✓) the appropriate check boxes according to your level of computer literacy.

(Panuto: Pakipunan ang mga kinakailangang impormasyon sa ibinigay na patlang at lagyan ng tsek (✓) ang naaangkop na kahon ayon sa inyong antas ng kaalaman sa paggamit ng kompyuter.)

Name (optional):_____ Date:_____

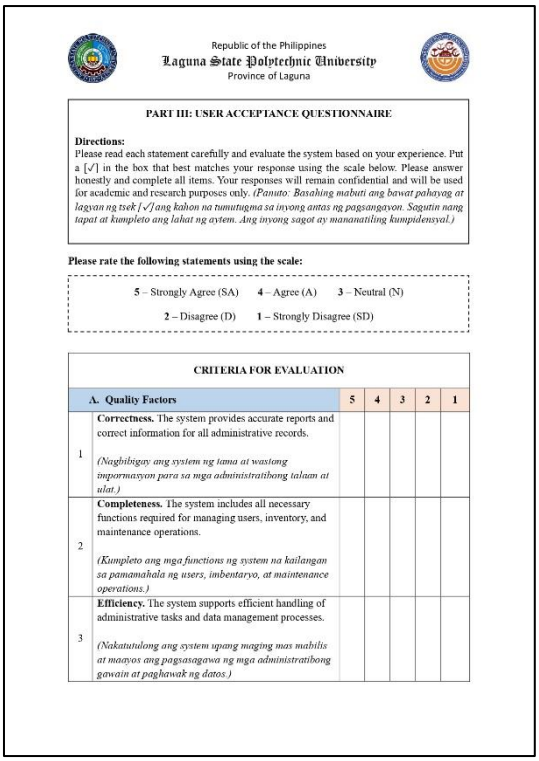
Role/Position :_____ Age:_____

Level of Computer Literacy:

☐ **Basic** – Able to perform computer tasks such as browsing websites, opening mobile applications, and basic typing, but may still need assistance. *(Marunong sa paggamit ng kompyuter tulad ng pagba-browse ng website, pagbubukas ng mobile application, at simpleng pagta-type, nguni't nangangailangan pa rin ng gabay.)*

☐ **Intermediate** – Able to independently use and navigate websites and mobile applications and solve minor technical issues. *(Marunong gumamit at mag-navigate ng website at mobile application nang mag-isa at kayang solusyonan ang simpleng teknikal na problema.)*

☐ **Advanced** – Highly skilled in using websites and mobile applications, including advanced features and troubleshooting. *(Marunong at bihasa sa paggamit ng website at mobile application kabilang ang advanced features at pagresolba ng teknikal na problema.)*



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Province of Laguna

PART III: USER ACCEPTANCE QUESTIONNAIRE

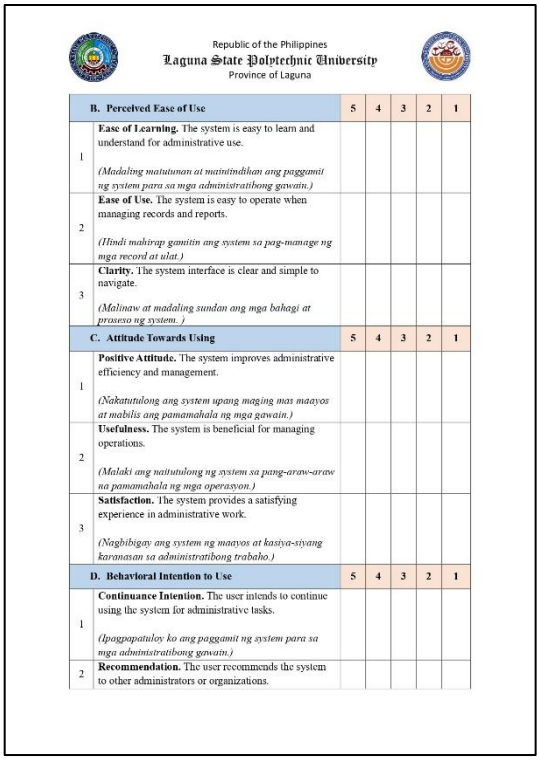
Directions:
Please read each statement carefully and evaluate the system based on your experience. Put a [✓] in the box that best matches your response using the scale below. Please answer honestly and complete all items. Your responses will remain confidential and will be used for academic and research purposes only. *(Panuto: Basahing mabuti ang bawat pahayag at lagyan ng tsek [✓] ang kahon na tumutugma sa inyong antas ng pagsangayon. Sagutin nang tapat at kumpleto ang lahat ng aytem. Ang inyong sagot ay mananatiling kumpidensyal.)*

Please rate the following statements using the scale:

5 – Strongly Agree (SA) **4** – Agree (A) **3** – Neutral (N)
2 – Disagree (D) **1** – Strongly Disagree (SD)

CRITERIA FOR EVALUATION

	A. Quality Factors	5	4	3	2	1
1	**Correctness.** The system provides accurate reports and correct information for all administrative records. *(Nagbibigay ang system ng tama at wastong impormasyon para sa mga administratibong talaan at ulat.)*					
2	**Completeness.** The system includes all necessary functions required for managing users, inventory, and maintenance operations. *(Kumpleto ang mga functions ng system na kailangan sa pamamahala ng users, imbentaryo, at maintenance operations.)*					
3	**Efficiency.** The system supports efficient handling of administrative tasks and data management processes. *(Nakatutulong ang system upang maging mas mabilis at maayos ang pagsasagawa ng mga administratibong gawain at paghawak ng datos.)*					



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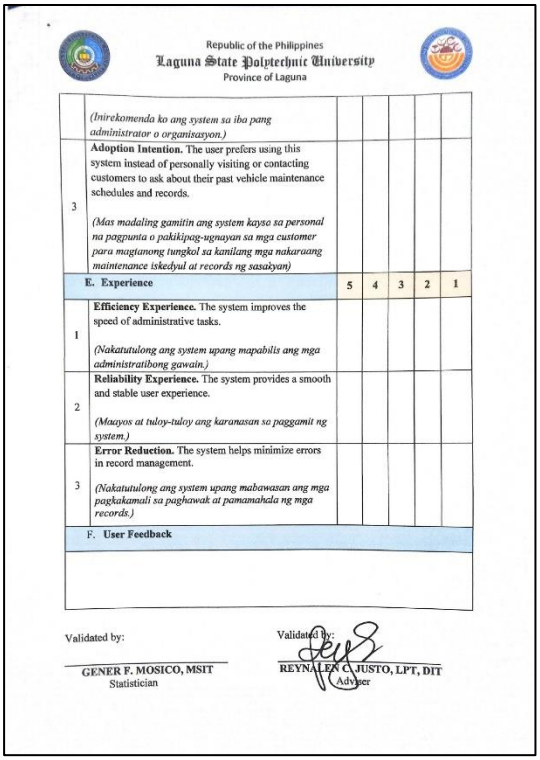
Province of Laguna

	B. Perceived Ease of Use	5	4	3	2	1
1	**Ease of Learning.** The system is easy to learn and understand for administrative use. *(Madaling matutunan at mainitindihan ang paggamit ng system para sa mga administratibong gawain.)*					
2	**Ease of Use.** The system is easy to operate when managing records and reports. *(Hindi mahirap gamitin ang system sa pag-manage ng mga record at ulat.)*					
3	**Clarity.** The system interface is clear and simple to navigate. *(Malinaw at madaling sundan ang mga bahagi at proseso ng system.)*					
	C. Attitude Towards Using	5	4	3	2	1
1	**Positive Attitude.** The system improves administrative efficiency and management. *(Nakatutulong ang system upang maging mas maayos at mabilis ang pamamahala ng mga gawain.)*					
2	**Usefulness.** The system is beneficial for managing operations. *(Malaki ang naitutulong ng system sa pang-araw-araw na pamamahala ng mga operasyon.)*					
3	**Satisfaction.** The system provides a satisfying experience in administrative work. *(Nagbibigay ang system ng maayos at kasiya-siyang karanasan sa administratibong trabaho.)*					
	D. Behavioral Intention to Use	5	4	3	2	1
1	**Continuance Intention.** The user intends to continue using the system for administrative tasks. *(Ipagpapatuloy ko ang paggamit ng system para sa mga administratibong gawain.)*					
2	**Recommendation.** The user recommends the system to other administrators or organizations. *(Inirekomenda ko ang system sa iba pang administrator o organisasyon.)*					



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3	**Adoption Intention.** The user prefers using this system instead of personally visiting or contacting customers to ask about their past vehicle maintenance schedules and records. *(Mas madaling gamitin ang system kaysa sa personal na pagpunta o pakikipag-ugnayan sa mga customer para magtanong tungkol sa kanilang mga nakaraang maintenance iskedyul at records ng sasakyan)*					
	E. Experience	5	4	3	2	1
1	**Efficiency Experience.** The system improves the speed of administrative tasks. *(Nakatutulong ang system upang mapabilis ang mga administratibong gawain.)*					
2	**Reliability Experience.** The system provides a smooth and stable user experience. *(Maayos at tuloy-tuloy ang karanasan sa paggamit ng system.)*					
3	**Error Reduction.** The system helps minimize errors in record management. *(Nakatutulong ang system upang mabawasan ang mga pagkakamali sa paghawak at pamamahala ng mga records.)*					
	F. User Feedback					

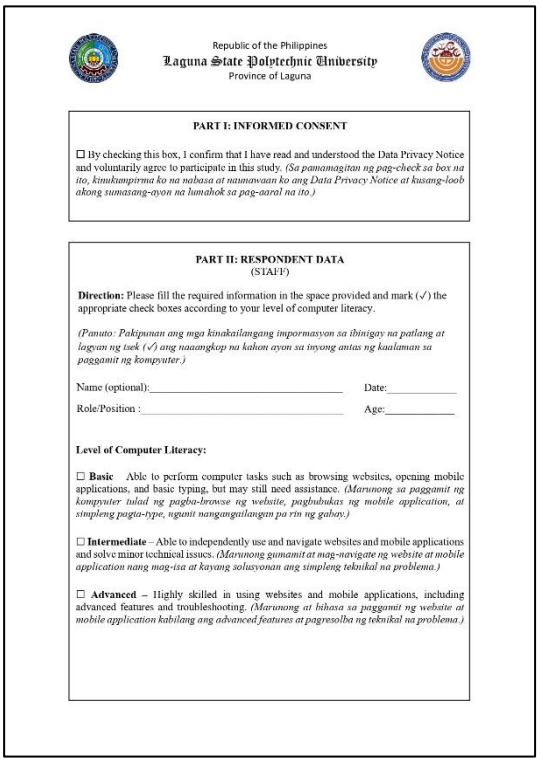
Validated by:

GENER F. MOSICO, MSIT
Statistician

Validated by:

REYNALEN C. JUSTO, LPT, DIT
Adviser

Questionnaire for Staff



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PART I: INFORMED CONSENT

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PART II: RESPONDENT DATA
(STAFF)

Direction: Please fill the required information in the space provided and mark (✓) the appropriate check boxes according to your level of computer literacy.

(Panuto: Pakipunan ang mga kinakailangang impormasyon sa ibinigay na patlang at lagyan ng tsek (✓) ang naaangkop na kahon ayon sa inyong antas ng kaalaman sa paggamit ng kompyuter.)

Name (optional):_____ Date:_____

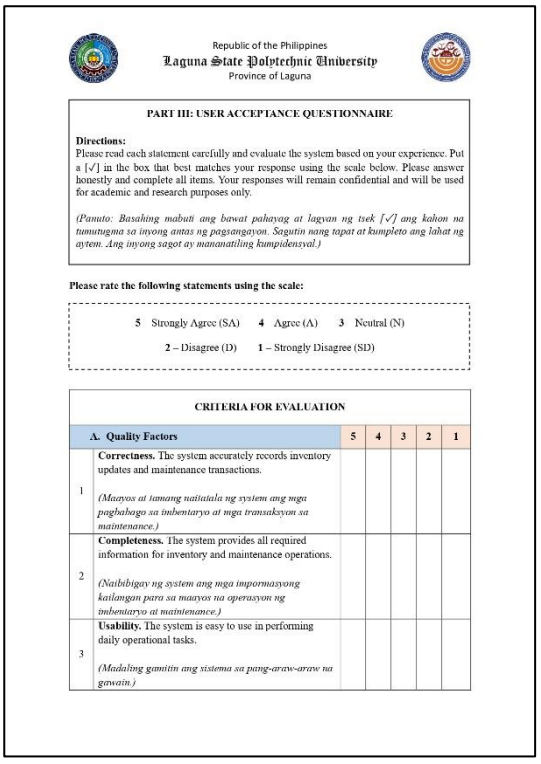
Role/Position :_____ Age:_____

Level of Computer Literacy:

☐ **Basic** Able to perform computer tasks such as browsing websites, opening mobile applications, and basic typing, but may still need assistance. *(Marunong sa paggamit ng kompyuter tulad ng pagba-browse ng website, pagbubukas ng mobile application, at simpleng pagta-type, ngunit nangangailangan pa rin ng gabay.)*

☐ **Intermediate** – Able to independently use and navigate websites and mobile applications and solve minor technical issues. *(Marunong gumamit at mag-navigate ng website at mobile application nang mag-isa at kayang solusyonan ang simpleng teknikal na problema.)*

☐ **Advanced** – Highly skilled in using websites and mobile applications, including advanced features and troubleshooting. *(Marunong at bihasa sa paggamit ng website at mobile application kabilang ang advanced features at pagresolba ng teknikal na problema.)*



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PART III: USER ACCEPTANCE QUESTIONNAIRE

Directions:
Please read each statement carefully and evaluate the system based on your experience. Put a [✓] in the box that best matches your response using the scale below. Please answer honestly and complete all items. Your responses will remain confidential and will be used for academic and research purposes only.

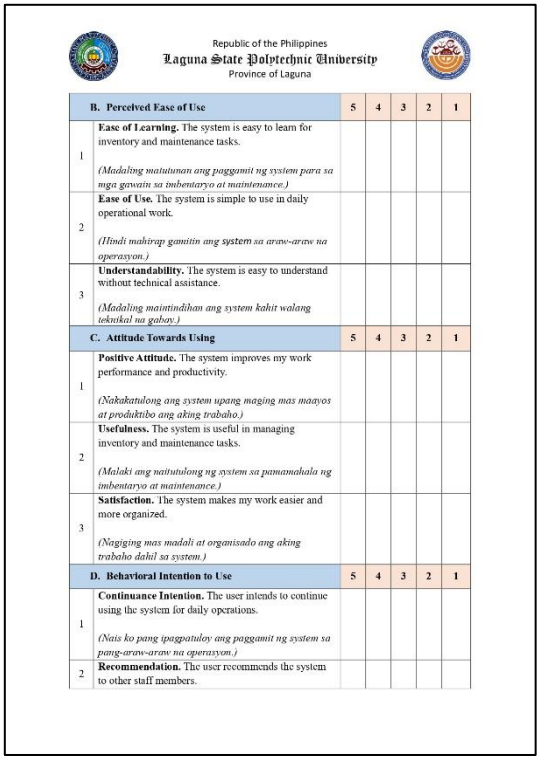
(Panuto: Basahing mabuti ang bawat pahayag at lagyan ng tsek [✓] ang kahon na tumutugma sa inyong antas ng pagsangayon. Sagutin nang tapat at kumpleto ang lahat ng aytem. Ang inyong sagot ay mananatiling kumpidensyal.)

Please rate the following statements using the scale:

5 Strongly Agree (SA) **4** Agree (A) **3** Neutral (N)
2 – Disagree (D) **1** – Strongly Disagree (SD)

CRITERIA FOR EVALUATION

	A. Quality Factors	5	4	3	2	1
1	**Correctness.** The system accurately records inventory updates and maintenance transactions. *(Maayos at tamang naitatala ng system ang mga paghahago sa imbentaryo at mga transaksyon sa maintenance.)*					
2	**Completeness.** The system provides all required information for inventory and maintenance operations. *(Naibibigay ng system ang mga impormasyong kailangan para sa maayos na operasyon ng imbentaryo at maintenance.)*					
3	**Usability.** The system is easy to use in performing daily operational tasks. *(Madaling gamitin ang sistema sa pang-araw-araw na gawain.)*					



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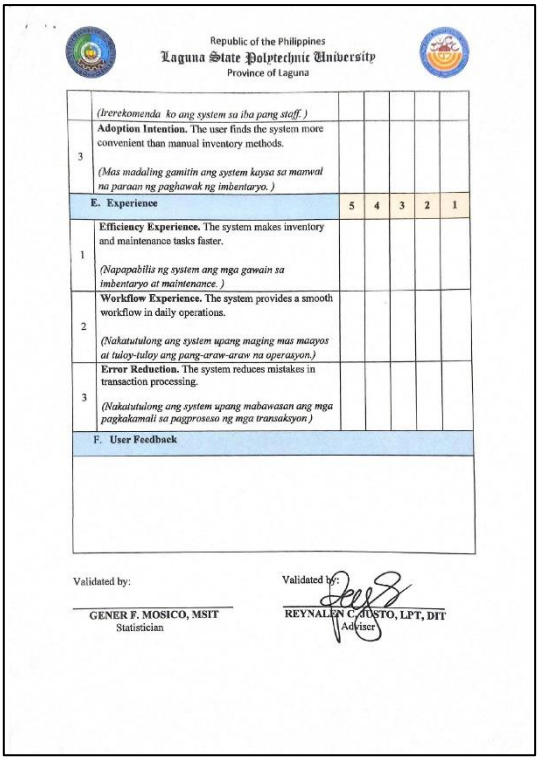
	B. Perceived Ease of Use	5	4	3	2	1
1	**Ease of Learning.** The system is easy to learn for inventory and maintenance tasks. *(Madaling matutunan ang paggamit ng system para sa mga gawain sa imbentaryo at maintenance.)*					
2	**Ease of Use.** The system is simple to use in daily operational work. *(Hindi mahirap gamitin ang system sa araw-araw na operasyon.)*					
3	**Understandability.** The system is easy to understand without technical assistance. *(Madaling maintindihan ang system kahit walang teknikal na gabay.)*					
	C. Attitude Towards Using	**5**	**4**	**3**	**2**	**1**
1	**Positive Attitude.** The system improves my work performance and productivity. *(Nakakatulong ang system upang maging mas maayos at produktibo ang aking trabaho.)*					
2	**Usefulness.** The system is useful in managing inventory and maintenance tasks. *(Malaki ang naitutulong ng system sa pamamahala ng imbentaryo at maintenance.)*					
3	**Satisfaction.** The system makes my work easier and more organized. *(Nagiging mas madali at organisado ang aking trabaho dahil sa system.)*					
	D. Behavioral Intention to Use	**5**	**4**	**3**	**2**	**1**
1	**Continuance Intention.** The user intends to continue using the system for daily operations. *(Nais ko pang ipagpatuloy ang paggamit ng system sa pang-araw-araw na operasyon.)*					
2	**Recommendation.** The user recommends the system to other staff members. *(Irerekomenda ko ang system sa iba pang staff.)*					
3	**Adoption Intention.** The user finds the system more convenient than manual inventory methods. *(Mas madaling gamitin ang system kaysa sa manwal na paraan ng paghawak ng imbentaryo.)*					
	E. Experience	**5**	**4**	**3**	**2**	**1**
1	**Efficiency Experience.** The system makes inventory and maintenance tasks faster. *(Napapabilis ng system ang mga gawain sa imbentaryo at maintenance.)*					
2	**Workflow Experience.** The system provides a smooth workflow in daily operations. *(Nakatutulong ang system upang maging mas maayos at tuloy-tuloy ang pang-araw-araw na operasyon.)*					
3	**Error Reduction.** The system reduces mistakes in transaction processing. *(Nakatutulong ang system upang mabawasan ang mga pagkakamali sa pagproseso ng mga transaksyon.)*					
	F. User Feedback					



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Province of Laguna

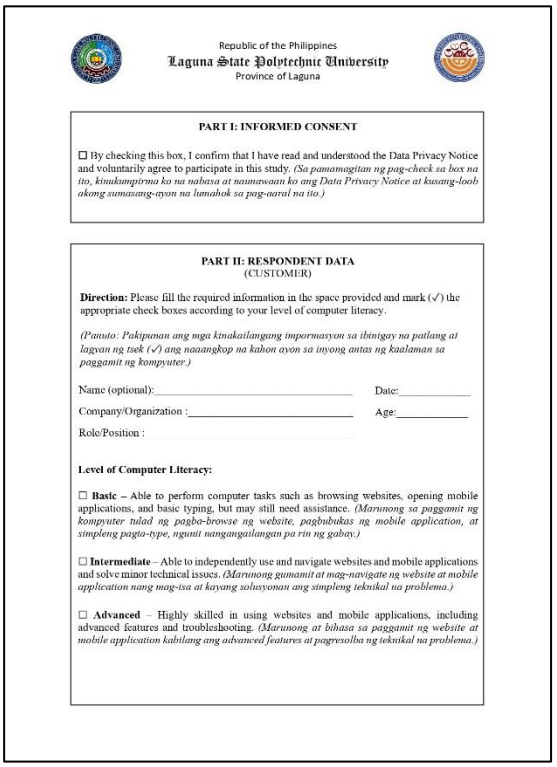
Validated by:

GENER F. MOSICO, MSIT
Statistician

Validated by:

REYNALEN C. JUSTO, LPT, DIT
Adviser

Questionnaire for Customer



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PART I: INFORMED CONSENT

☐ By checking this box, I confirm that I have read and understood the Data Privacy Notice and voluntarily agree to participate in this study. *(Sa pamamagitan ng pag-check sa box na ito, kinukumpirma ko na nabasa at naunawaan ko ang Data Privacy Notice at kusang-loob akong sumasang-ayon na lumahok sa pag-aaral na ito.)*

PART II: RESPONDENT DATA
(CUSTOMER)

Direction: Please fill the required information in the space provided and mark (✓) the appropriate check boxes according to your level of computer literacy.

(Panuto: Pakipunan ang mga kinakailangang impormasyon sa ibinigay na patlang at lagyan ng tsek (✓) ang naaangkop na kahon ayon sa inyong antas ng kaalaman sa paggamit ng kompyuter.)

Name (optional):_____ Date:_____

Company/Organization :_____ Age:_____

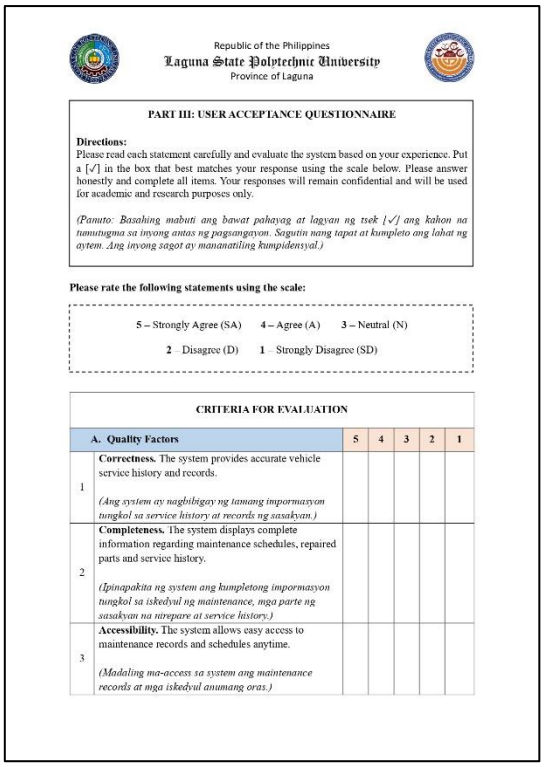
Role/Position :

Level of Computer Literacy:

☐ **Basic** – Able to perform computer tasks such as browsing websites, opening mobile applications, and basic typing, but may still need assistance. *(Marunong sa paggamit ng kompyuter tulad ng pagba-browse ng website, pagbubukas ng mobile application, at simpleng pagta-type, ngunit nangangailangan pa rin ng gabay.)*

☐ **Intermediate** – Able to independently use and navigate websites and mobile applications and solve minor technical issues. *(Marunong gumamit at mag-navigate ng website at mobile application nang mag-isa at kayang solusyonan ang simpleng teknikal na problema.)*

☐ **Advanced** – Highly skilled in using websites and mobile applications, including advanced features and troubleshooting. *(Marunong at bihasa sa paggamit ng website at mobile application kabilang ang advanced features at pagresolba ng teknikal na problema.)*



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PART III: USER ACCEPTANCE QUESTIONNAIRE

Directions:
Please read each statement carefully and evaluate the system based on your experience. Put a [✓] in the box that best matches your response using the scale below. Please answer honestly and complete all items. Your responses will remain confidential and will be used for academic and research purposes only.

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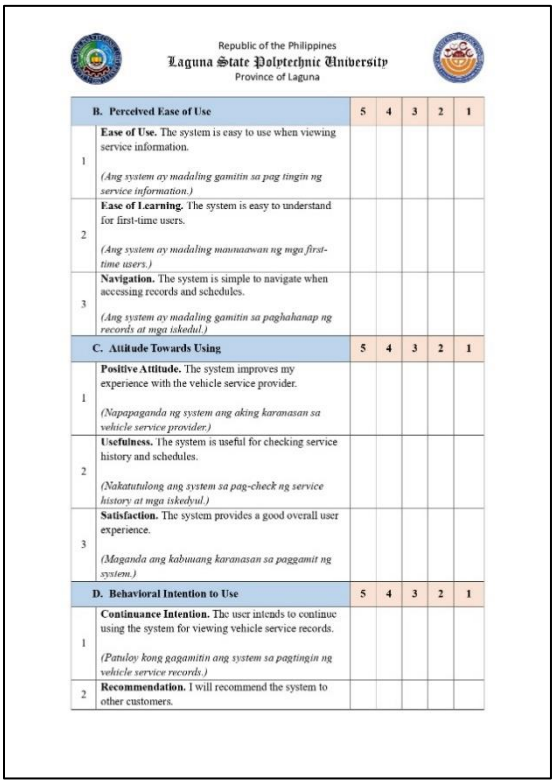
Please rate the following statements using the scale:

5 – Strongly Agree (SA) **4** – Agree (A) **3** – Neutral (N)
2 – Disagree (D) **1** – Strongly Disagree (SD)

CRITERIA FOR EVALUATION

	A. Quality Factors	5	4	3	2	1
1	**Correctness.** The system provides accurate vehicle service history and records. *(Ang system ay nagbibigay ng tamang impormasyon tungkol sa service history at records ng sasakyan.)*					
2	**Completeness.** The system displays complete information regarding maintenance schedules, repaired parts and service history. *(Ipinapakita ng system ang kumpletong impormasyon tungkol sa iskedyul ng maintenance, mga parte ng sasakyan na nirepare at service history.)*					
3	**Accessibility.** The system allows easy access to maintenance records and schedules anytime. *(Madaling ma-access sa system ang maintenance records at mga iskedyul anumang oras.)*					



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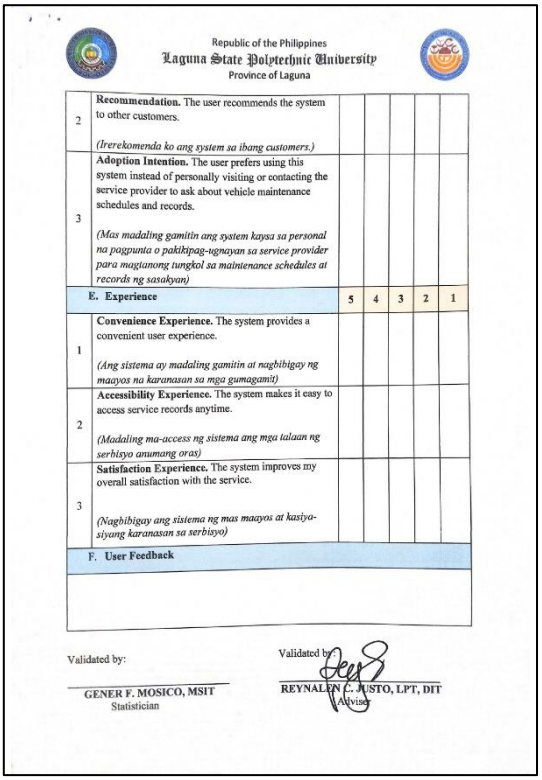
	B. Perceived Ease of Use	5	4	3	2	1
1	**Ease of Use.** The system is easy to use when viewing service information. *(Ang system ay madaling gamitin sa pag tingin ng service information.)*					
2	**Ease of Learning.** The system is easy to understand for first-time users. *(Ang system ay madaling maunawaan ng mga first-time users.)*					
3	**Navigation.** The system is simple to navigate when accessing records and schedules. *(Ang system ay madaling gamitin sa paghahanap ng records at mga iskedul.)*					
	C. Attitude Towards Using	**5**	**4**	**3**	**2**	**1**
1	**Positive Attitude.** The system improves my experience with the vehicle service provider. *(Napapaganda ng system ang aking karanasan sa vehicle service provider.)*					
2	**Usefulness.** The system is useful for checking service history and schedules. *(Nakatutulong ang system sa pag-check ng service history at mga iskedyul.)*					
3	**Satisfaction.** The system provides a good overall user experience. *(Maganda ang kabuuang karanasan sa paggamit ng system.)*					
	D. Behavioral Intention to Use	**5**	**4**	**3**	**2**	**1**
1	**Continuance Intention.** The user intends to continue using the system for viewing vehicle service records. *(Patuloy kong gagamitin ang system sa pagtingin ng vehicle service records.)*					
2	**Recommendation.** I will recommend the system to other customers.					



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2	**Recommendation.** The user recommends the system to other customers. *(Irerekomenda ko ang system sa ibang customers.)*					
3	**Adoption Intention.** The user prefers using this system instead of personally visiting or contacting the service provider to ask about vehicle maintenance schedules and records. *(Mas madaling gamitin ang system kaysa sa personal na pagpunta o pakikipag-ugnayan sa service provider para magtanong tungkol sa maintenance schedules at records ng sasakyan)*					
	E. Experience	**5**	**4**	**3**	**2**	**1**
1	**Convenience Experience.** The system provides a convenient user experience. *(Ang sistema ay madaling gamitin at nagbibigay ng maayos na karanasan sa mga gumagamit)*					
2	**Accessibility Experience.** The system makes it easy to access service records anytime. *(Madaling ma-access ng sistema ang mga talaan ng serbisyo anumang oras)*					
3	**Satisfaction Experience.** The system improves my overall satisfaction with the service. *(Nagbibigay ang sistema ng mas maayos at kasiya-siyang karanasan sa serbisyo)*					
	F. User Feedback					

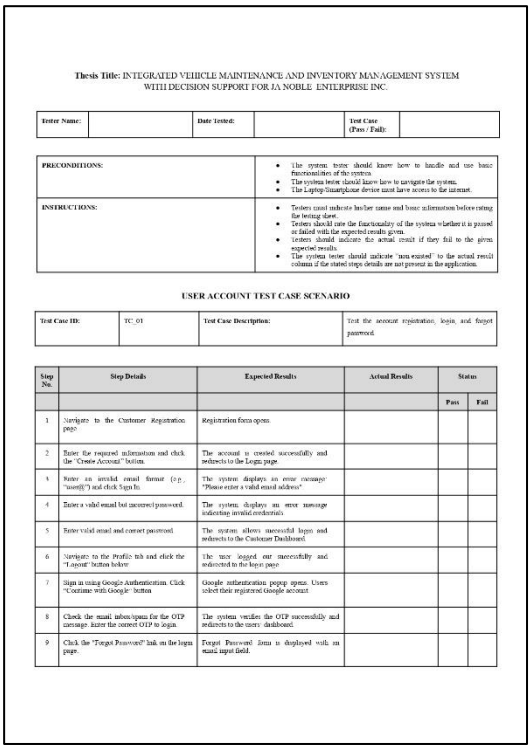
Validated by:

GENER F. MOSICO, MSIT
Statistician

Validated by:

REYNALEN C. JUSTO, LPT, DIT
Adviser

Blackbox Form Testing Questionnaire



Thesis Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.

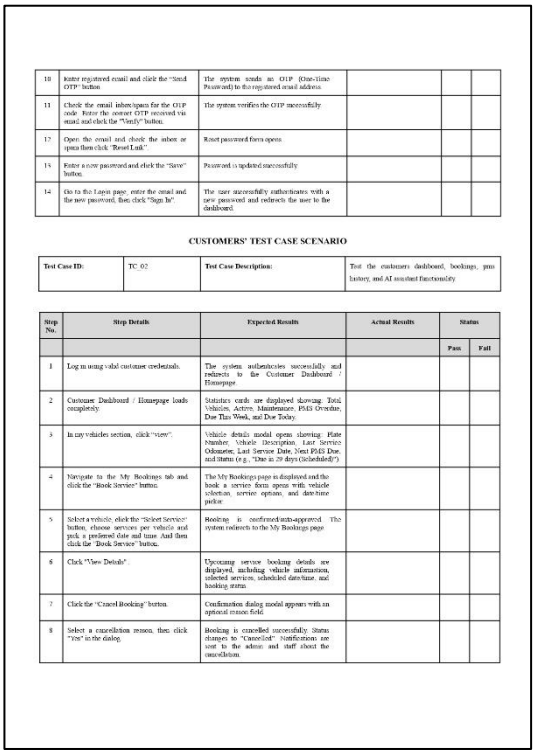
Tester Name:		**Date Tested:**		**Test Case (Pass / Fail):**	

PRECONDITIONS:	• The system tester should know how to handle and use basic functionalities of the system • The system tester should know how to navigate the system. • The Laptop/Smartphone device must have access to the internet.
INSTRUCTIONS:	• Testers must indicate his/her name and basic information before using the testing sheet. • Testers should rate the functionality of the system whether it is passed or failed with the expected results given. • Testers should indicate the actual result if they fail to the given expected results. • The system tester should indicate "non-existed" to the actual result column if the stated steps details are not present in the application.

USER ACCOUNT TEST CASE SCENARIO

Test Case ID:	TC_01	**Test Case Description:**	Test the account registration, login, and forget password.

Step No.	**Step Details**	**Expected Results**	**Actual Results**	**Status**	
				Pass	**Fail**
1	Navigate to the Customer Registration page.	Registration form opens.			
2	Enter the required information and click the "Create Account" button.	The account is created successfully and redirects to the Login page.			
3	Enter an invalid email format (e.g., "user@") and click Sign In.	The system displays an error message "Please enter a valid email address."			
4	Enter a valid email but incorrect password.	The system displays an error message indicating invalid credentials.			
5	Enter valid email and correct password.	The system allows successful login and redirects to the Customer Dashboard.			
6	Navigate to the Profile tab and click the "Logout" button below.	The user logged out successfully and redirected to the login page.			
7	Sign in using Google Authentication. Click "Continue with Google" button	Google authentication popup opens. Users select their registered Google account.			
8	Check the email inbox/spam for the OTP message. Enter the correct OTP to login.	The system verifies the OTP successfully and redirects to the users' dashboard.			
9	Click the "Forgot Password" link on the login page.	Forgot Password form is displayed with an email input field.			

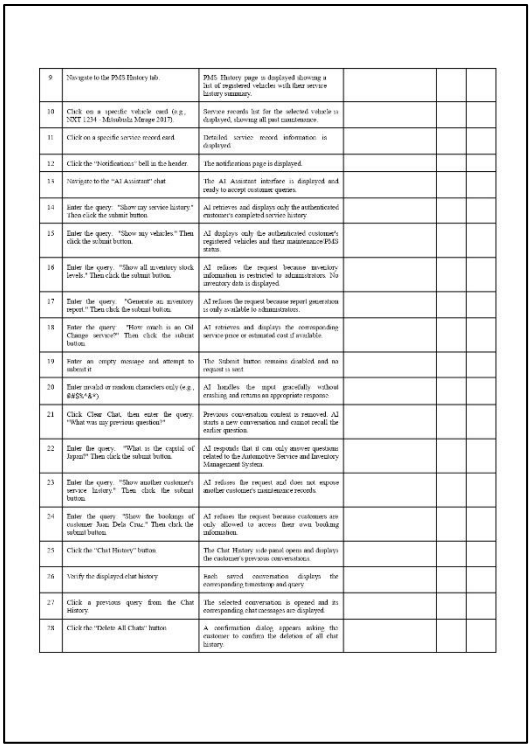


10	Enter registered email and click the "Send OTP" button	The system sends an OTP (One-time Password) to the registered email address.			
11	Check the email inbox/spam for the OTP code. Enter the correct OTP received via email and click the "Verify" button.	The system verifies the OTP successfully.			
12	Open the email and check the inbox or spam then click "Reset Link".	Reset password form opens.			
13	Enter a new password and click the "Save" button.	Password is updated successfully.			
14	Go to the Login page, enter the email and the new password, then click "Sign In"	The user successfully authenticates with a new password and redirects the user to the dashboard.			

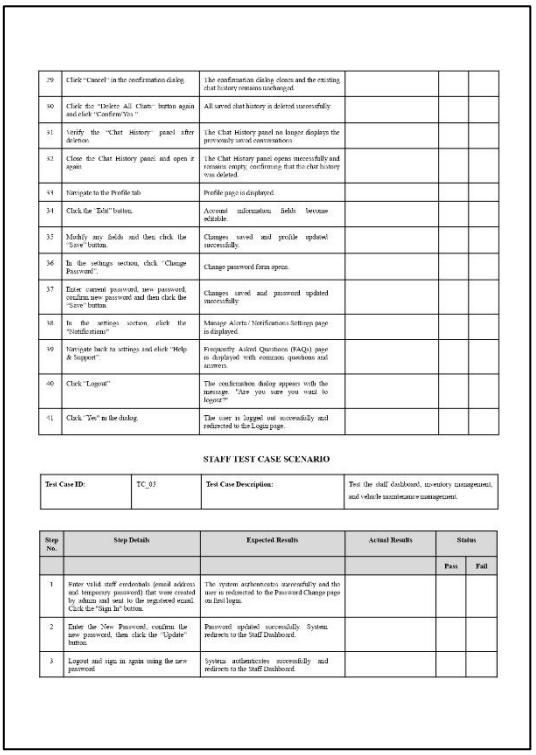
CUSTOMERS' TEST CASE SCENARIO

Test Case ID:	TC_02	**Test Case Description:**	Test the customers dashboard, bookings, pms history, and AI assistant functionality.

Step No.	**Step Details**	**Expected Results**	**Actual Results**	**Status**	
				Pass	**Fail**
1	Log in using valid customer credentials.	The system authenticates successfully and redirects to the Customer Dashboard / Homepage.			
2	Customer Dashboard / Homepage loads completely.	Statistics cards are displayed showing: Total Vehicles, Active, Maintenance, PMS Overdue, Due This Week, and Due Today.			
3	In my vehicles section, click "view".	Vehicle details modal opens showing: Plate Number, Vehicle Description, Last Service Odometer, Last Service Date, Next PMS Due, and Status (e.g., "Due in 29 days (Scheduled)").			
4	Navigate to the My Bookings tab and click the "Book Service" button.	The My Bookings page is displayed and the book a service form opens with vehicle selection, service options, and date/time picker.			
5	Select a vehicle, click the "Select Service" button, choose services per vehicle and pick a preferred date and time. And then click the "Book Service" button.	Booking is confirmed/auto-approved. The system redirects to the My Bookings page.			
6	Click "View Details".	Upcoming service booking details are displayed, including vehicle information, selected services, scheduled date/time, and booking status.			
7	Click the "Cancel Booking" button.	Confirmation dialog modal appears with an optional reason field.			
8	Select a cancellation reason, then click "Yes" in the dialog.	Booking is cancelled successfully. Status changes to "Cancelled". Notifications are sent to the admin and staff about the cancellation.			



9	Navigate to the PMS History tab.	PMS History page is displayed showing a list of registered vehicles with their service history summary.			
10	Click on a specific vehicle card (e.g., NNT 1234 - Mitsubishi Mirage 2017).	Service records list for the selected vehicle is displayed, showing all past maintenance.			
11	Click on a specific service record card.	Detailed service record information is displayed.			
12	Click the "Notifications" bell in the header.	The notifications page is displayed.			
13	Navigate to the "AI Assistant" chat	The AI Assistant interface is displayed and ready to accept customer queries.			
14	Enter the query: "Show my service history." Then click the submit button.	AI retrieves and displays only the authenticated customer's completed service history.			
15	Enter the query: "Show my vehicles." Then click the submit button.	AI displays only the authenticated customer's registered vehicles and their maintenance/PMS status.			
16	Enter the query: "Show all inventory stock levels." Then click the submit button.	AI refuses the request because inventory information is restricted to administrators. No inventory data is displayed.			
17	Enter the query: "Generate an inventory report." Then click the submit button.	AI refuses the request because report generation is only available to administrators.			
18	Enter the query: "How much is an Oil Change service?" Then click the submit button.	AI retrieves and displays the corresponding service price or estimated cost if available.			
19	Enter an empty message and attempt to submit it.	The Submit button remains disabled and no request is sent.			
20	Enter invalid or random characters only (e.g., @@$%^&*).	AI handles the input gracefully without crashing, and returns an appropriate response.			
21	Click Clear Chat, then enter the query: "What was my previous question?"	Previous conversation context is removed. AI starts a new conversation and cannot recall the earlier question.			
22	Enter the query: "What is the capital of Japan?" Then click the submit button.	AI responds that it can only answer questions related to the Automotive Service and Inventory Management System.			
23	Enter the query: "Show another customer's service history." Then click the submit button.	AI refuses the request and does not expose another customer's maintenance records.			
24	Enter the query: "Show the bookings of customer Juan Dela Cruz." Then click the submit button.	AI refuses the request because customers are only allowed to access their own booking information.			
25	Click the "Chat History" button.	The Chat History side panel opens and displays the customer's previous conversations.			
26	Verify the displayed chat history	Each saved conversation displays the corresponding timestamp and query.			
27	Click a previous query from the Chat History.	The selected conversation is opened and its corresponding chat messages are displayed.			
28	Click the "Delete All Chats" button.	A confirmation dialog appears asking the customer to confirm the deletion of all chat history.			



29	Click "Cancel" in the confirmation dialog.	The confirmation dialog closes and the existing chat history remains unchanged.			
30	Click the "Delete All Chats" button again and click "Confirm/Yes."	All saved chat history is deleted successfully.			
31	Verify the "Chat History" panel after deletion	The Chat History panel no longer displays the previously saved conversations.			
32	Close the Chat History panel and open it again	The Chat History panel opens successfully and remains empty, confirming that the chat history was deleted.			
33	Navigate to the Profile tab	Profile page is displayed.			
34	Click the "Edit" button.	Account information fields become editable.			
35	Modify any fields and then click the "Save" button.	Changes saved and profile updated successfully.			
36	In the settings section, click "Change Password".	Change password form opens.			
37	Enter current password, new password, confirm new password and then click the "Save" button.	Changes saved and password updated successfully.			
38	In the settings section, click the "Notifications"	Manage Alerts / Notifications Settings page is displayed.			
39	Navigate back to settings and click "Help & Support".	Frequently Asked Questions (FAQs) page is displayed with common questions and answers.			
40	Click "Logout"	The confirmation dialog appears with the message, "Are you sure you want to logout?"			
41	Click "Yes" in the dialog.	The user is logged out successfully and redirected to the Login page.			

STAFF TEST CASE SCENARIO

Test Case ID:	TC_03	**Test Case Description:**	Test the staff dashboard, inventory management, and vehicle maintenance management.

Step No.	**Step Details**	**Expected Results**	**Actual Results**	**Status**	
				Pass	**Fail**
1	Enter valid staff credentials (email address and temporary password) that were created by admin and sent to the registered email. Click the "Sign In" button.	The system authenticates successfully and the user is redirected to the Password Change page on first login.			
2	Enter the New Password, confirm the new password, then click the "Update" button.	Password updated successfully. System redirects to the Staff Dashboard.			
3	Logout and sign in again using the new password	System authenticates successfully and redirects to the Staff Dashboard.			

4	Navigate to the Inventory Management tab	Stock inventory page is displayed.			
5	Click on a specific stock item card to view details	Stock item details modal opens.			
6	Navigate to the Receiving Items tab and click the "Plus/Add" button then click the "Create Purchase Order".	A purchase order form opens, and the list of items that need replenishment is displayed with suggested order quantities.			
7	Select the items that need replenishment and click the "Save" button.	Purchase order created successfully and appears in the list. Status is "For Admin Review"			
8	Click the "Plus/Add" button then click the "Edit Purchase Order" button	The Edit Purchase Order form opens, then select the cards to edit			
9	Modify Order Quantity and click the "Update" button.	Purchase order updated successfully.			
10	Wait for the admin to approve the Purchase Order request.	Status changes from "For Admin Review" to "Pending"			
11	Click the "Plus/Add" button then click the "Scan Item" button.	Item is identified successfully via barcode/QR code scan. Receiving item form opens with item details pre-filled.			
12	Enter Actual Qty, select Condition (OK/Defective/Shortage), add Remarks (optional), and click the "Submit" button.	Item submitted successfully. Status changes to "Awaiting Review".			
13	Wait for the admin to review and verify the received items	Status changes from "Awaiting Review" to "Approved" or "Rejected" based on the admin's decision.			
14	Navigate to the Stock Inventory tab and verify stock inventory is updated after admin approval	The received quantity has been added to the current stock.			
15	Navigate to the Vehicle List tab and click the "Add" button.	Add Vehicle form opens.			
16	Enter Plate Number, Description, select Vehicle Type, Owner Name or Company Name, Current Odometer (km), check "Vehicle was just serviced", enter Last Service Date, Last Service Odometer (km), Service Frequency (months), and click "Save" button.	Vehicle added successfully and appears in the list			
17	Click the card then the "Edit" button.	Vehicle form becomes editable.			
18	Modify any fields and click the "Update" button.	Vehicle updated successfully.			
19	Navigate to Vehicle Maintenance tab and click "Add"	New Service form opens			
20	Enter Plate Number, Mechanic Name, Service Date, Odometer Reading (km), select Vehicle Issues (e.g., Engine Problem), Broke Down, add custom issue if needed, add new in Services Rendered with Service Name, add new in Materials Used with Material Name, and click "Save" button	Service record added successfully. Total Cost is automatically calculated and a record appears in the list with status "Ongoing"			
21	Click the card then the "Edit" button.	Service form becomes editable.			
22	Modify any fields and then click the "Update" button	Service record updated successfully.			

23	Click "Mark as Completed" button	The confirmation dialog appears			
24	Click "Yes" in the dialog	Service marked as completed successfully. Status changes from "Ongoing" to "Completed". The quantity of materials used is automatically deducted from the Stock Inventory.			
25	Navigate to the Service Bookings tab	Service Bookings page is displayed showing all customer bookings.			
26	Click the card then click the "Start Service" button for a booking with today's scheduled date	The confirmation dialog appears			
27	Click "Yes" in the dialog	System redirects to the Vehicle Maintenance tab Service form opens with booking information pre-filled (plate number, requested services).			
28	Complete the service form and click the "Save" button.	Service record is created successfully.			
29	Return back to the Service booking tab and click another booked service card then click the "Reschedule" button	Reschedule Booking modal opens with current date and time.			
30	Select new date and time and click "Confirm" button.	Booking is rescheduled to the new date. Customers are automatically notified about the schedule change.			
31	Select another Booked Service card, then click the "Cancel" button.	The confirmation dialog appears			
32	Click "Yes" in the dialog	Booking is cancelled successfully. The customer is notified about the cancellation.			
33	Return back to the Staff Dashboard and click the "Notifications" bell in the header.	The notifications page is displayed.			
34	Navigate to the Profile tab	Profile page is displayed.			
35	Click the "Edit" button.	Account information becomes editable.			
36	Modify any field and then click the "Save" button.	Changes saved and profile updated successfully.			
37	In settings section, click the "Change Password"	Change password form opens.			
38	Enter current password, new password, and confirm new password and then click the "save" button.	Changes saved and password updated successfully.			
39	In settings section, click the "Notifications"	Manage Alerts / Notifications Settings page is displayed.			
40	Return back to the settings and click "Help & Support".	Frequently Asked Questions (FAQs) page is displayed with common questions and answers			
41	Return back and click the "Logout" button below	The confirmation dialog appears.			
42	Click "Yes" in the dialog.	The user is logged out successfully and redirected to the Login page			

SCANNING BARCODE / QR CODE TEST CASE SCENARIO

Test Case ID:	TC_04	Test Case Description:	Test the scanning barcode / qrcode functionality

Step No.	Step Details	Expected Results	Actual Results	Status	
				Pass	Fail
1	Click the Scanner icon/button located in the center of the bottom navbar	Upon clicking, the system requests camera permission (if first time) the scanner camera interface opens with a live camera feed.			
2	Point the camera at a valid barcode or QR code on an item. Ensure the code is clearly visible within the scanning frame.	Item is identified successfully. An Action Modal opens showing the item details.			
3	Click the "View Stock" button in the action modal.	View Stock Details modal opens.			
4	Close the View Stock modal, scan the item again, and click "Receiving Items" (for items with pending delivery).	Receiving Items modal opens with the scanned item pre-selected. Order details and quantity fields are displayed for pending deliveries.			
5	Enter Actual Quantity Received, select Condition (OK/Defective/Shortage), add Remarks(optional), and click the "Submit" button.	Receiving submission is successful Status changes to "Awaiting Review"			
6	Close modals, scan another item, and click "Use in Maintenance".	System redirects to the Vehicle Maintenance tab. The service form opens with the scanned material/item automatically pre-filled in the Materials Used section.			
7	Complete the service form with required details and click the "Save" button.	Service record is created successfully with the scanned material included.			

ADMIN TEST CASE SCENARIO

Test Case ID:	TC_05	Test Case Description:	Test the admin dashboard, inventory management, vehicle maintenance management, stock replenishment and pos scheduling, user management, domain management, archive, and AI assistant functionality

Step No.	Step Details	Expected Results	Actual Results	Status	
				Pass	Fail
1	Log in using valid admin credentials	The system authenticates successfully and redirects to the Admin Dashboard.			

2	Observe the Admin Dashboard in the main screen/ homepage.	Dashboard Overview page loads and displays all dashboard widgets, statistics, and charts.			
3	Navigate to the Item Master tab and click the "Add" button	The Item master page is displayed and the Item form opens.			
4	Enter the Supplier Item Code, OEM Item Name, Description, Commodity Group, UOM, Cost, and Type, then click the "Save" button	Item added successfully. The system redirects to the Stock Inventory page and adds stock item form opens.			
5	Enter the Current Quantity, Minimum Level, and Maximum Level. Verify that the Reorder Quantity is automatically calculated, then click the "Save" button.	Stock items added successfully.			
6	Click the "Edit" button of the current added item.	Edit Item form becomes editable.			
7	Modify any field and click the "Update" button.	Item updated successfully.			
8	Click the "Archive" button	The confirmation dialog appears.			
9	Click "Ok" in the dialog	Items archived successfully. Item is moved to the Archive section /tab.			
10	Navigate to the Stock Inventory tab	The Stock inventory page is displayed.			
11	Click the "Edit" button.	Stock Item form becomes editable.			
12	Modify any field and click the "Update" button.	Stock item updated successfully.			
13	Navigate to the Receiving Items tab and click the "Plus/Add" button then click the "Create Purchase Order" button.	Purchase Order form opens, and the list of items that need replenishment is displayed.			
14	Select items that need replenishment and click "Save" button.	Purchase order created successfully and appears in the list.			
15	Click the "Edit" button	Edit form opens.			
16	Modify Order Quantity and click the "Update" button.	Quantity is updated successfully.			
17	Click the "Plus/Add" button then click the " Review Purchase Order" button.	The Review Purchase Order form opens, then select the cards to review.			
18	Review and Verify the item and then click the "Approve" button	Purchase order approved successfully. Status set to "Pending".			
19	Click the "View" button to open the item details, then click the "Receive Item" button.	Receive item modal opens.			
20	Enter Actual Qty, select Condition (e.g., OK), add Remarks, and click the "Submit" button.	Item submitted successfully. Status changes to "Awaiting Review".			
21	Click the "Review & Verify" button.	Review & verify modal opens.			
22	Verify details are correct and click the "Approve & Update Stock" button.	Delivery item approved successfully. Status changes to "Approved". Stock inventory is automatically updated with the received quantity.			

23	Navigate to the Inventory Transactions tab.	All Transactions are displayed.			
24	Filter the records by Type, Date Range, or Commodity Group.	Filtered results appear.			
23	Select Sort by: Date (Newest First) / Date (Oldest First) / Item Name A→Z / Item Name Z→A / Qty (Highest) / Qty (Lowest)	Data is sorted according to the selected option.			
24	Enter the search keyword in the search box.	Filtered results matching search criteria are displayed.			
25	Click the "Print" button, set Print settings, and click Print button again	Print preview opens and documents that can be printed will display successfully.			
26	Click the "Excel" button to export the file in excel format.	The file is downloaded in selected format with correct data.			
27	Navigate to the Vehicle List tab and click the "Add" button.	Vehicle list form opens.			
28	Enter Plate Number, Description, select Vehicle Type, Owner Name or Company Name, Current Odometer (km), check "Vehicle was just serviced", enter Last Service Date, Last Service Odometer (km), Service Frequency (months), and click "Save" button.	Vehicle added successfully and appears in the list.			
29	Click the "Edit" button.	Vehicle form becomes editable.			
30	Modify any fields and click the "Update" button.	Vehicle updated successfully.			
31	Click the "Archive" button.	The confirmation dialog appears.			
32	Click "Ok" in the confirmation dialog.	Vehicle archived successfully.			
33	Navigate to Vehicle Maintenance tab and click "Add"	Service form opens.			
34	Enter Plate Number, Mechanic Name, Service Date, Odometer Reading (km), select Vehicle Issues (optional)(e.g., Engine Problem, Brake Issue), add custom issue if needed, add *one* in Services Rendered with Service Name, add row in Materials Used with Material Name, and click "Save" button.	Service record added successfully. Total Cost is automatically calculated. The record appears in the Vehicle Maintenance list with status "Ongoing".			
35	Click the "Edit" button.	Service form becomes editable.			
36	Modify any fields and click "Update" button	Service record updated successfully.			
37	Click the "View" button to open the details modal, then click the "Mark as Completed" button.	The confirmation dialog appears.			
38	Click "OK" in the dialog.	Service marked as completed successfully. Record appears in the Issuances list and Transactions list, and the quantity of materials used is automatically deducted from the Stock Inventory.			
39	Navigate to the Service Bookings tab.	Service Bookings page is displayed showing all customer bookings.			

40	Click the "Start Service" button for a booking with today's scheduled date.	The confirmation dialog appears.			
41	Click "Ok" in the dialog.	System redirects to the Vehicle Maintenance tab. Service form opens with booking information pre-filled (plate number, requested services).			
42	Complete the service form and click the "Save" button.	Service record is created successfully. Booking status is set to "Ongoing".			
43	Return back to the Service booking tab and click the "Reschedule" button.	Reschedule Booking modal opens with date picker.			
44	Select new date and time, then click the "Confirm" button.	Booking is rescheduled to the new date. Customers are automatically notified about the schedule change.			
45	Click the "Cancel" button.	The confirmation dialog appears.			
46	Click "Ok" in the dialog.	Booking is cancelled successfully. The customer is notified about the cancellation.			
47	Navigate to the Issuances Tab.	All Issuances are displayed.			
48	Select filter by type / date from to / commodity group.	Filtered results appear.			
49	Select Sort by: Date (Newest First) / Date (Oldest First) / Item Name A→Z / Item Name Z →A / Qty (Highest) / Qty (Lowest) / Cost (Highest) / Cost (Lowest)	Data is sorted according to the selected option.			
50	Enter the search keyword in the search box.	Filtered results matching search criteria are displayed.			
51	Click the "Print" button.	Print preview opens and documents can be printed successfully.			
52	Click the "Export" button (Excel).	The file is downloaded in selected format with correct data.			
53	Navigate to the Stock Replenishment tab	Stock Replenishment page is displayed.			
54	Navigate to the PMS Scheduling tab	PMS Scheduling page is displayed.			
55	Click the "Notification" bell in the header	Notification dropdown shows the alerts for stock replenishment, pms scheduling, booking service, and new account registration.			
56	Navigate to the User Management tab and click the "Add user" button.	User add form opens.			
57	Enter first name, last name, email address, and select the role. If the role is a customer, enter the company name and click the "Save" button.	The user account is created successfully and a temporary password is automatically generated and sent to the user's registered email.			
58	Click the "Edit" button.	Edit User form opens with all current information pre-filled and editable.			
59	Modify any user details and click the "Update" button.	Changes saved and user updated successfully.			
60	Click the "Deactivate" button.	The confirmation dialog appears.			

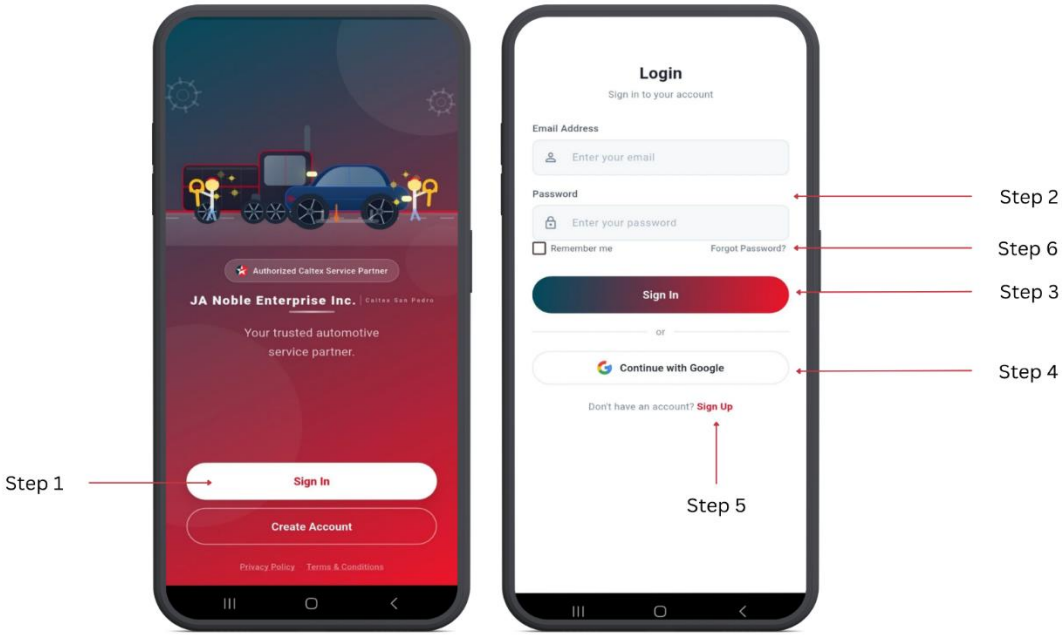
61	Click "OK" in the dialog.	The user is deactivated successfully. Status changes from "Active" to "Inactive". Users can no longer log in to the system.			
62	Click the "Activate" button.	The confirmation dialog appears.			
64	Click "OK" in the dialog.	The user is activated successfully. Status changes from "Inactive" to "Active". Users can log in to the system.			
64	Click the "Archive" button.	The confirmation dialog appears.			
65	Click "Ok" in the confirmation dialog.	User is archived successfully.			
66	Navigate to the Domain Management tab.	Domain Management page is displayed.			
67	Click on any card (e.g., Commodity Groups / Unit of Measurements / Vehicle Types) and then click the "Add" button.	Form opens with Name input field.			
68	Enter Name and click the "Save" button.	Added successfully and appears in the list.			
69	Click the "Edit" button.	Edit form becomes editable.			
70	Modify the name and click the "Update" button.	Updated successfully.			
71	Click the "Delete" button.	The confirmation dialog appears.			
72	Click "OK" in the dialog.	Deleted successfully.			
73	Navigate to the Archive tab.	The archive page is displayed.			
74	Click the "Restore" button.	The confirmation dialog appears.			
75	Click "OK" in the dialog.	Success message is displayed and it reappears in its respective list. When the item is restored to the Item Master page it will be redirected to the Stock Inventory page to set stock level			
76	Navigate to the respective module (e.g., Item Master, Vehicle Maintenance) and verify the restored item.	The restored item is visible in the active list with all original information preserved.			
77	Return to the Archive tab and click the "Delete Permanently" button	Confirmation dialog appears with strong warning.			
78	Confirm permanent deletion by clicking "Yes."	Permanently deleted from the archive.			
79	Navigate to the AI Assistant chat bubble on the dashboard.	The AI Assistant modal is displayed and ready to accept queries.			
80	Enter a valid report query (e.g., "Generate the issuance report for the month of July 2026 in PDF.") then click the send button.	AI retrieves the requested data from the database, confirms that the report is ready, and displays the appropriate Download PDF or Download Excel button(s).			
81	Click the Download PDF or Download Excel button.	The selected report is successfully generated and downloaded in the requested format.			
82	Enter an empty message and attempt to submit it.	The Submit button remains disabled and no request is sent.			

83	Enter invalid or random input (e.g., @#$%^&*). Then click the send button.	The AI Assistant handles the input gracefully without crashing and provides an appropriate response.			
84	Enter a query outside the automotive management system (e.g., "Who won the FIFA World Cup in 2018?") then click the send button.	AI responds that it can only answer questions related to the Automotive Service and Inventory Management System.			
85	Request a report with filters that return no records (e.g., "Generate the issuance report for December 3030") then click the send button.	AI indicates that the report cannot be generated because no matching records were found. No download button is displayed.			
86	Click the Clear Chat button.	The conversation history is cleared and a new conversation session starts.			
87	Enter the query: "Show all low-stock inventory items." then click the send button.	AI retrieves and displays the current low-stock inventory items from the database.			
88	Enter the query: "How many registered customers are in the system?." Then click the send button.	AI retrieves and displays the current number of registered customers from the database.			
89	Enter the query: "Show all pending service bookings." Then click the send button.	AI retrieves and displays the list of pending service bookings from the database.			
90	Click the "Chat History" button.	A Chat History side panel opens and displays the Delete All Chats button at the top and a list of previous conversations showing their timestamp and query.			
91	Verify the displayed chat history.	Previous conversations are displayed with the corresponding timestamp and query.			
92	Click the "Delete All Chats" button.	A confirmation dialog appears asking the user to confirm whether they want to delete all chat history.			
93	Click "Cancel" in the confirmation dialog.	The confirmation dialog closes, and all chat history remains unchanged.			
94	Click the "Delete All Chats" button again and click "Confirm/Yes."	All saved chat history is permanently deleted, and the Chat History panel displays no previous conversations.			
95	Close the Chat History side panel and open it again.	The Chat History panel opens successfully and remains empty, confirming that the chat history was deleted.			
96	Click the "Logout" button below in the sidebar menu.	The confirmation dialog appears.			
97	Click "Ok" in the dialog.	The user logged out successfully. The system is redirected to the Landing Page.			

User's Manual

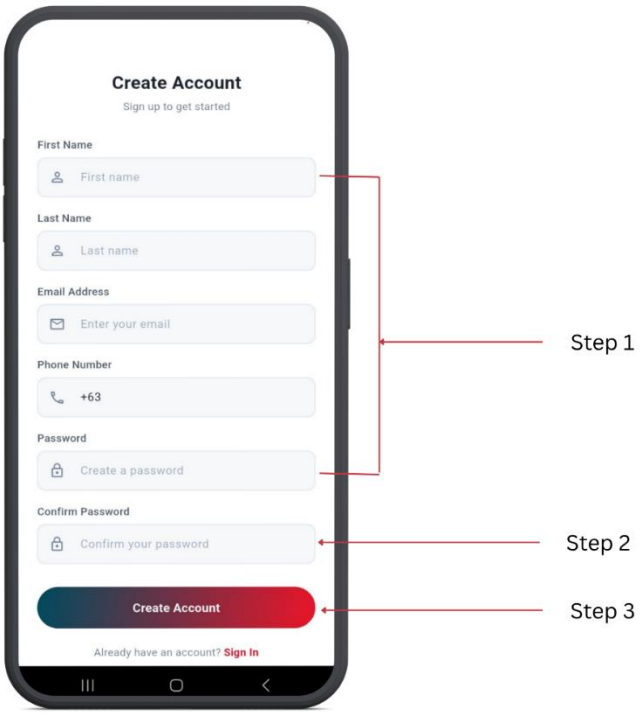
This section shows the User Manual of the study entitled **"**Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc."

Mobile – Customer Account



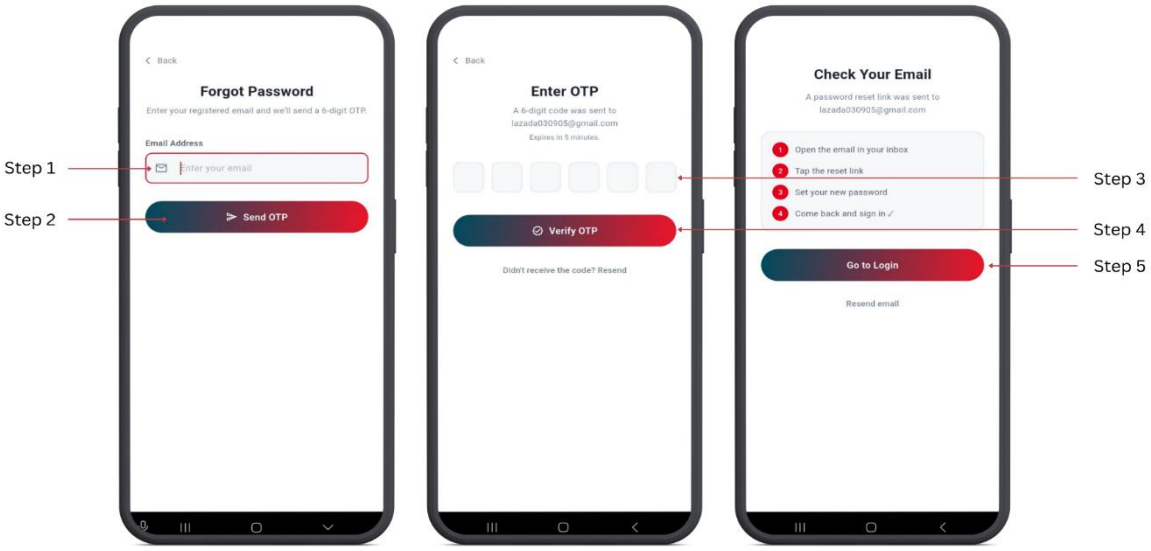
This shows the Welcome Screen and Login Page

- **Step 1:** Open the app and click "Sign In".
- **Step 2:** Enter your valid email and password.
- **Step 3:** Click "Sign in" to access your dashboard.
- **Step 4:** Alternatively, click "Continue with Google" to use Google account.
- **Step 5:** Click" Sign Up" if Customer doesn't' have an account.
- **Step 6:** Click "Forgot Password" if users forget their password.



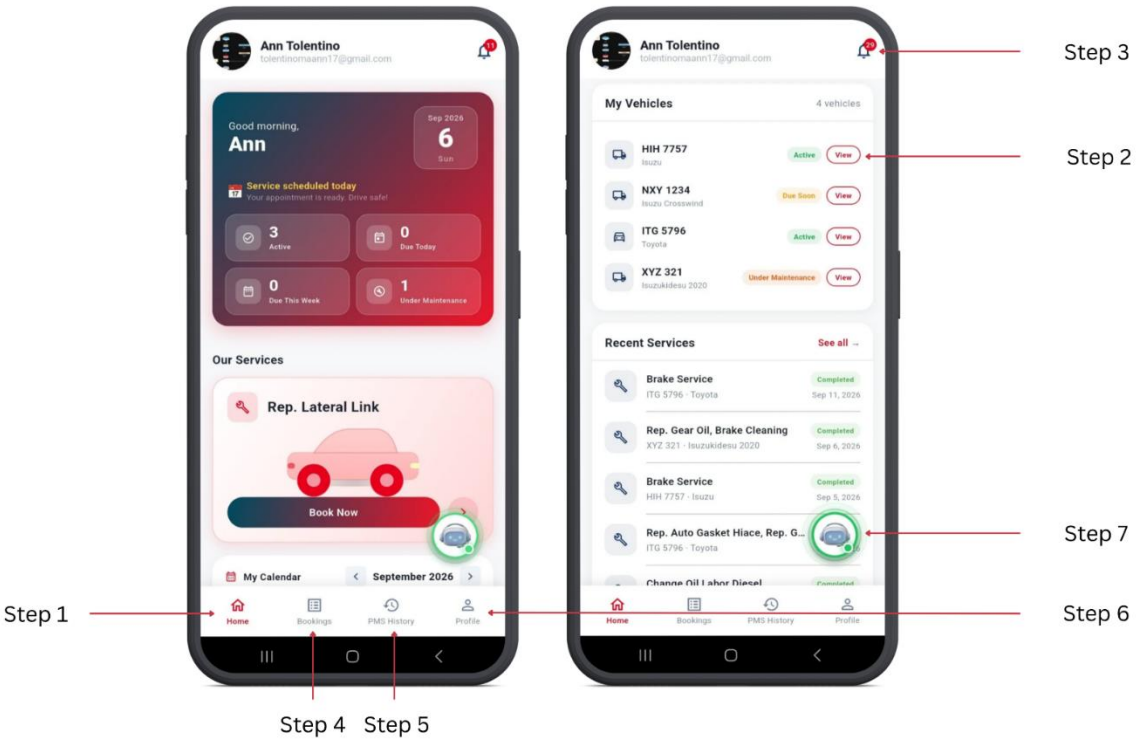
This shows the Creating an Account (Customer Registration) Page

- **Step 1:** Enter personal details including first name, last name, email address, phone number and password.
- **Step 2:** Confirm password.
- **Step 3:** Click “Create Account”.
- **Step 4:** Your account will be created, and you can log in immediately.



This shows the Forgot Password Page.

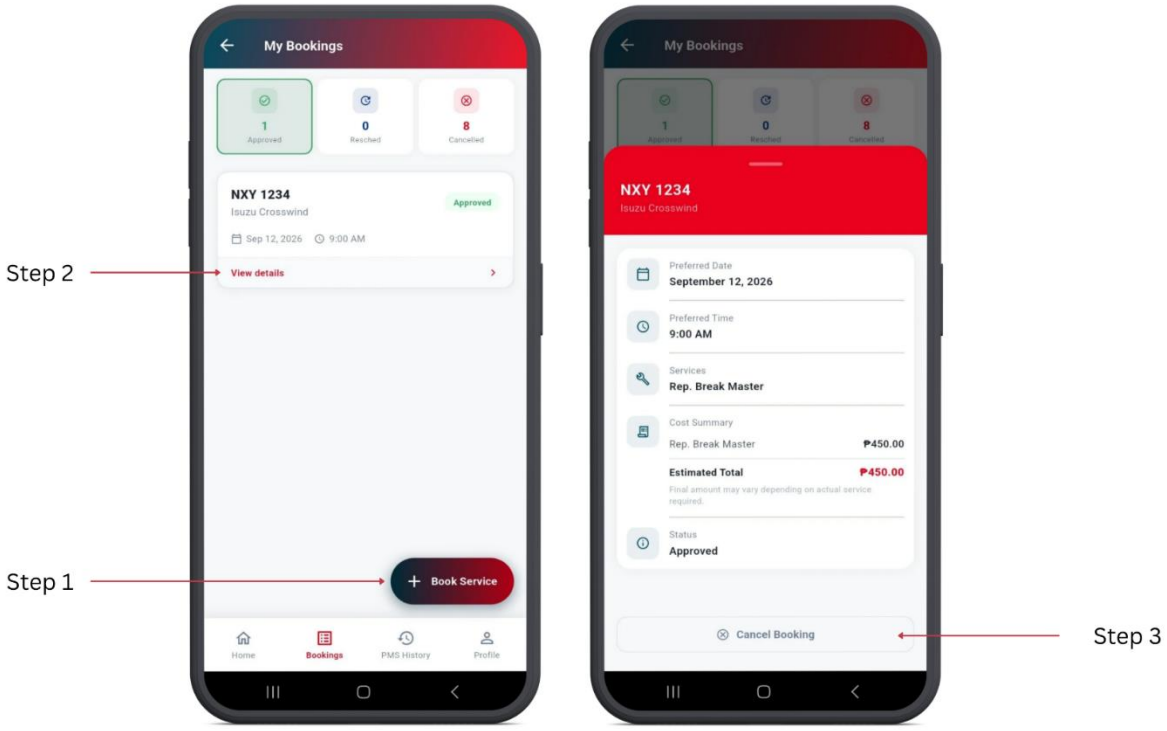
- **Step 1:** Enter registered email address.
- **Step 2:** Click the “Send OTP” button. A 6-digit OTP code will be sent to your email address.
- **Step 3:** Enter correct OTP code.
- **Step 4:** Click the “Verify OTP” button to confirm the code. A password reset link will be sent to your email. Click the link to set a new password.
- **Step 5:** Click “Go to login” page



This shows the Customer ‘s Dashboard Page.

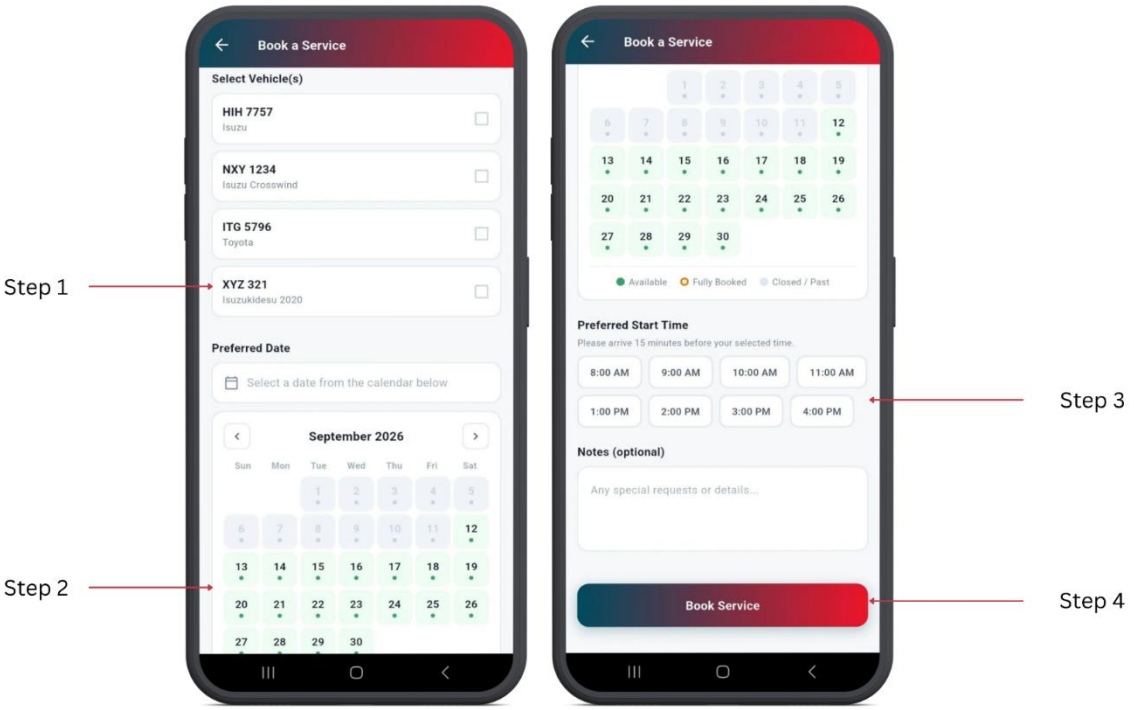
- **Step 1:** Click” Homepage / Dashboard” on the bottom navigation.
- **Step 2:** Click “View” button in the My Vehicles section to view your registered vehicles / status.
- **Step 3:** Click “Notifications” at the top right corner of the screen.
- **Step 4:** Click “Bookings” tab to book a service.
- **Step 5** Click “PMS History” tab to view service history.

- **Step 6:** Click “Profile” tab to view account information.
- **Step 7:** Click “AI Assistant” to ask questions about services / history or anything that related to their vehicles / booking service.



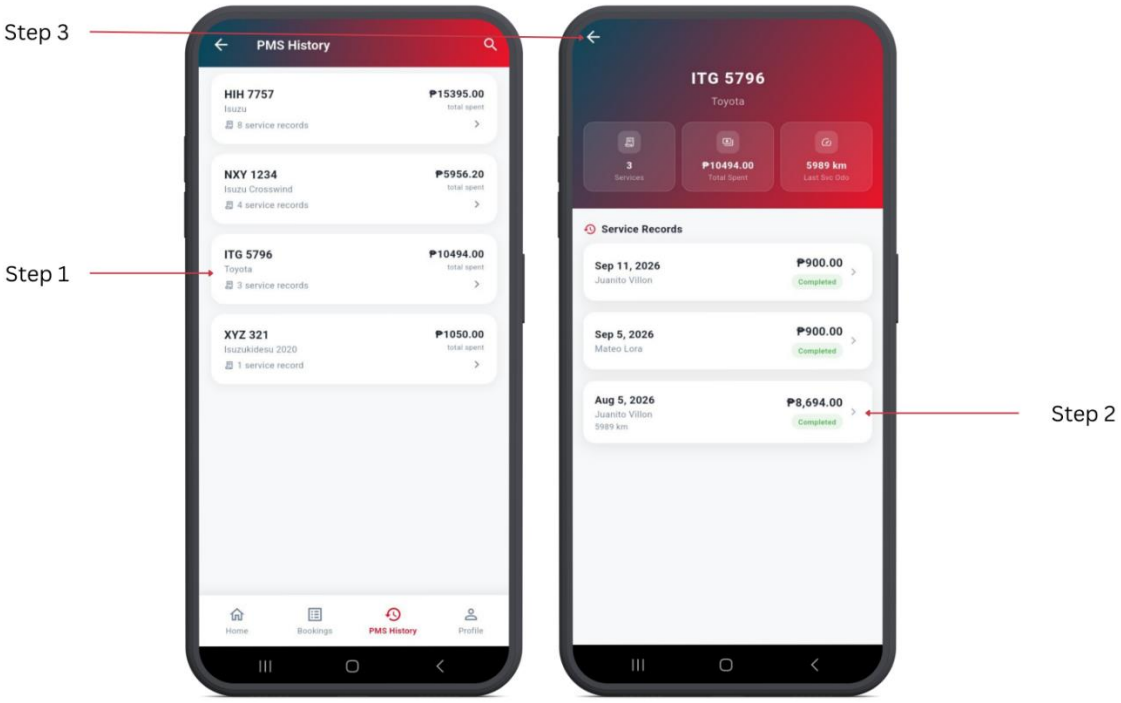
This shows the Bookings Page.

- **Step 1:** Click “Book Service” button to book a service.
- **Step 2:** Click “View details” to view booking details.
- **Step 3:** Click “Cancel booking” if users want to cancel their bookings schedule.



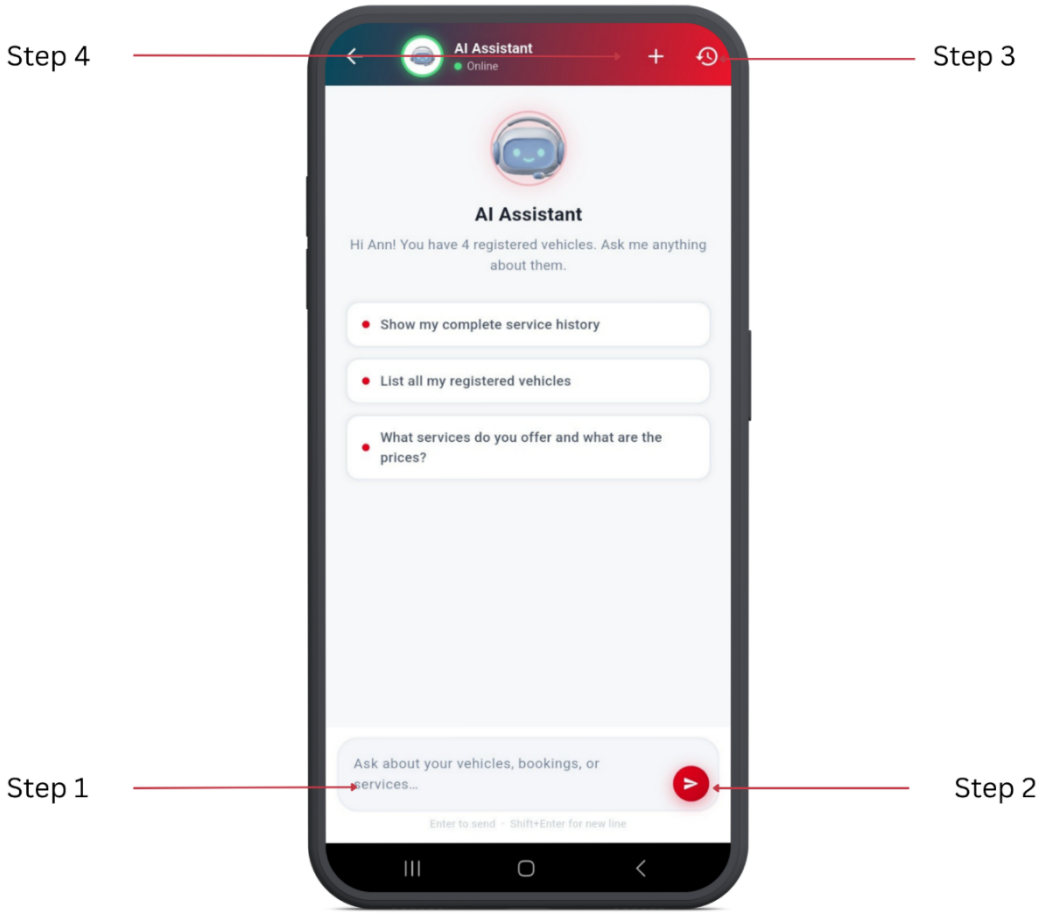
This shows the Book Service Page.

- **Step 1:** Select vehicle(s) and services per vehicle.
- **Step 2:** Choose preferred date, select in the calendar below.
- **Step 3:** Choose availability time slots for their bookings schedule.



This shows the PMS History Page.

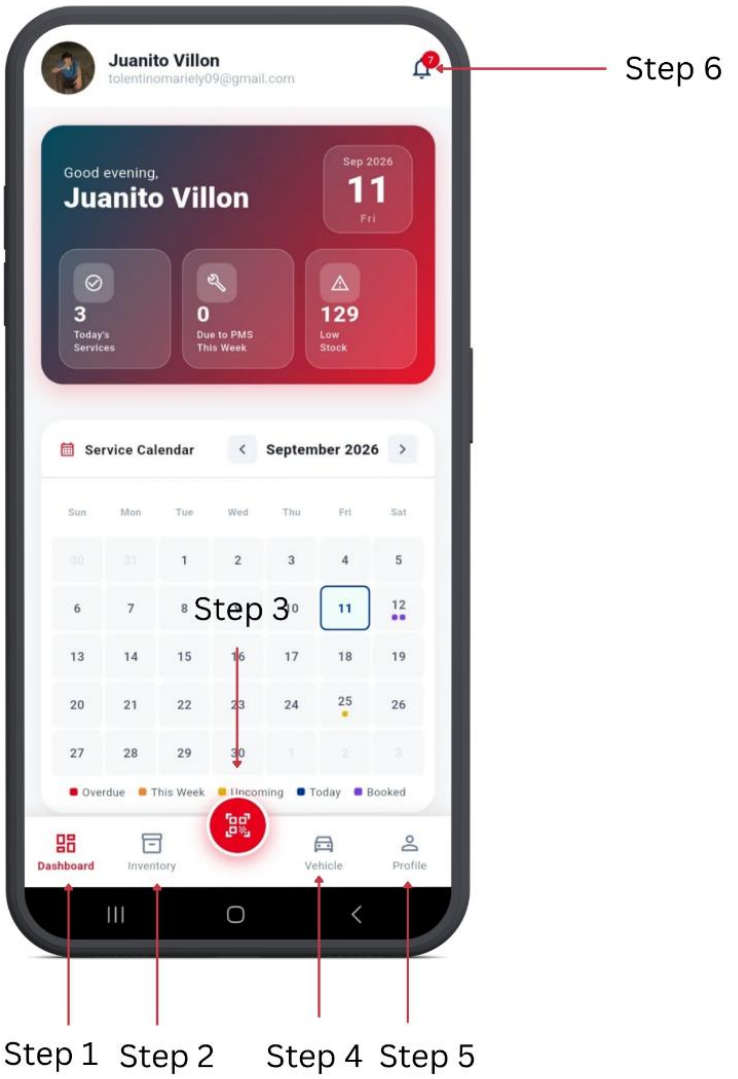
- **Step 1:** Select a Vehicle from the PMS History List
- **Step 2:** Tap the card to view a Specific Service Record
- **Step 3:** Click the back button to return to the previous page.



This shows the AI Assistant Page.

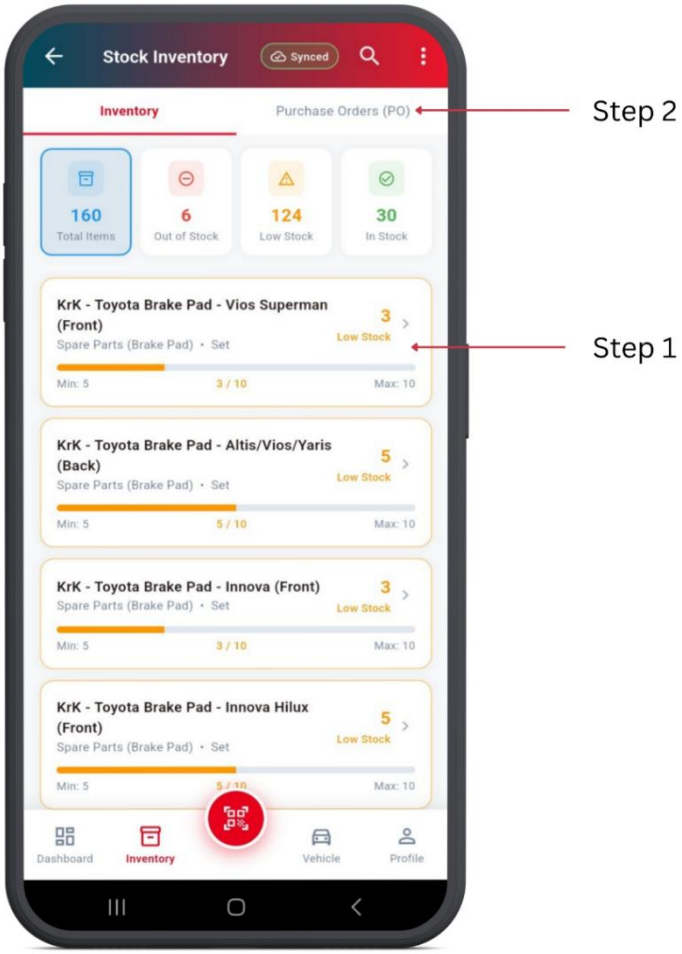
- **Step 1:** Type a message or questions.
- **Step 2:** Click the “red send” button. The AI Assistant will process your question and provide a relevant response instantly.
- **Step 3:** Click “Add” to start a new chat session. This clears the current conversation and allows you to begin a fresh discussion with the AI Assistant.
- **Step 4:** Click the History icon to view your previous chat conversations.

Mobile - Staff Account



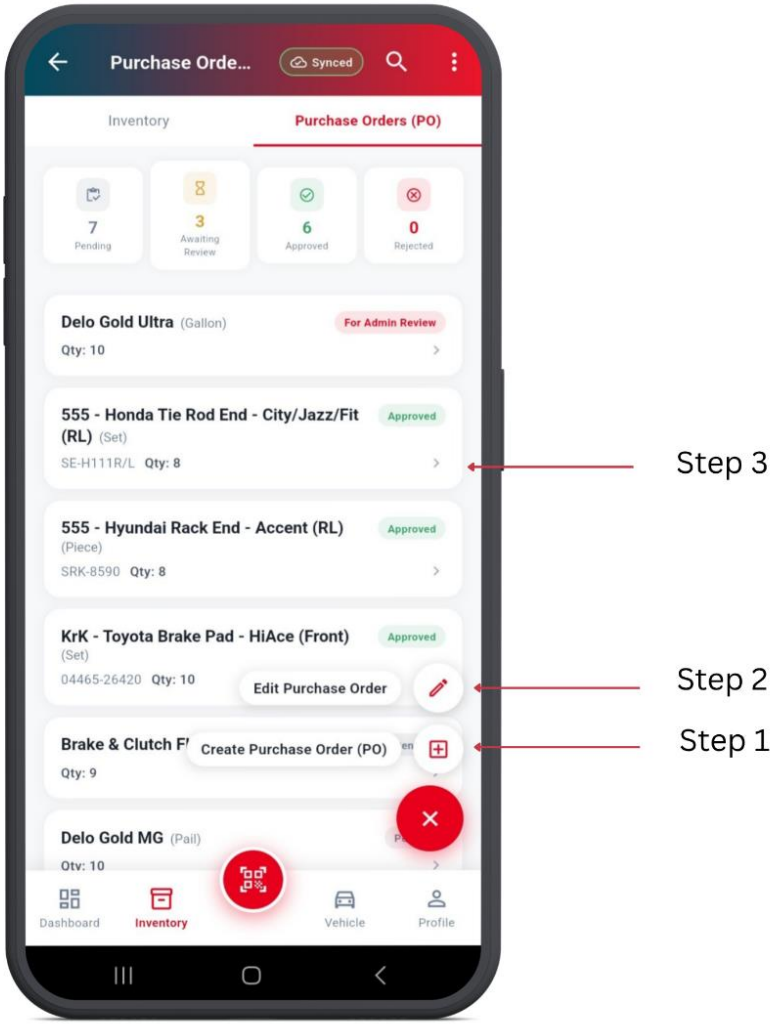
This shows the Staff Dashboard Page.

- **Step 1:** Click the “Dashboard “button to return to the main Staff Dashboard from any other page.
- **Step 2:** Click “Inventory” tab to manage inventory.
- **Step 3:** Click “Scanner” to scan barcode / qr code of the items.
- **Step 4:** Click “Vehicles” tab to manage vehicles.
- **Step 5:** Click “Profile” tab to manage account information.
- **Step 6:** Click “Notifications” bell button to view alerts / notifications.



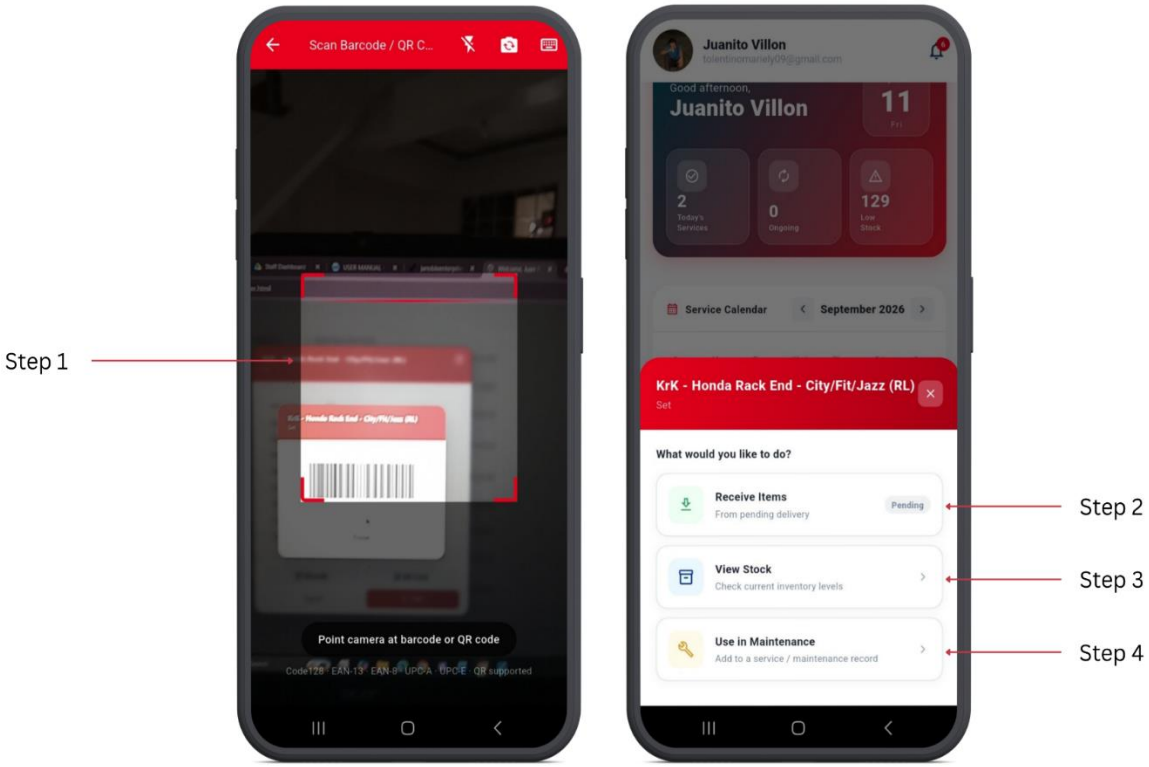
This shows the Staff Inventory Page.

- **Step 1:** Click specific card to view stock item details.
- **Step 2:** Click “Purchase Orders” tab to manage purchase orders for receiving



This shows the Staff Purchase Orders Page.

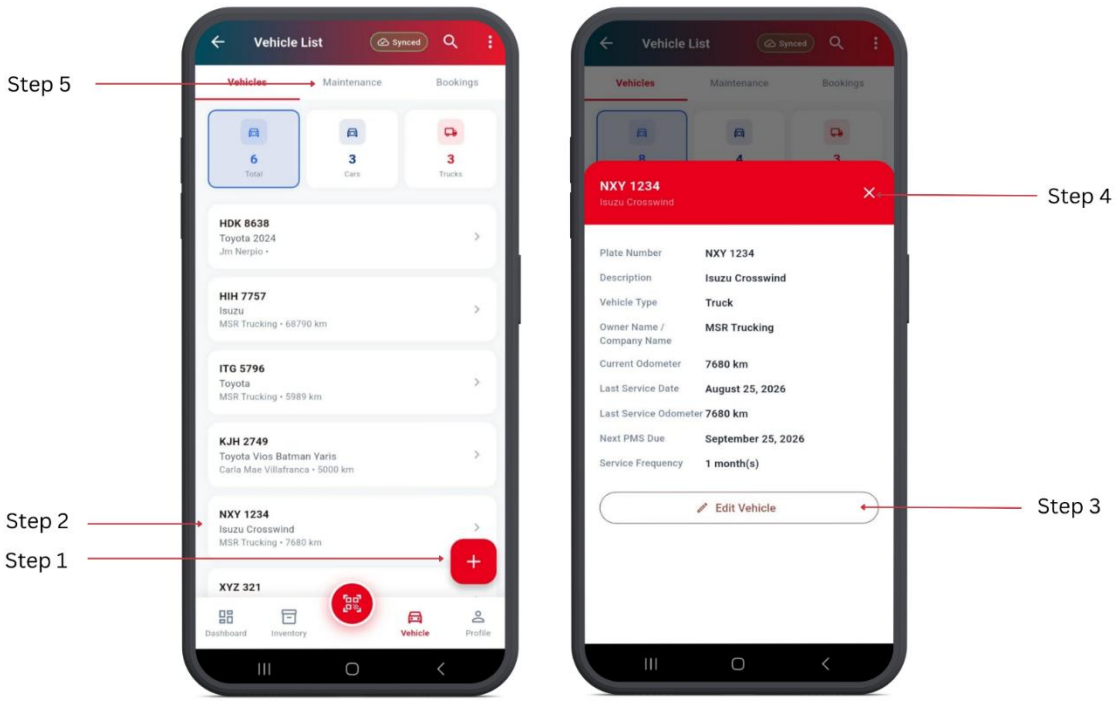
- **Step 1:** Click the "+ Add" button, then click the "Create Purchase Order" button to add an item for stock replenishment.
- **Step 2:** Click the "Edit Purchase Order" button to modify order quantity or items.
- **Step 3:** Click the "View" button to view item details.



This shows the Scanner Functionality.

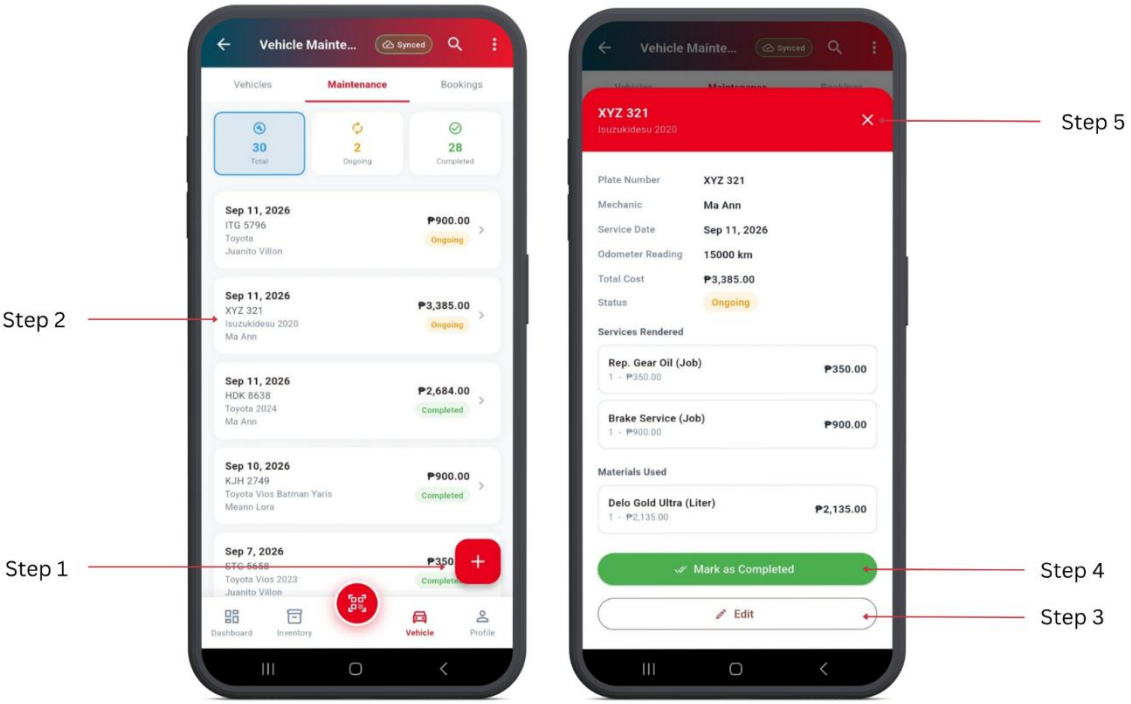
- **Step 1:** On the Scanner Page, the camera will automatically activate and display the scanning interface.
- **Step 2:** After a successful scan, the action menu will appear with the message *"What would you like to do?"* Click the "Receive Items" option. This allows you to Confirm receipt of the scanned item. Record the receiving quantity and confirm. And Wait for the admin to review and verify to update quantity from the inventory.

  Step 3: On the action menu, locate and click the "View Stock" option. This will display the current stock information for the scanned item.
- **Step 4:** On the action menu, locate and click the "Add to Maintenance" option. This allows you to link the scanned item to a vehicle maintenance record.



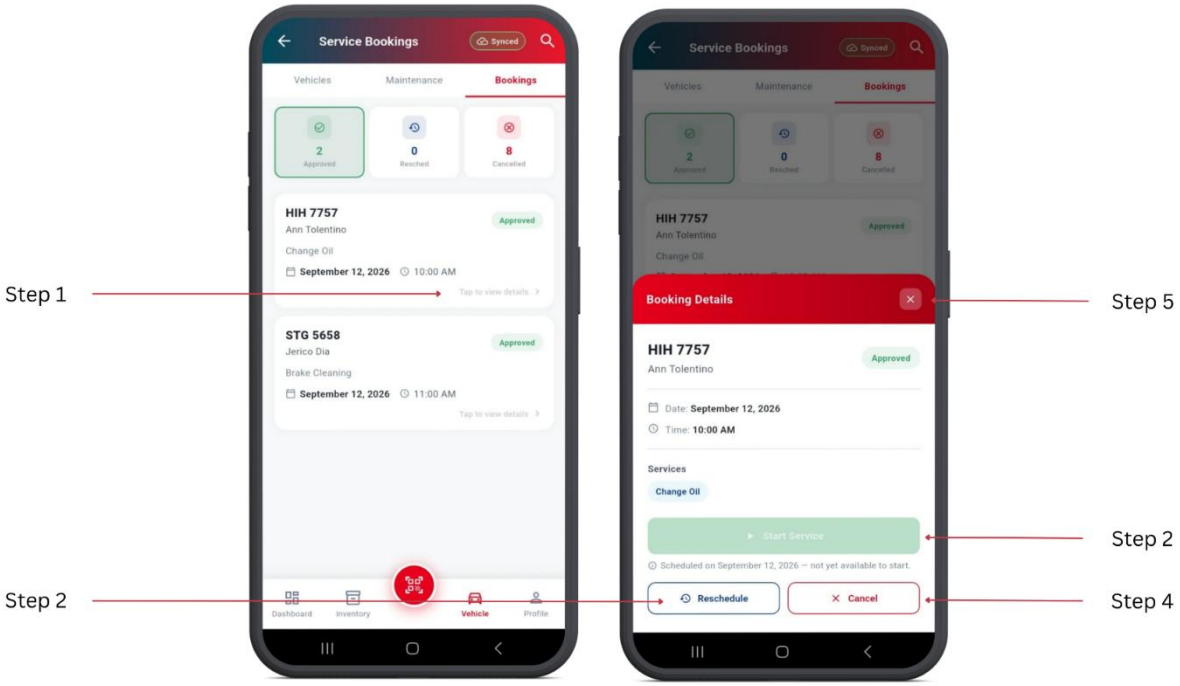
This shows the Staff Vehicle List Page.

- **Step 1:** Click "+Add" button to add new vehicle.
- **Step 2:** Click specific card to view vehicle details/ status.
- **Step 3:** Click "Edit vehicle" button to modify vehicle info.
- **Step 4:** Click "Close" button to close the modal.



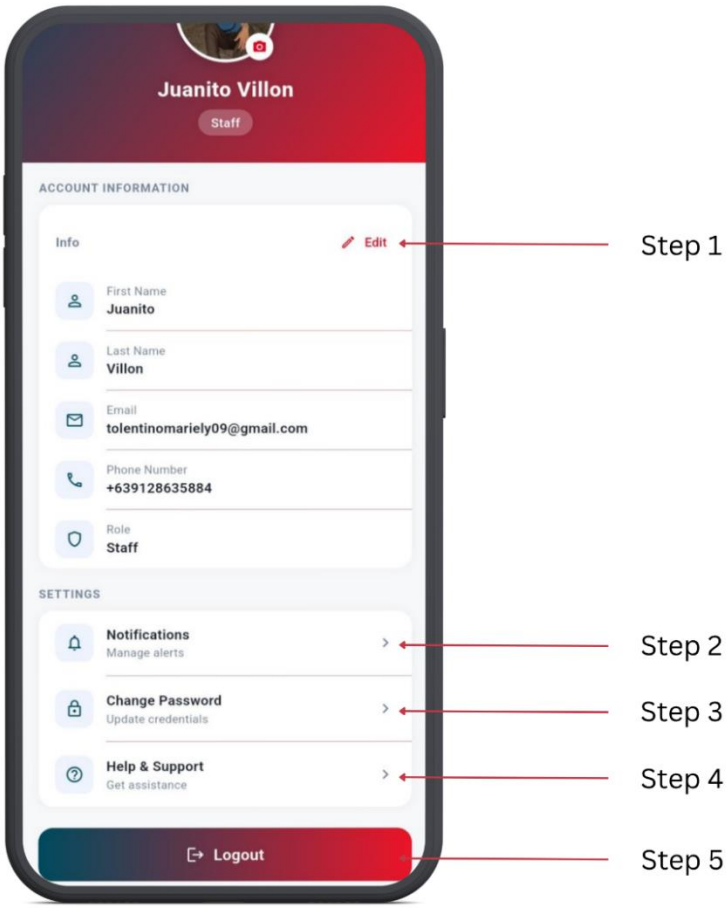
This shows the Staff Vehicle Maintenance Page.

- **Step 1:** Click “+Add” button to add new service.
- **Step 2:** Click specific card to view service record details.
- **Step 3:** Click “Edit” button to modify service record details.
- **Step 4:** Click the “Mark as completed” button to update the service status to Completed.
- **Step 5:** Click “Close” button to close the modal.



This shows the Staff Service Booking Page.

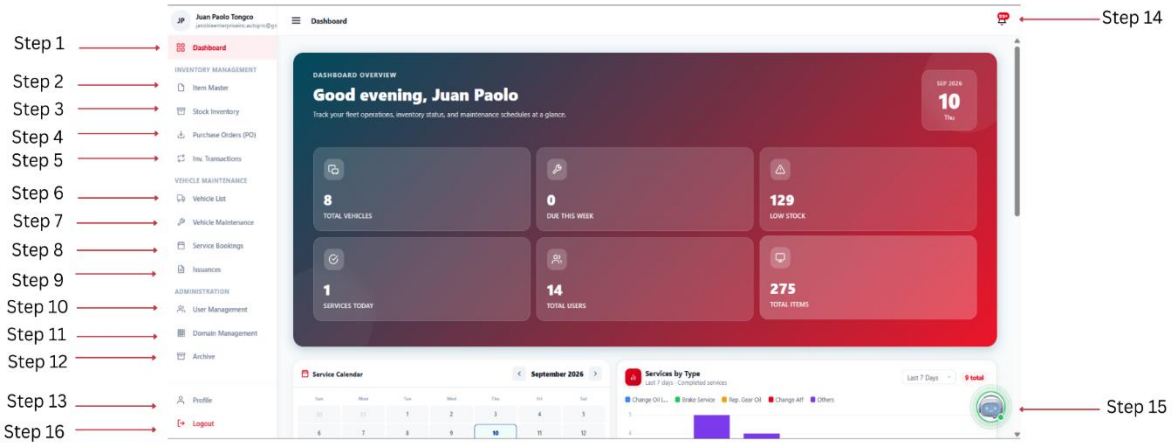
- **Step 1:** Click the specific card to view the booking details.
- **Step 2:** Click the **"Start Service"** button when the scheduled service date is today and you are ready to begin the service.
- **Step 3:** Click "Reschedule" Button to Change the Booking Date.
- **Step 4:** Click “Cancel “button to cancel booking service.
- **Step 5:** Click “Close” button to close the modal.



This shows the Staff Profile Page.

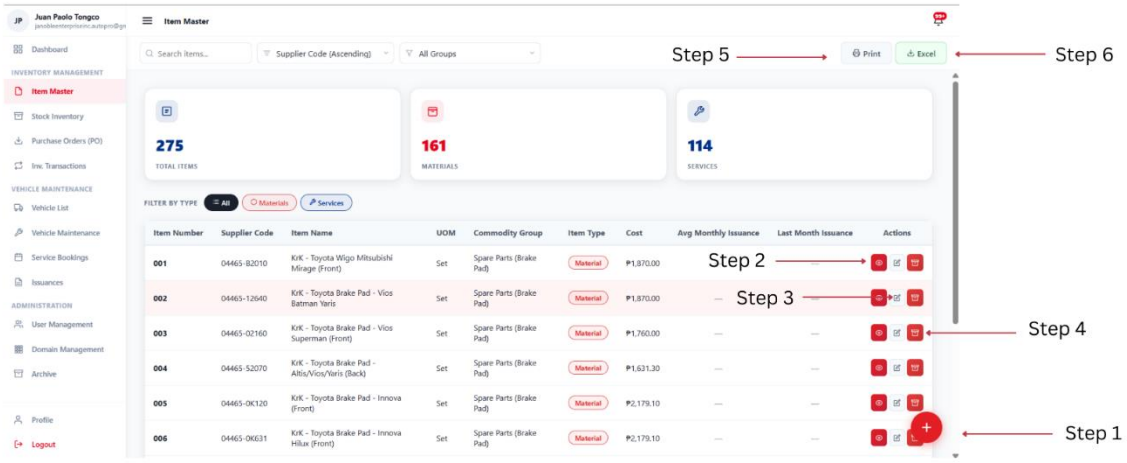
- **Step 1:** Click “Edit” button to modify account information.
- **Step 2:** In the Settings section, click the "Notifications" option. This will open the Notification Preferences Page.
- **Step 3:** Click “Change Password” to change password.
- **Step 4:** Click “Help and Support” to access assistance.
- **Step 5:** Click “Logout” to securely sign out of your account.

Website – Admin Account



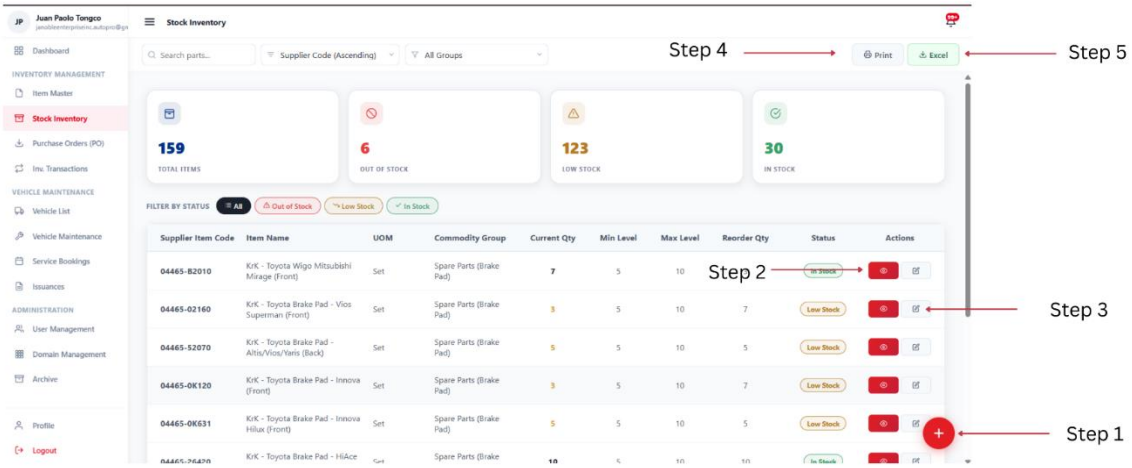
This shows the Admin Dashboard Page.

- **Step 1:** Click “Dashboard” to redirect to the Dashboard page.
- **Step 2:** Click “Item Master” to access the Item Master page for managing materials and services.
- **Step 3:** Click “Stock Inventory” to manage inventory levels.
- **Step 4:** Click “Purchase Orders” to manage purchase orders.
- **Step 5:** Click “Inv. Transactions” to review inventory transactions.
- **Step 6:** Click “Vehicle List” to manage vehicle details.
- **Step 7:** Click “Vehicle Maintenance” to manage vehicle maintenance.
- **Step 8:** Click “Service Bookings” to manage vehicle service appointments.
- **Step 9:** Click "Issuances”" to process item issuances.
- **Step 10:** Click “User Management” to manage system users and permissions.
- **Step 11:** Click “Domain Management" to configure system domain settings.
- **Step 12:** Click “Archive” to redirect to the Archive page.
- **Step 13:** Click “Profile” to view and edit user profile.
- **Step 14:** Click “Notifications” to view notifications and alerts.
- **Step 15:** Click "AI - Assistant" to access AI-powered help and assistance.
- **Step 16:** Click “Logout” to securely sign out of your account.



This shows the Item Master Page.

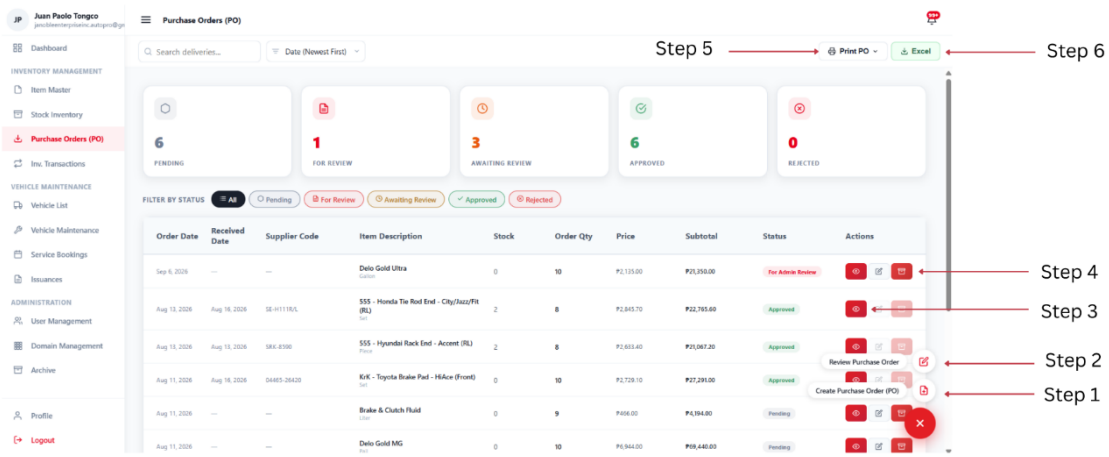
- **Step 1:** Click the "+ Add" button to add new item
- **Step 2:** Click the "View" button to view item details.
- **Step 3:** Click the "Edit" button to modify item info.
- **Step 4:** Click the "Archive" button to remove from the list and move to archive page
- **Step 5:** Click the "Print" button to print an item record.
- **Step 6:** Click "Export as excel" to export all items.



This shows the Stock Inventory Page.

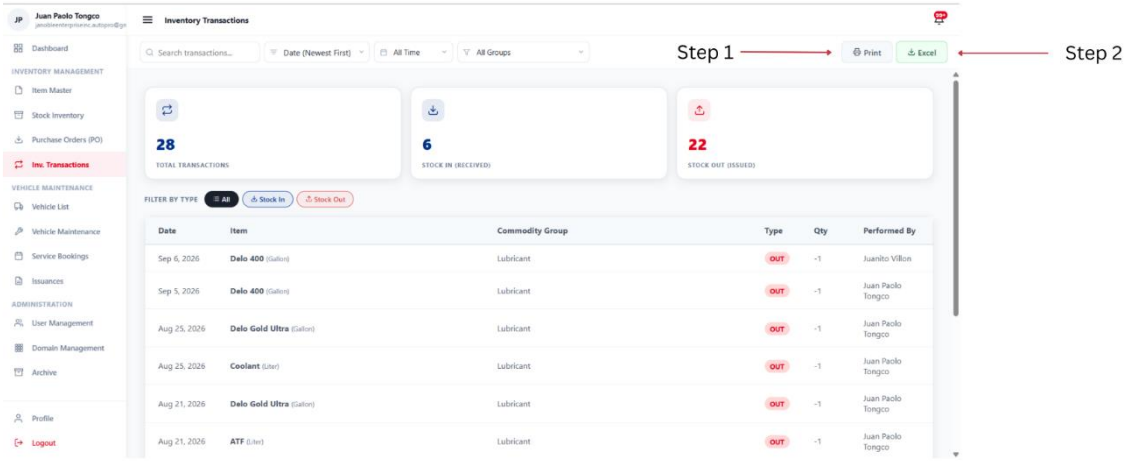
- **Step 1:** Click the "+ Add" button to add a new stock item.

- **Step 2:** Click the “View” button to view item details.
- **Step 3:** Click the “Edit” button to modify item info.



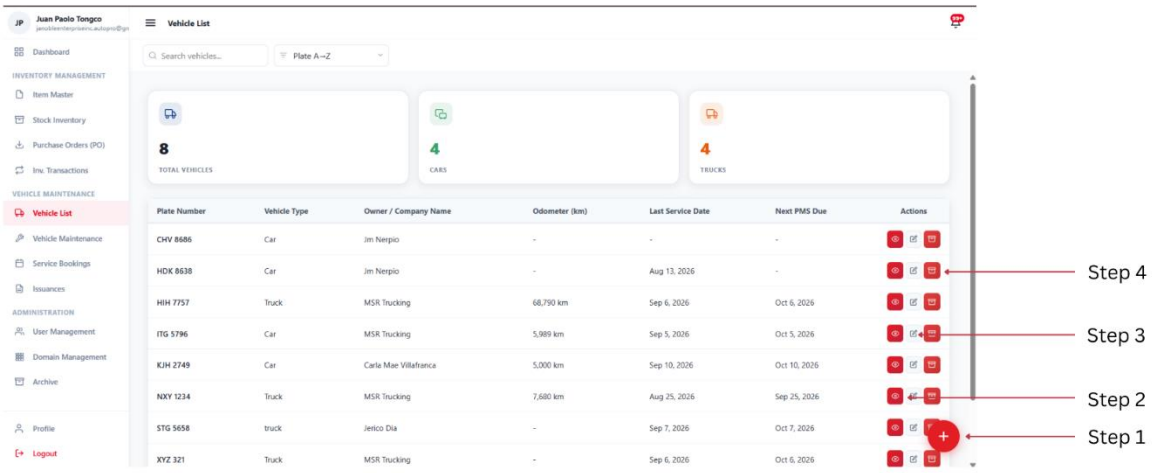
This shows the Purchase Orders Page.

- **Step 1:** Click the “+ Add” button, then click the “Create Purchase Order” button to add an item for stock replenishment.
- **Step 2:** Click the “Review Purchase Order” button to verify and modify order quantity.
- **Step 3:** Click the “View” button to view item details.
- **Step 4:** Click “Edit” to modify the order quantity of the item.
- **Step 5:** Click the “Print” button to print purchase orders.
- **Step 6:** Click “Export as excel” to export purchase orders.



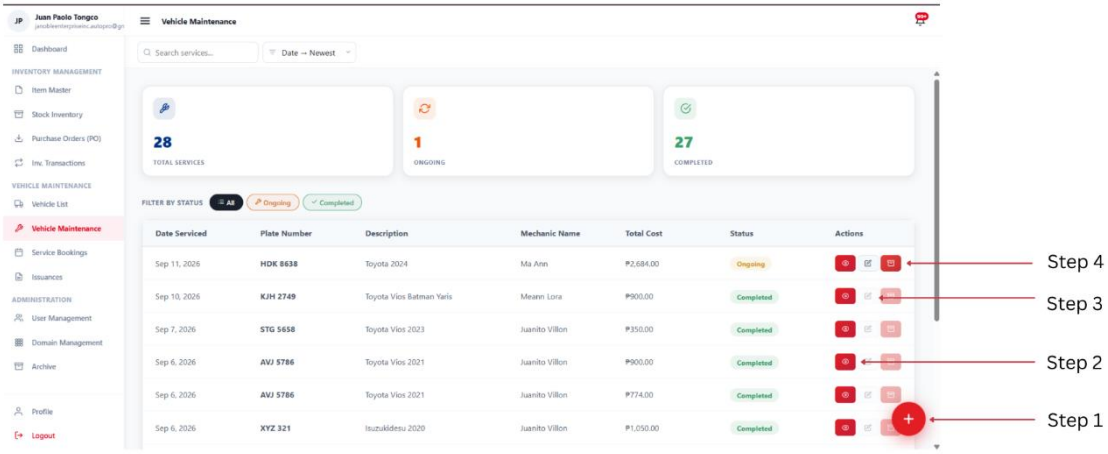
This shows the Inventory Transactions Page.

- **Step 1:** Click the “Print” button to print transactions.
- **Step 2:** Click “Export as excel” to export transactions.



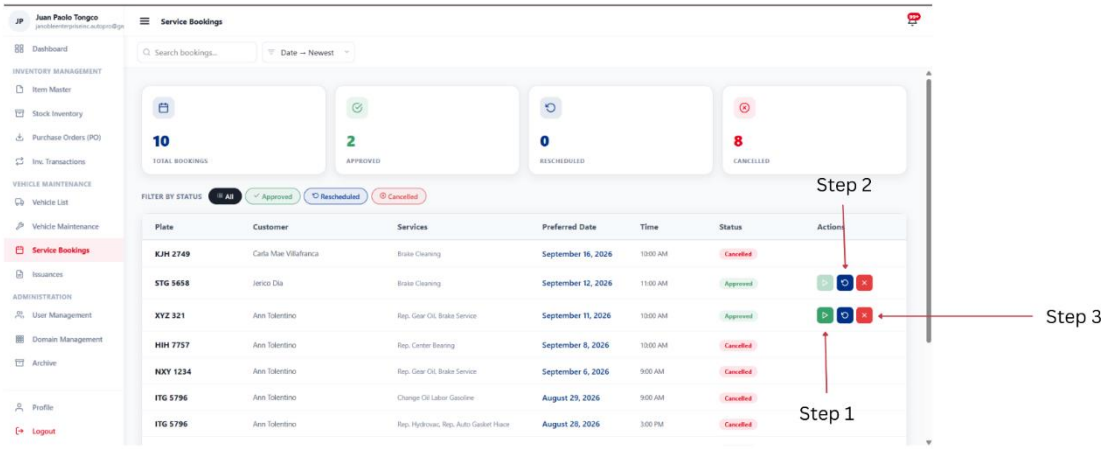
This shows the Vehicle list Page.

- **Step 1:** Click the “+ Add” button to add vehicle.
- **Step 2:** Click the “View” button to view vehicle details.
- **Step 3:** Click the “Edit” button to modify vehicle info.
- **Step 4:** Click the “Archive” button to remove from the list and move to archive



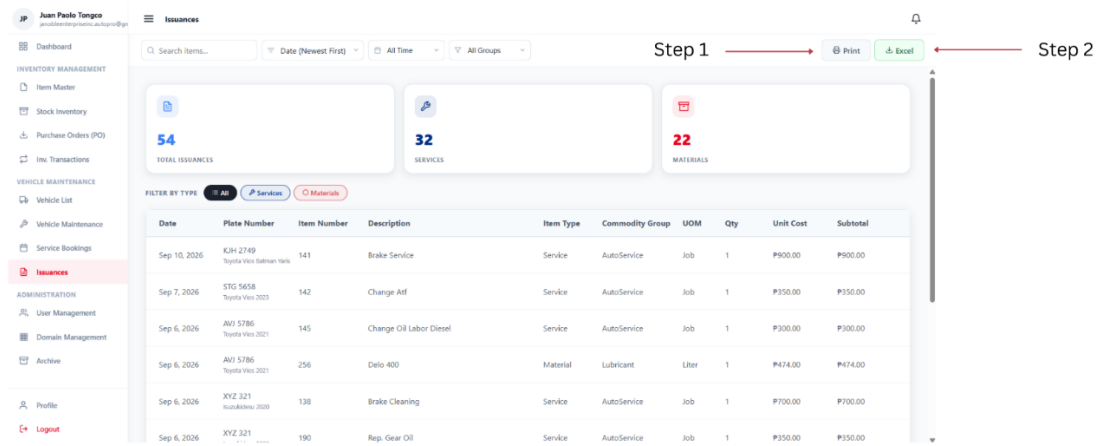
This shows the Vehicle Maintenance Page.

- **Step 1:** Click the "+ Add" button to add new service.
- **Step 2:** Click the "View" button to view service record details.
- **Step 3:** Click the Edit" button to modify service record info.
- **Step 4:** Click the "Archive" button to remove from the list and move to archive page.



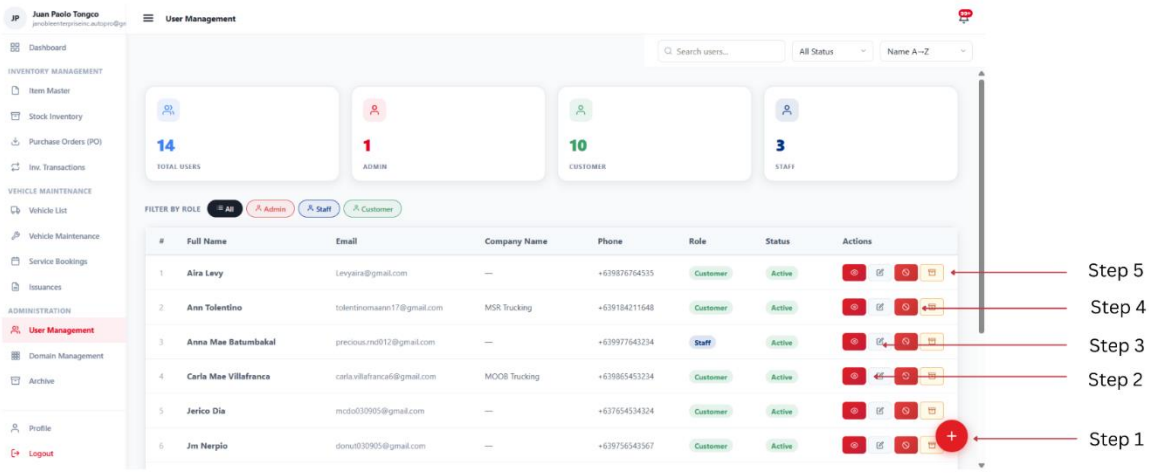
This shows the Service Booking Page.

- **Step 1:** Click the "Start Service" button when the scheduled service date is today and you are ready to begin the service.
- **Step 2:** Click "Reschedule" Button to Change the Booking Date.
- **Step 3:** Click "Cancel "button to cancel booking service.



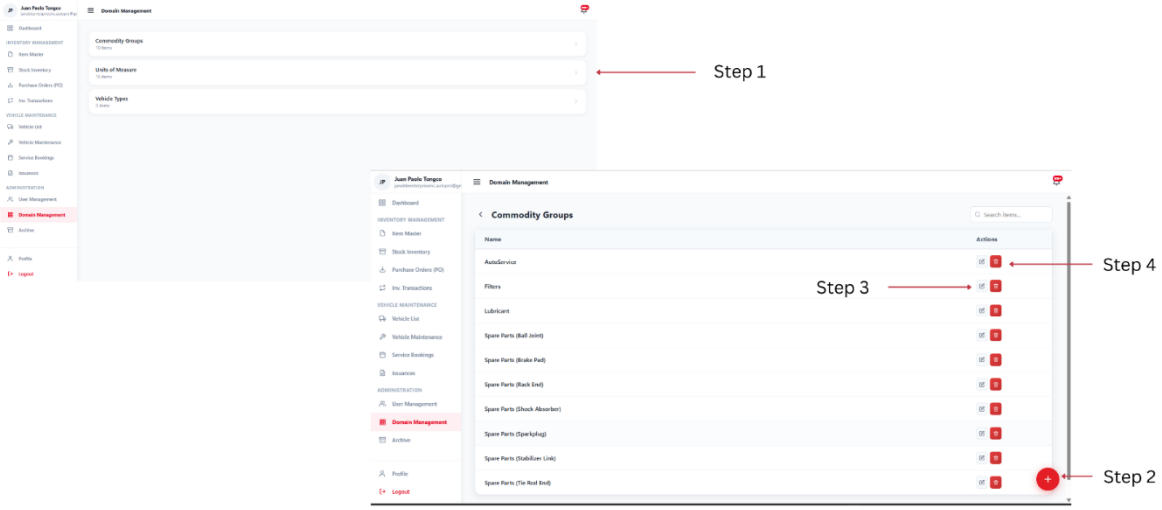
This shows the Issuances Page.

- **Step 1:** Click the “Print” button to print issuances.
- **Step 2:** Click “Export as excel” to export issuances.



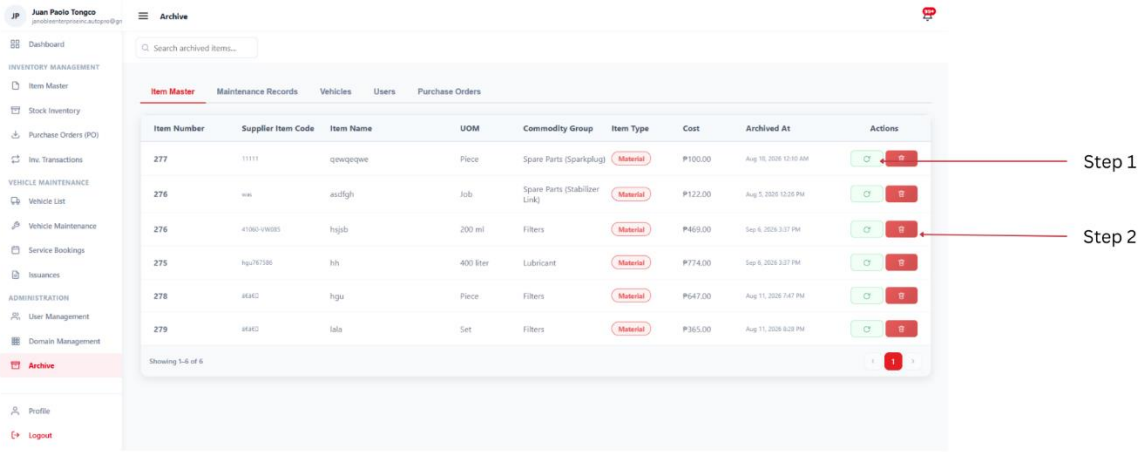
This shows the User Management Page.

- **Step 1:** Click the “+ Add” button to add new users.
- **Step 2:** Click the “View” button to view users.
- **Step 3:** Click the “Edit” button to modify users’ info.
- **Step 4:** Click the “Deactivate” button to deactivate the users’ account.
- **Step 5:** Click the “Archive” button to remove from the list and move to archive



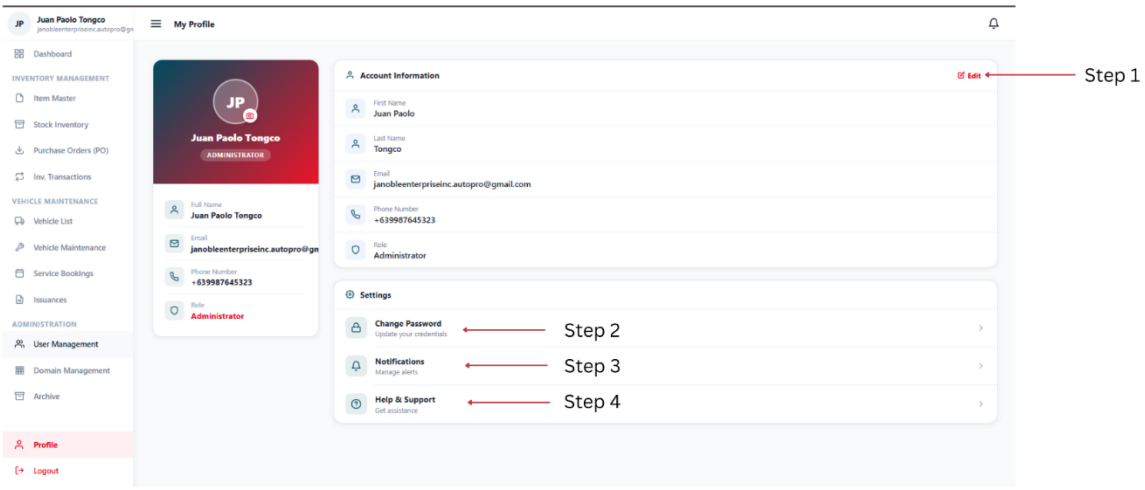
This shows the Domain Management Page.

- **Step 1:** Click the specific card.
- **Step 2:** Click the "+ Add" button to add new item name.
- **Step 3:** Click the "Edit" button to modify item name.
- **Step 4:** Click the "Delete" button to delete item name.



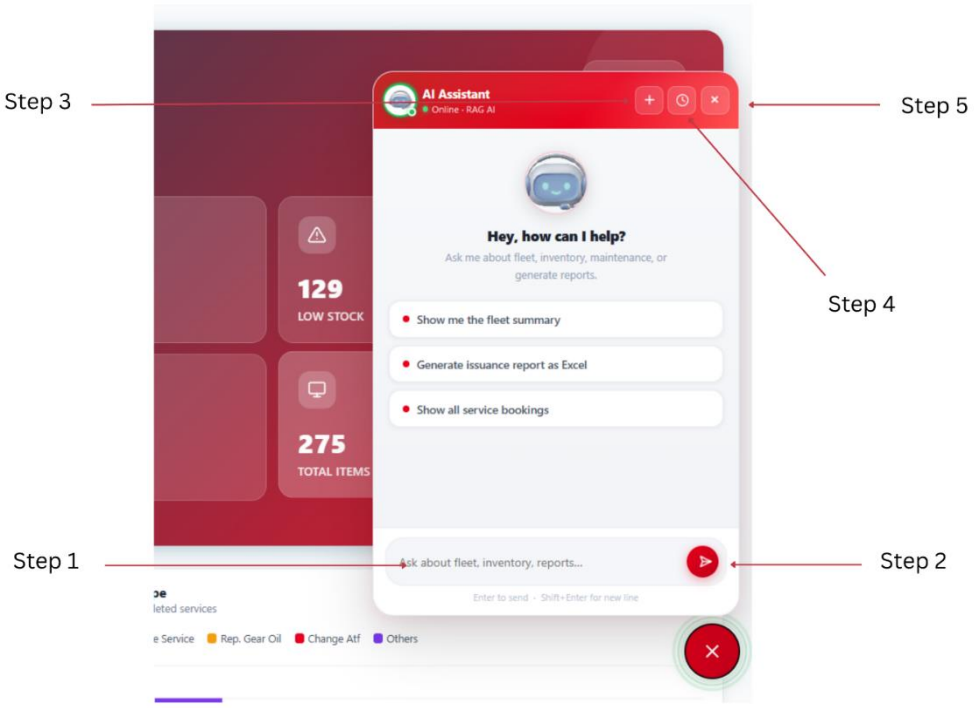
This shows the Archive Page.

- **Step 1:** Click the "Restore" button to restore from the respective module.
- **Step 2:** Click the "Delete" button to permanently deleted.



This shows the Admin Profile Page.

- **Step 1:** Click “Edit” button to modify account information.
- **Step 2:** In the Settings section, click the "Notifications" option. This will open the Notification Preferences Page.
- **Step 3:** Click “Change Password” to change password.
- **Step 4:** Click “Help and Support” to access assistance



This shows the AI- Assistant.

- **Step 1:** Type a message or questions.
- **Step 2:** Click the “red send” button. The AI Assistant will process your question and provide a relevant response instantly.
- **Step 3:** Click “Add” to start a new chat session. This clears the current conversation and allows you to begin a fresh discussion with the AI Assistant.
- **Step 4:** Click the History icon to view your previous chat conversations.
- **Step 5:** Click the "X" button to close the AI Assistant modal.

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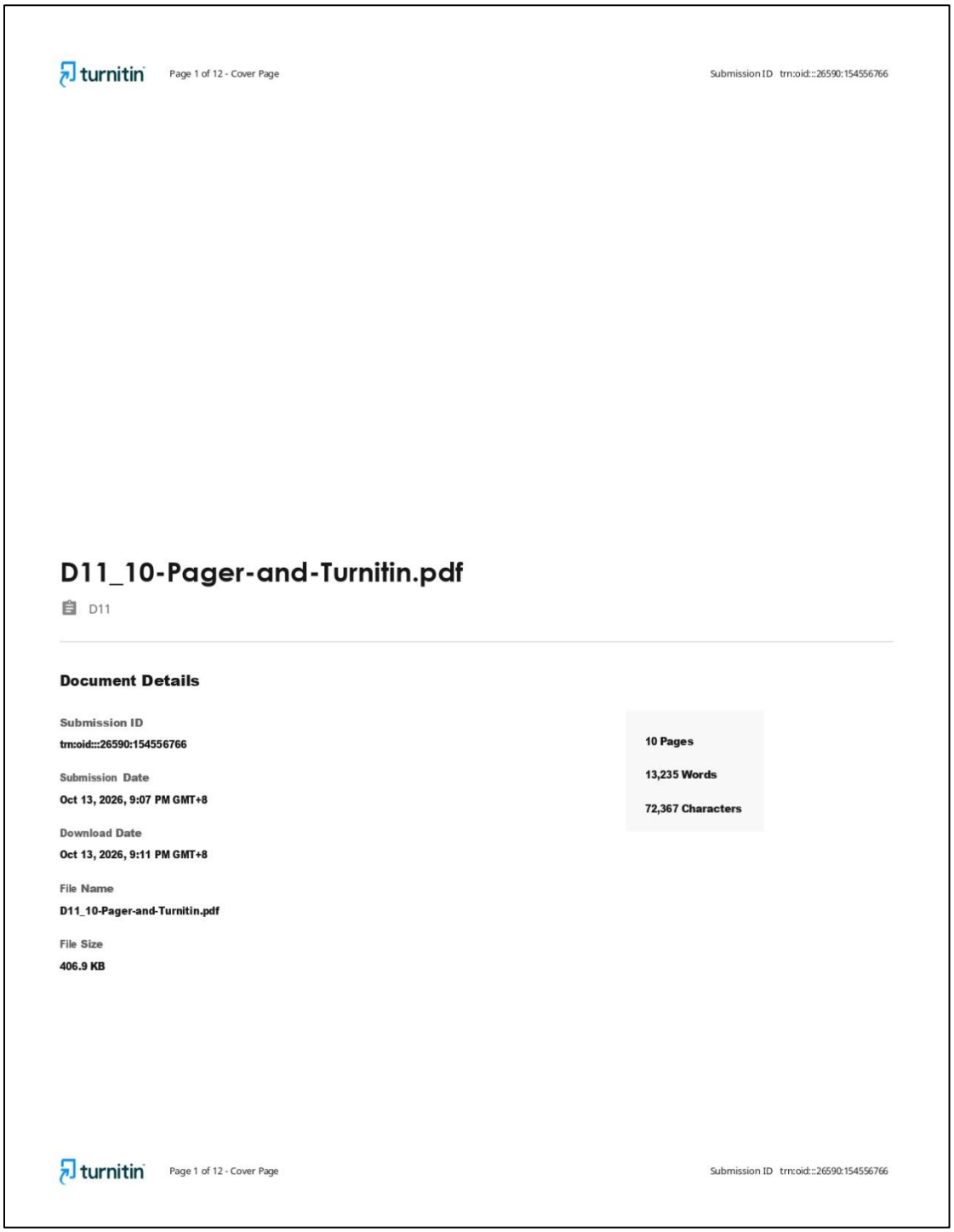

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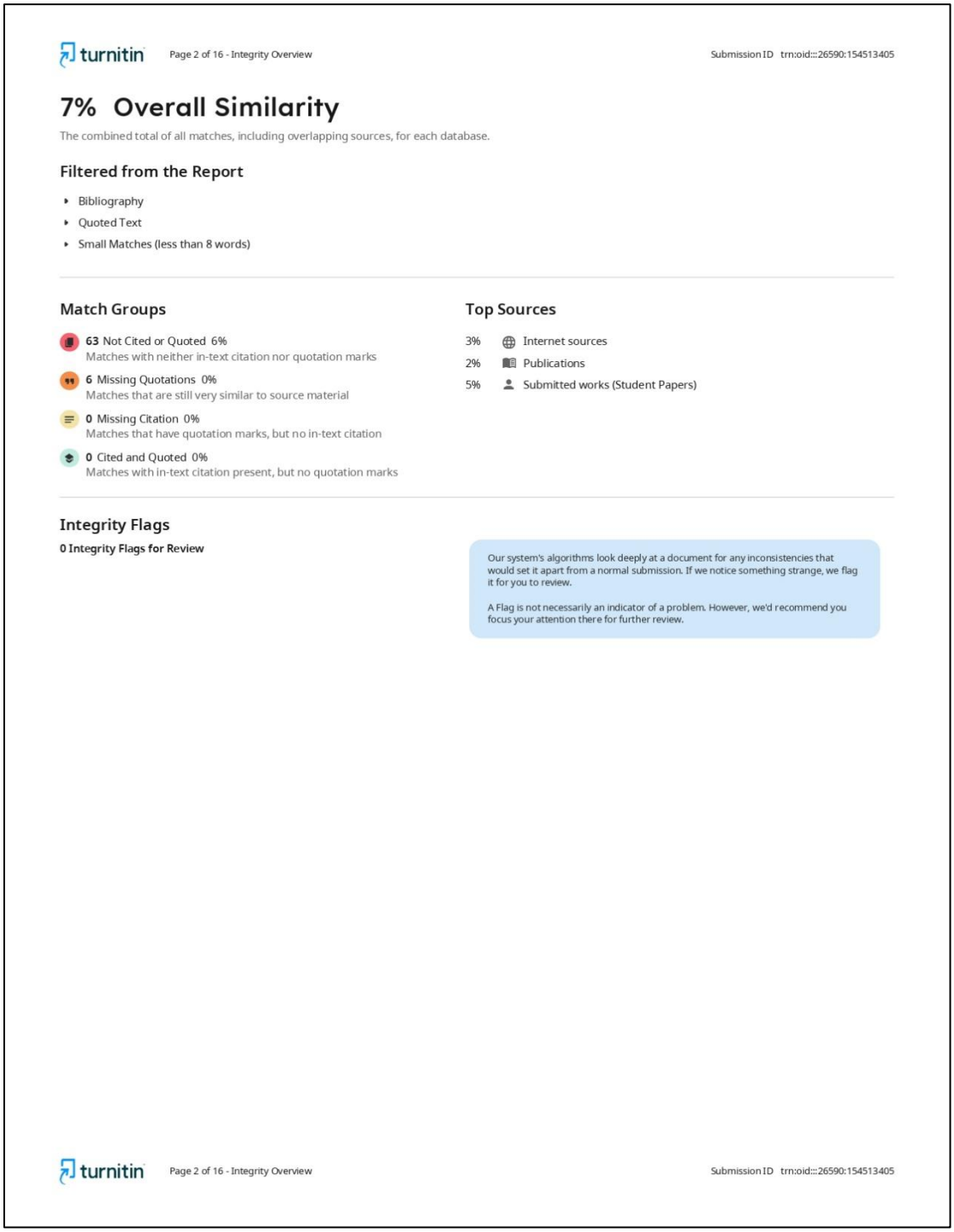

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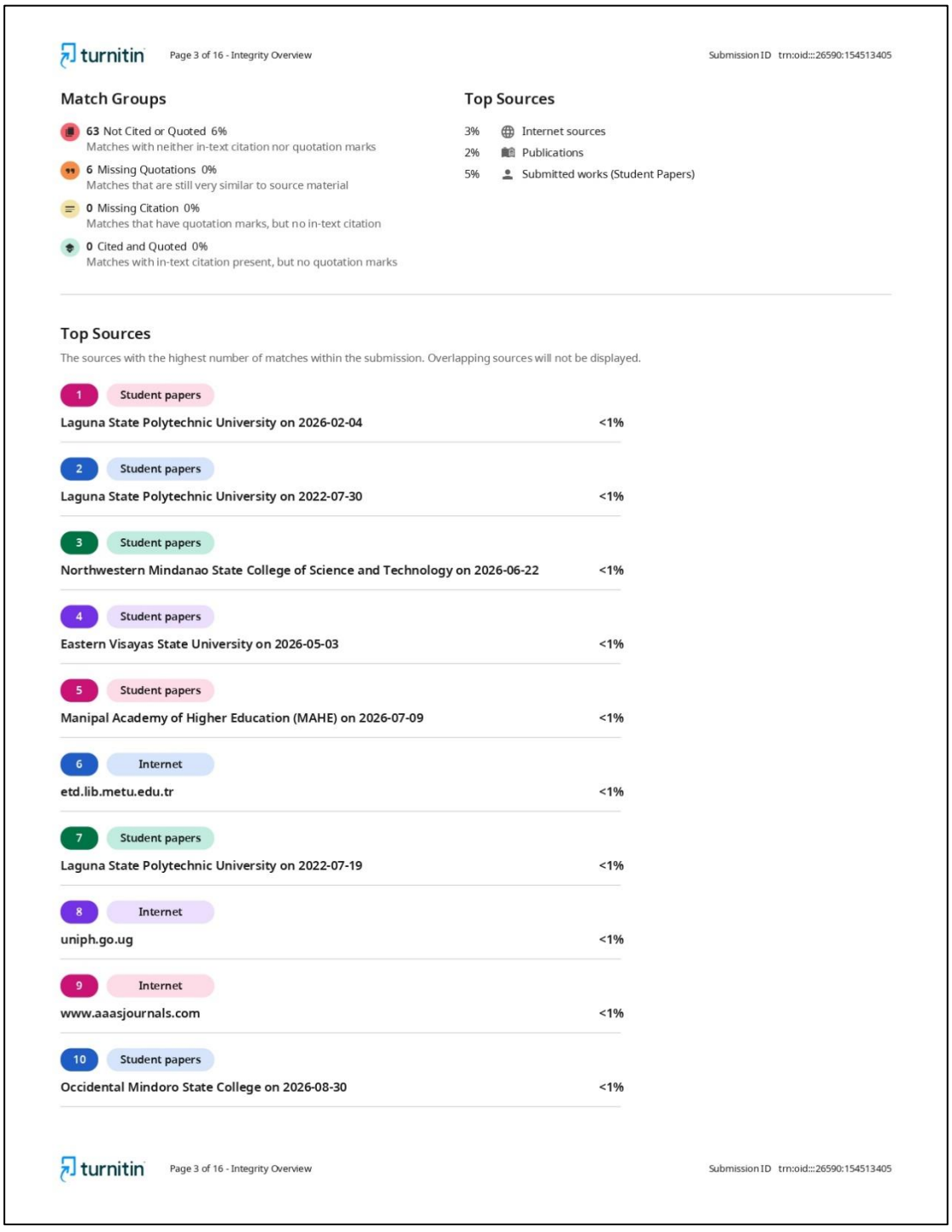

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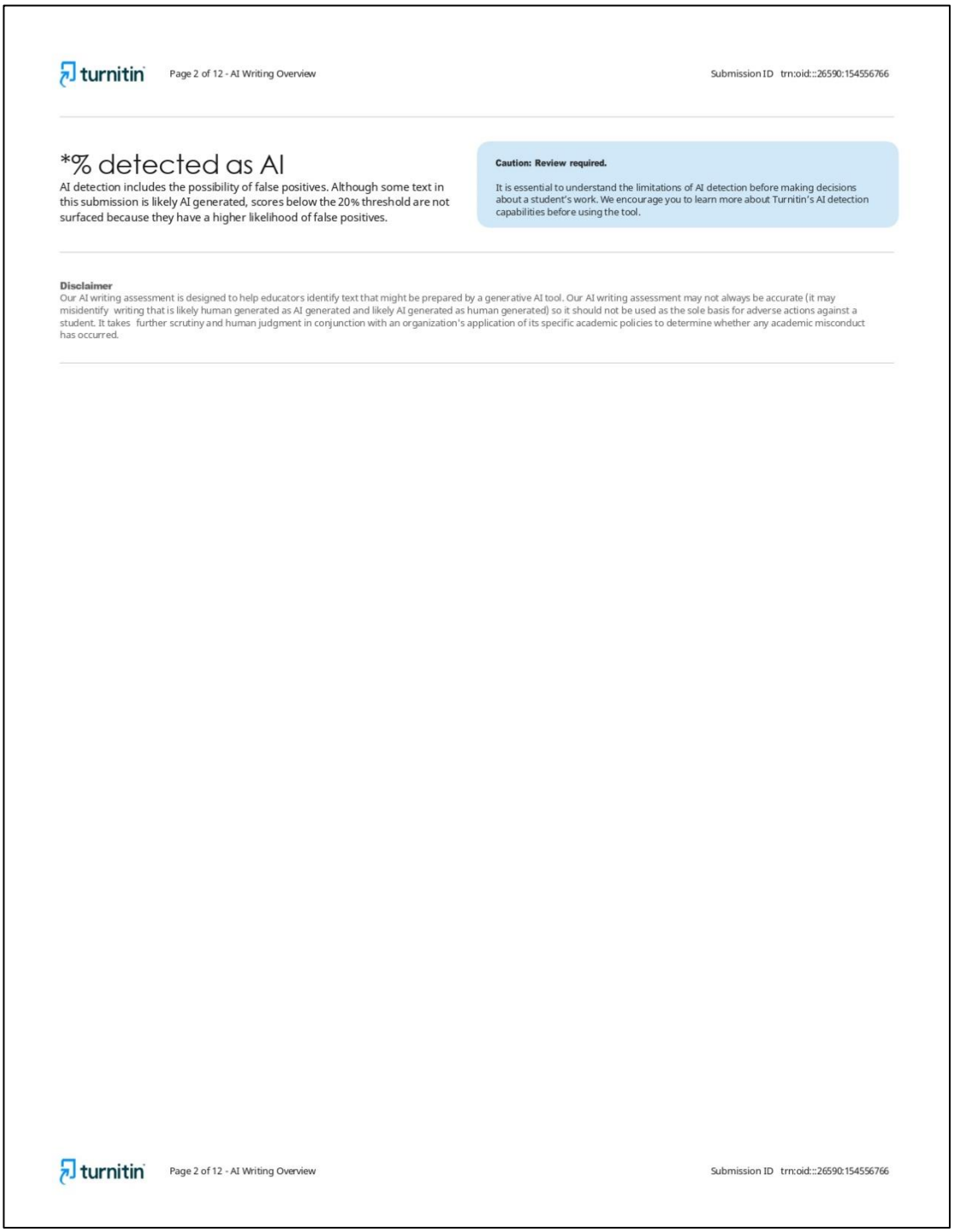

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INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC

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ABSTRACT

In this study, the development and evaluation of the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., in San Pedro, Laguna. The project resolved issues like slow stock monitoring, delayed reporting etc. related to business. manual maintenance scheduling. Agile Scrum Methodology is used, scanners for staff, and an inventory system. The researchers created a web platform for admins, a mobile app with barcode/QR scanners for staff, and an inventory system. Staff will be scanned by code, and a Retrieval-Augmented Generation (RAG) AI will also be utilized. Assistant which generates text responses and generates PDF or Excel reports. The Ten (10) IT experts reviewed the system in a system testing. Accumulated 100% passing rate of and approved the BlackBox test cases Showing that the system is suitable for further testing. The researchers then used a Purposive Sampling design. The number of users was confirmed using actual company records, a final sample size of 144 was the resulting respondents. This number represents 3 administrators, 16 staff and 125 customers chosen through systematically random sampling. The user of the questionnaire was a combination of the Acceptance Evaluation Questionnaire and Technology Acceptance Quality metrics used in Model (TAM) and ISO/IEC 25010. The statistical analysis made the following outstanding results in the system, where an overall average weighted mean was obtained of 4.86 and interpreted as Highly Acceptable. It also got a very low standard deviation of 0.35, which proves that the answers of the respondents were highly consistent and shared positive assessment feedback from the intended users. In conclusion, these findings prove that the developed system successfully met all technical requirements and is completely ready for real-world deployment.

Keywords: Inventory Management; Vehicle Maintenance; Decision Support System; Agile Methodology; QR Code Technology; Retrieval-Augmented Generation; ISO/IEC 25010 Software Quality Metrics.

I. INTRODUCTION

In today's industry, vehicle maintenance and inventory matter a lot, for operational efficiency and reliability. Integrated solutions can be used to manage fleets and spare parts, while gathering service data that improves performance (IBM, n.d.). But manual monitoring can result in inaccurate data, delays in service, and higher costs. Based on interviews conducted with the operations head at JA Noble Enterprise Inc., an automotive service center in San Pedro, Laguna and direct observation of their current workflows, the company faced problems with their paper-based processes and spreadsheets that were not linked. The operations head stated that their current POS system only records sales and could not keep track of a vehicle's service history, could not record the materials used for each job and could not provide real-time visibility of stock levels. Workflow observations showed that the lack of a central system led to inefficient communication, frequent stockouts of important parts, and an inability for managers to pull useful reports to make decisions.

To overcome these problems, this study proposed the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc. The system is a web and mobile-based application that uses QR code and barcode technology in monitoring services and inventory in real-time. It has a Decision Support System (DSS) to provide low-stock alerts and reorder recommendations, and automated notifications for Preventive Maintenance Scheduling (PMS). A secure mobile portal allows customers to view their vehicles service history and check maintenance due dates. The system also uses Retrieval-Augmented Generation (RAG) to produce useful management reports. This proved to be a way to increase transparency, accuracy and efficiency, at JA Noble Enterprise Inc. It consolidates all these tasks into one place, making Operations smoother and more reliable.

Research Objective

This study aimed to develop, test, and evaluate an Integrated Vehicle Maintenance and Inventory Management System with Decision Support System for JA Noble Enterprise Inc.,with regard to functional correctness using systematic testing and user acceptance through the Technology Acceptance Model (TAM).

Specifically, this study aimed to achieve the following objectives:

1. Design and develop a web and mobile-based Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., featuring core modules for vehicle maintenance tracking, QR/barcode-enabled inventory management, automated low-stock and preventive maintenance alerts (DSS), a RAG-powered AI assistant for intelligent queries and reporting, and a secure customer portal for service history access.

2. Evaluate the functional performance of the developed system across its major modules, including the Vehicle Maintenance Management, Mobile-Based Inventory Management with QR/Barcode Scanning, Decision Support System (DSS) for stock replenishment and PMS scheduling, and Retrieval-Augmented Generation (RAG) for the AI Assistant, by measuring relevant indicators to confirm that all key features are working as intended.

3. Determine the perceived software quality and user acceptance of the developed system after system demonstration and hands-on use, based on the Technology Acceptance Model (TAM) in terms of Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience.

II. REVIEW OF RELATED LITERATURE

Vehicle Preventive Maintenance Management System

For a fleet to operate smoothly and for machines to have a long life, it is essential to plan and manage the preventive maintenance of vehicles in a way that prevents any trouble on the road. In the past, vehicle service was handled by more conventional means: paper logs and physical cards. But such methods are prone to human error and can be slow, with records going missing and regular part repairs or inspections being overlooked. As a result, an increasing number of companies that run cars and move goods are turning to centralized systems and mobile tools. With these at their disposal they can organize jobs and put out reminders so as to get ahead of potential issues.

The shortcomings of the old ways have made the case for computerized maintenance systems all the more compelling. Dwipa et al. (2024) examined the upkeep of emergency coarsers and aircraft and found the paper checklists in use to be wanting; the inspection process was inconsistent and locating problems was a slow business, not to mention the missing records. To address this, they put together PMCSS, a web application that puts checklists in the form of digital documents requiring a sign off from personnel, thus ensuring

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emergency vehicles are ready for duty and errors are kept to a minimum. Then there is the matter of data fragmentation in repair shops. Babu and Narayani (2026) noted that keeping to paper makes it difficult to pull together all the information, leading to slow booking and trouble with old files. Their solution is a system for managing those records and tracking car status that has proven to make shop operations far more efficient.

In the field of motorbike and car repairs Kana and Aji (2024) The use of paper for service orders has been noted as a source of by-have pointed. Confusion and lack of sales records and slow report filing. To put an end to this, they put together a web and phone application to automatically Confirm sales and ensure service orders are delivered timely. There's also the issue of record keeping. Veena et al. (2024) noted how paper can be an obstacle when it is only in making reservations and registering that it becomes apparent. Their solution is an app that does away with keeping track of spares and streamline the process, not to be used as a paper of allocation of time and workers to wash and repair cars.

Ramadhan et al. (2026) have also been interested in the bus rentals operation where things can go awry if one is not on top of driver schedules or road time. They devised a system so that managers could have everything in one place to check the log of the drivers and available buses for hire. In a separate QR code into their project. Ramadhan and Falevi (2025) took their project one step further by adding QR codes. The full service comes with a scan that's provided by either the owner or the mechanic history of the vehicle is at hand, which preempts any delay in locating paper easily handles and expedites maintenance. What these studies have in common is to avoid putting these things on paper and digital records are the way to go. To prevent errors and to ensure proper maintenance. Digital systems are also put to use for planning and the dispatch.

Based on the records of reminders. One is Anisa (2025) who has Created a web system along with WhatsApp to look at car files when called and issue a warning; this will alert both drivers, and the man-in-charge was to them. Besides this, there is the work of Muharom (2024), who established a system for workers to assign work and to record when it was done, and this, in tum, accelerates things and helps minimize errors. The importance of real-time data.The importance of real-time data.

This is clearly visible in the phone app developed, which includes and alerts automation for improved planning This was written by Balia and Ab Ghaffar (2026). Through the use of a computer and car tool,The app can calculate battery level and distance travelled and deduce oil used Adjust intervals and give an alert, without the risk of late maintenance or battery change-over. trouble. To put it simply, these studies suggest that apps that alert users to the potential dangers of a phone call can help them avoid those risks.can change the maintenance strategy for vehicles.

It's more of a problem when it involves coming more. To fleet planning, where it becomes necessary to manage the availability of the cars with the creation of a plan.workforce and budget. Wang et al. (2022) have overcome this by creating an Apply model to make predictions of part-lives to inform service scheduling decisions at reasonable cost and time effective manner. A study from Crespo del Castillo et al. (2024) provides some insights. another approach. They rely on a model that is based on daily work plans and forecasts to see. How to allocate resources, both human, financial and spares, to ensure that they are used optimally to maintain cars in operation.

Planning the modern fleet is no longer a matter of the schedule, it's a matter of. can be used to prevent problems before they occur through tracking tools, sensors and mathematical models. gets out of hand. Abidin and Fadzil (2024) in their review of tracking systems noted that GPS and real-time car sensor information are extremely useful for this type of work. Prediction, of course, is another thing. Foké et al. (2023) put together a system using real data from city buses and after filtering the data, researchers found that there were some problems that are red flags for what may emerge as a big problem. problem. In short, their research indicates that one can use models to put Find out, just before they break. But some machine-driven approaches to problem solving require a great deal of pre-labeled data; this is not always practical. Instead, Giannoulidis and Gounaris (2023) suggested that: really detected in the environment. Invented a technique which could be applied to sensor data to group cars according to what they actually sensed. are doing and detecting faults without the need for any specific data. It is a way to keep vehicles running and costs down. Such innovations look toward a change in the product life cycle. Switching from one set of established schedules to more creative, flexible plans for maintenance.

Yet for all these digital advances, there are still obstacles. Babu and Narayani (2026) have identified the cost of implementation as the need for internet access and data security are big challenges. The human element cannot be ignored: overlooked either. The tools need to be simple, and workers will have to be trained in the use of. Otherwise, they can fall back into their old habits. Another issue is data quality. Abidin and Fadzil, in 2024 said that the data used must be good. If it is not the plans can be wrong and maintenance might be done soon or too late.

Inventory Management System for SMEs (Small and Medium Enterprises)

The growing complicated ways of working in medium businesses, especially those that depend on inventory and the car industry, have made it more important to have good, reliable and tech-based systems to manage things. In this case, the parts of the process that deal with inventory and keeping things in shape are very important to make sure the business can keep running well and make customers happy. According to Alam, Thakur and Islam (2023) small and medium businesses often have trouble keeping the amount of inventory because they do not have enough resources and still use old ways to check on inventory. Also, Madamidola et al. (2024) said that there has been a change in the ways inventory is managed over time. This change went from doing things by hand to using computer-based systems to make sure everything is done right and fast. Furthermore, adding decision support tools has become very important for companies that want to make the way they get more automatic and also try to predict what will happen next.

At first the way inventory was managed used manual methods that were based on looking at things, writing things down on paper and physically counting the stock. This way of doing things led to not working, having incorrect numbers of items and making decisions about when to get more stock late. According to Cabungcal et al. (2023). These old ways really hurt the money side of a business. Also, Gaborno et al. (2025) said that owners of small businesses often used informal and experience-based ways to track things, which made it hard to grow and be accurate. So, Ekron, Ongori and Akwee (2023) said that not having ways to control the stock directly made customers unhappy, which was another problem with manual inventory management.

With the advancement of technology, there has been a tendency to put computers to work on inventory in a more orderly fashion. Odasco and Saong (2023) would argue that computerised systems are of real value in how services are rendered; they simplify access to information and help avoid errors in the process.In line with this Gabriel (2025) said that using organized ways to manage inventory in stores helped make the inventory more accurate and the money side of things better. Owajege and Saif (2025) also showed that a good way to keep track of inventory can help small and medium businesses do better by making operations work better and making choices. At this stage the main thing was to get everything even though data was organized better, making choices was still mostly up to people.

Then the inventory management systems turned into systems that had some parts that worked on their own using tools that helped make decisions. Alam et al. (2023) Said that now a lot of inventory systems for medium businesses are done on digital platforms that help with procurement and tracking. In the way Rodrguez, Lopez and Vera (2025) talked about new ways to manage inventory that help make more money for small and medium businesses by keeping a close eye on things and making them better. When it comes to the car industry, Yang (2025) showed how managing inventory in the parts supply chain in the car industry needs to be done with systems that work together and are connected to handle needs and complicated supply issues. Zvinowanda, Ncube and Mafini (2025) explained how the car industry uses ways to manage inventory with knowledge from systems and organized steps even though the systems still are not fully automatic.

Advanced ways have also tried to deal with smart inventory systems. These kinds of systems can have ways to predict things and

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help with making decisions. Sayal et al. (2025) Looked at clear inventory systems in the car industry and said that managing uncertainty makes the choices about inventory better. In addition, Puiu, Petre and Boscoianu (2025) used Six Sigma to make car inventory. This means looking at data to make things work better. Johnson (2025) studied how to plan for making cars and said that better planning systems can help reduce the number of cars that are made and stored. Lastly, Ta and Le (2025) found that managing inventory well has an effect on how much money a store makes which helps small and medium businesses do better with better inventory systems.

Even though there has been a lot of progress in making inventory systems there are still many problems with the way these systems work. Madamidola et al. (2024) said that in most of the inventory systems there is still not full connection between different parts like getting things, tracking them and helping with decisions. Also, Alam et al. (2023) And Gabriel (2025) said that small and medium businesses still use old ways to do inventory tasks. Rodrguez et al. (2025) said that even the newer systems do not have the ability to predict what will happen which is needed to make getting more stock automatic. Sayal et al. (2025) also said that with the new smart models, it is still hard for many companies, especially small and medium businesses, to use them because of how complicated they are.

The studies reviewed in this literature show that the inventory system has changed a lot, going from doing things by hand to using computers and then to systems that have a lot of ways to help make choices. One finds that the system's various components are not as well integrated as they should be, and there are issues with the way predictions are made and with decisions on inventory and maintenance. It is clear from this that small to medium-sized enterprises in the car sector, for instance, require a properly functioning system that can tie everything together.

What the studies make clear is that vehicle maintenance is in the process of changing. The old way of putting pen to paper is being left behind for digital systems that put all the relevant data in one spot and can be accessed from a phone. These tools will even handle the scheduling on their own and keep an eye on things and that can predict when a car needs work. These things are often done on their own, which shows the need for a system that brings all the parts together like having one place, for car maintenance records and a way to plan when the work should be done for car service businesses.

Decision Support System (DSS)

For JA Noble Enterprise Inc, the vehicle maintenance and inventory management system is of considerable import. It is what allows the company to act in a proactive and data-driven manner, instead. Doing something by rote. The result is equipment that can be relied on, with less risk, and a reduction in the costs associated with downtime. This review is intended to provide the groundwork for a system Often designed specifically for JA Noble Enterprise Inc and based on appropriate research focused on decision Vehicle support, inventory management and vehicle maintenance.

Historically, such work was left to man's observation and judgment.This sometimes requires excessive maintenance, and low stock levels. Sayal et al. (2025) states that the size of the study groups ranged from 1 to 10 participants. Older inventory systems were not sufficiently flexible to meet the needs of the industry. On the other side, companies have resorted to management systems that can process operational data. Kozin (2023) put forward a decision support system to illustrate the value of using data to make better decisions, let's look at service workshops.

Web and mobile tech have come a long way, and interactive products like these are becoming more popular. SUV Maintenance Prediction System by Naga Raju & Nikhitha (2025); and it employs machine learning to give recommendations in line with technical specification of the manufacturer and driving behavior. They've grown into platforms that infuse intelligence into the overall supply chain and performance. JA Noble Enterprise Inc is a case in point.

Also, a newly emerging literature is exploring the possibility of: The replacement of the traditional time-based maintenance with predictive and condition-based maintenance. Implementation of predictive and condition-based maintenance instead of the traditional time-based maintenance. schedules. Sala et al. (2022) have put forth a framework for factories to ensure the health of assets can be monitored, and Bhat and Aravind (2025) argue that for the betterment of India's economy.

Mixing techniques to order and optimize maintenance jobs. All this suggests an increased need for predictive analytics to maintain as more relies on it operating costs down and the work in balance. The goal of inventory management scientists is to put in place systems capable of withstanding the vagaries of an unpredictable market. For example, Sayal et al. (2025) tested the fuzzy models on the Tested other inventory systems and concluded that they are not just as effective but also more economical. In addition, the work of Sá et al. (2025) developed a decision support system that included safety stock and time buffers as a means of uncertainty management on supply and demand side. Khalil et al. (2026) have focused their work in the area of fleet maintenance applied programming to figure out part requirements, thereby reducing inventory costs while maintaining service reliability.

Despite the improvements in technology, they are still open WILLING TO VETO any study of JA Noble Enterprise Inc.'s vehicle questions of relevance. A maintenance and inventory management system. Dwivedy et al. (2025) point This finding was that most readiness models for Industry 4.0 are too generic to be applicable to businesses in developing countries. Crespo del. Castillo et al. (2024) have drawn a conclusion. Attention to lack of models such as predictive maintenance Balancing workload and/or reducing inventory costs. To address this, the aim of the current work is to design a framework specific to the SME, one that combines live scheduling and automatic reorder points and prioritization.

The literature review indicates that there is a definite trend towards processes that become more 'data driven' with the backing of machine learning and AI to better Manage vehicle and inventory at JA Noble Enterprise Inc. While the shift manual to automated has accelerated decisions with greater speed and precision. Integrating such tools is not an easy task for a medium-size company with finite resources. In the end, the results indicate that the most effective way to increase is to: efficiency and contain long term costs is to integrate spare parts management with maintenance scheduling.

Retrieval-Augmented Generation (RAG) for AI Assistants

The older vehicle maintenance and inventory computer systems were developed using a rule-based process and a static database with little modification. evolution. While such methods were good at the business of data entry and in the case of storage, they were unable to provide the depth and impact of analysis needed. For effective decision making.

"As Gupta et al. (2024) noted, these traditional" setups "were content to just save and retrieve data; they didn't do much to synthesize information from various sources or make sense of a dynamic Faced with the situation in real time, decisions to be made slowly and by hand. But with the introduction of artificial intelligence, all that has changed. AI models can now access knowledge bases and use their knowledge in a manner that is now possible. systems a greater sense of context and adaptability. An example of this is the rise of retrieval-augmented generation (RAG). Where in the past, retrieval and generation were two discrete activities but now they are combined. single, integrated process.

Huang and Huang (2024) expressed it in two steps: documents are extracted from the available knowledge, and used as the foundation for a response, which serves to limit errors and keep the output right. But the current RAG systems have gone beyond their basic beginnings, as Sharma (2025) explains. pipeline model. With improved ranking and reliability, they are designed to deliver superior results. This has been supported by preliminary studies on passages. RQs that can comprehend actual queries (retrieval and dense retrieval), and Keyword matching never could have done this. And while long-context language models are able to take in vast quantities of text, Li et al. (2024) make the You will find a variety of cases that for an enterprise system that has to deliver accurate answers from its databases, a retrieval-based approach is still indispensable.

In a nutshell, when someone is putting a data structure, there is no room for error in data handling. Apply the same process to the system and this applies just as much in business. Khan et al. (2024) point out that RAG systems make frequent use of PDFs since they are where you will find operational details such as inventory

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records, maintenance reports and manuals. However, it is certainly not easy to work with these files. The design of the tables, and the intermingling of text with them, is simple. When using other materials, it can be difficult to extract the information. Researchers have put forward text segmentation and chunking as a means to retrieve documents more easily without affecting the semantic content. There's also the issue of a government. Kesuma (2025) states that an enterprise RAG system should enable every output to be traced back to its source. That traceability is what makes facts credible on which to make decisions about restocking or maintenance schedules, without relying on conjecture. More recent research has aimed to improve the capabilities of RAG systems Agile and adaptable to variability. Yan et al. (2024) have come up with a Corrective Retrieval Augmented Generation model that will re-run retrieval if the quality of the documents in hand is lacking, thus nipping any errors from No irrelevant or missing information sources in the bud before a response is released. Li et al. (2024) have explored combining LCRLMs with Retrieval for the purposes of reasoning and speed. Yet Yu et al. (2024) would argue that it can be a difficult task to evaluate, especially in the case of Evaluating uniformity and correctness of the type of conditions encountered in actual business operations.

Evaluation and research have been given more attention, Benchmarking with regard to the challenges. Narita et al. (2026) have Emphasized that in today's context, today's assessments are not suited to what enterprises need, Many wanting multi step action, or the ability to track evidence. To address this, Walden et al. (2026) have put forward automated tools for the intent of report checking. The lack is for a scalable and reliable method for measuring performance. One can also make the case that a pressing One of the problems that arises with RAG systems is whether or not the error is a result of the RAG system's failures. retrieval or on the generation side.

There is an evolution in literature: One had simple There are now some context-aware decision-making tools available in retrieval-based systems. making. Combining retrieval and generation has been the trend, and this is serving to improve the precision and value of a company. But it has its drawbacks. difficulties. Even ensuring reliability in real time is not a trivial matter and data structures are ill-suited for this purpose. Are likely to be complex, and decisions could be more transparent. Governance and evaluation need to be tightened to develop truly 1.4.2. integrated systems. Transitions to the next generation would do much to raise the bar on to be more efficient and accountable, and to be available and traceable.

Mobile QR/Barcode Scanning

The inventory has undergone a significant change in how it's managed. Where one was able to use hand labels, it's now digital. Solutions such as barcodes and QR codes are now the norm. The latter concept was introduced by Denso Wave of Toyota in the year 1994; Chikkamanju and Ambika. Covered by the other two mirrors, it was designed to monitor car parts during their manufacturing process. made. This was in a way an innovation for product and inventory control, providing companies with the ability to store and access information with a small device. digital tag.

Historically, warehousing and inventory management were a problem of. Manual counting and paper records. These are considered "old school" methods. Today, they're not only time-consuming but they're also prone to errors and the potential for documents to be misplaced and operations to be put on hold (Al Battashi, 2024).

A manual system is simply not able to keep up when inventory In high-volume sectors like, for example, sectors with a high number of web pages, levels rise too slowly and are not reliable. As auto parts or equipment. The solution has been automation, with barcode systems, using mechanical methods to accomplish tasks. As Punith et al. (2024) put it, this kind of automation reduces the amount of work that needs to be done in order to get huge quantities of things moving, accounting to an organization's need for efficiency and data that is accurate.

Then there's mobile technology, that has sped up the speed of product identification and tracking. With mobile apps one can Scan barcodes or QR codes – use a dedicated scanner for inventory control. Patil et al. (2022) have an app that will showcase an item's service history and other information from a central database is revealed after scanning its QR code.Noguerra does a similar thing with an inventory system linked to a database (2023) Through QR codes, real-time stock visibility. Patel et al. (2022) note that these centralized databases are faster in everything they do, from selling to Buying but preventing the loss of assets. From the findings that Kevin (2022) found, he noted that, for medium-sized enterprises, it is enough to use a scanning app in the workshop on the smartphone. There is no need for the hardware of a traditional scanner, making the whole process more accessible and cost effective.

The current fashion is to combine web and mobile for cloud storage and time sync. Wairooy et al., (2024) have built together an integrated warehouse solution using Flutter and Firebase. The former allows cross-platform apps and the latter for live databases, and in combination they serve to prevent losses and cut down on errors in management.

Research by Yaqin et al. (2024) shows that a Django Python QR system can put an end to 20% of inventory discrepancies. In a separate Data entry rate was 43% lower using barcode systems compared to undertaking. The accuracy of 98.5%, which further reduced the chance of error (Sahetapy and Suhirman) 2025\). The applications of such technology have increased, and now it's not only for keeping an eye on the usage, maintenance and condition, tracking items are used. inventory. In workshops, Sermonia and Pabelona (2025) verified this. Cloud-based solutions and QR tagging for repair & time inventory.

Yet Traditional methods are not an asset in specialized automotive applications. As per Saputra et al. (2024), there are numerous inventory applications that are lacking in. user features, but there is no means to cope with return requests or to keep people. Responsible for items borrowed equipment. Even if the system is fast in recording. Integration is frequently desired and/or required. Maintenance histories will not be available in one. Correlated with stock order or repair history.

Mobile and Web Push Notifications

There has been considerable change in the automotive repair and maintenance sector this century. With the advent of digital tools, customers expect to have a line of communication with companies in real time. Luna Alexander has pointed out that such dialogue is essential for customer satisfaction and a company's operation requires it. Therefore, service providers are avoiding the old approaches and opting for automated systems, looking to eliminate and control interactions more effectively and prevent errors.

One can observe how the system of automotive services has developed from simple record-keeping to mobile integrated platforms. In the old days, centers would still rely on traditional methods for some things such as confirmation, etc. or to give a service update. This was a method that would have been subject to delays, misunderstandings, not to mention the extra work it put on administrative staff. Those earlier systems would have been able to process reservations and calculate the fees, they faced challenges of lack of real-time tracking and specific dedicated applications. Garud et al. have noted a recent move in the direction of Vehicle Service Automation App ecosystems, which make use of Firebase and Android Studio to provide a more user-friendly interface for both parties. But there's the issue of whether the customer will be notified of the push notification, Maintains tasks on the schedule and in the loop.

Web Push Notifications are a good example of that; they allow one to send an individual message to a certain user even if the site is closed, browser instance will remain. Carboneri et al. report that they can get higher engagement than SMS or email. But timing is everything. Gavilan and Martinez-Navarro warn that a bad time or the "reminder" feature that is already enabled, is only going to irritate the person taking the reminder. And as Nguyen Tran et as demonstrated by al., a delay between some action and the onset of the change of state, such as 15-30 minutes, is enough to trigger the change of state. However, there is a difference when it comes to getting a response. Li et al also discovered that immediate can decrease engagement with notifications ranging from 1 to 6 minutes can make users more receptive.

The modern design of automotive service systems is based on twofold. The goal of being helpful not only to the user but the administrator. A customer, for instance, will be able to put in a

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registration for his vehicle, book a service and keep an eye on its progress as it happens; once the work is done, he can leave some feedback. The administrator has other tools available for them to use. handle appointments, change service statuses or pull up customer data for a more tailored experience. Then of course there are the automated functions. Alexander has pointed out that reminders reduce no shows by 25-60%. The number, cent. translates to more reliable revenue projections and more efficient use of resources. More advanced systems will even be self-monitored and include maintenance targets to remind users when maintenance is needed. As forThat restocking alerts can be triggered using the inventory and decision support systems can be programmed to do so, that parts are available for all planned jobs.

However, the current selections come with their drawbacks, hence the need for greater integration at JA Noble Enterprise Inc. the employers, and the students' readiness to meet that requirement.The level of digital literacy required by the employers and the students' preparedness to meet that requirement, coverage, language options. There may be some platforms that can be a barrier to access. Carboneri et al. have drawn the following conclusions: Also, attention to session mismanagement is important, as this type of issue. are not properly handled between browsers or sessions, opening the door for privacy reasons or attempted delivery failure. There is also a digital divide in the use of digital resources for there are differences in digital literacy.

For the big dealership networks, inventory is managed while the big dealership networks have their CRM. Smaller and middle-sized companies will typically remain in the older processes. In most cases, research on maintenance scheduling and service decisions can be found that are approached from either a maintenance or a service perspective. discovery on their own, little consideration was thought to how they might be Together with the inventory reminder structure.

The automotive service industry is heading towards an automated A system based on real-time data and optimized notification strategies. The aim of the proposed system for JA Noble Enterprise Inc is to solve the existing problems by integrating the two features of customer reminder for Remind the contractor to replenish the inventory on a maintenance and administrator reminder, or single platform. The System will provide timing optimization techniques, backed by research, which will help to minimize fatigue and enhance service compliance. Through this holistic solution, efficiency and customer retention can be enhanced via a data-based communication system.

The literature reviewed has collectively pointed to the evolution in the field of automotive and SME vehicle maintenance, inventory management, decision support systems and supporting digital technologies. Initial research focuses on time-based maintenance and inventory processes, which can be ineffective, unreliable and time-consuming, and in most cases, these processes are not in real time. These have evolved over time into condition and predictive maintenance systems, with analytics, optimization and new technologies like machine learning and IoT monitoring. The same applies to inventory management systems where there is a progression from tracking to computerized and semi-automated systems, and increased focus on accuracy, efficiency and demand forecasting. Despite all these developments, many systems are still deployed as standalone systems, with the maintenance schedule and inventory replenishment processes not being integrated.

The integration of decision support systems for optimizing efficiency and uncertainty minimization has been a common subject in the literature for automotive and SME applications. The studies demonstrate that there is a clear benefit to combining models, predictive algorithms and optimization techniques to optimize maintenance scheduling, minimize downtime and control inventory. Furthermore, technologies like QR/barcode systems and cloud platforms enhance the ability to track in real-time and ensure that data is readily accessible and operations can be coordinated. But there is a gap in creating a unified platform that integrates the maintenance data, inventory levels and intelligent decision making. Factors like automation, interoperability, and real-time decision support are still a major challenge in achieving operational efficiency, particularly for SMEs.

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III. RESEARCH METHODOLOGY

Research Design

The research design describes the plan used to look into the problem being studied and check if the goals of this project are met. It serves as a guide for getting the data needed, choosing the people to study, doing tests to see if users accept the product and looking at the data in a way. For this study the researchers used a Developmental Research Design that follows a way and also uses both numbers and words to get the information. The developmental approach controls the steps of finding problems in a company, checking the software that was made and making sure it works well for JA Noble Enterprise Inc. In this context, descriptive and inferential statistics are terms that mean the same. Statistics are the main tools used to summarize, describe and analyze. The facts, figures, and other findings obtained from the user acceptance testing process. This approach guarantees a clear separation between research validation and the development of the learning materials. The framework and technical development execution of the software.

Population of the Study

The population of the study is the population of people that is relevant to the evaluation of the capstone project. The target population in this situation would be the administrators, the staff and the customers of JA Noble Enterprise Inc. This shop is located at their Caltex service station in San Pedro, Laguna. These groups are the best fit for the study, since they're the people who deal with the business' vehicle service and inventory on a daily basis. They are very active in their daily use and thus are excellent subjects for some of the best assessments of the system in the real world.

The group selected for the study is similar to the user the system is designed to serve and the people who will gain from the system. Administrators and Shop staff are critical in that they must have tools to help them with data tracking inventory levels and alerts for maintenance schedules. Their contribution makes sure that the system is able to support the core functions that are needed for the business. However, the customers are also important as they will employ the mobile capabilities of the system. This also involves checking and receiving notification of their car's condition. The feedback assists in assessing end user usability and ease of use with the system.

There are 344 people in the population. For the evaluation, the researchers chose 144 participants. There are 3 administrators and 16 staff members in this sample. These were selected by enumeration; everyone was selected because they play an important role for the system to be successful. There are 325 services in total, with the remaining 125 being customers. This will guarantee that the sample is representative of the users and that the system can be fairly and accurately tested. Feedback from this group can be used to gauge the performance, speed, and acceptance of the system, before it goes live. This step is important to ensure that the system is effective in use and to satisfy the needs of all of the users.

Sampling design

The structural framework and techniques that were used to systematically select the participants to evaluate the Integrated Vehicle Maintenance and Inventory Management System with Decision Support are discussed. The researchers used Purposive Sampling Design to get the empirical validity of the study. The researchers used this non-probability sampling method because they were able to consciously select the sample subjects according to their predetermined conditions, professional qualifications and certain operational characteristics that are directly related to the central issue of the study. The design provided a set of data points that were collected from only the key stakeholders actively involved with the JA Noble Enterprise Inc. ecosystem, including: station administration, workshop staff, floor mechanics and active customers. This methodology focuses on cases and users that have high information content and that have first-hand knowledge of the context of the software being measured, so that the information collected is useful and taken from real operational knowledge.

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To mitigate the differences in population size of the stakeholder groups, a Mixed Sampling Approach was undertaken under this scheme. Total Enumeration was employed by the administrative staff who are internal users of the workshop as they form a small, census-viable user group as verified from the official human resource roster. A verified baseline population of 325 services in the company's central service log was obtained, and a Systematic Random Sampling technique was used to subjectively obtain a final sample size of 125 customers. The researchers found the sampling interval by dividing the total population of customers by the sample frame that was obtained from Slovin's Formula with a 7% margin of error. Every subsequent customer profile was picked at this exact step interval from a random starting point until the desired 125 distinct customers had been reached, thereby eliminating researcher selection bias.

Data Collection Instrument

This section provides the details of the instruments and the systematic approach used by the researchers to collect the qualitative and technical data for the requirements analysis, the structural design and initial development stage of the proposed system. Structured fact-finding mechanisms were heavily used to make sure the data collected were empirically grounded, logically consistent and adequate to answer the specific operational objectives of the study. These instruments were carefully developed to collect more granular data from the primary stakeholders and everyday consumers about the legacy workflows of the automotive service, identified transactional bottlenecks, back-end system requirements, and structural database relationships of JA Noble Enterprise Inc. The data collection framework was effectively designed to focus on the initial phase of data collection and requirement elicitation so that the system developed later is closely connected to the real environment of the enterprise.

The first step in this study was the fact-finding program to gather primary raw data on the actual preventive maintenance scheduling procedures and the autohouse and warehouse operations of Caltex service station and automotive service center (ASC) of JA Noble Enterprise Inc., San Pedro, Laguna. These base line activities were the key empirical foundations for defining the exact touch points and documents handled when customers check-in, the baseline functional limits of the proposed mobile and web platform, and the active service lifecycle.

They collected key functional information by conducting synchronous video conferencing discussions and thorough interviews with the business operations manager using a semi-structured approach. The researchers did not use this interview as a "quick overview", but rather carefully tracked how services orders move and how they get to the point where components are issued from the customer's arrival to the final issuance of the components. Special attention was given to mechanical tracking loops, service history logging, managing low stocks and allocation of resources. This narrative exploration enabled the researchers to capture the recording, cross-referencing and sharing of critical data fields (part serial numbers, fluid capacities, vehicle plate metrics, and previous vehicle history markers) throughout the management dashboard and the shop floor.

These fact-finding consultations also thoroughly examined the key operational issues, difficulties and constraints faced within the legacy framework. The researchers noted systemic weaknesses, such as high levels of real-time stock reporting issues, a high rate of human mistakes as a result of manual appointment scheduling, easily spotted discrepancies between parts inventories, and a large communications gap between administrative encoders and active mechanics in the shop. Data gathered from the mapping of these operational risks were mathematically and logically analyzed to identify manual subprocesses capable of being structurally centralized, automated or augmented. This directly supports the use of an algorithmic Decision Support System (DSS) and the notifications when stock levels are low.

In addition, the researchers conducted in-depth on-site observations directly at the Caltex service station, to verify the workflows claimed during the remote interviews. The team physically identified the type of documents and parameters used by the staff, with a specific focus on work orders, manual ledger pages and legacy spreadsheet architecture. These are direct observations of the physical and used directly to design the schema of the system's database, rules for input validation, and the relational structures of tables. By learning about the practical use of hardware barcodes and vehicle identification tags, researchers could design the particular hardware-level QR and barcode scanning functions of the mobile app. Finally, these story-driven approaches allowed the team to get a true and reality-based estimate of the automotive repair experience, making the data processing, Retrieval-Augmented Generation (RAG) report formats and workflows for the user actual.

System Development Methodology

The researchers used the Agile Development Methodology, the Scrum Framework to build the Integrated Vehicle Maintenance and Inventory Management System. This was chosen because it supports iterative development. The team is considering building the entire system at once. Delivered work in short, concentrated periods called "sprints. This allowed continuous enhancements that are needed in inventory tracking and maintenance features, based on feedback from JA Noble Enterprise Inc.

Figure 2. The Agile Software Development Life Cycle (SDLC) Model

The specific Agile SDLC model deployed throughout this project. During the execution of this model, the development stages were directly translated into actual project activities to fulfill the operational demands of the enterprise:

Phase 1 – Requirements: The actual project activities in this initial stage involved gathering detailed workflow specifications from the managers and workshop staff of JA Noble Enterprise Inc. The researchers focused on finding the structural goals of the project. They looked closely at where manual tasks were slowing things down. They also made sure to list all the details needed for system connections. These included automated maintenance scheduling, barcode scanning and a read-only AI chatbot dashboard.

Phase 2 – Design: In this stage, the actual activities focused on building the platform's architecture and designing the user interface layouts. The researchers created relational database schemas to manage data relationships such as linking spare parts to specific vehicle models. They also developed UI wireframes with care to make the system easy to use. The goal was to reduce confusion, for floor mechanics, administrators and customers by keeping the interface simple and intuitive.

Phase 3 – Development: During the development phase, the researchers executed active programming sprints to build and integrate system features piece by piece. The actual project activities involved coding the backend infrastructure developing the portal, for QR/Barcode scanning features and embedding Retrieval-Augmented Generation (RAG) engines to handle conversational AI query processing and automated management report generation.

Phase 4 – Testing: The actual activities, in this stage focused on using a black-box testing methodology to check the functional integrity of each newly added module. The researchers carefully examined how accurate the barcode scanning module was. They also looked at how fast data synchronization worked across platforms. At the time they tested the reliability of the RAG-powered AI Assistant when comparing it with authorized corporate records. All of this was done to

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make sure the system behaved exactly as described in the Functional Specification Document (FSD).

Phase 5 – Deployment and Feedback: Once a stable Release Candidate version was stable the project moved to launching a pilot deployment, at the Caltex service station of JA Noble Enterprise Inc. Selected staff, mechanics and administrators used the web and mobile apps every day, which let researchers gather real-world usage feedback and record runtime behaviour for adjustments.

Phase 6 – Maintenance: After the pilot deployment the work shifted to improving the software and stabilizing the platform. The researchers looked closely at system logs to find and fix runtime errors as soon as they appeared. They also made changes to how indexes were processed which helped the AI Assistant respond faster. At the time they strengthened the rules in the database to make sure transactions worked smoothly and that data stayed consistent across every web and mobile device.

IV. RESULTS AND DISCUSSION

Research Objective 1. To design and develop a web and mobile-based vehicle maintenance and inventory management system, for JA Noble Enterprise Inc. The web and mobile-based vehicle maintenance and inventory management system will use automation and decision support features to monitor stock, schedule maintenance and reduce mistakes that come from manual data entry. The proponents conducted requirements analysis to identify the functional and non-functional needs of the intended users. The identified requirements served as the basis for designing the system architecture, database, user interface, and major system modules. The system was developed using HTML, CSS and JavaScript for the front-end interface, Python and Flask Framework for the back-end processing and application services, Flutter for the mobile application and Cloud Firestore using Firebase for database management and data storage. The modules were. Tested with a defined functional test case. This functional test was used to confirm that each feature of the modules functioned in line with the specified requirements. It also ensured that the outcomes of the modules matched the expected results

The development process led to a working web and mobile application system. It includes three user roles: Admin, Staff and Customers. Each role has functions and access levels based on their job responsibilities. The admin has access to the Dashboard, Item Master, Stock Inventory, Receiving Items, Inventory Transactions, Vehicle List, Vehicle Maintenance, Service Bookings, Issuances, Stock Replenishment, PMS Scheduling, User Management, Domain Management, Archive and AI Assistant. The Staff have access to the Dashboard, Stock Inventory, Receiving Items, Vehicle List, Vehicle Maintenance, Service Bookings and Barcode/QR code Scanning. The Customers have access to the Dashboard, My Bookings, PMS History, AI Assistant, and Notifications. The functionality of each page and module was verified using actual screenshots of the implemented system.

Research Objective 2. aims to test the functionalities of the developed system based on its features. These features include Vehicle Maintenance Management, Mobile-Based Inventory Management with QR/Barcode Scanning, Decision Support System (DSS) for Stock Replenishment and Preventive Maintenance Scheduling and Retrieval-Augmented Generation (RAG) for AI Assistants. The completed system was tested using Software Quality Assurance (SQA) through black-box testing. The testing was carried out by ten (10) IT experts. The process followed the test cases and testing tools described in Chapter III. A pass-or-fail evaluation method was used. The focus was on checking whether each function of the system produced the expected result and performed according to the system requirements without looking at the code structure.

The system went through black-box testing conducted by ten (10) IT experts using defined test cases and testing instruments. The testing included the features of the system such as user authentication and password recovery, customer and profile management, vehicle and maintenance management, service booking, inventory and stock control receiving items and managing inventory transactions, barcode and QR code scanning, PMS scheduling and history user management, report generation, archive handling and AI assistant functions. All test cases received a Passed status. This shows that the tested functions gave the expected output and met the system requirements. The detailed results of the black-box testing, including expected outcomes, actual results and test status, for each case are given in Section 4.3.

Research Objective 3. To evaluate user acceptance and behavioral intention to use of the developed system, which include Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. The developed system was evaluated by the intended users through a user acceptance survey to determine their acceptance and behavioral intention to use the system. The evaluation covered the following criteria: Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. A five-point Likert scale was used to measure the respondents' level of agreement and acceptance of the system, and the responses were analyzed using the weighted mean.

The developed system obtained an overall weighted mean of 4.86, which is verbally interpreted as Highly Acceptable. This result indicates that the users highly accepted the developed system based on its quality factors, perceived ease of use, attitude toward using the system, behavioral intention to use, and overall user experience.

Table 44. Summary of User Acceptance Evaluation Result

Quality Factors	Weighted Mean	Standard Deviation	Verbal Interpretation
Quality Factors	4.89	0.31	Highly Acceptable
Perceived Ease of Use	4.78	0.40	Highly Acceptable
Attitude Towards Using	4.83	0.38	Highly Acceptable
Behavioral Intention to Use	4.88	0.33	Highly Acceptable
Experience	4.90	0.30	Highly Acceptable
Average	4.86	0.35	Highly Acceptable

Table 44 above shows that all five criteria received a Highly Acceptable verbal interpretation, with weighted means ranging from 4.83 to 4.90. This shows that the respondents generally provided favorable evaluations across all criteria. Among the five criteria, Criterion 5 obtained the highest weighted mean of 4.90 with a standard deviation of 0.30, indicating a consistently high level of agreement among the respondents. Criteria 1 gained an average weighted mean of 4.89, while Criterion 3 obtained 4.83 and Criterion 4 with 4.88. Although Criterion 2 received the lowest weighted mean of 4.78, it was still interpreted as Highly Acceptable, indicating that the respondents continued to view this aspect of the system positively.

The standard deviations ranged from 0.30 to 0.40, showing relatively low variation in the respondents' evaluations. This suggests that the ratings were generally consistent, with respondents having similar perceptions of the system across the five criteria. Overall, the system obtained an average weighted mean of 4.86 and a standard deviation of 0.35, resulting in a verbal interpretation of Highly Acceptable. These findings demonstrate that the completed system was positively evaluated across the assessed criteria and was generally perceived by the respondents as acceptable for its intended use.

V. SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study aimed to design, develop, test, and evaluate the Integrated Vehicle Maintenance and Inventory Management System with Decision Support for JA Noble Enterprise Inc., a Caltex Service Station and Automotive Service Provider in San Pedro, Laguna.

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The study addressed the issues associated with the vehicle service center, particularly related to the ability to track the history of vehicles serviced, monitor material requirements for each job, view stock levels in real-time, and utilize manual processes and/or spreadsheets which are slow, error-prone and disorganized resulting in stock shortages, poor staff communications and difficult-to-produce analytical reporting. The major findings of the study, aligned with the specific research objectives, are summarized as follows:

- System Development (Objective 1): The study successfully designed and developed the Integrated Vehicle Maintenance and Inventory Management System with Decision Support as a multi-platform system utilizing a Web Management Panel, Mobile App, and an Integrated AI Assistant. The developed system featured automated check-in processes using QR Code and Barcode Scanning, a Decision Support System (DSS) to provide Low-Stock Alerts and Preventive Maintenance Scheduling (PMS), and an Interactive AI Assistant utilizing Retrieval-Augmented Generation (RAG) to facilitate Chat-style Data Queries and Automatic Context Summaries. The system utilized a Relational Database to ensure availability for Administrators, Staff, and Customers. An Agile Development Methodology employing the Scrum Framework was utilized to manage the team's work through short sprints covering Requirement Planning, Design, Development, Testing, Deployment, and Feedback/Maintenance.
- System Testing and Functionality (Objective 2): The developed system underwent rigorous Black-Box testing across five (5) distinct test cases covering all major features, including the Vehicle and Mobile Inventory Management modules (with QR/Barcode Scanning), the Decision Support System (DSS) for Low-Stock Alerts and Preventive Maintenance Scheduling (PMS), and the Interactive AI Assistant built on Retrieval-Augmented Generation (RAG). The testing resulted in a 100% functional success rate, marked as passed with zero (0) failed results, confirming that all core functionalities of the system are working as intended.
- User Acceptance Evaluation (Objective 3): The system's acceptance was evaluated using the Technology Acceptance Model (TAM) and the ISO/IEC 25010 software quality standards. The evaluation measured Quality Factors, Perceived Ease of Use, Attitude Toward Using, Behavioral Intention to Use, and Experience. A questionnaire that was validated by the research adviser was administered to 144 respondents following system demonstrations, comprising 3 Administrators (2.08%), 16 Staff (11.11%), and 125 Customers (86.81%) of JA Noble Enterprise Inc. The results demonstrated very high scores for all of the metrics associated with quality, functionality, and usability. A weighted mean of 4.89 (on a scale from 1–5) was achieved for Quality Factors; 4.78 for Perceived Ease of Use; 4.83 for Attitude Toward Using; 4.88 for Behavioral Intention to Use; and 4.90 for Experience. Since all metrics are well above the minimum acceptable of 4.21 and into the High acceptable area (between 4.21 and 5.00), the overall result indicates that users viewed the system as reliable, simple, efficient, and useful to support inventory management, improve business processes, facilitate better-informed decisions by leveraging data, and simplify maintenance workflows. Overall, these findings indicate users believe the system is reliable, easy to use, and simple. The system has also been able to improve the way users manage their inventory, improve operational efficiencies, and provide them with tools to make more informed data-driven decisions. In addition, the findings demonstrate that the system has met its purpose in solving the issues that currently exist in the operation areas and meet the needs of those who will be utilizing the system.

Conclusions

Based on the development, testing, and evaluation of the proposed system, the following conclusions are drawn:

- With the successful development of the web and mobile Integrated Vehicle Maintenance and Inventory Management System with all functional requirements for JA Noble Enterprise. Inc. have been met. The system has provided a centralized, automated solution to replace the disjointed paper and spreadsheet processes of the transition to digital operations which was much less burdensome than it was in the past, thanks to the move. Integrating vehicle maintenance, inventory and the customer portal on one platform.
- The testing of the system's core modules has been performed, and the results show that the system is consistent. with technical specifications. Actually, 100% pass rate was achieved in all test cases. This applies to Vehicle Maintenance Management as well the Mobile-Based Soundscape project. The module has been engaged in recording service histories. Inventory Management module which is able to update stock via QR/Barcode scanning. Similarly, Decision Support System (DSS) is proven to put out the right low-stock alerts and PMS notifications, while the RAG based AI Assistant has shown it can pull the necessary. To generate responses and reports for management from database records.
- End-users are open to making use of the system. Based on ISO/IEC 25010 standards and the Technology Acceptance Model, the system came in at a weighted mean of 4.86 or "Highly Acceptable." It is evident that the tool is dependable and trustworthy to administrators, staff and customers. Help them with their work.
- In putting these tools together, we have created a framework to cut down on manual entry and open up lines of communication between the staff and their customers. There continues to be a need for post-deployment. The following month monitoring was performed to observe the actual impact of running in terms of eradication of data. The high acceptance and successful testing leave few errors or stockouts, there was never any doubt about the preparedness of the system. When time is up, the platform will be set to improve decision making at the Caltex level to be more efficient and data driven In San Pedro, Laguna, you will find franchise stores.

Recommendations

Based on the findings and limitations of this study, the following recommendations are proposed to enhance the system and guide future research:

- Future researchers implement a pilot in all Caltex service stations in Laguna. The goal is to test the system's viability in a multi-branch auto care environment and identify any limitations of scalability before expanding to other regions of the country and to continue to guide our infrastructure upgrades.
- Future researchers ought to consider SMS gateway or bulk messaging options so that automated alerts can be sent out for maintenance. Having real-time payment gateways like PayMaya, GCash or credit card processors will also be useful because it will allow customers to make an advance payment on their reservations, making the billing process much easier and reducing the chances of no-shows.
- Future researchers need to create API integrations that can be connected with supplier networks and POS to facilitate automatic inventory restocking. These can be integrated into ERP modules for HR, accounting and become a true business management system.
- Future researchers should train the NLP model and should be taught to identify slang & Tagish, as well as the ability to use voice commands. Authorized users may also be provided with the ability to add or edit database records without risk using role based transactional AI.

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APPENDICES B

(COMMUNICATION LETTERS & FORMS)

Request Letter

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: Feb 16, 2026

JUAN PAOLO TONGCO
Director/Operations Head
JA Noble Enterprise Inc.
San Pedro, Laguna

SUBJECT: Request for Initial Interview for Capstone Project System Development

Dear Mr. Tongco,

Warm greetings!

We are third-year students of the Bachelor of Science in Information Technology under the College of Computer Studies. As part of the academic requirements for our Capstone Project, we respectfully request permission from your good office to conduct an initial interview and requirements gathering with your organization.

The purpose of this interview is to understand your current system or processes and to identify possible areas where an information system may be designed and developed to help improve operational efficiency. The information gathered will be used strictly for academic purposes only and will be treated with utmost confidentiality.

Should you grant our request, the interview may be conducted at your most convenient date and time, either face-to-face or through an online platform.

We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time and cooperation.

Respectfully yours,

Rondilla, Precious C.
BSIT – 3D

Tolentino, Mariely Ann L.
BSIT – 3D

Villafranca, Carla Mae R.
BSIT – 3D

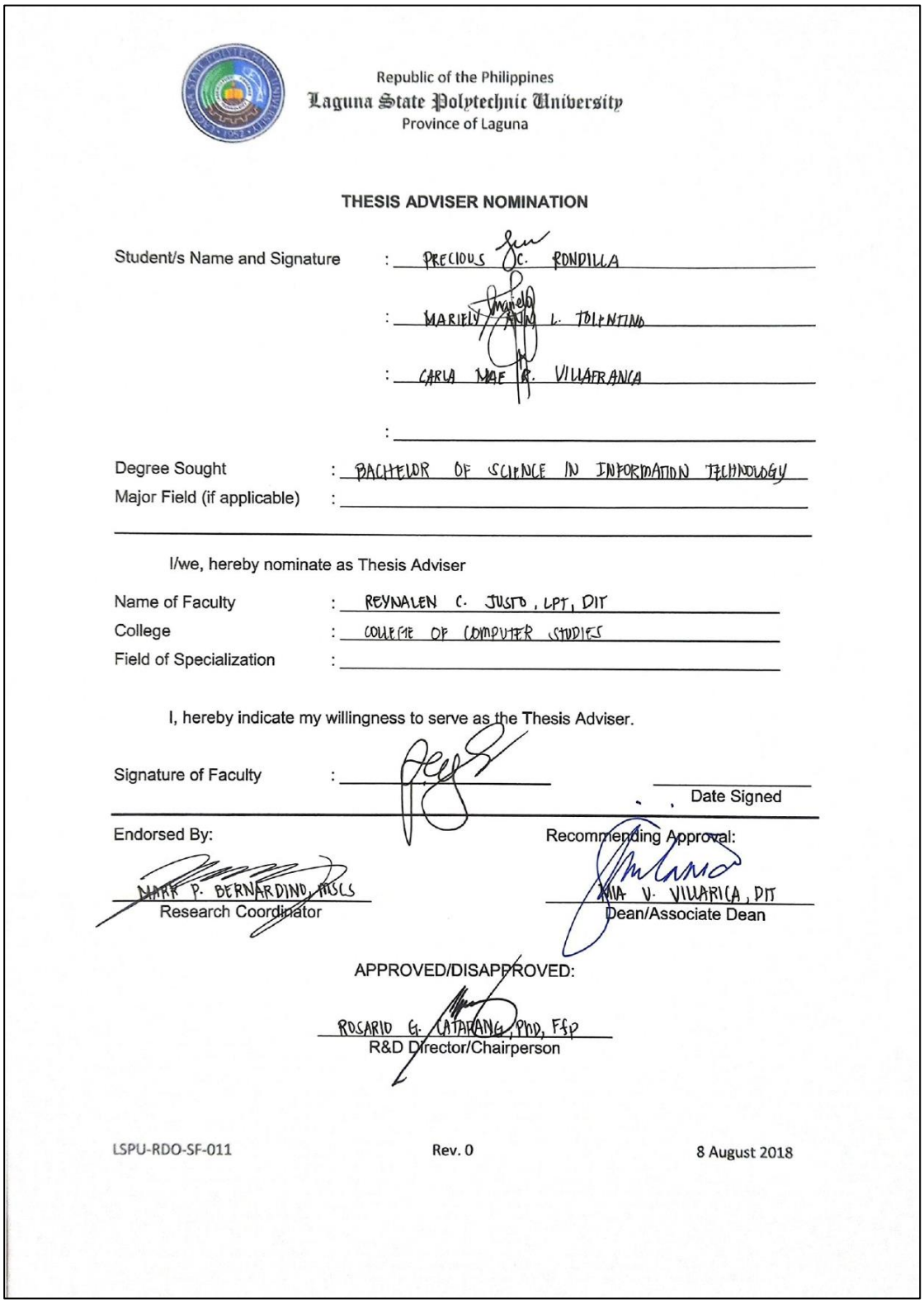
Noted by:

Maria Laureen B. Miranda, LPT, MIT
Capstone Project Subject Instructor
College of Computer Studies

Approved by:

Juan Paolo F. Tongco
Director/Operations Head

ISO Forms



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Student/s Name and Signature : PRECIOUS C. RONDILLA

: MARIELY ANN L. TOLENTINO

: CARLA MAE R. VILLAFRANCA

:

Degree Sought : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major Field (if applicable) :

I/we, hereby nominate as Thesis Adviser

Name of Faculty : REYNALEN C. JUSTO, LPT, DIT

College : COLLEGE OF COMPUTER STUDIES

Field of Specialization :

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty : Date Signed

Endorsed By:

MARK P. BERNARDINO, MSCS
Research Coordinator

Recommending Approval:

ANA V. VILLARICA, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

ROSARIO G. CATAPANG, PhD, FSP
R&D Director/Chairperson

LSPU-RDO-SF-011 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : PRECIOUS C. RONDILLA

: MARITHY ANN L. TOLENTINO

: CARLA MAE P. VILLAFRANCA

:

Degree Sought : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
Major Field (if applicable) : WEB AND MOBILE APPLICATION DEVELOPMENT

I/we, hereby nominate as Panel Member (SYSTEM EXPERT)

Name of Faculty : REYMART JOSEPH P. PIELAGO, MIT
College : COLLEGE OF COMPUTER STUDIES
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

MARK P. BERNARDINO, MSCS
Research Coordinator

Recommending Approval:

MIA V. VILLARICA, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

ROLANDO B. CATALOMO, PhD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-012 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : PRECIOUS O. ROBILLA

: MARIELLE [illegible] L. TOLENTINO

: CARLA MAE R. VILLAFRANCA

:

Degree Sought : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (TECHNICAL EDITOR)

Name of Faculty : RACHIEL B. RIVAND

College : COLLEGE OF COMPUTER STUDIES

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

MARK P. BERDARDINO, MCCS
Research Coordinator

Recommending Approval:

MIA V. VILLARICA, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

EDGARDO O. CATAPANG, PhD, FAP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : PRECIOUS C. PONDILLA

: MARIELY ANN L. TOLENTINO

: CARLA MAE R. VILLAFRANCA

:

Degree Sought : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (STATISTICIAN)

Name of Faculty : GENCE F. MOSICO, MSIT

College : COLLEGE OF COMPUTER STUDIES

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

MARK P. BERNARDINO, MSCS
Research Coordinator

Recommending Approval:

MA. VI VILLARICA, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

ROGARIO G. CATAPANG, PhD, FSP
R&D Director/Chairperson

LSPU-RDO-SF-012 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : PRECIOUS RONDILLA

: MARIELY ANN L. TOLENTINO

: CARLA MAE R. VILLAFRANCA

:

Degree Sought : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (LANGUAGE CRITIC)

Name of Faculty : MA. CEZANNE DIMACULANGAN

College : COLLEGE OF ARTS AND SCIENCE

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

MARK P. BERNARDINO, MSCS
Research Coordinator

Recommending Approval:

MIA, V. VILLARICA, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

ROSARIO G. CATAPANG, PhD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-012 Rev. 0 8 August 2018

Title Defense Rating Sheet

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. D11	Date: March 26, 2026	Time:
Name of Proponents: Rondilla, Precious C., Tolentino, Mariely Ann L., Villafranca, Carla Mae R.		
Research Title: Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Stock Replenishment and Preventive Maintenance Scheduling for JA Noble Enterprise Inc.		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	95.2

VERDICT

[ ] **APPROVED**. Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS**. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED**. The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Panel

Role on the Defense

Reynaldo C. Justo

Signature over Printed Name of Evaluator

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. D11	Date: March 26, 2026	Time:
Name of Proponents: Rondilla, Precious C., Tolentino, Mariely Ann L., Villafranca, Carla Mae R.		
Research Title: Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Stock Replenishment and Preventive Maintenance Scheduling for JA Noble Enterprise Inc.		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	90

VERDICT

[ ] **APPROVED**. Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS**. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED**. The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Role on the Defense

Signature over Printed Name of Evaluator

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. D11	Date: March 26, 2026	Time:
Name of Proponents: Rondilla, Precious C., Tolentino, Mariely Ann L., Villafranca, Carla Mae R.		
Research Title: Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Stock Replenishment and Preventive Maintenance Scheduling for JA Noble Enterprise Inc.		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	90

VERDICT

[ ] **APPROVED**. Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS**. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED**. The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

M. Miranda

Role on the Defense | **Signature over Printed Name of Evaluator**

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. D11	Date: March 26, 2026	Time:
Name of Proponents: Rondilla, Precious C., Tolentino, Mariely Ann L., Villafranca, Carla Mae R.		
Research Title: Integrated Vehicle Maintenance and Inventory Management System with Decision Support for Stock Replenishment and Preventive Maintenance Scheduling for JA Noble Enterprise Inc.		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	92

VERDICT

[✓] **APPROVED**. Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS**. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED**. The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

PANEL
Role on the Defense

REYMART JOSEPH P. PIELAGO
Signature over Printed Name of Evaluator

Form 3

Research Colloquium Rating Sheet

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: D11	Date: MAY 19, 2026	Time:
Name of Proponents:		
RONDILLA PRECIOUS C.	**TOLENTINO MARIELY ANN L.**	**VILLAFRANCA CARLA MAE R.**
Research Title: **INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR STOCK REPLENISHMENT AND PREVENTIVE MAINTENANCE SCHEDULING FOR JA NOBLE ENTERPRISE INC.**		

MANUSCRIPT (20%)	**WEIGHT (%)**	**EVALUATED SCORE**
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	**20**	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	**50**	
OVER ALL TOTAL	**100**	94

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[✓] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as PPPPFSNM. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

PANEL
Role on the Defense

REYMART JOSEPH P. PIEDAD, MIT
Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: DII	Date: MAY 10 2026	Time:
Name of Proponents:		
RONDILA, PRECIOUS C.	TUMPIAN, MARIEY ANN L.	VILLAFRANCA, CARLA MAE R.
Research Title:		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	**20**	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	**50**	
OVER ALL TOTAL	**100**	

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: 95-100.

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: 85-94.

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: 75-84.

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: 70-74.

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: BELOW 70.

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: D11	Date: MAY 19, 2026	Time:
Name of Proponents:		
RONDILLA PRECIOUS C.	TOLENTINO MARIELY ANN L.	VILLAFRANCA CARLA MAE R.
Research Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR STOCK REPLENISHMENT AND PREVENTIVE MAINTENANCE SCHEDULING FOR JA NOBLE ENTERPRISE INC.		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	50	
OVER ALL TOTAL	100	94

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[✓] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered or, REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

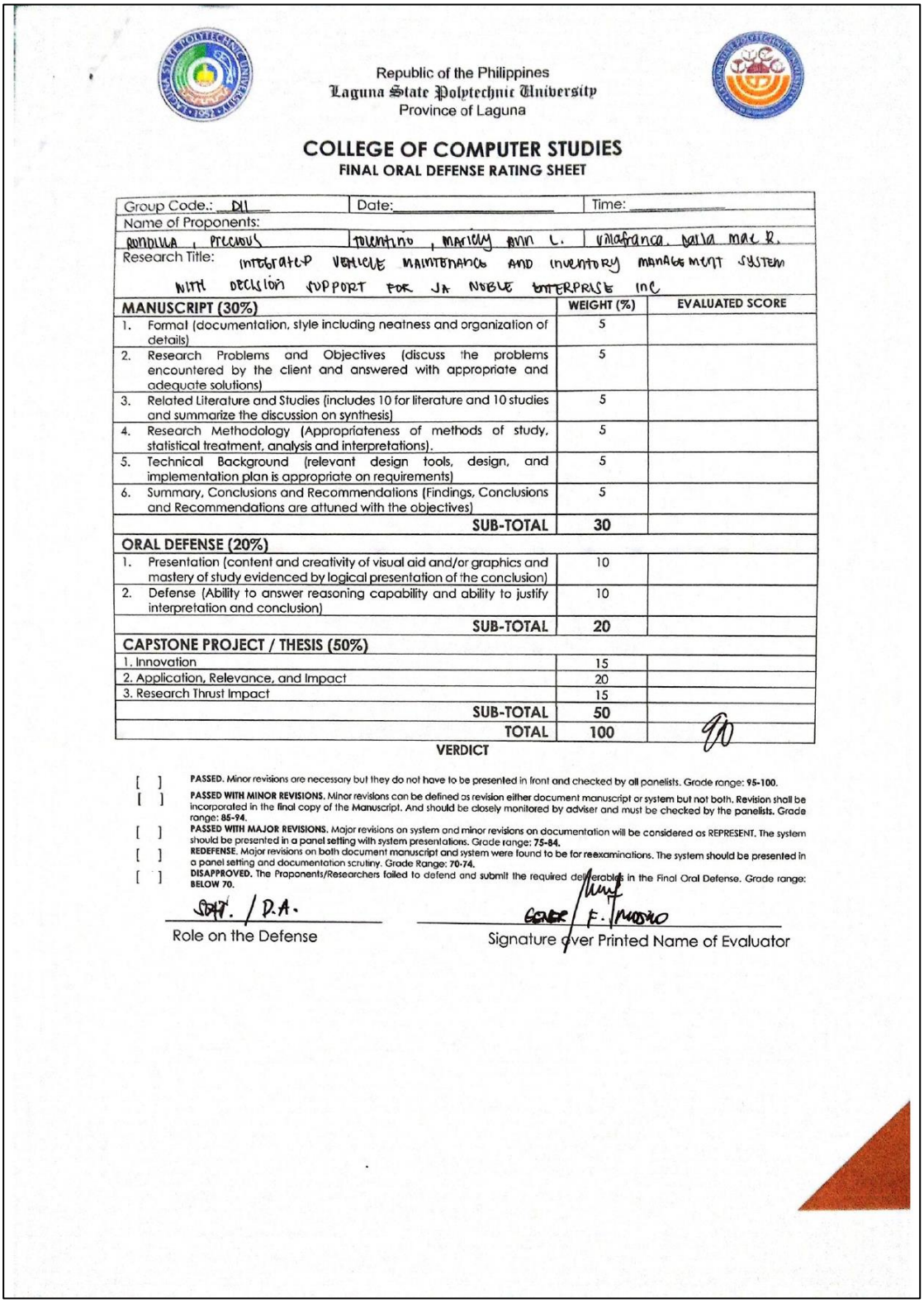
[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and coordinated in virtual. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Role on the Defense

Signature over Printed Name of Evaluator

Final Oral Defense Rating Sheet



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: DII	Date:	Time:
Name of Proponents:		
BONDILLA, PRECIOUS	TOLENTINO, MARIELY ANN L.	VILLAFRANCA, KAILA MAE R.
Research Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC		

	WEIGHT (%)	EVALUATED SCORE
MANUSCRIPT (30%)		
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	90

VERDICT

[ ] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

SOFT. / D.A.
Role on the Defense

GENER F. MUSNO
Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: D11	Date:	Time:
Name of Proponents:		
Bandilla, Precious C.	Tolentino, Mariely Ann L.	Villafranca, Carla Mae E.
Research Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	97

VERDICT

[✓] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

PANEL
Role on the Defense

REYMART JOSEPH P. PIELAGO, MIT
Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: P11	Date:	Time:
Name of Proponents:		
Bandilla, Precious C.	Talentino, Maricly Ann L.	Villafranca, Carla Mae R.
Research Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Formal (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	84

VERDICT

[ ] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as **REPRESENT.** The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

MIA K. VILLAN CPT

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: DII	Date:	Time:
Name of Proponents:		
BONDILLA, PRECIOUS C.	TOLENTINO, MARIELYN ANN L.	VILLARRANCA, CARLA MAE R.
Research Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	

VERDICT 94.7

[ ] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[✓] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

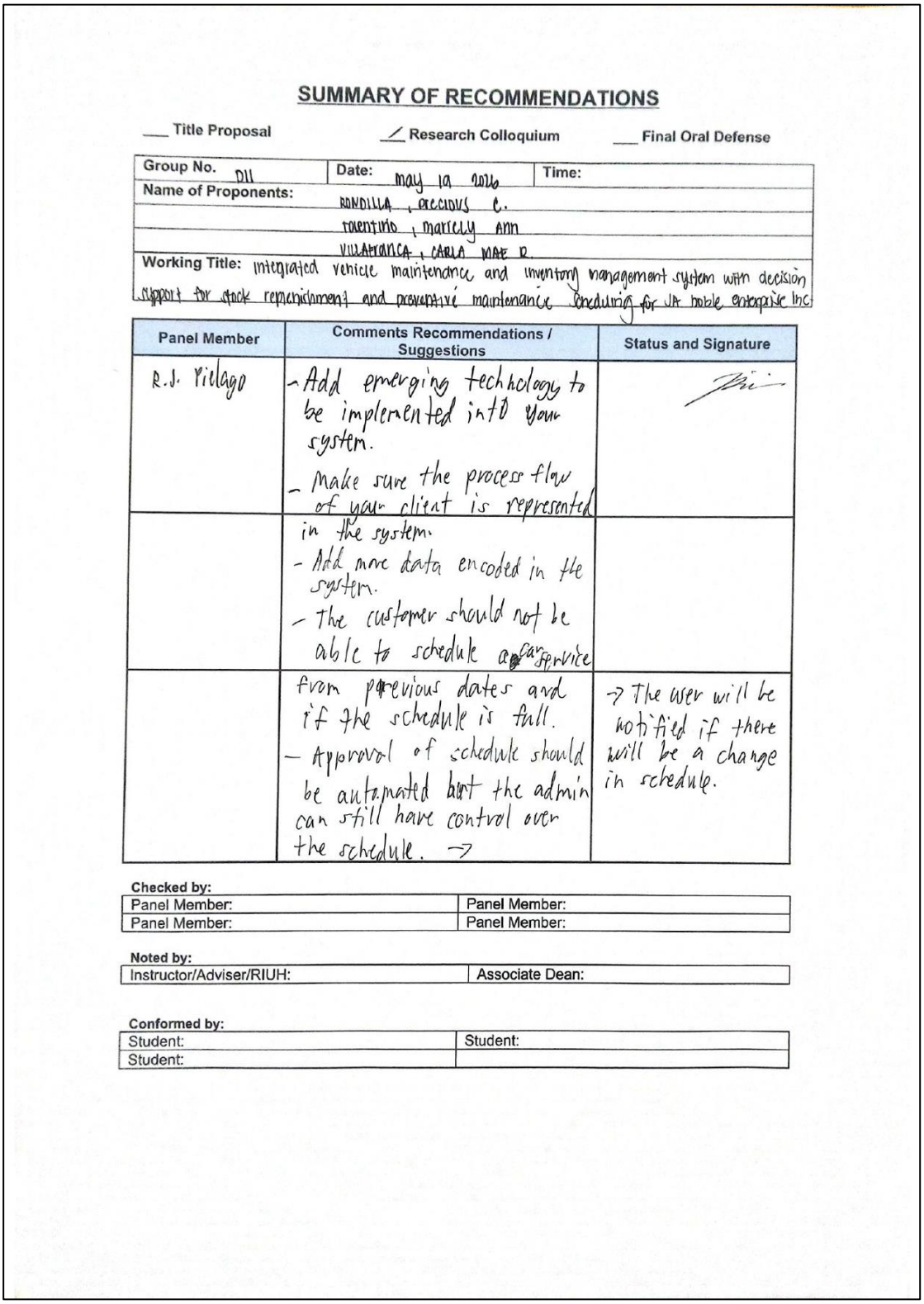
[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Role on the Defense

Signature over Printed Name of Evaluator

Summary of Recommendations



SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___/___ Research Colloquium ___ Final Oral Defense

Group No. DII	Date: May 19 2026	Time:
Name of Proponents:	BONDILLA, PRECIOUS C.	
	TOLENTINO, MARICEL ANN	
	VILLAFRANCA, CARLA MAE R.	
Working Title:	Integrated vehicle maintenance and inventory management system with decision support for stock replenishment and preventive maintenance scheduling for JA mobile enterprise inc	

Panel Member	Comments Recommendations / Suggestions	Status and Signature
R.J. Pitlago	- Add emerging technology to be implemented into your system. - Make sure the process flow of your client is represented	
	in the system. - Add more data encoded in the system. - The customer should not be able to schedule a car service	
	from previous dates and if the schedule is full. - Approval of schedule should be automated but the admin can still have control over the schedule. →	→ The user will be notified if there will be a change in schedule.

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. DII	Date: may 19, 2026	Time:
Name of Proponents:	RONDILLA, PRECIOUS C.	
	TOLENTINO, MARIELY ANN	
	VILLAFRANCA, CARLA MAE.	
Working Title: Integrated vehicle maintenance and inventory management system with decision support for stock replenishment and preventive maintenance scheduling for JA mobile enterprise inc.		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
mia villarica	- decision support system needs clearer attributes and data source - Remove caltex branding if the client is an independent distributor	
	- DSS needs clearer attributes and data sources - consider integrating existing pms booklet records as data input - make the title shorter	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. p11	Date: may 19, 2026	Time:
Name of Proponents:	ANDALLA, PRECIOUS C.	
	TOLENTINO, MARIELY ANN L.	
	VILLAFRANCA, CARLA MAE R.	
Working Title: Integrated vehicle maintenance and inventory management system with decision support for stock replenishment and preventive maintenance scheduling for JA noble enterprise inc.		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
mam bernardino	- Research more about JA noble entrepreneurial operations beyond cultex affiliation - clearly define if the type of business is B2B or B2C	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

Republic of the Philippines

Laguna State Polytechnic University

Province of Laguna

SUMMARY OF RECOMMENDATIONS

☐ Title Proposal ☐ Research Colloquium ☑ Final Oral Defense

Group Code: BII	Date and Time of Defense:
Working Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.	
Student Researchers: Rancillo, Precious C. Tolentino, Mariely Ann L. Villafranca, Carlo Marc R.	

Instructions: *The Adviser shall record the consolidated comments, suggestions, and recommendations discussed during the oral defense. Comments should specify the relevant page, chapter, section, table, figure, or component of the manuscript to facilitate the revision process*

Page / Chapter / Section	Panel's Comments, Suggestions, and Recommendations
	- edit the title of chapter 3 - do not cut titles
	- [redacted] discussion

Panel of Examiners Name and Signature:	
Revision Submission Deadline:	
Panel Chair Name and Signature:	

Conforme: We acknowledge that we have received, understood, and shall comply with the foregoing recommendations within the prescribed revision period.

Student Research Name and Signature:	
Student Research Name and Signature:	
Student Research Name and Signature:	

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

SUMMARY OF RECOMMENDATIONS

☐ **Title Proposal** ☐ **Research Colloquium** ☑ **Final Oral Defense**

Group Code: DII	Date and Time of Defense:
Working Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA MOBILE ENTERPRISE INC.	
Student Researchers: RONDILLA, PRECIOUS VALENTINO, MARILOU ANN VILLARONGA, KARLA MAE	

Instructions: *The Adviser shall record the consolidated comments, suggestions, and recommendations discussed during the oral defense. Comments should specify the relevant page, chapter, section, table, figure, or component of the manuscript to facilitate the revision process*

Page / Chapter / Section	Panel's Comments, Suggestions, and Recommendations
System	- Improve loading time of AI chatbot reply. - Include customer name and other info for the car history, and create a report.

Panel of Examiners Name and Signature:	Reynaldo Jose M. Villegas
Revision Submission Deadline:	
Panel Chair Name and Signature:	

Conforme: We acknowledge that we have received, understood, and shall comply with the foregoing recommendations within the prescribed revision period.

Student Research Name and Signature:	
Student Research Name and Signature:	
Student Research Name and Signature:	

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

SUMMARY OF RECOMMENDATIONS

☐ Title Proposal ☐ Research Colloquium ☑ Final Oral Defense

Group Code: D11	Date and Time of Defense:
Working Title: INTEGRATED VEHICLE MAINTENANCE AND INVENTORY MANAGEMENT SYSTEM WITH DECISION SUPPORT FOR JA NOBLE ENTERPRISE INC.	
Student Researchers: RONDILLA, PRECIOUS C. TOLENTINO, MARIELY ANN L. VILLAFRANCA, CARLA MAE R.	

Instructions: *The Adviser shall record the consolidated comments, suggestions, and recommendations discussed during the oral defense. Comments should specify the relevant page, chapter, section, table, figure, or component of the manuscript to facilitate the revision process*

Page / Chapter / Section	Panel's Comments, Suggestions, and Recommendations
	- Improve the manuscript the doc w/ comment will be uploaded to the gdrive & the checklist for correction as well

MIA V. VILLAMOR

Panel of Examiners Name and Signature:	
Revision Submission Deadline:	
Panel Chair Name and Signature:	

Conforme: We acknowledge that we have received, understood, and shall comply with the foregoing recommendations within the prescribed revision period.

Student Research Name and Signature:	
Student Research Name and Signature:	
Student Research Name and Signature:	

APPENDICES C

(CURRICULUM VITAE OF SYSTEM TESTERS)

Kevin Adlaon

Analyst Programmer

Summary

Analyst Programmer with 10+ years of experience in analyzing and developing software solutions. Adept in working with other highly competent colleagues.

Experience

2022-11 - 2024-07

Data Scientist

Kasatria Technologies Sdn Bhd, Kuala Lumpur, Malaysia

Responsibilities:

- Built pipelines using Google Cloud Platform and Amazon Web Services
- Analyzed data and developed software solutions

2017-10 - 2022-11

Analyst Programmer

iTAc MSC Outsourcing Sdn Bhd, Kuala Lumpur, Malaysia

Responsibilities:

- Developed systems as innovative visualization tools for different data representation and analysis
- Created Google Ads using Google's platform

2016-07 - 2017-10

Systems Developer and Administrator

Management Information Systems Office, Provincial Government of Laguna, Philippines

2012-04 - 2016-07

Computer Programmer I

Management Information Systems Office, Provincial Government of Laguna, Philippines

2011-04 - 2011-05

On-the-job Training

Management Information Systems Office, Provincial Government of Laguna, Philippines

Personal Info

Address
3773 P. Zamora Street, Brgy. Santo Angel Sur, Santa Cruz, Laguna

Phone
+63 965 427 1050

E-mail
adlaon.kevin@gmail.com

Place of birth
Calamba City, Laguna, Philippines

Date of birth
1992-07-30

LinkedIn
https://linkedin.com/in/kevin-adlaon

Facebook
https://facebook.com/adlaon.kevin

Marital status
Widowed

Citizenship
Filipino

Skills

PHP
MySQL
CSS
jQuery
Ajax
JavaScript
Java
TypeScript
Python
Git
Google Analytics
Google Cloud Platform
Amazon Web Services
Salesforce Marketing Cloud

Education

2008 - 2012
ACTS Computer College
Sta. Cruz, Laguna, BS in Computer Science

2004 - 2008
Masico National High School
Pila, Laguna, Salutatorian

2004 - 1998
Mangumit Elementary School
Calamba City, Laguna

Certificates

- Google Cloud Certified Professional Data Engineer
- Salesforce Marketing Cloud Developer
- Salesforce Marketing Cloud Email Specialist

Languages

Filipino

English

References

Adoree Ramos
Professor
ACTS Computer College
Sta. Cruz, Laguna
(049) 501 1484

Shing Nien Fong
Director
Research & Development
Kasatria Technologies Sdn Bhd
+60 16 261 6060

MHARJUN V. TOBIAS

Graphic Artist / Technical Support

EDUCATION

2008 - 2013
BS INFORMATION TECHNOLOGY
Laguna State Polytechnic University

2004 - 2008
SECONDARY EDUCATION
Pedro Guevara Memorial National High School

SKILLS

- Networking Fundamentals
- Hardware Diagnostics
- Graphic Arts
- Microsoft Office Savvy
- GIS Mapping
- Technical Support

CONTACT

09171302395
tobiasmharjun666@gmail.com
Patimbao, Santa Cruz, Laguna
www.reallygreatsite.com

PROFILE INFO

Highly skilled professional with extensive experience in graphic design, technical support, and administrative operations. Proven ability to handle complex networking tasks and digital asset creation with precision and efficiency. Dedicated to delivering high-quality visual solutions and reliable technical troubleshooting.

EXPERIENCE

2025 - Present
CASUAL CLERK
MIS Office - MGO Santa Cruz, Laguna
Handled management information systems and administrative support for the local government unit.

2023 - 2025
CASUAL CLERK
Assessor's Office - MGO Santa Cruz, Laguna
Assisted in data management and office operations for tax assessment and land records.

2020 - 2022
GRAPHIC ARTIST / TECH SUPPORT
Mayor's Office - MGO Santa Cruz, Laguna
Provided digital creative solutions and technical maintenance for municipal operations.

2019 - 2019
ADMINISTRATIVE AIDE
PIO - MGO Santa Cruz, Laguna
Supported public information dissemination and media relations.

ACHIEVEMENTS

2022
CIVIL SERVICE PROFESSIONAL
Passed with a rating of 80.56.

2019
CIVIL SERVICE SUB-PROFESSIONAL
Passed with a rating of 82.26.

JONAS T. SANTOS

Barangay. Jasaan, Santa Cruz, Laguna, Philippines, 4009
Jonassantos122113@gmail.com | 09675977572

- IT professional with three years of experience in technical support, system administration, and IT operations in the Municipal Government of Santa Cruz, Laguna. Skilled in troubleshooting hardware, software, printers, and network issues, providing IT support across multiple municipal offices, and maintaining systems to ensure efficient operations. Known for strong problem-solving abilities, adaptability, and delivering reliable technical solutions that improve productivity and service delivery.

Work Experience

- **MUNICIPAL GOVERNMENT OF SANTA CRUZ**

Management Information Systems office | MIS IT Staff **2025 – Present**

- Provide technical support for hardware, software, printers, and network issues across municipal offices.
- Install, configure, and maintain computer systems, peripherals, and user accounts.
- Perform system administration, preventive maintenance, and software updates.
- Troubleshoot network connectivity issues and support IT projects and system implementations.
- Maintain IT asset inventory, service records, and technical documentation.

Municipal Administrator's Office / Municipal ICT Office | Administrative Aide **2024 – 2025**

- Performed administrative and clerical functions to support daily office operations.
- Prepared and maintained records, reports, and official correspondence.
- Assisted with document processing, office coordination, and administrative support.

Municipal Government of Santa Cruz | DOLE - Government Intern **2023 – 2024**

- Municipal Administrator's Office / Municipal ICT Office

- **PROVINCIAL GOVERNMENT OF LAGUNA | ON-THE-JOB TRAINING** **2022**
  - Laguna Parole and Probation Office - R4A

Key responsibilities

- Provide technical and systems support for multiple LGU information systems, including the ***General Services Office Management System, Motorized Tricycle Operators Permit System***, and ***EBOSS Queueing System.***
- Troubleshoot and resolve system issues, perform database record updates and corrections, manage user accounts, and generate operational reports.
- Perform source code modifications, including layout adjustments (margins and page breaks), to improve system-generated reports and document formatting.
- Administer and maintain EBOSS Queueing System kiosks, ensuring reliable operation and resolving system-related concerns.
- Perform daily backups of LGU system databases to ensure data integrity, security, and disaster recovery readiness.
- Provide technical support for municipal offices by installing, maintaining, and troubleshooting computers, printers, network connectivity, and other IT equipment.
- Prepare and process Purchase Requests (PRs) for MIS office supplies, IT equipment, maintenance requirements, and other operational needs.
- Manage inventory and monitoring of office supplies, computer equipment, and IT tools to ensure availability and accountability.

Technical Skills & Abilities

- IT Technical Support
- Hardware Installation, Configuration & Troubleshooting
- Backup and recovery deleted files
- Software Installation & Maintenance
- Printer Installation & Troubleshooting
- Network Troubleshooting
- System Administration
- System User Account Creation and Management
- Operating System Installation & Configuration (Windows)
- Technical Documentation
- Computer System Maintenance
- Problem Solving & Troubleshooting
- Analytical Thinking
- Customer Service & End-User Support
- Communication Skills & Team Collaboration
- Time Management
- Attention to Detail
- Adaptability & Continuous Learning
- Organizational Skills
- Ability to Work Under Pressure

Education

- **TERTIARY LEVEL:** **2018 – 2022**
  Bachelor of Science in Information Technology
  Major in Service Management Program
  Laguna State Polytechnic University – Santa Cruz Campus
- **SENIOR HIGH SCHOOL:** **2016 – 2018**
  Science, Technology, Engineering and Mathematics – STEM
  AMA Computer Collage – Santa Cruz Campus

Civil Service Eligibility

- **Career Service Subprofessional Eligibility**
  Passed
  March 2026

Personal Information

- Age: 26 years old
- Birthdate: May 11, 2000
- Birthplace: Santa Cruz, Laguna
- Sex: Male
- Height: 4'11"
- Weight: 40kg
- Citizenship: Filipino
- Civil status: Single
- Religion: Roman Catholic
- Languages: English/Tagalog

Seminars & Training

• **Introduction to Networking**
Provincial Government of Laguna - Miso Training Room
April 30, 2026

• **E-LGU User's Training**
DICT- CALABARZON
April 23, 2026

• **Web Development (Laravel & MySQL) & Server Administration**
Provincial Government of Laguna - Miso Training Room
January 5 – 16, 2026

• **Data Analytics Essentials**
December 5 – 11, 2025
DICT-ITU DTC (Net. Acad. Program)

• **Gender Sensitivity and Safe Space Act of the Philippines**
LGU Santa Cruz
December 2, 2025

• **E-LGU Admin Training and System Set-Up**
DICT- CALABARZON
November 25 – 28, 2025

• **Google Workspace for Educators/LGU Employees**
DICT – CAR
October 27, 2025

• **Cybersecurity and PNPKI Awareness at LGU - Santa Cruz**
DICT - Calabarzon
August 11, 2025

• **Basic & Intermediate Geographic Information System (GIS) Training**
Uplink Integrated Solution Inc.
January 30 – February 1, 2025

• **City/Municipal-Level Training Coordination, and**
Field Visit of LGU on the Adoption of the Philippines Standard Industrial Classification
PHILLIPPINES STATISTICS AUTHORITY
July 6, 2024

• **Computer System Servicing Training**
Provincial Government of Laguna - Miso Training Room
October 16 – 20, 2023

• **Installing and Configuring Computer System & Introduction to Computer System Servicing**
TESDA Online Program
July 2022

Character Reference

Reference available upon request.

I hereby certify that the statement above are true and correct.

Jonas T. Santos

ROBIN PAOLO BONA

Bachelor of Science in Information Technology

To utilize experience and qualification in different and challenging position that professionally growth and enhance my future opportunities.

EDUCATION

2018

UNION COLLEGE OF LAGUNA

- Bachelor of Science in Information Technology

CERTIFICATE

2022

NATIONAL CERTIFICATION II

- Computer System Servicing NC II

SKILL

- Troubleshoot 100%
- Technical Support 90%
- Microsoft Office 100%
- Creative 90%
- Team Work 95%

EXPERIENCE

2019 - PRESENT

Municipal Government Santa Cruz, Laguna

Administrative Aide

- Information Technology

2018 - 2019

DEP-ED LAGUNA

Administrative Aide

- Information Technology

2018

SM Calamba

Stock Clerk

- Warehouse Stock Clerk

CONTACT

09650499411

Hushbona@gmail.com

229 Jamaica Subdivision, Santo Angel Central, Santa Cruz, Laguna

REFERENCE

BRYAN BOUVIER BONZA
Municipal Government Santa Cruz, Laguna
HRMO V
09177059405

MELVIN O. BONZA
Municipal Government Santa Cruz, Laguna
ADMINISTRATOR
09967079115

John Marvin A. Tabo

3192 Biya Street
Santo Angel Norte,
09754068759
itsotavs0099@gmail.com

Birthdate

May 29, 2001

EDUCATION

Santo Angel Norte Elementary School

June 2006 - April 2013

Academic Achiever (Grade 6)

Pedro Guevarra Memorial National High School (P.Gueverra Street, Sta Cruz Laguna. Junior High School)

June 2013- April 2017

Academic Achiever

Saint Therese Martin of Lisieux School and Business High School. - Sto Angel Sur, Santa Cruz, Laguna.(Information and Communication Technology)

June 2013- April 2017

Supreme Student Government President

With Honor

Laguna State Polytechnic University - Santa Cruz Campus (Bachelor of Science in Information Technology)

June 2019- July 2023

Dean`s Lister Second Semester AY 2022-2023

CSE PROFESSIONAL PASSER

SKILLS

Computer Literate

Knowledge in Microsoft Office

Hardware Configuration

Installation and Configuration

Computer Troubleshooting

AWARDS

Academic Achiever - Pedro Guevara Memorial National High School

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2017)

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2018)

Dean`s Lister - Laguna State Polytechnic University (2022)

LANGUAGES

JANSSEN REYES ESPINO

Committed graduate of Bachelor of Science in Information Technology in Laguna State Polytechnic University who has experience in the field of telecommunication activation and data encoding skills that I have learned in the program and internship to elevate self-growth and to contribute to future organizations. Passionate about precision and quality of work.

ACHIEVEMENTS AND PROFESSIONAL AFFILIATIONS

- Civil Service Exam Professional Passer, March 2025
- Cisco Networking Academy Operating Systems Basics - Completer, December 2024
- eLGU Admin Training and System Setup
- System Administrator in phpMyAdmin completer
- Graduated First Honorable Mention, 2015

EXPERIENCES

- **MUNICIPALITY OF SANTA CRUZ IT STAFF**
  JULY 2025-PRESENT
- **NESTLE BRAND AMBASSADOR**
  JULY 2022 - OCTOBER 2022
- **NATIONAL ELECTION SUPPORT STAFF**
  MAY 2022
- **BARANGAY ELECTION THIRD MEMBER**
  OCTOBER 2023
- **PLDT DATA ENCODER AND SERVICE ACTIVATION**
  JANUARY 2025-JUNE 2025
- **SENATORIAL ELECTION SUPPORT STAFF**
  MAY 2025

INTERNSHIP

CENTEGRA TECHNOLOGY INC. PLDT HOME DISTRIBUTOR
Banca-Banca, Victoria, Laguna

- Took internship in the company for 600 hours as data encoder and service activation.
- Validated customer's application before passing their service order to main office.
- Calling out customers to confirm the status of their internet and voice data.
- Data activation using third party apps.
- Manual service activation.

SKILLS

- FOUNDATIONAL KNOWLEDGE IN MICROSOFT EXCEL
- DATA ENTRY AND ENCODING
- PROFICIENT IN COMPUTER APPLICATIONS
- ADOPTABLE TO NEW TECHNOLOGIES
- BASIC UNDERSTANDING OF COMPUTER MAINTENANCE AND REPAIR
- DETAIL ORIENTED
- COMMUNICATION SKILLS
- TIME MANAGEMENT
- ENTRY-LEVEL EXPERIENCE IN COMPUTER HARDWARE SERVICES
- COMPUTER TROUBLESHOOTING

EDUCATION

Laguna State Polytechnic University, 2021-2025
Bachelor of Science in Information Technology

Laguna Senior High School, 2019-2021
Senior High School
Science, Technology, Engineering, and Mathematics

Pedro Guevara Memorial National High School 2015-2019
Junior High School

Bagumbayan Elementary School, 2009-2015
Grade 1-6

CONTACT INFORMATION

0975-353-7625
ssen.jre@gmail. com
Sitio 1 Bagumbayan Santa Cruz, Laguna

CHARACTER REFERENCE

Asher H. Pasco
0925-301-6826
Education Program Supervisor
San Pedro City Laguna

WILMAR T. MOROTA

09202603909

Lynville 8, Bagumbayan, Sta Cruz, Laguna

EDUCATION

College

AMA Computer College
Computer Programming – Under Graduate
2007-2012

Secondary

Pedro Guevarra Memorial High School
Sta Cruz, Laguna
2003-2007

Primary

Our Lady of Maulawin
1998-2003

REFERENCE

RYAN R. DUÑGO
09166738313
Bagumbayan, Sta Cruz, Laguna

JUNE DAE L. MORGA
09686058086
Bubukal, Sta Cruz, Laguna

PROFILE

Birthdate : December 8, 1990
Place of Birth : Sta Cruz, Laguna, Philippines
Civil Status : Married
Religion : Roman Catholic

WORK EXPERIENCE

Management Information Systems Office — **2025 - Present**
Job-Order

Economic Enterprice Management Office — **2022 - 2025**
Job-Order

- Daily submission of Collection Report of Market Collectors
- Daily Printout of Market Collector Reciepts

Freelance Encoder — **2017-2020**

OTHER ATTAINMENTS

CAREER SERVICE EXAMINATION PROFESSIONAL PASSER
MAY 2024

JOHN MARIE P. NERPIO

APPLICANT

Brgy, San Luis, Luisiana, Laguna 4032
jmnerpio333@gmail.com
(+63)909-313-9350

SUMMARY

To begin a professional career where I can apply my skills, learn continuously, and contribute to the success of the organization. I am eager to grow, take on new challenges, and become a valuable part of a dynamic and forward-thinking team.

PROFESSIONAL EXPERIENCE

Administrative Aide I / Encoder | AUGUST 2025 - PRESENT | MUNICAL HEALTH OFFICE

- Encoded patient information and consultation records using iClinicSys. Updated and verified patient details, maintained accurate digital medical records, and ensured the confidentiality of sensitive information. Assisted in organizing patient data to support efficient clinic operations and quality healthcare services.

Virtual Assistant / Scheduling and Chat Support| DECEMBER 2025 - PRESENT | TRAPQUEEN SALON

- Managed client communication and customer support through Yelp and Gmail, responding to inquiries, assisting with bookings, providing service information, and maintaining professional and timely communication to enhance the overall customer experience.
- Handled appointment scheduling and calendar management using Fresha, including booking, confirming, rescheduling, and canceling appointments while coordinating with salon staff to ensure smooth daily operations and maximize scheduling efficiency.
- Monitored employee attendance through Homebase by checking time-in and time-out records, tracked sales data from Fresha, and organized the information in Microsoft Excel to help the business owner monitor stylist performance, expenses, sales, and overall profit.

On-The-Job Training | FEBRUARY 17 2025 - JUNE 13 2025 | LUISIANA, LAGUNA

- Assisted in data encoding and records management using Microsoft Excel for various municipal departments
- Provided basic computer maintenance and troubleshooting, including cleaning workstations and assisting staff with technical issues

EDUCATION

Bachelor of Science in Information Technology Major in Web and Mobile Application development | 2021-2025
Laguna States Polythecnic University Main Campus, Santa Cruz Laguna

- Relevant Coursework: Web Development, Mobile Application Development, Programming Fundamentals,

PROJECT

ByaHero: A Real-time Jeepney Tracking System for Lucban-StaCruz Commuters with GPS Tracking and Route Mapping
A Capstone Project, Third Year Second Semester to Fourth Year First Semester

- Developed a GPS-based mobile and web app to track jeepney routes in real time.
- Assisted with system documentation and analysis, including user requirements and data flow diagrams.

CERTIFICATE AND SEMINARS

DICT SPARK Technical Training for General Virtual Assistant 2.0: A Paradigm Shift in AI-Powered Interaction - June 8 - 19 2026

- Completed comprehensive training in virtual assistance, including client communication, branding and niching, online payment systems, and social media content creation.
- Gained hands-on knowledge of time management, productivity tools, AI-powered applications, and common virtual assistant services to improve business efficiency.
- Demonstrated proficiency in professional work ethics, dispute management, and delivering reliable administrative support in a virtual work environment.

CISCO Networking Academy: Operating Systems Basics - Dec 07, 2024

- Gained essential knowledge of operating systems, focusing on Windows, Linux, and mobile platforms, including system processes, file management, security, and basic troubleshooting.

SKILL HIGHLIGHTS: TECHNICAL SKILLS

VIRTUAL ASSISTANCE | ADMINISTRATIVE SUPPORT

- Experienced in providing reliable virtual administrative support, including calendar management, appointment scheduling, email management, customer communication, and data organization. Skilled at handling multiple tasks efficiently while maintaining professionalism, accuracy, and confidentiality.

CUSTOMER SUPPORT | CHAT & EMAIL MANAGEMENT

- Proficient in delivering responsive customer support through chat and email platforms such as Yelp and Gmail. Skilled in addressing customer inquiries, managing bookings, resolving concerns professionally, and ensuring a positive client experience.

SCHEDULING & CALENDAR MANAGEMENT | FRESHA & HOMEBASE

- Experienced in managing appointments using Fresha and monitoring employee attendance through Homebase. Skilled in coordinating schedules, maintaining organized calendars, and ensuring smooth day-to-day business operations.

DATA MANAGEMENT & REPORTING | MICROSOFT EXCEL

- Proficient in organizing, tracking, and analyzing business data using Microsoft Excel. Experienced in maintaining sales records, monitoring employee attendance, tracking expenses, and creating accurate reports to support business decisions.

GRAPHIC DESIGN | ADOBE PHOTOSHOP & CANVA

- Experienced in creating engaging visual content for digital and print media, including social media graphics, marketing materials, apparel designs, branding assets, and promotional content. Skilled in photo editing, typography, layout design, and visual storytelling.

UI/UX DESIGN | FIGMA

- Proficient in designing intuitive user interfaces and interactive prototypes using Figma. Skilled in wireframing, responsive design, and applying user-centered design principles to create functional and visually appealing web and mobile interfaces.

COMPUTER LITERACY

- Proficient in Microsoft Office (Word, Excel, PowerPoint), Google Workspace, email communication, internet research, file management, and cloud-based collaboration tools. Able to quickly learn and adapt to new software and digital platforms.

COMPUTER TROUBLESHOOTING

- Skilled in diagnosing and resolving basic hardware and software issues, installing drivers, configuring computer systems, and performing routine maintenance. Knowledgeable in basic networking, operating system installation, and system optimization.

ARTIFICIAL INTELLIGENCE TOOLS

- Experienced in using AI-powered tools to improve productivity, streamline administrative tasks, assist with content creation, generate ideas, and enhance workflow efficiency while maintaining quality and accuracy.

JOHN MARIE PEDRON NERPIO

APPLICANT

I hereby certify that the above information is true and correct to the best of my knowledge.

CONTACT

Phone
09958533476

Email
jerico.dia@lspu.edu.ph

Address
199 purok VI Bucal,
Calamba City, Laguna

CONTACT

- Web Systems
- Database Management
- Networking Management
- Computer Troubleshooting

JERICO D. DIA

PERSONAL INFORMATION:

Age:	**28**
Date of Birth:	**March 16, 1997**
Place of Birth:	**Bucal Calamba, Laguna**
Citizenship:	**Filipino**
Religion:	**Born Again Christian**
Status:	**Single**
Height:	**1.70 Meter**
Weight:	**88 Kilograms**
Language/Dialects Spoken:	**English / Tagalog**
Father's Name:	**Rey Noel P. Dia | Farmer/Fisherman**
Mother's Name:	**Maria Teresa D. Dia | Retired**
Siblings:	**Lexter D. Dia**

Work Experience

Information System Analyst 1 – Information and Communication Technology Services
Laguna State Polytechnic University - Los Baños Campus
March 2024 - PRESENT

Administrative Aide VI - RECORDS OFFICE
Laguna State Polytechnic University - Los Baños Campus
June 2023-2024

Network Administrator – Information and Communication Technology Services
Laguna State Polytechnic University - Los Baños Campus
August 2017 - 2023

EDUCATION

Graduate Studies (2023-2025)	**Master in Information Technology** **Laguna State Polytechnic University - Los Baños Campus** **Brgy. Mayondon-Malinta Los Baños, Laguna**
Tertiary (2013-2017)	**Bachelor of Science in Information Technology** **Laguna State Polytechnic University - Los Baños Campus** **Brgy. Mayondon-Malinta Los Baños, Laguna**
Secondary (2009-2013)	**Calamba Institute** **Brgy. Halang Calamba, Laguna**
Primary (2003-2009)	**Bucal Elementary School** **Brgy. Bucal Calamba, Laguna**

Justine Jhay Lopez

ASPIRING QA ENGINEER |
WEB DEVELOPER |
IT SUPPORT TECHNICIAN

Contact

+63 976 238 5693

jhayjustinlopez@gmail.com

Santa Cruz, Laguna, 4009, Phillipines

Skills

- Leadership & Team Collaboration
- Computer Literacy (MS Office, Email, Digital Filing)
- Visual Design & Editing
- Photo & Video Editing
- Basic UI/UX & Web Content Management

Language

- English
- Filipino

Education

(2022 - 2026)
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bachelor of Science in Information Technology
1st - 4th Year College

(2020 - 2022)
LAGUNA SENIOR HIGH SCHOOL
Science, Technology, Engineering, and Mathematics strand
Grade 11 - Grade 12

(2016 - 2019)
PEDRO GUEVARA MEMORIAL NATIONAL HIGH SCHOOL
Grade 7 - Grade 10

Experience

(2025 - Present)
MANAGEMENT INFORMATION SYSTEMS OFFICE
- MIS Programmer

(2024 - 2026)
MUNICIPAL SOCIAL WORKER AND DEVELOPMENT OFFICE
- Data Entry Clerk

Projects

- **IKNOWLGRC - KNOWLEDGE MANAGEMENT SYSTEM**
  -2026 (Full Stack Developer)
- **EXELENCIA ACADEMY PORTAL**
  -2024 (Full Stack Developer)
- **SHE WEAR (E-COMMERCE WEB)**
  -2024 (Programmer)
- **BARANAGAY RECORDING SYSTEM**
  -2024 (Full Stack Developer)
- **JAMAICA RECORDING SYSTEM**
  -2023 (Full Stack Developer)

APPENDICES D

(CURRICULUM VITAE OF RESEARCHERS)

CARLA MAE R. VILLAFRANCA

09654656608 carla.villafranca6@gmail.com maeeeeee_v

ABOUT ME

Motivated and detail-oriented student with a strong passion for technology and problem-solving. Developing skills in software development, data analysis, and project management, with a growing ability to deliver quality work in academic and project-based settings. Committed to continuous learning and personal growth, with a collaborative mindset and strong communication skills.

EDUCATION

2023-present
Laguna State Polytechnic University

Bachelor of Science in Information Technology
IT student with a foundation in programming, data structures, and databases. Actively exploring web and mobile development, with hands-on project experience and a 3.7 GPA. Passionate about continuous learning and tech innovation.

2021-2023
Acts Computer College

Senior High School
Graduated with honors. Completed senior high school coursework in science, technology, engineering, and mathematics, including subjects like physics, calculus and research

2017-2021
Pagsanjan Integrated National High School

High School
Graduated with honors. Completed coursework in core subjects such as English, Mathematics, Science, Social Studies, and Communication.

2010-2017
Francisco Benitez Memorial School

Elementary School
Graduated with honors. Completed foundational subjects including English, Math, Science, Filipino, and values education.

EXPERIENCE

2023-Present
Laguna State Polytechnic University

IT Student – Technical Experience
As an IT student, I have developed software projects using Java, HTML, CSS, Android Studio, and SQLite. I have created responsive and user-friendly web interfaces and mobile application designs using Figma. I am also skilled in using Canva to create presentations, posters, graphics, and other visual materials. Additionally, I have experience in basic UI/UX design, system testing, debugging, and resolving common programming and interface issues, as well as database integration.

2023 (3months)
Office of the Mayor in Pagsanjan

Office Immersion – Front Desk Assistant
During my office immersion at the Office of the Mayor, I assisted visitors by providing basic information, directions, and support regarding office services and procedures. I also helped organize documents, accommodated inquiries, and performed basic administrative tasks. This experience helped me develop my communication, customer service, teamwork, and professional workplace skills.

SKILLS

- Java
- Android Studio
- Figma
- Basic UI/UX Design
- Basic Computer Troubleshooting
- HTML & CSS
- Database
- Canva
- System Testing & Debugging
- Microsoft Office

CONTACTS

Gmail: carla.villafranca6@gmail.com **Phone:** +639654656608 **Facebook:** Carla Mae R. Villafranca **Instagram:** maeeeeee_v

MARIELY ANN TOLENTINO

IT STUDENT
09184211648 | tolentinomariely09@gmail.com | Luisiana, Laguna

EDUCATION

Laguna State Polytechnic University (LSPU) — Santa Cruz, Laguna
Bachelor of Science in Information Technology — 2023 - present

WORK EXPERIENCE

SPES – Special Program for Employment of Students — June 2023
Encoder
- Served as an Encoder under the SPES program, assisting with data entry, document filing, and clerical tasks while maintaining accuracy and confidentiality in handling official records.

Sudent Administrative Assistant | On-the-Job Training — April 2023 - May 2023
Liceo de Luisiana Registrar's Office
- Assisted the Registrar in managing student records, processing enrollment documents, and handling transcript requests while ensuring accuracy and confidentiality. Provided administrative support through data entry, filing, and customer service to students and staff regarding scheduling and academic records.

PROJECTS

Liceo de Luisiana - Library Management System | 2025
Developer
- Developed a comprehensive library management system with user authentication, catalog search, and automated tracking using PHP.

San Benito - Barangay Health Information System | 2025
Developer
- Developed a digital health records system for patient management, medical history tracking, and community health statistics reporting using PHP.

MStyle - Ecommerce Mobile App and Website| 2026
Developer
- Built a full-stack ecommerce platform with product catalog, shopping cart, payment integration, and responsive design for web and mobile using Python (Flask) and Dart (Flutter).

Vehicle Maintenance and Inventory Management System| 2026
Developer
- Built a full-stack vehicle maintenance and inventory management system with QR/Barcode scanning, Decision Support System for stock alerts and maintenance scheduling, and an AI Assistant using RAG for data queries and report generation, using Python (Flask) and Dart (Flutter).

SKILLS

- Microsoft Office (Word, Excel, Powerpoint)
- Web and App Development (Python, PHP, Flutter, Html, CSS, Javacript)
- Data Management (Firebase, MySQL)
- Basic UI / UX Design (Figma, Canva)
- Software Testing & Debugging
- Basic Computer Troubleshooting
- Time Management
- Communication Skills
- Problem-Solving
- Data Entry

CERTIFICATIONS

- Python Essentials 1 | Cisco Networking Academy
- CS50's Introduction to Databases with SQL | CS50
- CS50's Introduction to Programming with Python | CS50
- Data Analytics Essentials | Cisco Networking Academy
- TOPCIT (Test of Practical Competency in ICT) Certificate

PRECIOUS RONDILLA

Luisiana, Laguna
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WORK EXPERIENCE & EXPOSURE

MUNICIPAL GOVERNMENT OF LUISIANA — *Luisiana, Laguna*
SPES Data Encoder (Special Program for the Employment of Students) — *May 2023 –2024*

- Conducted daily house-to-house data gathering to support municipal demographic profile data collection.
- Encoded and verified community records in Microsoft Excel and the government information system.
- Collaborated with municipal staff to process and update community information within project deadlines.

LICEO DE LUISIANA – REGISTRAR'S OFFICE — *Luisiana, Laguna*
Student Administrative Assistant — *March – April 2023*

- Assisted with daily administrative operations, including data encoding, document processing, and records management.
- Designed digital materials and office documents to support communication and administrative activities.
- Performed basic computer troubleshooting and provided technical assistance to office staff.

TECHNICAL SKILLS

- Microsoft Office (Word, Excel, PowerPoint)
- Google Workspace (Docs, Sheets, Drive)
- Graphic Design (Canva, Figma)
- Time Management & Problem Solving
- Data Encoding
- Document Management
- Basic Computer Troubleshooting
- Database Management

EDUCATION

LAGUNA STATE POLYTECHNIC UNIVERSITY — *Santa Cruz, Laguna*
Bachelor of Science in Information Technology — *2023 - present*

LEADERSHIP EXPERIENCE

- **COLLEGE OF COMPUTER STUDIES COMELEC**
  Auditor (2025 - 2026)
- **STUDENT ORGANIZATION OF THE COLLEGE OF COMPUTER STUDIES**
  Event Coordinator (2024 - 2025)

PROJECT

LICEO DE LUISISANA LIBRARY MANAGEMENT SYSTEM (LMS)

- Developed a Library Management System Website with QR code scanning for book borrowing, returns, and attendance.
- Managed book records and library data for accurate organization and easy access to resources.
- Implemented Dewey Decimal Classification and book cataloging features to help organize and manage library books.